

Introduction to CT Regulation

GOOD CLINICAL PRACTICE AND CLINICAL
TRIALS

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History

The concept of the 'good physician' dates back to the ancient world and it is evidenced by the Hippocratic Oath (460 BC).

In the United States, the first landmark in the regulation of drugs was the Food and Drugs Act of 1906. This was a result of harmful and lethal drugs that could be bought across the counter just like any other consumer product.

Some examples are 'Grandma's Secret' and 'Kopp's Baby's Friend' which contained large doses of morphine, as well as 'Dr King's Consumption Cure' and 'Dr Bull's Cough Syrup' which contained morphine and chloroform

460BC	Oath of Hippocrates
1930's	U.S. Food, Drugs and Cosmetic Act
1947	Nuremberg Code
Dec. 10th 1948	Declaration of Human Rights
1962	Kefauver-Harris Amendment
1964, revised 2000	Declaration of Helsinki
1979	The Belmont Report
1982	International Guidelines for Biomedical Research Involving Human Subjects
1996	ICH-GCP guidelines issued
1997	ICH-GCP guidelines becomes law in some countries

In 1938, the Federal Food, Drug and Cosmetic Act was enacted by the Food and Drug Administration (FDA) and for the first time, manufacturers were required to test drugs for safety and present the evidence of safety testing to the FDA prior to marketing

In 1947, the Nuremberg Code was created as a result of the unethical and horrific experiments carried out during World War II at Nazi war camps by German physicians, who were subsequently tried and charged at the Nuremberg Military Tribunal. This code states the need for a scientific basis in research on human subjects and voluntary consent and protection of participants.

The Universal Declaration of Human Rights (December 10th 1948) was also adopted and proclaimed by the United Nations after the atrocities of World War II and it further reiterated the human factor involved in medical experiments.

In 1964, the Declaration of Helsinki was developed by the World Medical Association, forming the basis for the ethical principles that underlie the ICH-GCP guidelines we have today. The focus of this declaration is the protection of the rights of human subjects and this is clear in its introduction:

“The World Medical Association has developed the Declaration of Helsinki as a statement of ethical principles to provide guidance to physicians and other participants in medical research involving human subjects. It is the duty of the physician to promote and safeguard the health of the people. The physician’s knowledge and conscience are dedicated to the fulfilment of this duty”

In 1962 the world was once again shocked by the severe foetal limb deformities linked to the use of maternal thalidomide. In fact this drug reaction was only discovered after 10,000 infants were born in over 20 countries worldwide. In response to this, the Kefauver-Harris Amendments were passed which required the FDA to evaluate all **new drugs for safety and efficacy**

Another important milestone in the formation of the ICH-GCP guidelines was The Belmont Report which was issued in April 1979 by the National Commission for Protection of Human Subjects of Biomedical and Behavioural Research. The principles of this report are as follows:

Respect for Persons: This principle acknowledges the dignity and freedom of every person. It requires obtaining informed consent from research subjects (or their legally authorised representatives)

Beneficence: This principle requires that researchers maximise benefits and minimise harms associated with research. Research-related risks must be reasonable in light of the expected benefits.

Justice: This principle requires equitable selection and recruitment and fair treatment of research subjects.

In 1982, the World Health Organization (WHO) and the Council for International Organizations of Medical Sciences (CIOMS) issued a document entitled 'International Guidelines for Biomedical Research Involving Human Subjects'. This document was released to help developing countries apply the principles of the Declaration of Helsinki and the Nuremberg Code

ICH

Organized to provide opportunity for global harmonization of drug development regulated by requirements

Concerned with harmonization of technical requirements for registration of pharmaceutical products among 3 regions:

- United states, the European Union, and Japan

One of the goals of harmonization is to **identify and then reduce differences in technical requirements** for drug development among regulatory agencies.

FDA heavily involved and committed to harmonization initiatives

Many medical device sponsors typically follow ICH when conducting activities for clinical trials, but modify to device standards.

ISO 14155: Clinical Investigation of medical devices for Human subjects– Good Clinical Practices

ICH TOPICS Four major categories.

In an effort to overcome international GCP inconsistencies throughout the countries, the International Conference for Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) issued the ICH Guidelines: Topic E6 Guideline for GCP.

This guideline was approved on 17 July 1996 and implemented for clinical trials from 17 January 1997. The participants of these guidelines were representatives of authorities and pharmaceutical companies from the EU, Japan and the United States as well as those of Australia, Canada, the Nordic countries and WHO

ICH Topic Codes assigned according to these categories

Q = “Quality” Topics, i.e., relating to chemical and Quality assurance

S = “safety” Topics, i.e., relating to in vitro and in vivo pre- clinical studies

M = “Multidisciplinary” topics, i.e., cross-cutting topics which do not fit uniquely into one of other categories

E = “Efficacy” Topics, i.e., relating to clinical studies in human subject

Q1A - Q1F Stability	▼
Q2 Analytical Validation	▼
Q3A - Q3E Impurities	▼
Q4A - Q4B Pharmacopoeias	▼
Q5A - Q5E Quality of Biotechnological Products	▼
Q6A- Q6B Specifications	▼
Q7 Good Manufacturing Practice	▼
Q8 Pharmaceutical Development	▼
Q9 Quality Risk Management	▼
Q10 Pharmaceutical Quality System	▼
Q11 Development and Manufacture of Drug Substances	▼
Q12 Lifecycle Management	▼
Q13 Continuous Manufacturing of Drug Substances and Drug Products	▼
Q14 Analytical Procedure Development	▼

ICH “Q” Topics

S1A - S1C Carcinogenicity Studies



S2 Genotoxicity Studies



S3A - S3B Toxicokinetics and Pharmacokinetics



S4 Toxicity Testing



S5 Reproductive Toxicology



S6 Biotechnological Products



S7A - S7B Pharmacology Studies



S8 Immunotoxicology Studies



S9 Nonclinical Evaluation for Anticancer Pharmaceuticals



S10 Photosafety Evaluation



S11 Nonclinical Paediatric Safety



S12 Non-clinical Biodistribution Considerations for Gene Therapy Products



ICH “S” Topics

M1 MedDRA Terminology



M2 Electronic Standards



M3 Nonclinical Safety Studies



M4 Common Technical Document



M5 Data Elements and Standards for Drug Dictionaries



M6 Gene Therapy



M7 Mutagenic Impurities



M8 Electronic Common Technical Document (eCTD)



M9 Biopharmaceutics Classification System-based Biowaivers



M10 Bioanalytical Method Validation and Study Sample Analysis



M11 Clinical electronic Structured Harmonised Protocol (CeSHarP)



M12 Drug Interaction Studies



M13 Bioequivalence for Immediate-Release Solid Oral Dosage Forms



M14 Use of real-world data for safety assessment of medicines



M15 General Principles for Model-Informed Drug Development



ICH “M” Topics

E1 Clinical Safety for Drugs used in Long-Term Treatment	▼
E2A - E2F Pharmacovigilance	▼
E3 Clinical Study Reports	▼
E4 Dose-Response Studies	▼
E5 Ethnic Factors	▼
E6 Good Clinical Practice	▼
E7 Clinical Trials in Geriatric Population	▼
E8 General Considerations for Clinical Trials	▼
E9 Statistical Principles for Clinical Trials	▼
E10 Choice of Control Group in Clinical Trials	▼
E11 - E11A Clinical Trials in Pediatric Population	▼
E12 Clinical Evaluation by Therapeutic Category	▼
E14 Clinical Evaluation of QT	▼
E15 Definitions in Pharmacogenetics / Pharmacogenomics	▼
E16 Qualification of Genomic Biomarkers	▼
E17 Multi-Regional Clinical Trials	▼
E18 Genomic Sampling	▼
E19 Safety Data Collection	▼
E20 Adaptive Clinical Trials	▼
E21 Inclusion of Pregnant and Breastfeeding Individuals in Clinical	▼

ICH “E” TOPICS

GOOD CLINICAL PRACTICE (GCP)

The ICH-GCP is a harmonised standard that protects the rights, safety and welfare of human subjects, minimises human exposure to investigational products, improves quality of data, speeds up marketing of new drugs and decreases the cost to sponsors and to the public.

Good Clinical Practice covers the way a clinical trial is designed, conducted, performed, monitored, audited, recorded, analyzed, and reported.

Rules of conducting clinical research that is scientifically sound and upholds ethical principles

Increased Ethical Awareness

Improved Trial Methods

Clinical Trial Concept Better Understood

Public/Political Concern over Safety Aspects

Frauds and Accidents during Trials

Growing Research and Development Costs

Increasing Competition

Mutual Recognition of Data

New Market Structure

Reasons of GCPs

Compliance with this standard provides public assurance that the rights, safety and well-being of trial subjects are protected and consistent with the principles of the Declaration of Helsinki, and that the clinical trial data is credible Protect the rights, safety and welfare of research subjects

Core principles of ICH-GCP

1. Clinical trials should be conducted in accordance with ethical principles that have their origin in the Declaration of Helsinki, and that are consistent with GCP and the applicable regulatory requirement(s).
2. Before a trial is initiated, foreseeable risks and inconveniences should be weighed against anticipated benefit for the individual trial subject and society. A trial should be initiated and continued only if the anticipated benefits justify the risks.
3. The rights, safety and well-being of the trial subjects are the most important considerations and should prevail over interest of science and society.
4. The available non-clinical and clinical information on an investigational product should be adequate to support the proposed clinical trial.
5. Clinical trials should be scientifically sound, and described in clear, detailed protocol.
6. A trial should be conducted in compliance with the protocol that has received prior institutional review board (IRB)/ independent ethics committee (IEC) approval/favourable opinion.

7. The medical care given to, and medical decisions made on behalf of subjects should always be the responsibility of a qualified physician or, when appropriate, of a qualified dentist.
8. Each individual involved in conducting a trial should be qualified by education, training, and experience to perform his or her respective task(s).
9. Freely given informed consent should be obtained from every subject prior to clinical trial participation.
10. All clinical trial information should be recorded, handled, and stored in a way that allows its accurate reporting, interpretation and verification.
11. The confidentiality of records that could identify subjects should be protected, respecting the privacy and confidentiality rules in accordance with the applicable regulatory requirement(s).
12. Investigational products should be manufactured, handled and stored in accordance with applicable Good Manufacturing Practice (GMP). They should be used in accordance with the approved protocol.
13. Systems with procedures that assure the quality of every aspect of the trial should be implemented.

In simplified terms:

All clinical trials should be conducted in accordance with ethical principles, sound scientific evidence and clear detailed protocols.

The benefits of conducting trials should outweigh the risks.

The rights, safety and well-being of trial participants are of paramount importance, and these should be preserved by obtaining informed consent and maintaining confidentiality.

The care must be given by appropriately qualified personnel with adequate experience.

Records should be easily accessible and retrievable for accurate reporting, verification and interpretation.

Investigational products should be manufactured according to Good Manufacturing Practice

ICH Guideline Draft

Published in the Federal register on
8/17/95 (60 FR 42948)

ICH Guideline: effort to define GCP, to
create and provide unified standard for
designing, conducting, recording and
reporting trials that involve the
participation of human subjects

Published in the Federal Register on
5/9/97 as an adopted guideline

Regulatory Authorities	Review submitted clinical data and conduct inspections
The sponsor	Company or institution/organization which takes responsibility for initiation, management and financing of clinical trial
The project monitor	Usually appointed by sponsor
The investigator	Responsible for conduct of clinical trial at the trial site. Team leader.
The pharmacist at trial location	Responsible for maintenance, storage and dispensing of investigational products eg. Drugs in clinical trials
Patients	Human subjects
Ethical review board or Committee for protection of subjects	Appointed by Institution or if not available then the Authoritative Health Body in that Country will be responsible
Committee to monitor large trials	Overseas Sponsors eg. Drug Companies

What constitutes GCPs?

THE CLINICAL RESEARCH TEAM

Objectives

- Identify the members of the clinical research team;
- Describe their primary roles and responsibilities;
- Recognize the types of sponsor/CRO Investigator site visits;
- Apply the roles and responsibilities of the Clinical Research Associate and the Clinical Research Coordinator

Clinical Research Associate (CRA) and Clinical Research Coordinator (CRC) play integral role in product development

CRA

- Sponsor's first line representative with investigative site personnel
- Monitors for sponsor that clinical trial being handled according to protocol and Good Clinical Practice

CRC

- Assists investigator to ensure that clinical trial successfully implemented and completed
- Key liaison between investigator, subject, ethics board and sponsor

Sponsor

Regulatory Agencies [Food & Drug Administration (FDA)]

Contract Research Organization (CRO)

Research Subjects/Patients/Participants

Investigational Sites/Institution/Investigator

Others: For example: Central Labs, Randomization services, safety Boards

Ethics Committee/Institutional Review Boards (IRBs)

Sponsors Responsibilities

CFR REGULATING SPONSOR RESPONSIBILITIES:

GENERAL [21 CFR 312.50 & 21 CFR 812.40];

Transfer of Obligations to a Contract Research Organization (CRO)
[21 CFR 312.52];

Selecting Investigators and Monitors [21 CFR 312.53 & 21 CFR
812.43];

Informing Investigators [21 CFR 312.55 & 21 CFR 812.45].

Sponsors OBLIGATIONS

Recruit qualified Investigators and obtain suitable documentation concerning their training and experience (therapeutic area & research);

Secure a signed statement of intention to comply with the Federal Regulations from the Investigators (Form FDA-1572 or Statement of Investigator)

Provide detailed information to investigators concerning the properties of and risks associated with the study product;

Develop a complete protocol for each trial including detailed descriptions of the study plan and specific procedures;

Select competent monitors to conduct the initiation and closeout visits and to periodically monitor the progress of the trial;

Monitor all aspects of study performance to assure compliance with the protocol and good clinical practice (GCP);

Conduct audits of recorded clinical data, preferably by direct comparison with original source documentation;

Ensure that the FDA and all participating Investigators are promptly informed of significant new adverse effects or risks with respect to the product;

Ship investigational product only to Investigators participating in the trial.

Obtain the following from each Investigator, prior to shipping any investigational product:

- Statement of Investigator
- IRB approval of the protocol and informed consent
- Protocol approval/signature page
- CTA: Clinical Trial Agreement (contract)

Sponsor responsibilities

DRUG: A signed Statement of Investigator (From FDA-1572)

Form FDA-1572 serves as an agreement between the Investigator and the sponsor outlining PI roles and responsibilities, and provides information about the site.

The original “1572” is usually sent to the sponsor by the site and copies are maintained at site.

The sponsor is not required to send the 1572 to the FDA.

- The sponsor may choose to send the 1572 to the FDA to provide required information for the IND.
- FDA does not require execution of a new 1572, but sponsors do in their SOPs.

DRUG: A signed Statement of Investigator (From FDA-15720)

For example updating the form when:

Addition of new study personnel or withdrawal of study personnel,

Change of address for: Principal Investigator, Location of research facility, Institutional Review Board, Laboratory (& other vendor addresses noted),

Change or addition of: Institutional Review Board, Sub- Investigators, protocol title change, locations where study patients are seen and records are maintained.

DEVICE: Statement of Investigator

Serves as a agreement between investigator and sponsor

From each participating investigator signed agreement includes:

- Investigator's CV
- Statement of Investigator's relevant experience, including the dates, location, extent, and type of experience, and
- If Investigator was involved in an investigation or other research that was terminated (if so, an explanation of the circumstances that led to termination).

A statement of the Investigator's commitment to:

- Conduct the investigation in accordance with the agreement, the investigational plan, other applicable FDA regulations, and conditions of approval imposed by the reviewing IRB or FDA;
- Supervise all testing of the device involving human subjects; and
- Ensure that the requirements for obtaining informed consent are met.

Informing investigators

Investigators are supplied with current product labeling before study begins;

- Investigator must be informed of new observations discovered or reported on product by:
- Revising investigational labeling to include new findings from ongoing pre-clinical research or earlier clinical trials;
- Providing Investigating with relevant reprints or published studies;
- Sending study report updates or letters to clinical Investigators.

FORM FDA-1572 Review

Two options, latest 2012 version and may use the 2006 version in case “if the newer version does not work for you” FDA website link

Form expiration date is not applicable to the investigator and sponsor. It is related to the agency’s administrative activities. Refer to 2010 FAQ guidance.

Investigator responsibilities

GENERAL QUALIFICATIONS

An Investigator must be:

- An appropriately qualified person;
- Trained and experienced in clinical research;
- Familiar with the background of the study drug and requirements of the study;
- Known to have high ethical standards and professional integrity.

General

- Obtain IRB approval of a study protocol and informed consent;
- Enroll the required number of qualified subjects and obtain informed consent from all subjects;
- Administer (or supervise the administration of) the study drug/device and maintain accurate accountability for all clinical supplies received b the site;
- Observe, measure, and record the effect of the drug/device;
- Record all other data pertinent to the study;

Evaluate, manage, and report adverse experiences;

Submit protocol changes to the IRB for approval;

Notify the IRB if any problems which pose a threat to the welfare of the subjects;

Retain and make study documentation available to sponsor representatives and FDA inspectors for data verification purposes, and

Comply with all other procedures as specified in the protocol, sponsor SOPs, and Good Clinical Practice (GCP).

Control of the Investigational drug/Device

[21 CFR 312.61 & 21 CFR 812.110
]

Investigator must administer the product only to subjects under the Investigator's personal supervision,

- Or under the supervision of a sub-investigator responsible to the investigator;

Investigator must not supply investigational product to any person unauthorized/unqualified.

Record keeping and record retention

[21 CFR 312.62 & 21 CFR
812.140]

Investigator must maintain the accurate, complete, and current records relating to investigator's participation in an investigation;

Study Correspondence:

- Investigator is required to keep all correspondence with another investigator, IRB, sponsor, monitor, or FDA regarding work on a clinical trial.

Disposition of Investigational Product:

Investigator required to maintain adequate records of receipt, use and disposition of investigational product

e.g. dates dispensed, quantity used by subject, etc.

Case Histories:

Examples, Source Documents and Case Report Forms;

Investigator required to prepare and maintain adequate and accurate records of all observations and other data pertinent to investigation on each subject participating.

Records retention:

Investigator must retain study related records for two years following date marketing application approved, or for two years after withdrawal off application,

e.g. investigator completing trial in 1980 must retain all study records until 1997, if application for approval such as NDA, not approved until 1996.

Investigator reports

[21 CFR 312.64 7 21 CFR 812.150]

Progress reports

- Investigator must furnish study progress reports to IRB and sponsor;
- Investigator progress reports are required annually by IRB, although some IRBs require more frequent reporting;
- Both sponsor and/or FDA may request more frequent reporting for special circumstances.

Safety Reports:

- Investigator must promptly report to sponsor any adverse effect that may **reasonably** be regarded as caused by or, **probably** caused by investigational product
- Timing of these reports usually spelled out in study protocol
- If investigator deems safety event to meet regulatory criteria for serious,
Investigator must report adverse event immediately to sponsor

Safety Reports:

- For device trials, unexpected effects must also be reported immediately to sponsor;
- Investigator must report safety events to the reviewing IRB as required by that IRB;

Typically, events reported immediately to sponsor are also reported promptly to IRB

Final Report:

- Investigator provides sponsor and IRB/TEC with final report shortly after completing participation in trial;

Usually includes number of subjects enrolled, completed, withdrawn, number of deaths, SAEs and protocol violations..

Templates

Assurance of IRB review

[21 CFR 312.66 & 21 CFR 812.110]

Investigator will not make changes in reassert without prior IRB approval, except where necessary to eliminate apparent immediate hazards to subjects;

Investigator must be familiar and compliant with the reviewing IRB's requirements for submission of information and reports;

Examples of information required by the IRB may include:

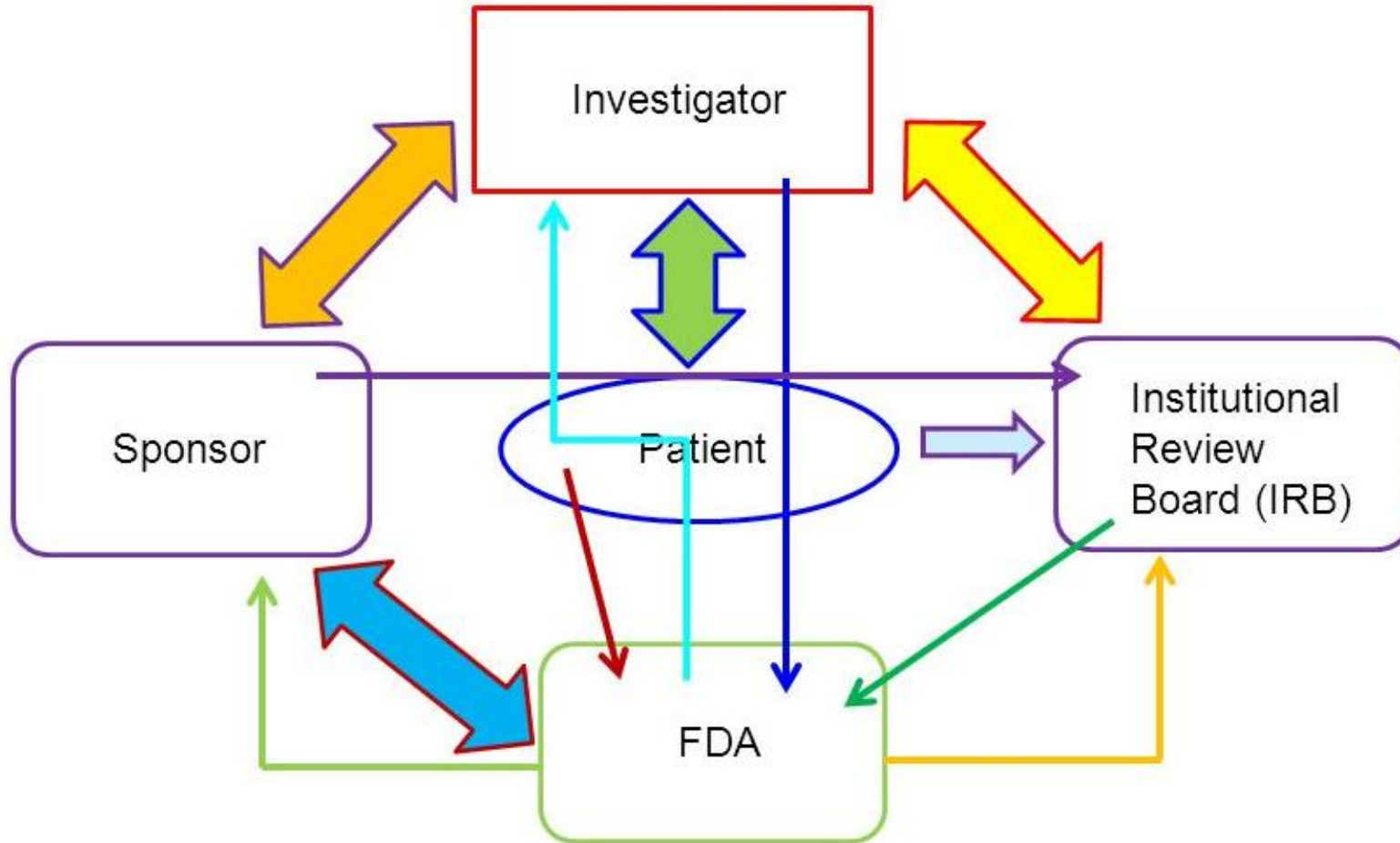
- Safety information and events, protocol deviations, and study progress reports.

INSPECTION OF RECORDS 7REPORTS

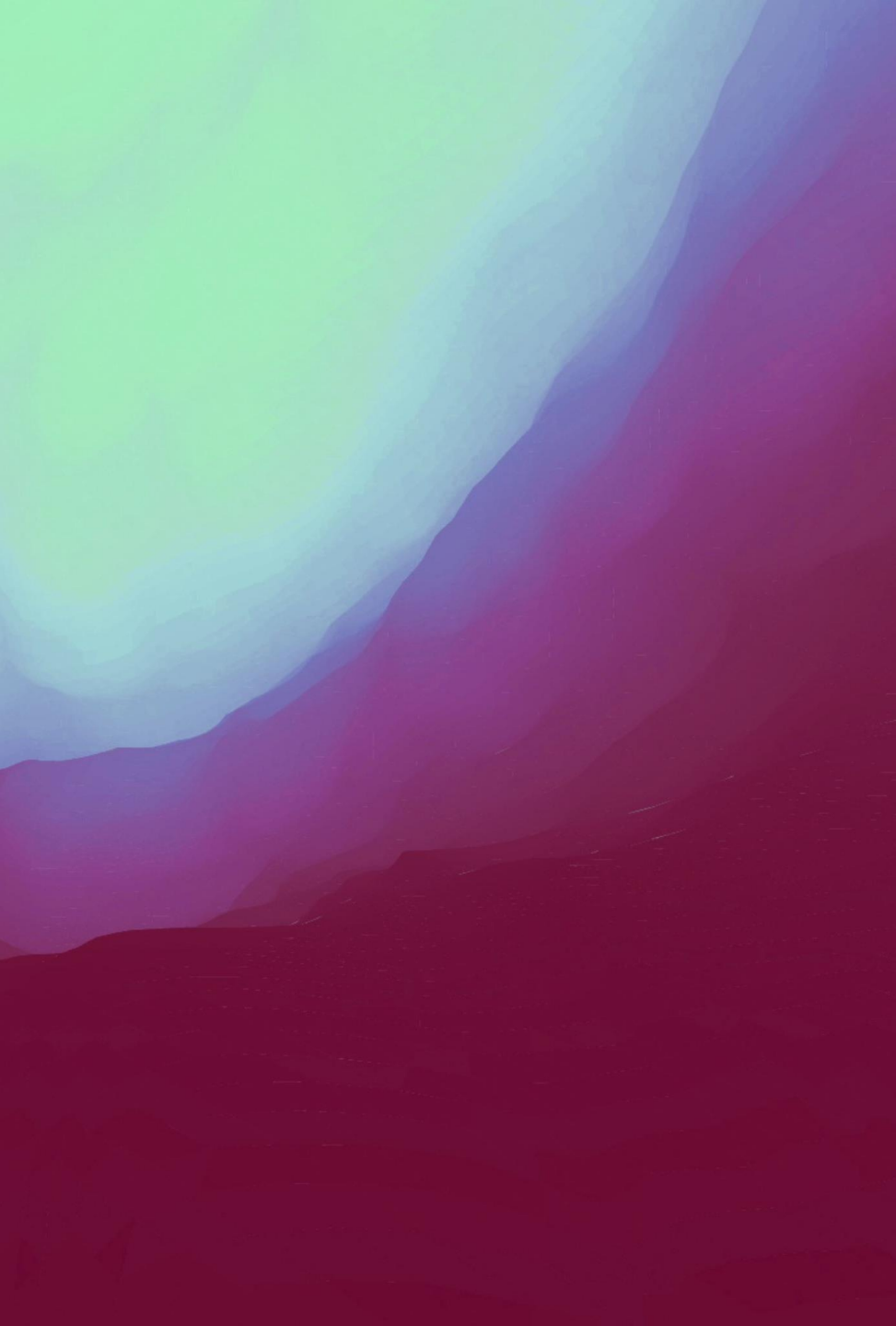
[21 CFR 312.68 & 21 CFR 812.45]

Investigator must allow properly authorized FDA representatives to inspect facilities and study records.

Good Clinical Practice



GCP Trial Flow



Overview of Site Study Monitoring Activities & Visits

Site Qualification (Pre-Study)

Technically, a site qualification visit is not required by FDA regulations, however the FDA Guideline from stated

- That the sponsor is responsible for assuring, through personal contact between the monitor and each investigator, that the investigator clearly understands and accepts the obligations incurred in undertaking a clinical trial.

The FDA monitoring guidance was deactivated in A replacement Draft was released in August of 2011 promoting a risk-based approach to determining how a site will be monitored by the sponsor.

- 2011 FDA Draft Guideline: Oversight of Clinical Investigations---A risk-Based Approach to Monitoring

INVESTIGATOR'S MEETINGS & STUDY INITIATION:

An initiation visit is performed to ensure the investigators and study staff understand the study protocol, that all the operational steps are in place, and that everyone is clear and well trained in their specific roles and responsibilities. A trial initiation visit is conducted prior to the first patient being recruited into a study. These meetings are important whether the trial is commercial, non-commercial or academic, and for any type of intervention.

The visit is usually conducted by the trial coordinator or often a monitor on behalf of the sponsor. Essentially their aim will be to work with the sites to ensure that the site's planned operational procedures fits with the requirements of the protocol and will ensure accurate data as well as safe and ethical conduct of the trial.

The initiation visit is held once sites have had their regulatory and Ethics approval in place and after all the essential documents have been retrieved for the site and sponsor files.

PERIODIC (INTERIM) MONITORING:

Periodic sponsor activities remote and/or onsite to research sites, the only required sponsor activity (periodic monitoring, adequate monitoring) required by regulations;---discussed in later chapter.

STUDY SITE TERMINATION (CLOSE-OUT)

The principal investigator is required to close out the study with the governing IRB when the following conditions have been met:

Enrollment is closed and research related activities, interventions or interactions with human subjects have been completed

All queries have been resolved

PHI data, and records are no longer required by study team or sponsor (data locked)

Sponsor has conducted a close out visit and issued a close out letter (sponsor grants permission to close the study with the IRB)

Reports should be submitted within 30 days of the completion of the study.

CRA vs. MONITOR

CRA is a specific level of “monitor”. Usually the job responsibilities focus on managing assigned sites.

The monitor could be a sponsor study manager, CRA supervisor, etc. The sponsor procedures usually list “monitor” to allow flexibility for personnel to conduct required activities.

A CRA is an industry term for an individual designated by a sponsor to monitor the progress of a clinical investigation.

Role of a CRA encompasses much more than the actual process of “monitoring”

Source document verification, assessing protocol adherence and compliance with GCP.

WHO ARE Monitors?

May be nurses, scientists, engineers, pharmacists, etc. (medical or non-medical backgrounds);

FDA requires a monitor to be qualified by “:training and experience”.

Need not be a person qualified to diagnose and treat disease or condition under investigation, like a physician,

Although somewhere in direct line of review of study data there should be a person so qualified (Medical Monitor).

Sometimes called CRA “Site Managers”; the role may be in-house and/or field based.

Overview of CRA Responsibilities

May be part of a team that conducts some or all of the following activities remotely and/or at site:

Protocol Development:

Prepare draft protocols

Collaborate with statisticians

Coordinate protocol review and approval processes

Confirm protocol submission to appropriate regulatory agencies

Prepare protocol amendments

May be part of a team that conducts some or all of the following activities remotely and/or at site:

Pre-Study Activities:

Identify and select investigators/sites

Prepare for and conduct site qualification activities/visits

Order test articles, supplies and request randomization schedules

Develop case report forms (CRFS) and study- specific documents

Identify and select CRO/centralized services (e.g., clinical laboratories)

Negotiate investigator budgets

Collaborate with data management

Develop subject tracking systems

May be part of a team that conducts some or all of the following activities remotely and/or at site:

Study initiation:

- Contribute to the conduct of investigator meetings
- Prepare for & conduct the initiation activities/visits

May be part of a team that conducts some or all of the following activities remotely and/or at site:

Study Management:

- Prepare for interim monitoring visits
- Conduct the monitoring visit
- Perform additional site management and administrative activities
- CRF/data management activities

May be part of a team that conducts some or all of the following activities remotely and/or at site:

Study Termination/ Site Closure:

- Prepare for & conduct the study termination visit
- Arrange for final investigator's payment
- Coordinate final study report review and approval process

Role of the CRC

CRC is a person responsible for conducting clinical trials using GCP under the auspices of a PI.

Federal regulations clearly define responsibilities of the Investigator and some members of the clinical research team.

Role of CRC is not specifically defined by the regulations.

CRC tasks/duties assigned must be delegated from the PI

Activities can be grouped according to the following general categories:

Study start-up

- Study approval
- Study initiation
- Subject recruitment, screening and enrollment
- Study implementation & data management
- Study termination activities

Study start-up:

Ensure a confidentiality agreement is signed by the investigator or authorized site personnel and returned to the sponsor (as requested), in order to receive a copy of the protocol;

Review the protocol outline or synopsis;

Assess feasibility of conducting the study;

Review and negotiate the contract;

Review proposed study budget, then assist with the development and negotiation;

Distribute protocol to relevant study team members;

Prepare list of questions to ask sponsor representatives;

Review investigational product listing;

Prepare & schedule sponsor site qualification (site selection);

Prepare list of potential subjects and/or methods that will be used to recruit subjects.

Prepare/provide examples of past study performance (e.g., sample source documents, audit findings, enrollment success rates, etc.);

Contact laboratory, pharmacy, and other departments to inform them of study requirements and secure their assistance and input;

Compile and submit pre-study documents to the sponsor and IRB

Prepare a draft informed consent form

Collect curriculum vitae and medical licenses for investigators and all sub-investigators

Obtain laboratory normal ranges and certification/accreditation

Obtain signatures from PI and sub-investigators for regulatory/study documents

Obtain a copy of IRB membership list or letter from IRB chairperson describing their compliance with FDA regulations.

Study Approval:

Review final protocol and modify all materials, study budget, and other documents accordingly;

Submit documents to the IRB for review and approval;

Respond to any requests for changes made by IRB & resubmit changes to the sponsor for approval

Resubmit any documents to IRB as necessary

Submit IRB approval letter to sponsor and file in investigator (regulatory) binder; Prepare recruitment materials and submit recruitment announcements to the sponsor and IRB approval.

Prepare/provide examples of past study performance (e.g., sample source documents, audit findings, enrollment success rates, etc.);

Contact laboratory, pharmacy, and other departments to inform them of study requirements and secure their assistance and input;

Compile and submit pre-study documents to the sponsor and IRB

Prepare a draft informed consent form

Collect curriculum vitae and medical licenses for investigators and all sub-investigators

Obtain laboratory normal ranges and certification/accreditation

Obtain signatures from PI and sub-investigators for regulatory/study documents

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Review final protocol and modify all materials, study budget, and other documents accordingly;

Submit documents to the IRB for review and approval;

Respond to any requests for changes made by IRB & resubmit changes to the sponsor for approval

Resubmit any documents to IRB as necessary

Submit IRB approval letter to sponsor and file in investigator (regulatory) binder; Prepare recruitment materials and submit recruitment announcements to the sponsor and IRB approval.

CRC Study Initiation:

Attend investigator meeting, if held by sponsor

Submit any revised documents/protocol amendments to the IRB for approval prior to study initiation

Create study documents. For example: study flow sheets, subject tracking logs, medication tracking logs

Organize the subject source documents to that information consistent and accessible by authorized practitioners for study participants

Place supplies in secure (locked) area under proper environmental conditions (e.g., temperature as per protocol) and ensure that all study personnel are familiar with the use of the equipment.

Study Initiation:

Organize the investigator study binder

Create site personnel signature list and ensure all personnel who are working on the study sign it

Meet with the sponsor during the initiation visit

Document initiation meeting attendees, as proof of study training

Conduct initial drug/device accountability

Check product and supplies received against the packing slip

Prepare calibration logs, maintenance logs, as per protocol.

CRC Subject Recruitment, Screening and Enrollment

Complete subject pre-screening activities

Recruit subjects per recruitment/advertising strategies

Schedule subjects for study visits/procedures

Conduct/assist in initial subject intake interview

Participate in and document the informed consent process

Document subjects participation in the source document

Maintain subject identification code list and other enrollment logs

Document subject eligibility

Study Implementation & Data Management

Collect study data, complete source documents and CRFs as soon as possible after each subject visit or as data becomes available

Ensure proper investigational product storage, administration and accountability

Solicit, record, and report adverse events

Maintain ongoing communication with CRA and respond to sponsor/CRO concerns

Prepare sponsor monitoring visits

Address any discrepancies identified by the CRA during the source document verification process/ site visit

File all correspondence in the study binder and maintain other study documentation

Follow-up on any other issues as requested by the CRA or other sponsor/CRO representatives

Document all important contacts with the sponsor/CRO (e.g., maintain a phone log, keep all written, fax and electronic correspondence), and file these in the investigator (regulatory) binder

Maintain ongoing communication with IRB

Notify the IRB if any safety reports, changes to protocol or consent form, and update IRB on the status of the study at least annually, or as requested by IRB

Follow-up with IRB on all activities regarding their recommendations for necessary action

Submit weekly updates as requested by sponsor

Prepare, process, store, and ship specimens as required by protocol

Encourage subject retention and compliance

Schedule patient follow-up visits and document all attempts to contact subject

Manage and communicate protocol amendments

Re-consent subjects when necessary

Equipment maintenance

Study Termination Activities Prepare for final sponsor visit

Complete all subject follow-up (CRFs, queries, etc.)

Maintain study records per final record keeping requirements

Perform final drug/device accountability

Return all equipment and supplies and unused CRFs as requested

Ensure all specimens have been sent out

Arrange for final payment

Request randomization information so subjects may be informed as to which treatment they received, when applicable

Query subject satisfaction regarding participation on trial

Prepare for FDA or quality assurance (QA) audits

Delegation of Study Tasks

Common for PI to delegate many study tasks to study site staff members

FDA requires documentation of delegation and supervision by the PI

Commonly a delegation listing is used in the form of a delegation log

The log template can be supplied by sponsor or site may use their own

PI must assure that delegation is to qualified individuals: with adequate training, education and/or experience

Including: direct staff, outside department, 3rd party

Refer to FDA Guidance, PI responsibility, OCT 2009

Delegation should be supported through documentation

PI should assess the adequacy of the delegation and supervise the staff

Refer to 2009 FDA Guidance for Investigator Responsibilities

CRA should monitor for adequate and accurate documentation of delegation

This includes review of credentials of delegates and of performance by the delegates, as well as supervision by the PI

Clinical Trials Regulation

European Union (EU) pharmaceutical legislation known as the Clinical Trials Regulation entered into application on 31 January 2022.

It aims to ensure the EU offers an attractive and favourable environment for carrying out clinical research on a large scale, with high standards of public transparency and safety for clinical trial participants.

On 31 January 2022, the Regulation repealed the Clinical Trials Directive (EC) No. 2001/20/EC and national implementing legislation in the EU Member States, which regulated clinical trials in the EU until the Regulation's entry into application.

A transition period applies to clinical trial submission under the Regulation.

Consult the Regulation:

Clinical Trials Regulation (Regulation (EU) No 536/2014)

Aims and benefits

The Clinical Trials Regulation harmonises the processes for assessment and supervision of clinical trials throughout the EU.

The evaluation, authorisation and supervision of clinical trials are the responsibilities of EU Member States and European Economic Area (EEA) countries.

Prior to the Regulation, clinical trial sponsors had to submit clinical trial applications separately to national competent authorities and ethics committees in each country to gain regulatory approval to run a clinical trial.

The Regulation enables sponsors to submit one online application via a single online platform known as the Clinical Trials Information System (CTIS) for approval to run a clinical trial in several European countries, making it more efficient to carry out such multinational trials.

The Regulation also makes it more efficient for EU Member States to evaluate and authorise such applications together, via the Clinical Trials Information System.

The purpose is to foster innovation and research in the EU, facilitating the conduct of larger clinical trials in multiple EU Member States/EEA countries.

Clinical Trials Information System

Under the Regulation, clinical trial sponsors can use the [Clinical Trials Information System](#) (CTIS) from 31 January 2022, but are not obliged to use it immediately, in line with a three-year transition period.

National regulators in the EU Member States and EEA countries will use CTIS from 31 January 2022.

The system:

enables sponsors to apply for clinical trial authorisation in up to 30 European countries with a single online application;

allows national regulators to collaboratively process clinical trial applications in more than one country, request further information, authorise or refuse a trial and oversee an authorised trial;

facilitates the expansion of trials to other EEA countries;

enables transparency and access to information for any party interested in clinical trials conducted in the EEA through a searchable public website.

CTIS went live on 31 January 2022 together with the public Clinical Trials website. For more information, see:

- [Clinical Trials Information System](#)
- [Development of the Clinical Trials Information System](#)
- [Clinical Trials website](#)

Under the Clinical Trials Regulation, EU Member States and EEA countries use the Clinical Trials Information System (CTIS) to carry out their legal responsibilities to assess and oversee clinical trials from 31 January 2022:

for the first year of implementation and until 30 January 2023, clinical trial sponsors could choose whether to apply to start a clinical trial via the Clinical Trials Information System or under the Clinical Trials Directive;

from 31 January 2023 onwards, clinical trial sponsors need to apply to start a clinical trial via the Clinical Trials Information System;

from 31 January 2025, any trials approved under the Clinical Trials Directive that continue running will need to comply with the Clinical Trials Regulation and their sponsors must have recorded information on them in CTIS.

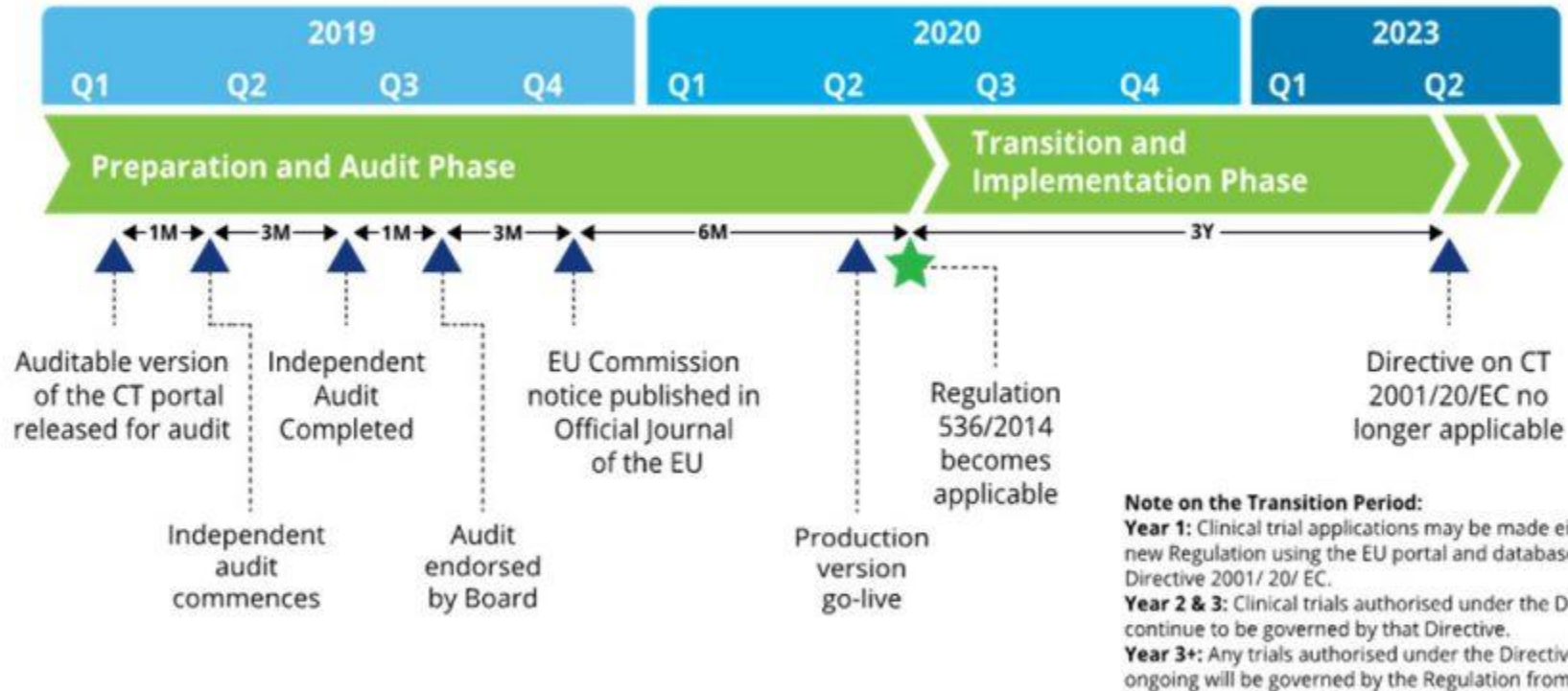
EMA encourages sponsors to use the transition period to ensure their information on clinical trials is recorded in CTIS in a timely manner.

The changing landscape of conducting clinical trials around the world and the intent to make the EU one of the hubs for clinical research served as an impetus for introducing the CTR, especially since the earlier Clinical Trial Directive (CTD) 2001/20/EU was found to potentially be tedious and deterrent to this exercise.

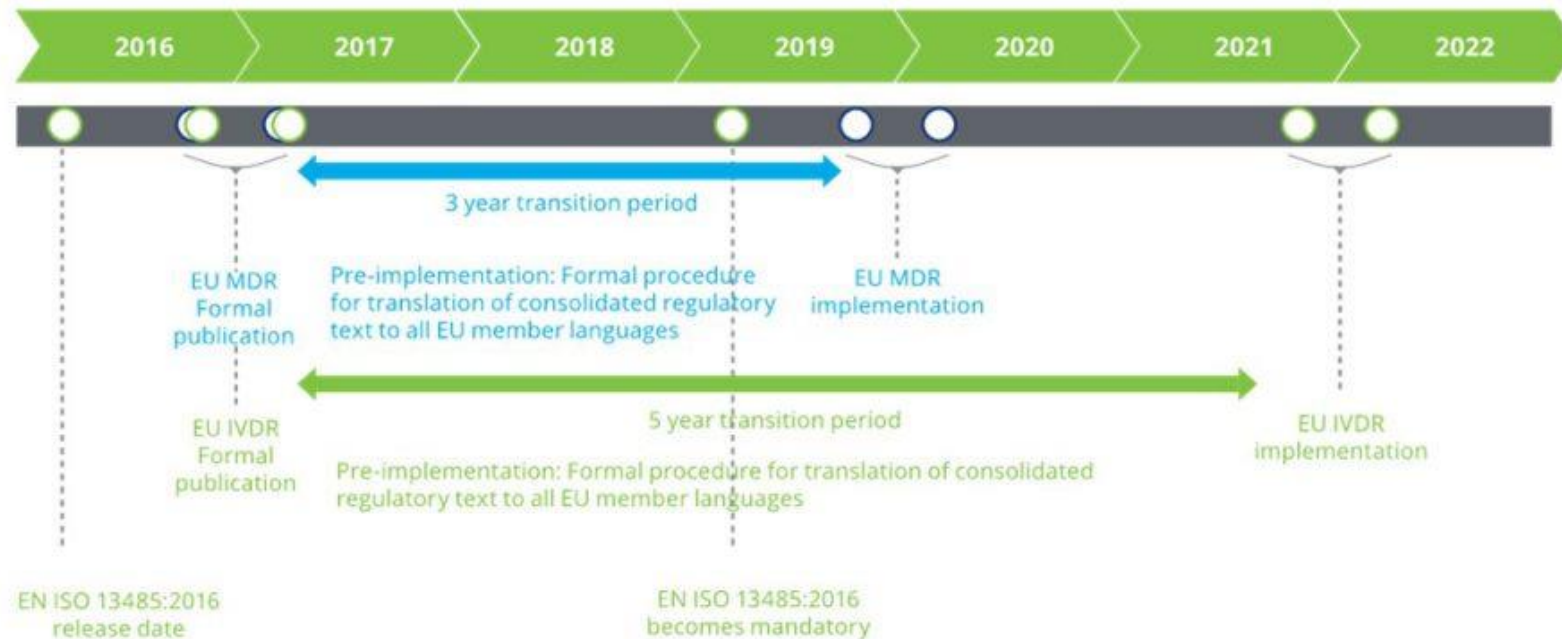
Thus, the main goals of CTR 536/2014 were –

- to ensure EU-wide harmonization of the approval process and with predefined time limits;
- to introduce a common evaluation process for multinational trials;
- to reduce the scope of regulatory autonomy at member state level;
- to make the EU competitive in research;
- to generate high-quality scientific data; and
- to ensure patient safety.

Timeline for CTR compliance



Timeline for MDR & IVDR compliance



The main aim behind drafting new regulations was to ensure conformity of medical devices and improve their safety profile across the EU with an integrated process. In the clinical context, perhaps the most impactful change that will be implemented by the MDR will be on the issue of 'clinical equivalence'.

Conclusion

The importance of GCP lies in the question 'why' and 'how' GCP trials came about. To know the answer to this, we have to look to the historical background that led to the formulation of GCP guidelines in the United States and Europe and also to the formation of the ICH.

The events that led up to the culmination of the ICH-GCP guidelines brought forth public awareness that there was a need to control and regulate clinical trials dealing with drugs and human subjects. The violation of human rights played a large role and that is why the Declaration of Helsinki and The Nuremberg Code remain as the framework of the present guidelines.

The ICH-GCP guidelines are therefore considered the 'bible' of clinical trials, and have become a global law which safeguards humanity as we know it today.

FDA regulations governing human subject protection and the conduct of clinical trials

- Electronic Records; Electronic Signatures (21 CFR Part 11)
- Regulatory Hearing Before the Food and Drug Administration (21 CFR Part 16)
- Protection of Human Subjects (Informed Consent) (21 CFR Part 50)
- Financial Disclosure by Clinical Investigators (21 CFR Part 54)
- Institutional Review Boards (21 CFR Part 56)
- Good Laboratory Practice for Nonclinical Laboratory Studies (21 CFR Part 58)
- Investigational New Drug Application (21 CFR Part 312)
- Applications for FDA Approval to Market a New Drug (21 CFR Part 314)
- Bioavailability and Bioequivalence Requirements (21 CFR Part 320)
- New Animal Drugs for Investigational Use (21 CFR Part 511)
- New Animal Drug Applications (21 CFR Part 514)
- Applications for FDA Approval of a Biologic License (21 CFR Part 601)
- Investigational Device Exemptions (21 CFR Part 812)
- Premarket Approval of Medical Devices (21 CFR Part 814)



EUROPEAN MEDICINES AGENCY

European regulatory system for medicines

A consistent approach to medicines regulation across the EU

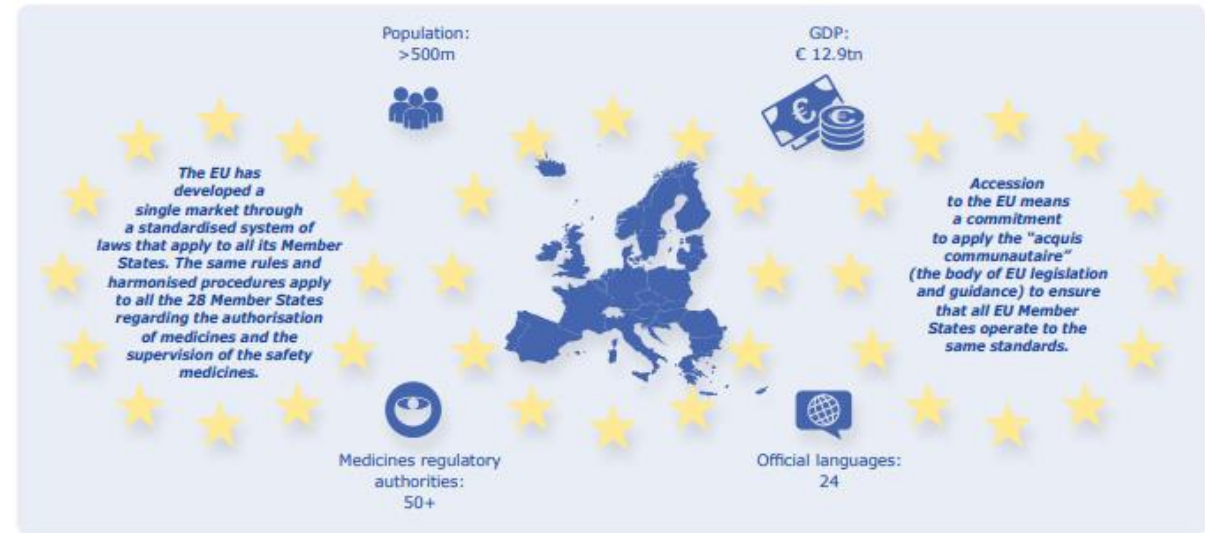
Ms. Sonali Correa

Ph.D., M.Sc.

Introduction

The European medicines regulatory system is based on a network of around 50 regulatory authorities from the 31 EEA countries (28 EU Member States plus Iceland, Liechtenstein and Norway), the European Commission and EMA.

The European Union—key facts



EU Member States: 28



The European Economic Area (EEA) is formed of the 28 EU Member States plus:



The logo for the European Medicines Agency (EMA) is displayed in white text on a dark blue background. The letters 'EMA' are in a bold, sans-serif font. Behind the text is a large, light blue circle that is partially obscured by the dark blue background.

EMA

The European Medicines Agency (EMA) is a decentralized agency of the European Union (EU)

The Agency is responsible for the scientific evaluation, supervision, and safety monitoring of medicines developed by pharmaceutical companies for use in EU

EMA protects public and animal health in 27 EU Member states, as well as the countries of the EEA, by ensuring that all medicines available on the EU market are **SAFE, EFFECTIVE and HIGH QUALITY**.

EMA serves a market of over 500 million people in the EU

- Founded in 1995, the European Medicines Agency (EMA) has worked across the European Union (EU) and globally to protect public and animal health by assessing medicines to rigorous scientific standards and by providing partners and stakeholders with independent, science-based information on medicines.
- EMA has a 25-year track record of ensuring efficacy and safety of human and veterinary medicines across Europe, and promoting research and innovation in the development of medicines.
- EMA's success is based on cooperation within the European medicines regulatory network – a unique partnership between the European Commission, the medicines regulatory authorities in the European Economic Area countries, and EMA. Working together has encouraged the exchange of knowledge, ideas and best practices, in order to ensure the highest standards in medicines regulation.
- Today, seven EMA scientific committees and more than 30 working parties provide scientific expertise for the regulation of medicines by drawing on a pool of several thousand European scientific experts from the network.
- EMA enables one application, one assessment, one market authorisation for the whole of the EU

Mission

The mission of the European Medicines Agency (EMA) is to foster scientific excellence in the evaluation and supervision of medicines, for the benefit of public and animal health in the European Union (EU).





What does the strapline in the Logo represent?

- The strapline in the EMA logo represents the three pillars the Agency's work is based on:
- **Science**, representing the scientific expertise that guides the EMA in its regulatory decision-making.
- **Medicines**, representing EMA's focus on assessing and monitoring medicines to ensure their quality, safety and efficacy.
- **Health**, representing the Agency's mission to protect and improve public and animal health.

Milestones and achievements

- EMA was set up **in 1995** to harmonise the work of existing national medicine regulatory bodies.
- EMA is also responsible for products developed in the specialised areas of medicines for **rare diseases (since 2000)**, **herbal medicines (since 2004)**, medicines for children **(since 2006)** and advanced-therapy **medicines (since 2007)**.
- With the establishment of the Committee for **Orphan Medicinal Products in 2000**, EMA opened its doors to patients and healthcare professionals.
- Today, their representatives take part in most of EMA's scientific committees as full members, adding their unique perspective and experiences to discussions. They play an increasingly important role in the assessment of the risks and benefits of medicines. **In 2014**, patients discussed the benefit-risk evaluation of a medicine within the Committee for **Medicinal Products for Human Use (CHMP)** for the first time.
- With the creation of the **Pharmacovigilance and Risk Assessment Committee (PRAC) in 2012**, EMA started to play an even more important role in monitoring the safety of medicines across Europe.
- As of **January 2015**, EMA has been implementing its landmark policy on publishing the clinical data that underpin European decision-making on medicines. This will provide an unprecedented level of transparency for patients, healthcare professionals, academia and industry.

EMA functioning within EU Regulatory system

EMA and the Member States cooperate and share expertise in the assessment of new medicines and of new safety information

They also rely on each other for exchange of information in the regulation of medicine, for example regarding the reporting of side effects of medicines, the oversight of clinical trials and the conduct of inspections of medicines' manufacturers and compliance with good clinical practice (GCP), good manufacturing practice (GMP), good distribution practice (GDP), and good pharmacovigilance practice (GVP)

IT systems which connect all parties in the network facilitate the exchange of information on aspects such as safety monitoring of medicines, authorisation and supervision of clinical trials or compliance with good manufacturing and distribution practices.

- The European system offers different routes for such an authorisation
- The **centralised procedure** allows the marketing of a medicine on the basis of a single EU-wide assessment and marketing authorisation which is valid throughout the EU
- Pharmaceutical companies submit a single authorisation application to EMA.
- The Agency's Committee for Medicinal Products for Human Use (CHMP) or Committee for Medicinal Products for Veterinary Use (CVMP) then carries out a scientific assessment of the application and gives a recommendation to the European Commission on whether or not to grant a marketing authorisation.
- Once granted by the European Commission, the centralised marketing authorisation is valid in all EU Member States.
- The use of the centrally authorised procedure is compulsory for most innovative medicines, including medicines for rare diseases.

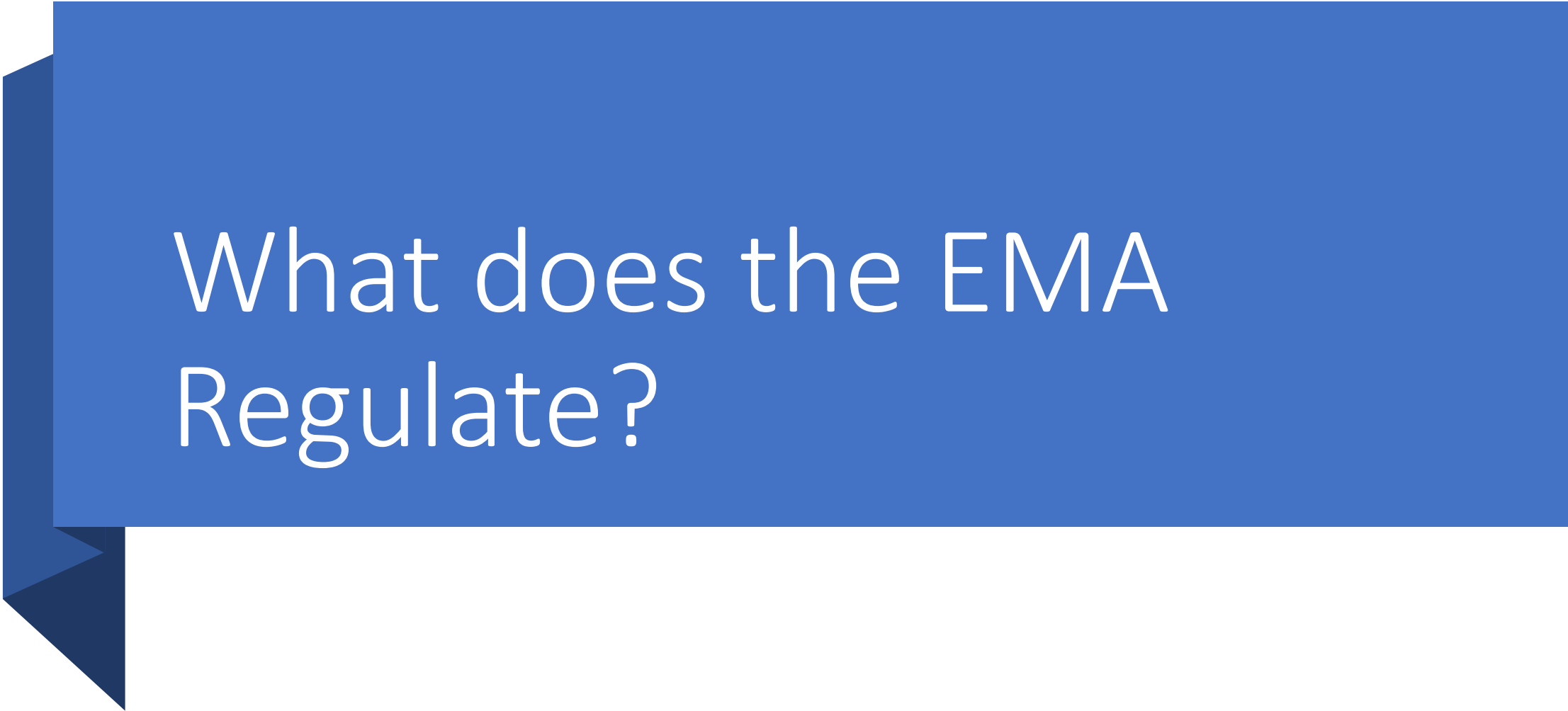
- When a company wants to authorise a medicine in several Member States, it can use one of the following procedures:
- the **decentralised procedure** where companies can apply for the simultaneous authorisation of a medicine in more than one EU Member State if it has not yet been authorised in any EU country and does not fall within the scope of the centralised procedure;
- the **mutual-recognition** procedure where companies that have a medicine authorised in one EU Member States can apply for this authorisation to be recognised in other EU countries. This process allows Member States to rely on each other's scientific assessments.
- A European Public Assessment Report, or EPAR, is published for every human or veterinary medicine that has been granted or refused a marketing authorisation following an assessment by EMA.
- For a medicine that is authorised by a Member State, details on the assessment of the medicine are also available in a Public Assessment Report

The role of the European Commission

- On the basis of scientific assessments carried out by EMA, it grants or refuses, changes or suspends marketing authorisations for medicines that have been submitted via the centralised procedure.
- It can also take EU-wide action when a safety issue has been identified for a nationally authorised product and when harmonised regulatory measures in all MSs are considered necessary following assessment by EMA's PRAC.
- The European Commission can also take action concerning other aspects of medicine regulation:
 - Right of initiative—it can propose new or amended legislation for the pharmaceutical sector;
 - Implementation—it can adopt implementing measures as well as oversee the correct application of EU law on pharmaceuticals;
 - Global outreach—it ensures appropriate collaboration with relevant international partners and promotes the EU regulatory system globally.

The role of EMA

- EMA is responsible for the **scientific evaluation**, primarily of innovative and high-technology medicines developed by pharmaceutical companies for use in the EU.
- Experts participate in the work of EMA as members of its scientific committees, working parties, scientific advisory groups and other ad hoc advisory groups, or as members of the national assessment teams that evaluate medicines.
- The experts are chosen on the basis of their scientific expertise and many of them are made available to EMA by the NCAs in Member States.
- Increasingly, patients and healthcare professionals are involved in the work of the Agency including in the evaluation of medicines.



What does the EMA
Regulate?

Guidelines and scientific advice

- EMA prepares **scientific guidelines** in cooperation with experts from its scientific committees and working groups.
- These guidelines reflect the latest thinking on developments in biomedical science.
- They are available to guide the development programmes of all medicine developers who wish to submit an application for a marketing authorisation in the EU, and to ensure that medicines are developed consistently and to the highest quality.
- EMA also gives **product-specific scientific advice** to companies for the development of medicines.
- This is an important tool to help develop and make available high-quality, effective and safe medicines, for the benefit of patients.
- Scientific advice can also be given by NCAs.

Authorisation & supervision of manufacturers

- Manufacturers, importers and distributors of medicines in the EU must be licensed before they can carry out those activities.
- The regulatory authorities of each Member State are responsible for granting licences for such activities taking place within their respective territories.
- All manufacturing and importing licenses are entered into EudraGMDP, the publiclyavailable European **database operated by EMA**.

Safety and Monitoring of Medicines

- The European regulatory system for medicines monitors the safety of all medicines that are available on the European market throughout their life span.
- EMA has a committee dedicated to the safety of medicines for human use—the Pharmacovigilance Risk Assessment Committee, or PRAC.
- If there is a safety issue with a medicine that is authorised in more than one Member State, the same regulatory action is taken across the EU and patients and healthcare professionals in all Member States are provided with the same guidance.
- All suspected side effects that are reported by patients and healthcare professionals must be entered into EudraVigilance, the EU web-based information system operated by EMA that collects, manages and analyses reports of suspected side effects of medicines.
- These data are continuously monitored by EMA and the Member States in order to identify any new safety information.

- 
- EMA provides public access to reports of suspected side effects for centrally-authorised medicines in the European database of suspected drug-reaction reports.
 - This website allows users to view all suspected side effect reports submitted to EudraVigilance
 - The PRAC has a broad remit covering all aspects of pharmacovigilance. In addition to its role in risk assessment, the committee provides advice and recommendations to the European medicines regulatory network on risk management planning and benefit-risk assessment for medicines after marketing.

International cooperation

- The European Commission and EMA, in close cooperation with Member States, work to forge close ties with partner organisations around the world.
- These activities aim to foster the timely exchange of regulatory and scientific expertise and the development of best practices in the regulatory field across the world
- The European Commission and EMA work with the World Health Organization (WHO) on a range of issues, including medicines intended for markets outside the EU (medicines reviewed under EMA's so-called '**Article 58 procedure**'), the quality of medicines and the development of international non-proprietary names.
- For the EU, one of the main forums for multilateral international cooperation is the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), which brings together medicines regulatory authorities and pharmaceutical industry from around the world.
- EMA and many NCAs are also involved in the Pharmaceutical Inspection Convention and the Pharmaceutical Inspection Co-operation Scheme (jointly referred to as PIC/S), a close international cooperation between pharmaceutical inspection authorities in the field of GMP.

What EMA does not Regulate?

- **Evaluation of the Initial Marketing Authorization application for all medicines in the EU**
 - The vast majority of medicines available in the EU are authorized at national level either because they were authorized before the EMAs inception or they not fall in the scope of the centralized procedure the Agency is responsible for.
- **Evaluate Authorization of CT**
 - Authorization of CT happens at the Member State level
- **Conducting R&/orD of Medicines**
 - EMA does not operate laboratories on its own premises or elsewhere and is not involved in conducting clinical trials.
- **Pricing and reimbursement**
 - Once a marketing authorisation has been granted, decisions about price and reimbursement take place at the level of each Member State considering the potential role and use of the medicine in the context of the national health system of that country

- **Development of treatment**

- National govt or the Health Authorities of individual EU Member States develop guidelines for decisions regarding diagnosis, management and treatment in specific areas of healthcare

- **Issues of Marketing Authorizations**

- The legal decision to grant, suspend, or revoke a marketing authorization for any medicine falls under the remit of the European Commission for centrally authorized products and the NCA of the EU Member States for nationally authorized products

How does EMA work?

EMA brings together **scientific experts** from across Europe by working closely with the national regulatory authorities in European Union (EU) Member States, in a partnership known as the European medicines regulatory network.

The network **pools resources and expertise** in the EU and gives EMA access to thousands of European scientific experts who take part in the regulation of medicines.

Ensuring the independence of its **scientific assessments** is a high priority for EMA. The Agency takes care to ensure that its scientific experts, staff and Management Board do not have any financial or other interests that could affect their impartiality.

EMA strives towards being **as open and transparent** as possible about how it reaches its scientific conclusions. EMA's European public assessment reports describe the scientific basis for EMA's recommendations on all centrally authorised medicines.

EMA also publishes a **large amount of information** in lay language about its work and about medicines. For more information, see Transparency.

The Agency **also seeks to publish clear and up-to-date information** on how it operates, including planning and reporting documents and information on funding, financial management and budgetary reporting.

Organizational Chart of the EMA

- EMA is governed by an independent Management Board. Its day-to-day operations are carried out by the EMA staff, overseen by EMA's **Executive Director**.
- EMA is a networking organisation whose activities involve thousands of experts from across Europe. These experts carry out the work of EMA's scientific committees.
- On **2 March 2020**, EMA implemented changes to its **organisational structure** to ensure that it operates as efficiently as possible to deliver high quality outputs for public and animal health.
- The main changes comprise:
 - integrating operations in the area of human medicines into one human medicines division;
 - establishing four mission-critical task forces to support the human and veterinary medicines divisions, bringing together expertise to drive transformational change in high-priority areas.
- The re-organisation exercise takes into account the rapidly evolving landscape for pharmaceutical research and development that requires regulators to keep up with advances in science and technology and prepare for future challenges at an ever-accelerating pace.
- The change is also driven by the need to recalibrate to a lower head count following the EMA's relocation to Amsterdam in 2019 whilst dealing with an increased workload due the implementation of various new legislation relating to clinical trials, veterinary medicines, medical devices and data protection.

Organizational Chart

Management Board

- The Management Board consists of 36 members, appointed to act in the public interest, who do not represent any government, organisation or sector.
- The Board sets the Agency's budget, approves the annual work programme and is responsible for ensuring that the Agency works effectively and co-operates successfully with partner organisations across the EU and beyond.

Executive Director

The Agency's Executive Director is the legal representative of the Agency. She is responsible for all operational matters, staffing issues and drawing up the annual work programme.

Agency staff

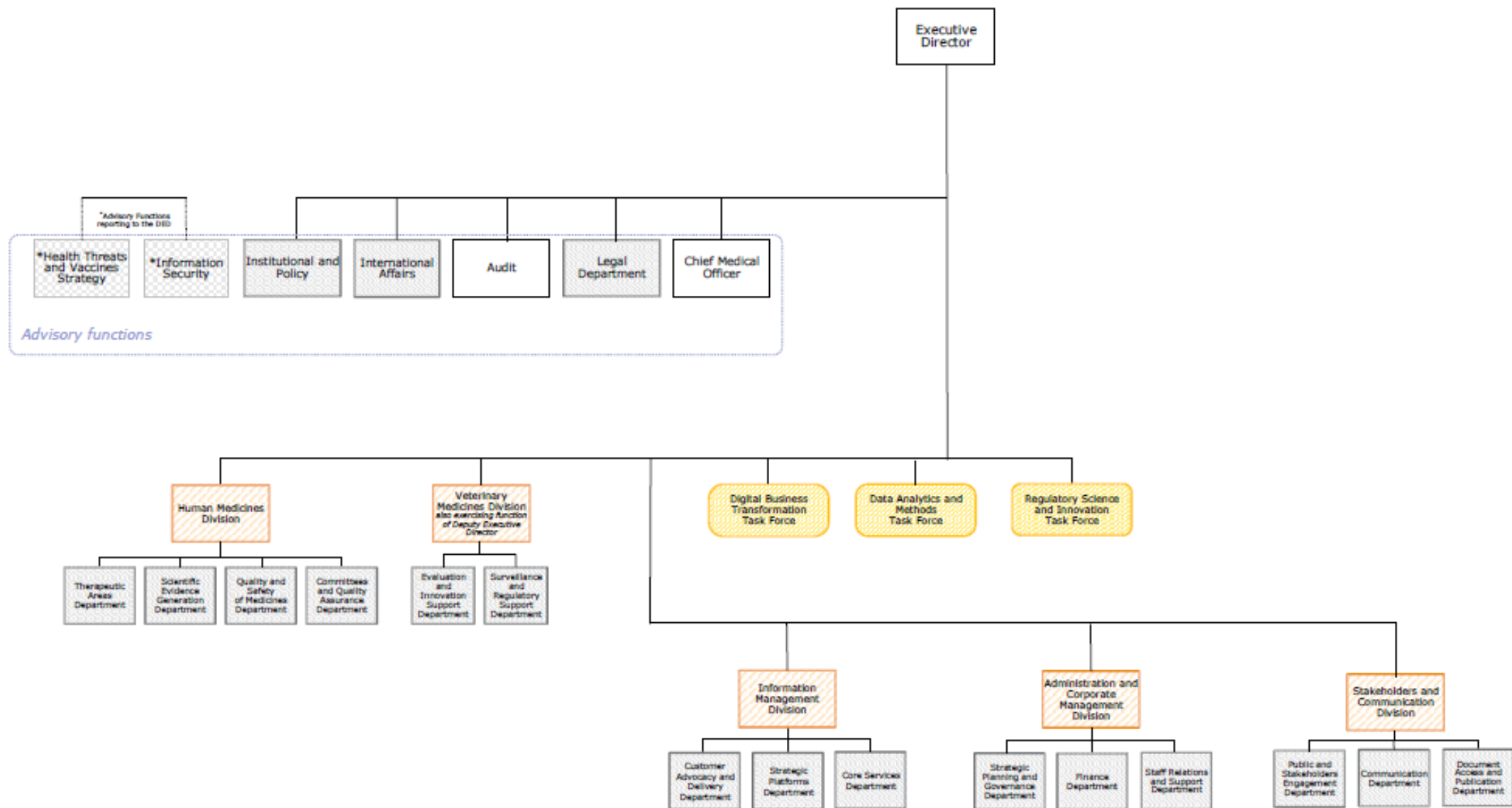
The Agency's staff support the Executive Director in carrying out their responsibilities, including administrative and procedural aspects of EU law related to the evaluation and safety-monitoring of medicines in the EU

Scientific Committees

EMA has seven scientific committees that evaluate medicines along their lifecycle from early stages of development, through marketing authorisation to safety monitoring once they are on the market.

The Agency has a number of working parties and related groups, which the committees can consult on scientific issues relating to their particular field of expertise.

These bodies are composed of European experts made available by NCA of the EU Member States, which work closely with EMA in the European medicines regulatory network.



The logo for the European Directorate for the Quality of Medicines & HealthCare (EDQM) is displayed in white text on a dark blue background. The letters 'EQDM' are in a clean, sans-serif font. To the left of the text is a large, semi-transparent circular graphic that overlaps the dark blue background.

The **European Directorate for the Quality of Medicines & HealthCare** is a Directorate of the Council of Europe that traces its origins and statutes to the Convention on the Elaboration of a European Pharmacopoeia

Vision

The EDQM is a leading organisation that protects public health by:

- enabling the development,
- supporting the implementation and
- monitoring the application
- of quality standards for safe medicines and their safe use which are recognised as a scientific benchmark and applied worldwide.

Mission

- EDQM contributes to the basic human right of access to good quality medicines and healthcare, and promotes and protects human and animal health

EudraLex - EU Legislation

- EudraLex is the collection of rules and regulations governing medicinal products in the European Union.
- It consists of 10 volumes concerning Medicinal products for Human use

- Volume 1 - Pharmaceutical Legislation.
- Volume 2 - Notice to Applicants.
- Volume 2A deals with procedures for marketing authorisation.
- Volume 2B deals with the presentation and content of the application dossier.
- Volume 2C deals with Guidelines.
- Volume 3 - Guidelines.

Concerning Medicinal Products for human use in clinical trials (investigational medicinal products).

- Volume 10 - Clinical trials.

Concerning Veterinary Medicinal Products:

- Volume 5 - Pharmaceutical Legislation.
- Volume 6 - Notice to Applicants.
- Volume 7 - Guidelines.
- Volume 8 - Maximum residue limits.

Concerning Medicinal Products for Human and Veterinary use:

- Volume 4 - Good Manufacturing Practices.
- Volume 9 - Pharmacovigilance.

Miscellaneous:

- Guidelines on Good Distribution Practice of Medicinal Products for Human Use (94/C 63/03)

EMA's scientific committees

The EMA has seven scientific committees that carry out its scientific assessments:

- Committee for Medicinal Products for Human Use (CHMP)
- Pharmacovigilance Risk Assessment Committee (PRAC)
- Committee for Medicinal Products for Veterinary Use (CMVP)
- Committee for Orphan Medicinal Products (COMP)
- Committee on Herbal Medicinal Products (HMPC)
- Committee for Advanced Therapies (CAT)
- Paediatric Committee (PDCO)

Committee for Medicinal Products for Human Use (CHMP)

- **The CHMP is the EMA committee responsible for human medicines.**
- The CHMP replaced the former Committee for Proprietary Medicinal Products (CPMP) in May 2004

Role

The CHMP plays a vital role in the authorisation of medicines in the EU.

In the **centralised procedure**, the CHMP is responsible for:

- conducting the initial assessment of EU-wide marketing authorisation applications;
- assessing modifications or extensions ('variations') to an existing marketing authorisation;

In Mutual-recognition procedure

- considering the recommendations of the Agency's Pharmacovigilance Risk Assessment Committee on the safety of medicines on the market and when necessary, recommending to the European Commission changes to a medicine's marketing authorisation, or its suspension or withdrawal from the market.

- The CHMP also evaluates medicines authorised at national level referred to EMA for a harmonised position across the EU. For more information, see Referral procedures.
- In addition, the CHMP and its working parties contribute to the development of medicines and medicine regulation, by:
- providing scientific advice to companies researching and developing new medicines;
- preparing scientific guidelines and regulatory guidance to help pharmaceutical companies prepare marketing authorisation applications for human medicines;
- cooperate with international partners on the harmonisation of regulatory requirements.

Assessments

- The CHMP's assessments are based on a comprehensive scientific evaluation of data. They determine whether the medicine meets the necessary quality, safety and efficacy requirements and that it has a positive risk-benefit balance.
- An internal peer-review system safeguards the accuracy and validity of the opinions of the committee.

Pharmacovigilance Risk Assessment Committee

- **The Pharmacovigilance Risk Assessment Committee (PRAC) is the European Medicines Agency's (EMA) committee responsible for assessing and monitoring the safety of human medicines.**
- The PRAC was formally established in line with the pharmacovigilance legislation which came into effect in 2012 to help strengthen the safety monitoring of medicines across Europe.

Role

The PRAC is responsible for assessing all aspects of risk management of human medicines, including:

- the detection, assessment, minimisation and communication of the risk of adverse reactions, while taking the therapeutic effect of the medicine into account;
- design and evaluation of post-authorisation safety studies;
- pharmacovigilance audit.

The PRAC provides recommendations on questions on pharmacovigilance and risk management systems, including the monitoring of their effectiveness, to the:

- Committee for Medicinal Products for Human Use (CHMP) for centrally authorised medicines and referral procedures;
- Coordination Group for Mutual Recognition and Decentralised Procedures – Human (CMDh) on the use of a medicine in Member States;
- the EMA secretariat, Management Board and European Commission, as applicable.

Committee for Veterinary Medicinal Products (CVMP)

The Committee for Veterinary Medicinal Products (CVMP) is the European Medicines Agency's (EMA) committee responsible for veterinary medicines.

- The CVMP was formerly known as the Committee for Medicinal Products for Veterinary Use before 28 January 2022.

Role

The CVMP plays a vital role in the authorisation of veterinary medicines in the European Union (EU). In the **centralised procedure**, the CVMP is responsible for:

- conducting the initial assessment of EU-wide marketing authorisation applications;
- post-authorisation and maintenance activities, including the assessment of any modifications or variations to an existing marketing authorisation;
- safety monitoring of veterinary medicines on the market and when necessary, recommending to the European Commission changes to a medicine's marketing authorisation, or its suspension or withdrawal from the market. For more information, see Veterinary pharmacovigilance.

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- The CVMP also evaluates veterinary medicines authorised at national level referred to EMA for a harmonised position across the EU. For more information, see Veterinary referral procedures.
 - The CVMP recommends safe limits for residues of veterinary medicines used in food-producing animals and biocidal products used in animal husbandry, for the establishment of maximum residue limits by the European Commission.
 - In addition, the CVMP and its working parties contribute to the development of veterinary medicines and medicine regulation, by:
 - providing scientific advice to companies researching and developing new veterinary medicines;
 - preparing scientific guidelines and regulatory guidance to help pharmaceutical companies prepare marketing authorisation applications for veterinary medicines;
 - cooperate with international partners on the harmonisation of regulatory requirements.

Assessments

- The CVMP's assessments are based on a comprehensive scientific evaluation of data. They determine whether the medicine meets the necessary quality, safety and efficacy requirements and that it has a positive risk-benefit balance in favour of the animal population they are intended for.

Committee for Orphan Medicinal Products (COMP)

- **The Committee for Orphan Medicinal Products (COMP) is the European Medicines Agency's (EMA) committee responsible for recommending orphan designation of medicines for rare diseases.**
- The COMP was established in 2000, in line with Regulation (EC) No 141/2000

Role

- The COMP is responsible for evaluating applications for orphan designation. This designation is for medicines to be developed for the diagnosis, prevention or treatment of rare diseases that are life-threatening or very serious. In the European Union (EU), a disease is defined as rare if it affects fewer than 5 in 10,000 people across the EU. The European Commission decides whether to grant an orphan designation for the medicine based on the COMP's opinion.
- An orphan designation allows a pharmaceutical company to benefit from incentives from the EU, such as reduced fees and protection from competition once the medicine is placed on the market.

The COMP also advises and assists the European Commission on matters related to orphan medicines, including:

- developing and establishing an EU-wide policy;
- drawing up detailed guidelines;
- liaising internationally.

Committee on Herbal Medicinal Products (HMPC)

The Committee on Herbal Medicinal Products (HMPC) is the European Medicines Agency's (EMA) committee responsible for compiling and assessing scientific data on herbal substances, preparations and combinations, to support the harmonisation of the European market.

- The HMPC replaced the Committee for Proprietary Medicinal Products' Working Party on Herbal Medicinal Products in September 2004.
- The Committee was established in accordance with Regulation (EC) No 726/2004 and the Herbal Directive, which introduced a simplified registration procedure for traditional herbal medicinal products in EU Member States.
- The HMPC is composed of scientific experts in the field of herbal medicines.
- Role
- The HMPC prepares the Agency's opinions on herbal substances and preparations, along with information on recommended uses and safe conditions.

- This work supports the harmonisation of the European market: national competent authorities are able to refer to one unique set of information on a herbal substance or preparation when evaluating marketing applications for herbal medicines.
- To support EU Member States, the HMPC focuses on two main tasks:
- establishing EU monographs covering the therapeutic uses and safe conditions of well-established and/or traditional use for herbal substances and preparations;
- drafting an EU list of herbal substances, preparations and combinations thereof for use in traditional herbal medicinal products.
- The HMPC and its working parties and other groups also:
- prepare scientific guidelines and regulatory guidance to help companies prepare marketing authorisation and registration applications for herbal medicines;
- prepare opinions on questions referred to EMA by the national competent authorities regarding the period and evidence of safe use for traditional herbal medicinal products;
- cooperate with the European Directorate for the Quality of Medicines and Healthcare on European Pharmacopoeia standards and EMA guidance on the quality of herbal medicines;
- coordinate with other scientific committees at the Agency on the regulation and safe use of herbal medicines;
- provide scientific and regulatory support to companies researching and developing herbal medicines;
- interact with interested parties;
- provide advice and training to herbal assessors of national competent authorities;
- cooperate with international partners on the harmonisation of regulatory requirements.

Committee for Advanced Therapies (CAT)

- **The Committee for Advanced Therapies (CAT) is the European Medicines Agency's (EMA) committee responsible for assessing the quality, safety and efficacy of advanced therapy medicinal products (ATMPs) and following scientific developments in the field.**
- It was established in accordance with Regulation (EC) No 1394/2007 on ATMPs as a multidisciplinary committee, gathering some of the best available experts in Europe.

Role

- The committee's main responsibility is to prepare a draft opinion on each ATMP application submitted to EMA, before the Committee for Medicinal Products for Human Use (CHMP) adopts a final opinion on the marketing authorisation of the medicine concerned.
- At the request of EMA's Executive Director or the European Commission, the CAT can also draw up an opinion on any scientific matter relating to ATMPs.

The CAT also:

- participates in certifying quality and non-clinical data for small and medium-sized enterprises developing ATMPs;
- participates in providing scientific recommendations on the classification of ATMPs;
- contributes to scientific advice, in cooperation with the Scientific Advice Working Party (SAWP);
- takes part in any procedure delivering advice on the conduct of efficacy follow-up, pharmacovigilance or risk-management systems for ATMPs;
- advises the CHMP on any medicinal product that may require expertise in ATMPs for the evaluation of its quality, safety or efficacy;
- assists scientifically in developing any documents relating to the objectives of the Regulation on ATMPs;
- provides scientific expertise and advice for any Community initiative related to the development of innovative medicines and therapies that requires expertise on ATMPs;
- supports the work programmes of the CHMP working parties.

Paediatric Committee (PDCO)

- **PDCO is the EMA scientific committee responsible for activities on medicines for children and to support the development of such medicines in the European Union by providing scientific expertise and defining paediatric needs.**
- The PDCO was established in line with the Paediatric Regulation, which came into effect in 2007, to improve the health of children in Europe by facilitating the development and availability of medicines for children aged 0 to 17 years.
- **Role**
- The PDCO's main role is to assess the content of paediatric investigation plans (PIPs), which determine the studies that companies must carry out in children when developing a medicine. This includes assessing applications for a full or partial waiver and for deferrals.
- The committee's other roles include:
- assessing data generated in accordance with agreed PIPs;
- adopting opinions on the quality, safety or efficacy of a medicine for use in the paediatric population, at the request of the Committee for Medicinal Products for Human Use (CHMP) or a medicines regulatory authority in a European Union (EU) Member State. The PDCO can give an opinion if the data have been generated in accordance with an agreed PIP;

- 
- advising Member States on the content and format of data to be collected through surveys on the uses of medicines in children;
 - advising and supporting the development of the European Network of Paediatric Research at the European Medicines Agency (Enpr-EMA);
 - providing advice on questions on paediatric medicines, at the request of the Agency's Executive Director or the European Commission;
 - establishing and regularly updating an inventory of paediatric medicine needs;
 - advising the Agency and the European Commission on how to communicate the arrangements available for conducting research into paediatric medicines.
 - The PDCO is not responsible for marketing authorisation applications for medicines for use in children, which is in the remit of the CHMP.

Drug Approval Process

There are two regulatory steps to go through before a drug is approved in the EU

- Clinical trials application
- Marketing Authorization application
 - ❑ CTA is approved at the Member State Level
 - ❑ MAA at the Member state or Centralized levels

Clinical Trial Approval Process

- EU directive 2001/20/EC (April 2001) set out new rules and regulations for the applications and conduct of CT in EU
- A sponsor submits a CT application to the Competent Authority in each Member State where the trials are conducted
 - 60 days to review and approve or reject application
- Application prescribed forms and covers proposed clinical trial protocol, manufacturing and quality drug controls and supporting data
 - Chemical and pharmaceutical and biological data
 - Non-chemical pharmacological and toxicological data
 - Clinical data and previous human experiences
- Supporting data in Common Technical Document (CTD) format
- Qualified personnel to certify that the IP is manufactured according to GMP
- Competent authority will audit and inspect facility for GMP compliance, preclinical for GLP and CT sites for GCP

Investigational Medicinal Product Dossier (IMPD)

- An Investigational Medicinal Product Dossier (IMPD) contains data on the quality, production and control of the medicinal product being researched.
- The IMPD consists of 2 parts:
 - The Quality section with information on the active medicinal product, placebo and reference medicine (if applicable).
 - The Safety and Efficacy section with a summary of data from all clinical and non-clinical studies, with an overall assessment of the risks and benefits. For this part, reference can also be made to the Investigator's Brochure (IB).
- If the IMP is authorised in the EU or has a marketing authorisation in an ICH country, a valid Summary of Product Characteristics (SmPC) at the time of application may be submitted. Table 1, paragraph G, of Annex I to the CTR gives an overview of which data should be included in a simplified IMPD.
- If the IMP has a marketing authorisation (MA) in several Member States concerned with different SmPCs, the sponsor should justify its selection of the most appropriate SmPC.

Marketing authorization (MAA)

<https://www.ema.europa.eu/en/human-regulatory/marketing-authorisation/obtaining-eu-marketing-authorisation-step-step>

- **The European Medicines Agency (EMA) is responsible for the scientific evaluation of centralised marketing authorisation applications (MAA).**
- Once granted by the European Commission, the centralised marketing authorisation is valid in all European Union (EU) Member States, Iceland, Norway and Liechtenstein.
- Following successful CTs, the sponsor may apply for MAA in EU

Submitting an MAA

To submit an MAA, applicant should notify the EMA at least 7 months before the submission to provide an estimate date of submission to the agency. MAA can be filed in the following ways:

Centralized Procedure – For submitting a single MAA

National Procedure – For obtaining market access in specific EU states

Mutual Recognition Procedure – For manufacturers who already have market access in one of the EU member states

Decentralized Procedure – For medicines which are not yet authorized in the EU

MAA is the most important part of placing a medicinal product in the EU market. Failing to submit a compliant MAA may lead to unwanted consequences. Therefore, it is advised to consult a Regulatory expert to ensure successful market-entry.

Marketing authorisation procedures in Europe



National procedure

Assessment time:
210 days

Mutual recognition procedure (MRP)

Assessment time:
90 days + 30 days (national phase)

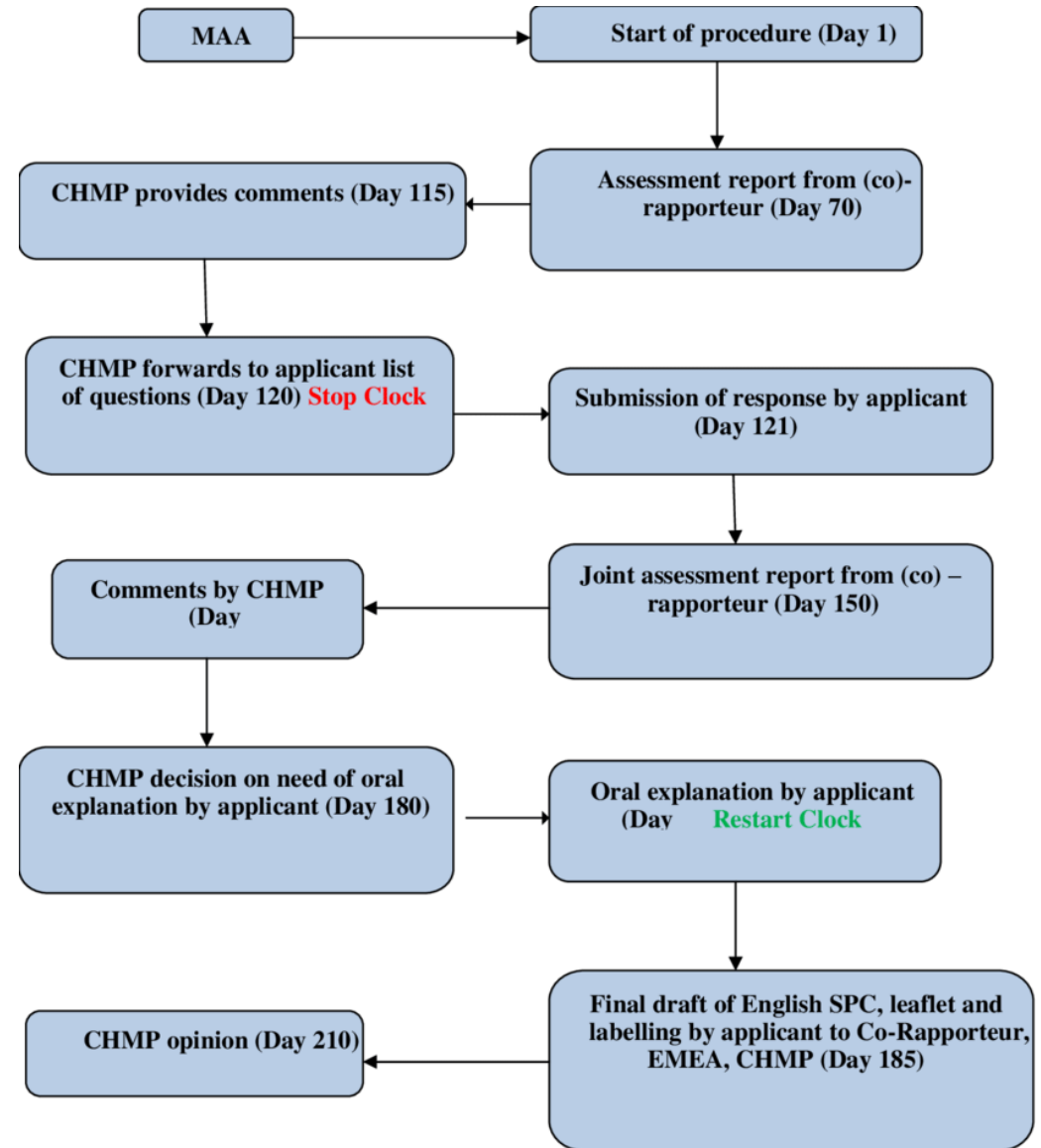
Decentralised procedure (DCP)

Assessment time:
210 days + 30 days (national phase)

Centralised procedure (CP)

Assessment time:
210 days + the time taken by
the European Commission's
decision-making process

Centralized MAA procedure



- The centralised procedure is compulsory for:
- human medicines containing a new active substance to treat:
- human immunodeficiency virus (HIV) or acquired immune deficiency syndrome (AIDS);
- cancer;
- diabetes;
- neurodegenerative diseases;
- auto-immune and other immune dysfunctions;
- viral diseases.
- medicines derived from biotechnology processes, such as genetic engineering;
- advanced-therapy medicines, such as gene-therapy, somatic cell-therapy or tissue-engineered medicines;
- orphan medicines (medicines for rare diseases);
- veterinary medicines for use as growth or yield enhancers.
- It is optional for other medicines:
- containing new active substances for indications other than those stated above;
- that are a significant therapeutic, scientific or technical innovation;
- whose authorisation would be in the interest of public or animal health at EU level.
- Today, the great majority of new, innovative medicines pass through the centralised authorisation procedure in order to be marketed in the EU.

National authorisation procedures

- While the majority of new, innovative medicines are evaluated by EMA and authorised by the European Commission in order to be marketed in the EU, most generic medicines and medicines available without a prescription are assessed and authorised at national level in the EU.
- In addition, many older medicines available today were authorised at national level because they were marketed before EMA was created.
- Most Member States maintain National registers of authorised medicines.
- Each EU Member State has its own national authorisation procedures. Information about these can normally be found on the websites of the national competent authorities:
- National competent authorities (human)
- National competent authorities (veterinary)
- If a company wishes to request marketing authorisation in several EU Member States for a medicine that is outside the scope of the centralised procedure, it may use one of the following routes:
- mutual-recognition procedure, whereby a marketing authorisation granted in one Member State can be recognised in other EU countries;
- decentralised procedure, whereby a medicine that has not yet been authorised in the EU can be simultaneously authorised in several EU Member States.

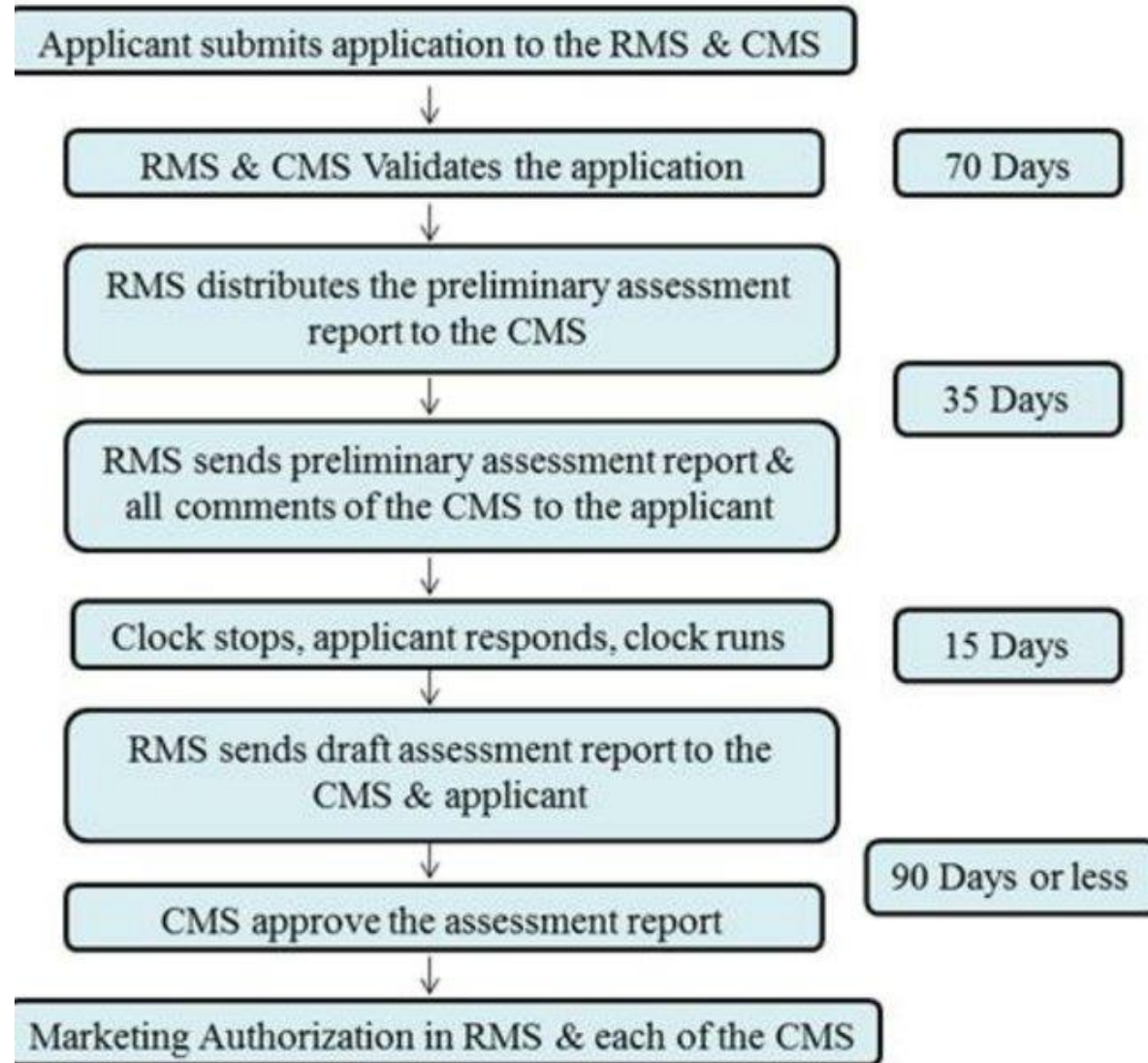
Mutual recognition procedure

- The mutual recognition procedure (MRP) is a European authorisation route resulting in a mutually recognised product.
- Mutual recognition must be used when a product is already authorised in at least one Member State on a national basis and the Marketing Authorisation Holder wishes to obtain a Marketing Authorisation (MA) for the same product in at least one other Member State.
- The Member State that has already authorised the product is known as the Reference Member State (RMS). The RMS submits their evaluation of the product to other Member State/s, these are known as Concerned Member States (CMS). The CMS is asked to mutually recognise the MA of the RMS.
- If the applicant is successful, the CMS will then issue a MA for that product permitting the marketing of that product in their country.

Decentralised procedure

- The decentralised procedure (DCP) is a European authorisation route resulting in a mutually recognised product (MRP).
- The difference between MRP and DCP is that a product must already be authorised in at least one Member State on a national basis in order for MRP to be used. DCP may be used if the product is not already authorised in any Member State, but does not want to use the centralised procedure, or the product is not eligible for the centralised procedure.
- One of the proposed Member States will be asked by the applicant company to act as Reference Member State (RMS). The RMS does the initial evaluation of the product and issues a draft assessment report. The other Member States, known as the Concerned Member States (CMS), either agree with the RMS's evaluation or they ask further questions/raise objections.
- If all the issues are resolved and the application is successful, each Member State will then issue a MA for that product permitting it to be marketed in their country.

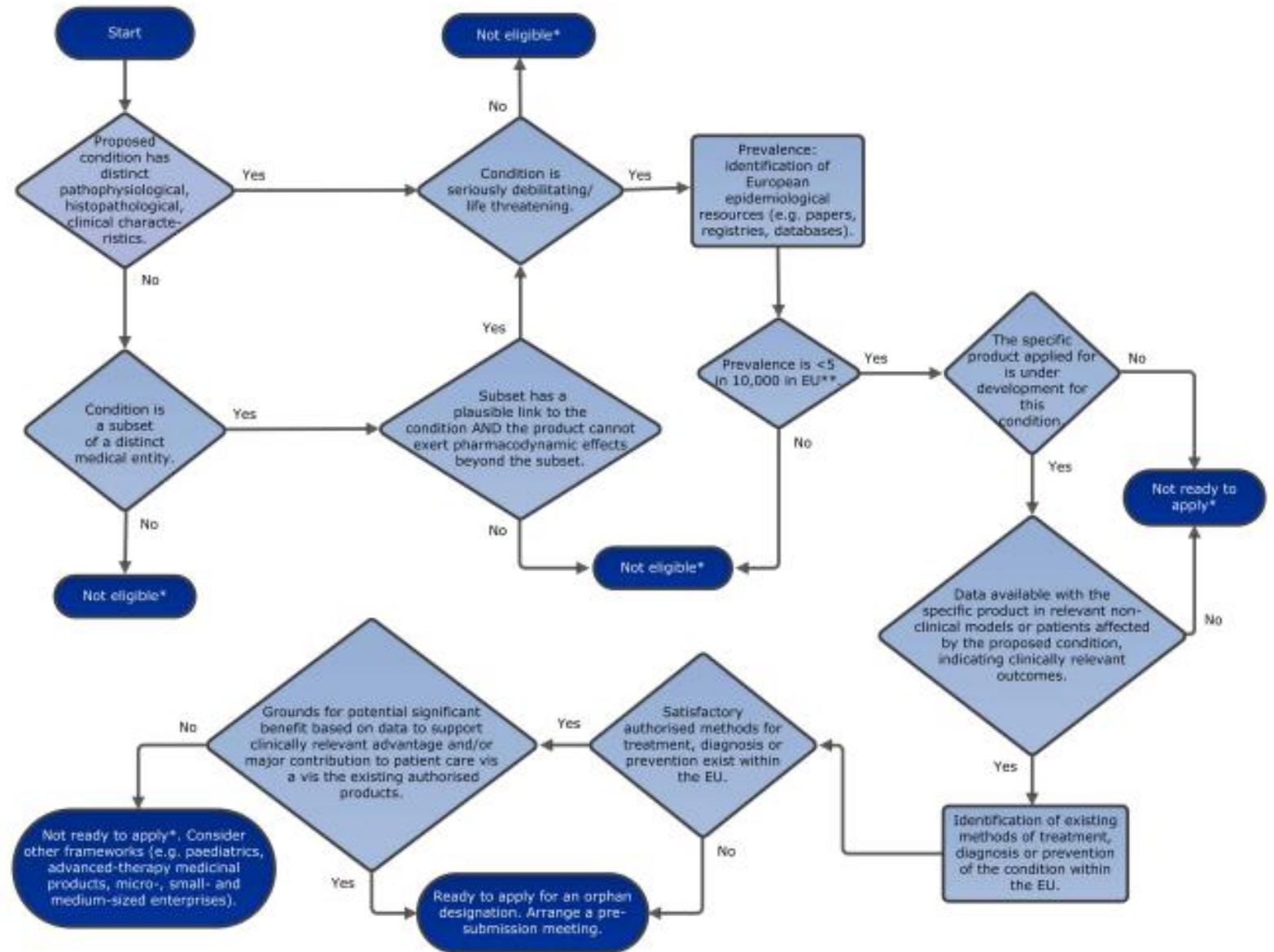
Decentralised procedure



Fees payable to EMA

Fee type	Human medicines	Veterinary medicines
Marketing-authorisation application (single strength, one pharmaceutical form, one presentation)	From €278,800	From €139,600
Extension of marketing authorisation (level I)	€83,700	€34,900
Type-II variation (major variation)	€83,700	€41,800
Scientific advice	From €41,800 to €83,700	From €13,800 to 41,800
Annual fee (level I)	€100,000	€33,400

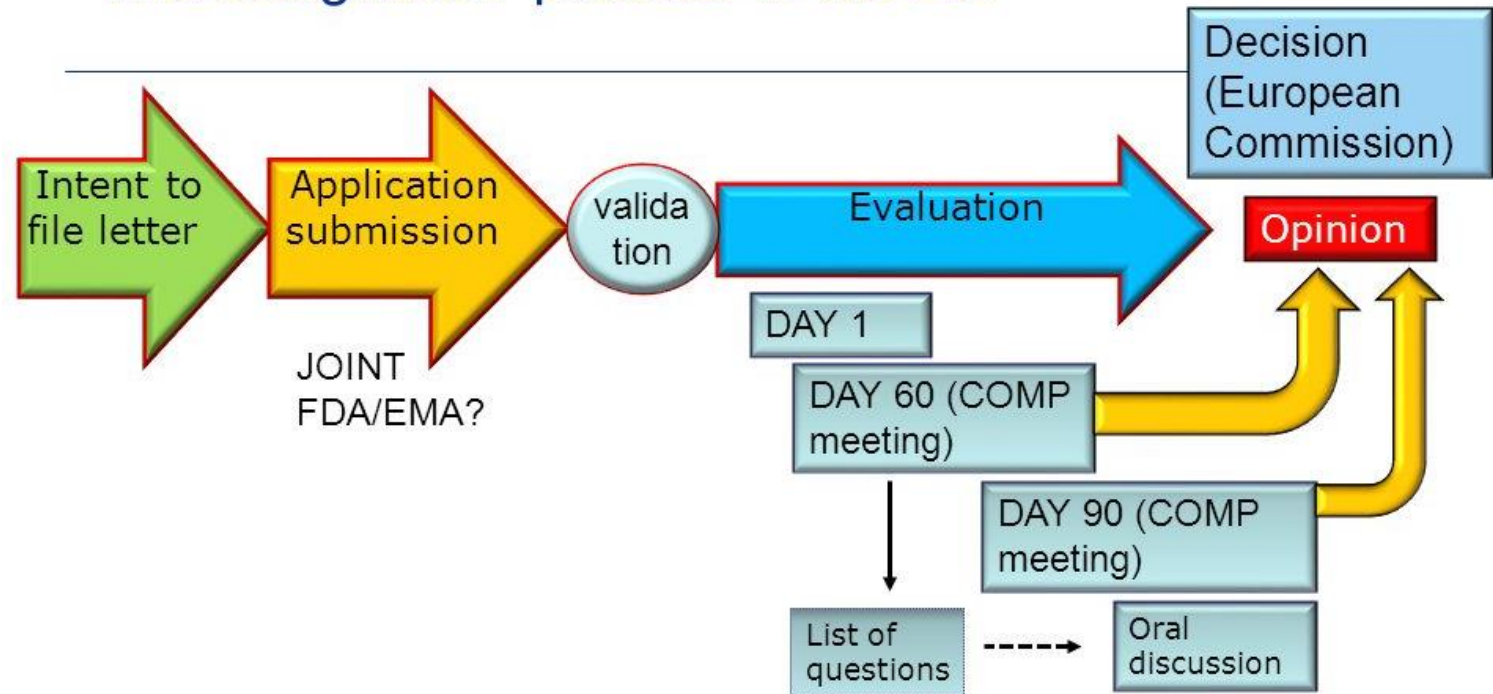
Orphan Drug procedure



- **Sponsors should follow one of the two options below to submit an application for orphan designation:**
- Submit directly an application to EMA, through the IRIS system:
- Pre-submission meetings are not mandatory and sponsors are welcome to send an application for orphan drug designation without notice.
- However, EMA would appreciate it if sponsors could send the application a few days before any of the published submission deadlines available on the EMA website to allow more time for the validation process and the possibility to intervene in case of technical problems.
- **Request a pre-submission meeting/teleconference:**
- EMA strongly encourages sponsors to request a pre-submission meeting with the Agency prior to filing an application.
- If a sponsor feels they could benefit from a preliminary discussion before the submission of an orphan drug application, they can request a pre-submission meeting/teleconference at least two months prior to their planned submission date via the IRIS portal.
- Sponsors should create an initial draft application for orphan designation before creating the application for the pre-submission discussion. These are two separate submissions to be made in the IRIS system. The draft submission for orphan designation should be populated with the relevant data and documents, but not submitted, at least one week before the pre-submission meeting date.
- Pre-submission meetings are useful since the evaluation process has a fixed duration of 90 days and cannot be lengthened to accommodate for the lack of data or other omissions in the application. Experience has shown that they have a positive impact on the success rate of the applications.

Evaluation of orphan drug

The designation process in the EU



Herbal MAA procedure

- The Herbal Directive (Directive 2004/24/EC) was adopted to facilitate the placing on the EU market of traditional herbal medicinal products.
- In particular, herbal products marketed in the form of food supplements should comply with Directive 2002/46/EC on food supplements and Regulation (EC) No 1924/2006 on nutrition and health claims made on foods.

SIMPLIFIED PROCEDURE:-

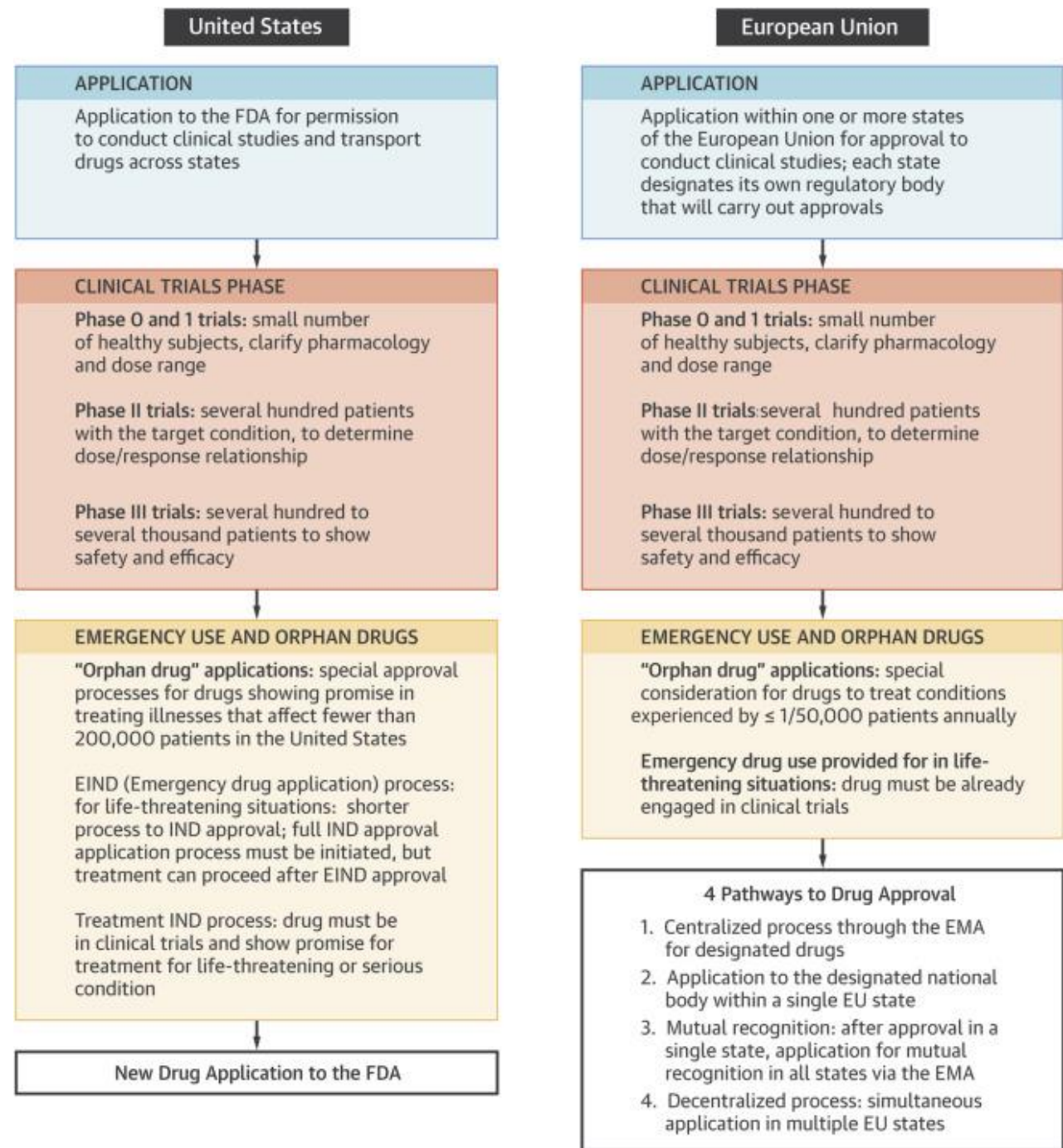
- The simplified procedure allows the registration of traditional herbal medicinal products, including Chinese or Ayurvedic herbal medicinal products or herbal medicinal products from any other tradition, without requiring tests and trials on safety and efficacy, which the applicant is normally obliged to provide.
- Instead for registration of traditional herbal medicinal products, the applicant has to only provide sufficient evidence of the medicinal use of the product throughout a period of at least 30 years, including at least 15 years in the European Union.

Active substance master file procedure – Scientific guideline

- The main objective of the Active Substance Master File (ASMF) procedure, formerly known as the European Drug Master File (EDMF) procedure, is to allow valuable confidential intellectual property or 'know-how' of the manufacturer of the active substance (ASM) to be protected,
- While at the same time allowing the Applicant or Marketing Authorisation (MA) holder to take full responsibility for the medicinal product and the quality and quality control of the active substance. National Competent Authorities/EMA thus have access to the complete information that is necessary for an evaluation of the suitability of the use of the active substance in the medicinal product

Difference between FDA and EMA

<https://www.sciencedirect.com/science/article/pii/S2452302X16300638>

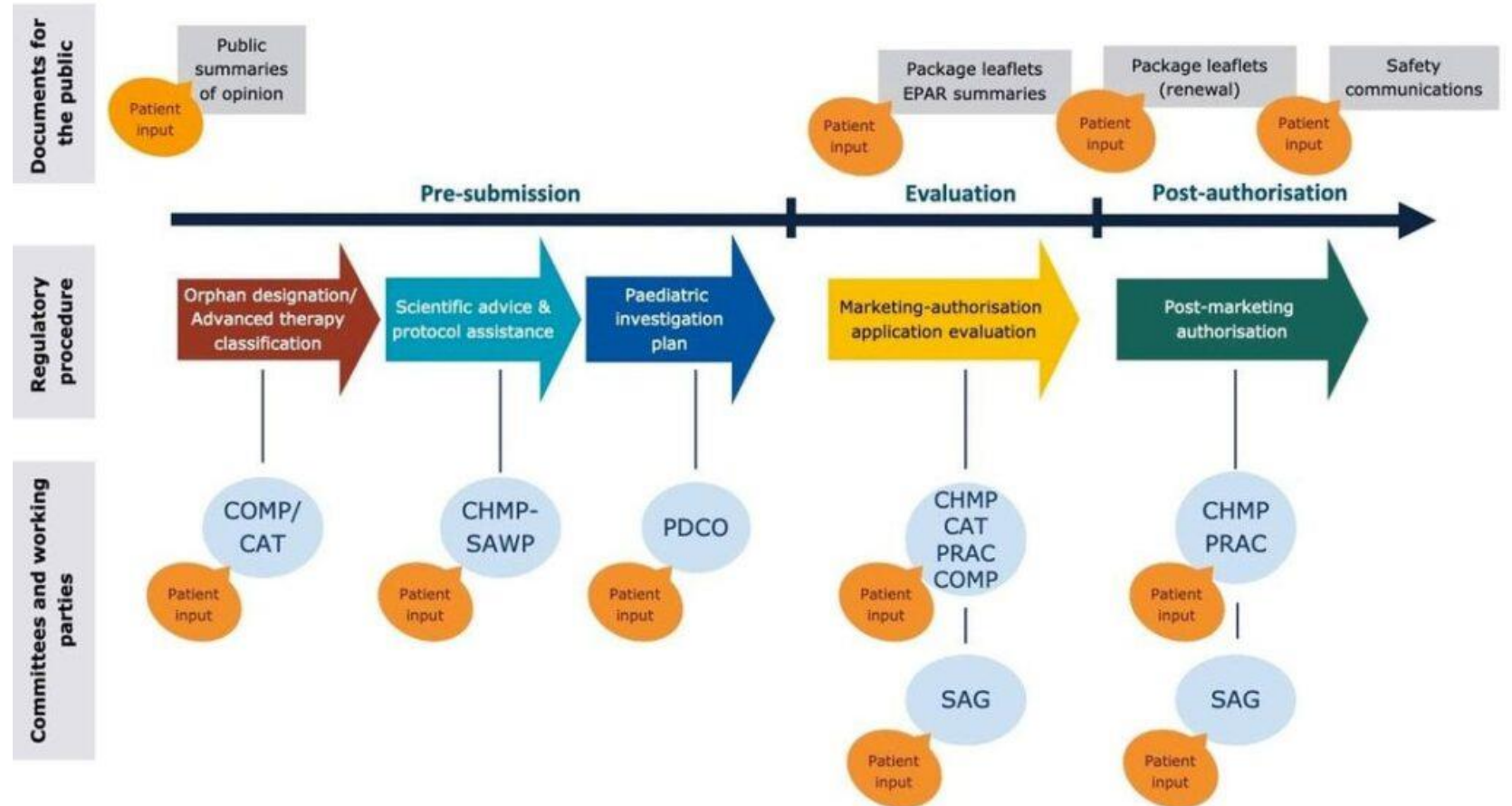


Patient involvement in Medicine lifecycle in EMA



EUROPEAN MEDICINES AGENCY
SCIENCE MEDICINES HEALTH

Overview of patient involvement along the medicines lifecycle at EMA



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COMP: Committee for Orphan Medicinal Products
CHMP: Committee for Medicinal Products for Human Use
EPAR: European Public Assessment Report

PRAC: Pharmacovigilance Risk Assessment Committee
CAT: Committee for Advanced Therapies
PDCO: Paediatric Committee

SAWP: Scientific Advice Working Party
SAG: Scientific Advisory Group
HCP: Healthcare professionals

Conclusion

- The European Medicines Agency (EMA) was formed in 1995 with funding from the EU, pharmaceutical industry, and member states.
- The EMA was charged with harmonizing processes in the member state regulatory agencies to reduce annual costs to drug companies (that previously were required to obtain separate approvals in each member state) as well as to eliminate competition-restricting regulation in sovereign states.
- However, the EMA does not oversee all drug approvals the way the FDA does in the United States.
- In Europe, there are 4 routes by which a drug can be approved, depending on the drug class and manufacturer preference
- The regulation of medical drugs and devices involves competing goals of assuring safety and efficacy while providing rapid movement of innovative therapies through the investigative and regulatory processes as quickly as possible.



U.S. Department of Health and Human Services
Food and Drug Administration



INTRODUCTION

- The Food and Drug Administration (FDA or USFDA) is a government agency of the United States Department of Health and Human Services.
- The FDA is responsible for regulating and supervising the safety of foods, tobacco products, dietary supplements, prescription and non-prescription medication, vaccines, biopharmaceuticals, blood transfusions, medical devices, electromagnetic radiation emitting devices (ERED), veterinary products, and cosmetics.
- The FDA also enforces other laws, notably Section 361 of the Public Health Service Act and the associated regulations. Many of these regulations are not directly related to food or drugs.
- These include sanitation requirements on interstate travel and control of disease on products ranging from certain household pets to sperm donation for assisted reproduction

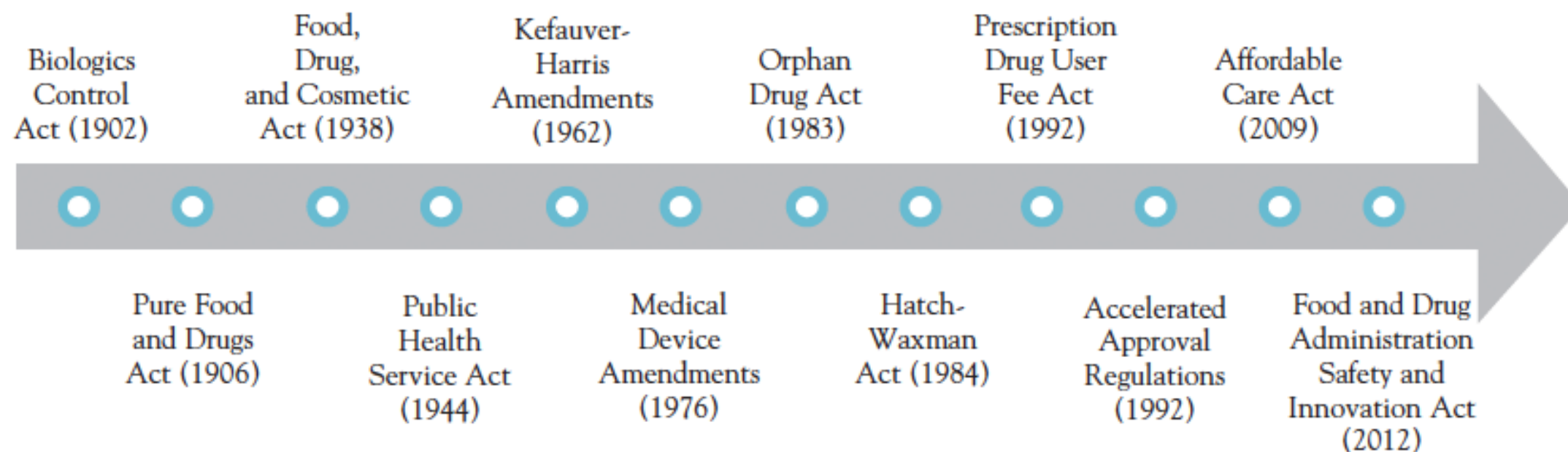


HISTORY

- **The Pure Food and Drug Act of 1906** was the first of more than 200 laws that constitute one of the world's most comprehensive and effective networks of public health and consumer protections. Here are a few of the congressional milestones:
- **The Federal Food, Drug, and Cosmetic Act of 1938** was passed after a legally marketed toxic elixir killed 107 people, including many children.
- The FD&C Act completely overhauled the public health system. Among other provisions, the law authorized the FDA to demand evidence of safety for new drugs, issue standards for food, and conduct factory inspections.
- **The Kefauver-Harris Amendments of 1962**, which were inspired by the thalidomide tragedy in Europe (and the FDA's vigilance that prevented the drug's marketing in the United States), strengthened the rules for drug safety and required manufacturers to prove their drugs' effectiveness.
- **The Medical Device Amendments of 1976** followed a U.S. Senate finding that faulty medical devices had caused 10,000 injuries, including 731 deaths. The law applied safety and effectiveness safeguards to new devices.
- **The Family Smoking Prevention and Tobacco Control Act** (Tobacco Control Act) was enacted in 2009, giving FDA authority to regulate the manufacture, distribution, and marketing of tobacco products



Figure History of the FDA



FUNDING

- The FDA regulates more than \$1 trillion worth of consumer goods, about 25% of consumer expenditures in the United States.
- This includes \$466 billion in food sales, \$275 billion in drugs, \$60 billion in cosmetics and \$18 billion in vitamin supplements. Much of the expenditures is for goods imported into the United States; the FDA is responsible for monitoring a third of all imports

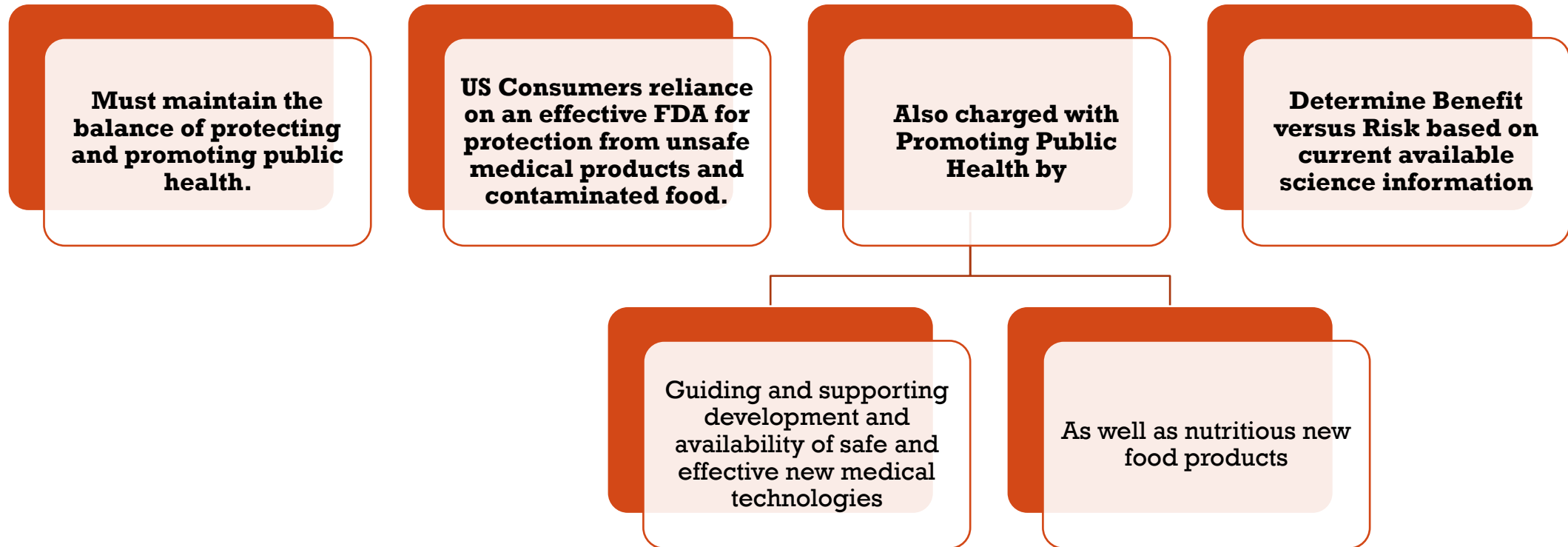


LEGAL AUTHORITY

- Most federal laws concerning the FDA are part of the **Food, Drug and Cosmetic Act**, (first passed in 1938 and extensively amended since) and are codified in **Title 21, Chapter 9 of the United States Code**.
- Other significant laws enforced by the FDA include the **Public Health Service Act**, **parts of the Controlled Substances Act**, the , as well as many others. in many cases these responsibilities are shared with other federal agencies.



CHALLENGES AND OPPORTUNITIES FACING FDA



ORGANIZATION

- FDA is an agency within the **Department of Health and Human Services**.
- Effective March 31, 2019, FDA began operational implementation of an agency reorganization.
- FDA's reorganization reflects the agency's commitment to modernizing its structure to advance its mission to protect and promote public health, and to meet the challenges of rapid innovation across the industries regulated by FDA.
- The FDA's reorganization will realign several entities across the agency to promote strategic priorities, and will elevate the role of the centers, offices and field forces.
- FDA consists of 6 product centers, one research centre, and two offices-
- Center for Biologics Evaluation and Research (CBER)- CBER is the Center within FDA that regulates biological products for human use under applicable federal laws, including the Public Health Service Act and the Federal Food, Drug and Cosmetic Act. CBER protects and advances the public health by ensuring that biological products are safe and effective and available to those who need them. CBER also provides the public with information to promote the safe and appropriate use of biological products.

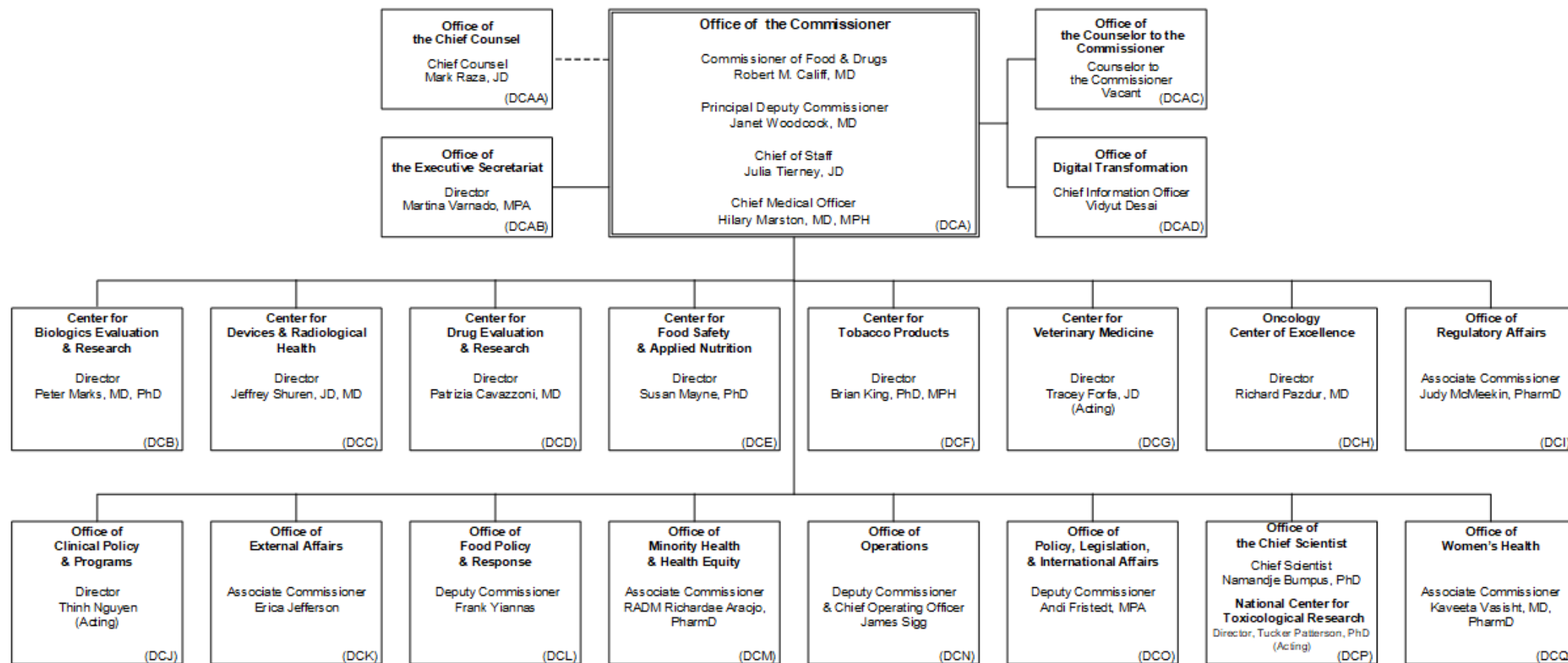


- Center for Devices and Radiological Health-- the Center for Devices and Radiological Health (CDRH) is responsible for protecting and promoting the public health. We assure that patients and providers have timely and continued access to safe, effective, and high-quality medical devices and safe radiation-emitting products.
- Center for Drug Evaluation and Research (CDER)--The Center for Drug Evaluation and Research (CDER) performs an essential public health task by making sure that safe and effective drugs are available to improve the health of people in the United States.
- As part of the U.S. Food and Drug Administration (FDA), CDER regulates over-the-counter and prescription drugs, including biological therapeutics and generic drugs.
- Center for Food Safety and Applied Nutrition (CFSAN)-- carry out the mission of the FDA. It provides services to consumers, domestic and foreign industry and other outside groups regarding field programs; agency administrative tasks; scientific analysis and support; and policy, planning and handling of critical issues related to food and cosmetics
- Center for Tobacco Products-- oversees the implementation of the Family Smoking Prevention and Tobacco Control Act.
- Center for Veterinary Medicine (CVM)-- Protecting Human and Animal Health.
- Office of Regulatory Affairs (ORA)-- inspects regulated products and manufacturers, conducts sample analyses of regulated products and reviews imported products offered for entry into the United States.
- Oncology Center of Excellence (OCE)--unites experts across the FDA to conduct expedited review of medical products for oncologic and hematologic malignancies



**Department of Health and Human Services
Food and Drug Administration**

January 2023



Legend:

--- Direct report to DHHS General Counsel





FDA'S REGULATORY RESPONSIBILITIES: LAWS AND REGULATIONS



MISSION

- The Food and Drug Administration is responsible for protecting the public health by ensuring the safety, efficacy, and security of human and veterinary drugs, biological products, and medical devices; and by ensuring the safety of our nation's food supply, cosmetics, and products that emit radiation.
- FDA also has responsibility for regulating the manufacturing, marketing, and distribution of tobacco products to protect the public health and to reduce tobacco use by minors.
- FDA is responsible for advancing the public health by helping to speed innovations that make medical products more effective, safer, and more affordable and by helping the public get the accurate, science-based information they need to use medical products and foods to maintain and improve their health.
- FDA also plays a significant role in the Nation's counterterrorism capability. FDA fulfills this responsibility by ensuring the security of the food supply and by fostering development of medical products to respond to deliberate and naturally emerging public health threats.



WHAT DOES FDA REGULATE?

- The scope of FDA's regulatory authority is very broad. FDA's responsibilities are closely related to those of several other government agencies.

In general, FDA regulates:

- **Foods, including:**
 - dietary supplements
 - bottled water
 - food additives
 - infant formulas
- other food products (although the U.S. Department of Agriculture plays a lead role in regulating aspects of some meat, poultry, and egg products)



▪ **Drugs, including:**

- prescription drugs (both brand-name and generic)
- non-prescription (over-the-counter) drugs

▪ **Biologics, including:**

- vaccines for humans
- blood and blood products
- cellular and gene therapy products
- tissue and tissue products
- Allergens

▪ **Medical Devices, including:**

- simple items like tongue depressors and bedpans
- complex technologies such as heart pacemakers
- dental devices

- surgical implants and prosthetics

▪ **Electronic Products that give off radiation, including:**

- microwave ovens
- x-ray equipment
- laser products
- ultrasonic therapy equipment
- mercury vapor lamps
- Sunlamps

▪ **Cosmetics, including:**

- color additives found in makeup and other personal care products
- skin moisturizers and cleansers
- nail polish and perfume

▪ **Veterinary Products, including:**

- livestock feeds
- pet foods
- veterinary drugs and devices

▪ **Tobacco Products, including:**

- cigarettes
- cigarette tobacco
- roll-your-own tobacco
- smokeless tobacco
- Cigars
- Hookah
- E-cigarettes (tobacco and non-tobacco derived nicotine)



PRODUCT APPROVAL

- Approval information by product type

Drugs

- Human drugs and therapeutic biologicals (proteins and other products derived from living sources used for therapeutic purposes)
- Development & Approval Process-- The main consumer watchdog in this system is FDA's Center for Drug Evaluation and Research (CDER).
- The center's best-known job is to evaluate new drugs before they can be sold. CDER's evaluation not only prevents quackery, but also provides doctors and patients the information they need to use medicines wisely. The center ensures that drugs, both brand-name and generic, work correctly and that their health benefits outweigh their known risks.
- FDA approval of a drug means that data on the drug's effects have been reviewed by CDER, and the drug is determined to provide benefits that outweigh its known and potential risks for the intended population.
 - Analysis of the target condition and available treatments—FDA reviewers analyze the condition or illness for which the drug is intended and evaluate the current treatment landscape, which provide the context for weighing the drug's risks and benefits.



- 
- **Assessment of benefits and risks from clinical data**—FDA reviewers evaluate clinical benefit and risk information submitted by the drug maker, taking into account any uncertainties that may result from imperfect or incomplete data.
 - **Strategies for managing risks**—All drugs have risks. Risk management strategies include an FDA-approved drug label, which clearly describes the drug's benefits and risks, and how the risks can be detected and managed. Sometimes, more effort is needed to manage risks. In these cases, a drug maker may need to implement a Risk Management and Mitigation Strategy (REMS)



RECALLS, MARKET WITHDRAWALS, & SAFETY ALERTS

- A drug recall is the most effective way to protect the public from a defective or potentially harmful product. A recall is a voluntary action taken by a company to remove a defective drug product from the market. Drug recalls may be conducted on a company's own initiative or by FDA request. FDA's role in a recall is to oversee a company's strategy, assess the adequacy of the recall and classify the recall.
- Alerting the Public
- Not all recalls are announced on FDA.gov or in the news media.
- Public notification is generally issued when a product that has been widely distributed or poses a serious health hazard is recalled.
- If a company does not issue public notification of a recall, FDA may do so if the agency determines it is necessary to protect patients. Patients also may learn that their medicine has been recalled through notification from the manufacturer, their health care professional or pharmacist.



- **Weekly Enforcement Report**
- **All recalls are posted weekly in the FDA enforcement report. Recalls that are classified will have a classification of Class I, Class II or Class III based on the level of hazard.**
 - **Recall Classification**
 - **Class I: A dangerous or defective product that could cause serious health problems or death.**
 - **Class II: A product that might cause a temporary health problem, or pose slight threat of a serious nature.**
 - **Class III: A products that is unlikely to cause any adverse health reaction, but that violates FDA labeling or manufacturing laws**
- **Determining the Effectiveness of the Recall**
 - **FDA evaluates the effectiveness of a recall by evaluating a company's efforts to properly notify customers and remove the defective product from the market. If a recall is determined to be ineffective FDA will request the company take additional actions.**



PAGES TO REFER

- **Drugs:** Additional safety information about human medical products can be found on FDA's **MedWatch** page.
- **Animal Health:** Additional safety information about products affecting animal health can be found on **FDA's Animal & Veterinary Recalls & Withdrawals** page.
- **Biologics:** A more complete listing of Biologic Recalls and Market Withdrawal information about can be found on **FDA's Biologics Recalls** page.
- **Medical Devices:** A more complete listing of Medical Device Recalls can be found on **FDA's Medical Device Recalls** page.
- **Cosmetics:** Additional safety information about cosmetic products can be found at **FDA's Cosmetics Recalls and Alert** page.



FDA REGULATIONS: 21 CFR, FOOD AND DRUGS

- The Code of Federal Regulations (CFR) annual edition is the codification of the general and permanent rules published in the Federal Register by the departments and agencies of the Federal Government.
- It is divided into 50 titles that represent broad areas subject to Federal regulation. The 50 subject matter titles contain one or more individual volumes, which are updated once each calendar year, on a staggered basis.
- The annual update cycle is as follows: titles 1-16 are revised as of January 1; titles 17-27 are revised as of April 1; titles 28-41 are revised as of July 1; and titles 42-50 are revised as of October 1.
- Each title is divided into chapters, which usually bear the name of the issuing agency. Each chapter is further subdivided into parts that cover specific regulatory areas. Large parts may be subdivided into subparts.
- All parts are organized in sections, and most citations to the CFR refer to material at the section level.



▼ Title 21	Food and Drugs	Part / Section
▼ Chapter I	Food and Drug Administration, Department of Health and Human Services	1 – 1299
Subchapter A	General	1 – 99
Subchapter B	Food for Human Consumption	100 – 199
Subchapter C	Drugs: General	200 – 299
Subchapter D	Drugs for Human Use	300 – 499
Subchapter E	Animal Drugs, Feeds, and Related Products	500 – 599
Subchapter F	Biologics	600 – 680
Subchapter G	Cosmetics	700 – 799
Subchapter H	Medical Devices	800 – 898
Subchapter I	Mammography Quality Standards Act	900
Subchapter J	Radiological Health	1000 – 1040
Subchapter K	Tobacco Products	1100 – 1150
Subchapter L	Regulations Under Certain Other Acts Administered by the Food and Drug Administration	1210 – 1299
► Chapter II	Drug Enforcement Administration, Department of Justice	1300 – 1399
▼ Chapter III	Office of National Drug Control Policy	1400 – 1499
	<i>Part 1400 [Reserved]</i>	
	Part 1401 Public Availability of Information	1401.1 – 1401.24
	Part 1402 Mandatory Declassification Review	1402.1 – 1402.7
	<i>Parts 1403-1499 [Reserved]</i>	



WHAT DOES THE FDA NOT REGULATE?

- **Advertising**

- The Federal Trade Commission is a federal agency that regulates many types of advertising. The FTC protects consumers by stopping unfair, deceptive or fraudulent practices in the marketplace.

- **Alcohol**

- The Department of the Treasury's Alcohol and Tobacco Tax and Trade Bureau (TTB) regulates aspects of alcohol production, importation, wholesale distribution, labeling, and advertising

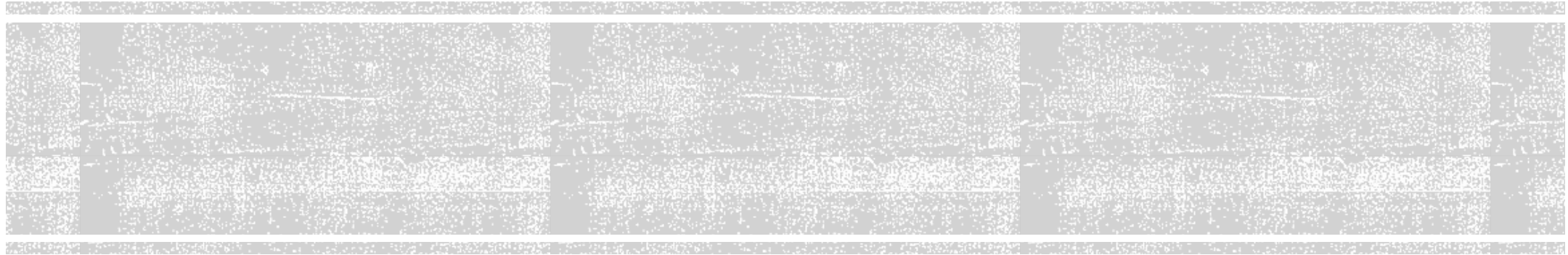
- **Consumer Products**

- The Consumer Product Safety Commission (CPSC) works to ensure the safety of consumer products such as toys, cribs, power tools, cigarette lighters, household chemicals, and other products that pose a fire, electrical, chemical or mechanical hazard

- **Drugs of Abuse**

- The Department of Justice's Drug Enforcement Administration (DEA) works to enforce the controlled substances laws and regulations of the United States, including as they pertain to the manufacture, distribution, and dispensing of legally produced controlled substances





- **Meat and Poultry**

- The U.S. Department of Agriculture's Food Safety and Inspection Service regulates aspects of the safety and labeling of traditional (non-game) meats, poultry, and certain egg products.

- **Pesticides**

- The Environmental Protection Agency (EPA) regulates many aspects of pesticides. EPA sets limits on how much of a pesticide may be used on food during growing and processing, and how much can remain on the food you buy

- **Vaccines for Animal Diseases**

- The U.S. Department of Agriculture's Animal and Plant Health Inspection Service (APHIS), Center for Veterinary Biologics, regulates aspects of veterinary vaccines and other types of veterinary biologics

- **Water**

- The Environmental Protection Agency (EPA) regulates aspects of drinking water. EPA develops national standards for drinking water from municipal water supplies (tap water) to limit the levels of impurities.



EMPLOYMENT IN THE FDA

	FY 2018 Actuals			FY 2019 Estimate			FY 2020 Estimate		
	Civilian	Military	Total	Civilian	Military	Total	Civilian	Military	Total
Center for Food Safety and Applied Nutrition	1,069	36	1,105	1,114	36	1,150	1,194	36	1,230
Center for Drug Evaluation and Research	4,697	506	5,203	4,996	506	5,502	5,118	506	5,624
Center for Biologics Evaluation and Research	1,142	60	1,202	1,103	60	1,163	1,131	60	1,191
Center for Veterinary Medicine	608	12	620	620	12	632	647	12	659
Center for Devices and Radiological Health	1,652	85	1,737	1,731	85	1,816	1,802	85	1,887
National Center for Toxicological Research	307	---	307	301	---	301	301	---	301
Office of Regulatory Affairs	4,449	338	4,787	4,601	338	4,939	4,659	338	4,997
Headquarters and Office of the Commissioner	1,094	75	1,169	926	75	1,001	943	75	1,018
Export Certification	22	---	22	26	---	26	26	---	26
Color Certification	36	---	36	37	---	37	37	---	37
Family Smoking Prevention and Tobacco Control Act....	773	36	809	896	36	932	956	36	992
Priority Review Vouchers (PRV) Pediatric Disease	---	---	---	---	---	---	---	---	---
MCMi - No Year.....	---	---	---	---	---	---	---	---	---
Opiods - No Year.....	---	---	---	8	---	8	---	---	---
21st Century Cures (BA Only).....	26	---	26	100	---	100	100	---	100
Total.....	15,875	1,148	17,023	16,459	1,148	17,607	16,914	1,148	18,062

Five Year History of GS/GM Average Grade

Year	Grade
FY 2016	13
FY 2017	13
FY 2018	13
FY 2019	13
FY 2020	13



WHAT IS THE DIFFERENCE BETWEEN THE FEDERAL FOOD, DRUG, AND COSMETIC ACT (FD&C ACT), FDA REGULATIONS, AND FDA GUIDANCE?

- The Federal Food, Drug, and Cosmetic Act (FD&C Act) is a federal law enacted by Congress. It and other federal laws establish the legal framework within which FDA operates. The FD&C Act can be found in the United States Code, which contains all general and permanent U.S. laws, beginning at 21 U.S.C. 301.
- FDA develops regulations based on the laws set forth in the FD&C Act or other laws under which FDA operates. FDA follows the procedures required by the Administrative Procedure Act, another federal law, to issue FDA regulations. This typically involves a process known as "notice and comment rulemaking" that allows for public input on a proposed regulation before FDA issues a final regulation. FDA regulations are also federal laws, but they are not part of the FD&C Act. FDA regulations can be found in Title 21 of the Code of Federal Regulations (CFR).
- FDA follows the procedures required by its "Good Guidance Practice" regulation to issue FDA guidance. FDA guidance describes the agency's current thinking on a regulatory issue. Guidance is not legally binding on the public or FDA. The Good Guidance Practice regulation can be found at 21 CFR 10.115.



FDA-REGULATED PRODUCTS AND FACILITIES

- FDA is responsible for the oversight of more than \$2.7 trillion in consumption of food, medical products, and tobacco.
- FDA-regulated products account for about 20 cents of every dollar spent by U.S. consumers.
- FDA regulates about 78 percent of the U.S. food supply. This includes everything we eat except for meat, poultry, and some egg products.
- There are over 20,000 prescription drug products approved for marketing.
- FDA oversees over 6,700 different medical device product categories.
- There are over 1,600 FDA-approved animal drug products.
- There are about 621 FDA-licensed biologics products.
- FDA oversees over 100,000 tobacco products, not including ENDS products.



FDA-REGISTERED FACILITIES

FDA-Registered Facilities

Program	Domestic	Foreign	Total
Animal Drugs	1,858	1,103	2,961
Animal Food	17,523	7,394	24,917
Biologics	4,944	556	5,500
Human Drugs	6,799	6,318	13,117
Human Food	81,588	116,344	197,932
Medical Devices	13,010	12,891	25,901
Tobacco	2,391	0	2,391
Total	128,113	144,606	272,719

- FDA regulations cover about 35,000 produce farms, 300,000 restaurant chain establishments, and 10,500 vending machine operators.
- FDA products are manufactured or handled at nearly 275,000 registered facilities, more than half of which are overseas.
- About 78 percent of active pharmaceutical ingredients manufacturers are located outside of the U.S.
- FDA-regulated products account for 15 percent of U.S. imports and 15 percent of U.S. exports.



FDA IMPORTS

U.S. Imports and Exports by FDA Program

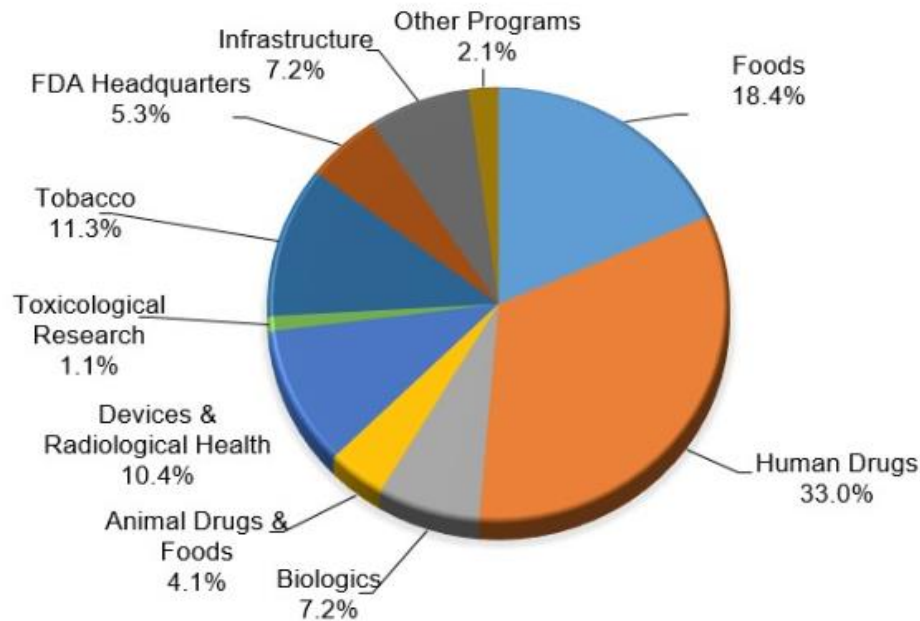
Program	Imports (\$Billions)	Exports (\$Billions)
Animal Food	\$2.79	\$5.49
Biologics	\$42.73	\$25.42
Human and Animal Drugs	\$93.62	\$31.67
Human Food and Cosmetics	\$158.90	\$122.90
Medical Devices	\$68.01	\$58.69
Tobacco	\$1.58	\$1.05
FDA Total	\$367.63	\$245.22
USA Total	\$2,493.74	\$1,642.82

- About 84 percent of biologics sales are imports.
- About 57 percent of fresh fruit, 34 percent of vegetables, and 93 percent of seafood consumption by volume are imports.
- About 44 percent of medical devices used in this country are imports.
- About 5 percent of animal product sales are imports.
- About 3 percent of tobacco product sales are imports.



FY 2021 FDA Budget by Program (Total = \$6.1 billion)

FY 2021 FDA Budget by Program (Total = \$6.1 billion)



Notes: Infrastructure includes rent, rent related activities, FDA buildings and facilities, and White Oak consolidation. Other programs includes Export Certification and Color Certification Fund.

PROGRAM FUNDING

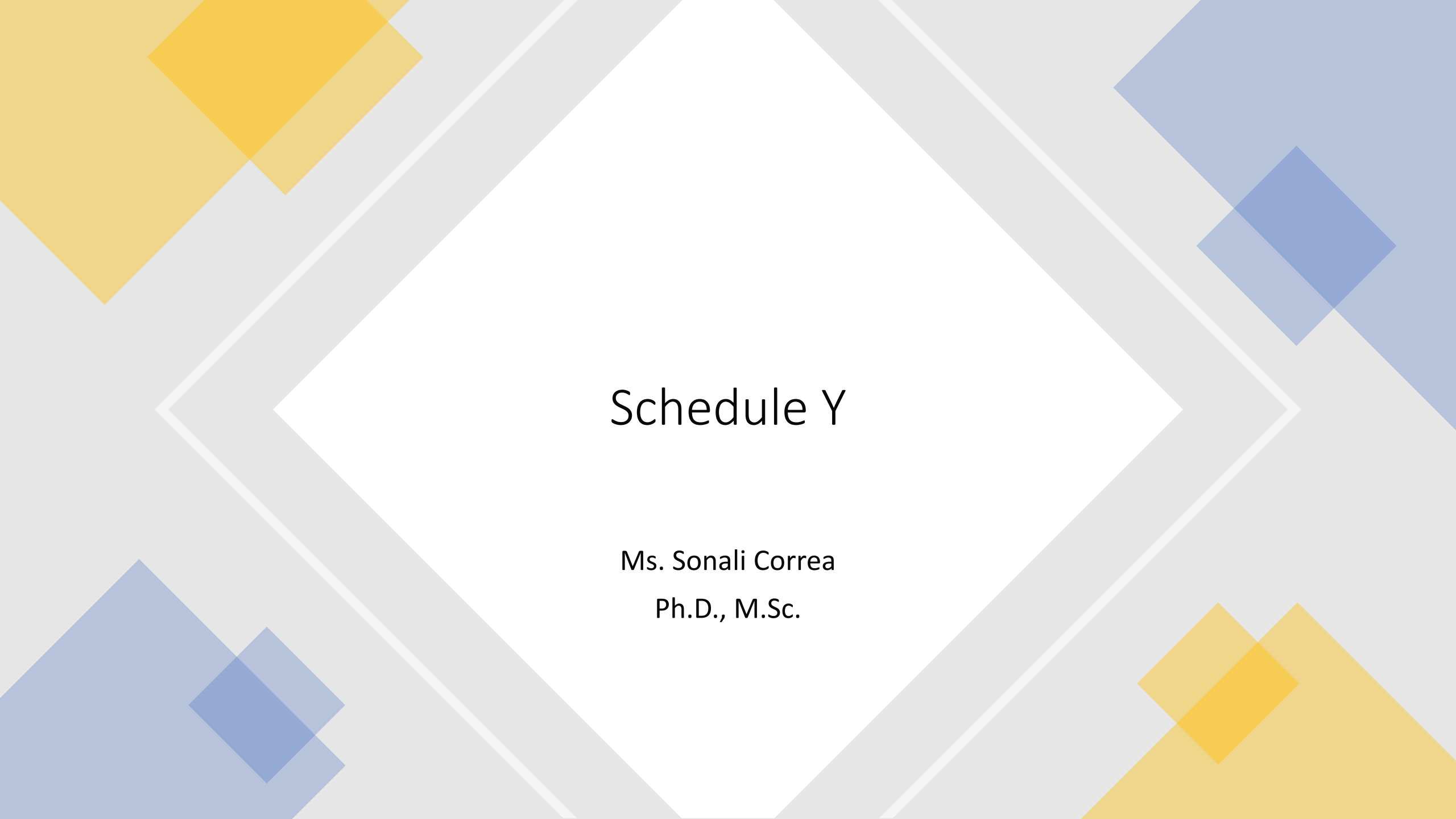
- FDA's budget for Fiscal Year 2021 is:
- approximately \$6.1 billion.
- provided by federal budget authorization (54%, or \$3.3 billion) and user industry fees (46%, or \$2.8 billion).
- equivalent to \$10.01 per American per year and includes 18,062 full time equivalents (FTEs). The share of a program's funding paid by industry user fees varies from about 1% for Foods to 100% of Tobacco regulatory activities.



CONCLUSION

- Today, the FDA regulates \$1 trillion worth of products a year. It regulates all foods and food ingredients introduced into or offered for sale in interstate commerce except for meat, poultry and some egg and catfish (which are regulated by USDA);
- ensures the safety and effectiveness of all drugs, biological products (including blood, vaccines and cellular and gene therapy products), medical devices, and animal drugs and feed; and makes sure that cosmetics and medical and consumer products that emit radiation do no harm.
- As FDA celebrates more than 100 years of service to the American people as the world's gold standard regulatory agency, it looks to the future. • FDA being a bridge, not a barrier
- The products of explosive progress in science and technology have made that future a possibility and not just a promise but the pathway requires FDA to look ahead to being a bridge and not a barrier to the delivery of safe and nutritious food and life-saving medical and health products to the people we serve.
- This strategic plan marks the path to achieve our vision for an organization that is dedicated to excellence as a science-based and science-led regulatory agency that provides global leadership in protecting public health



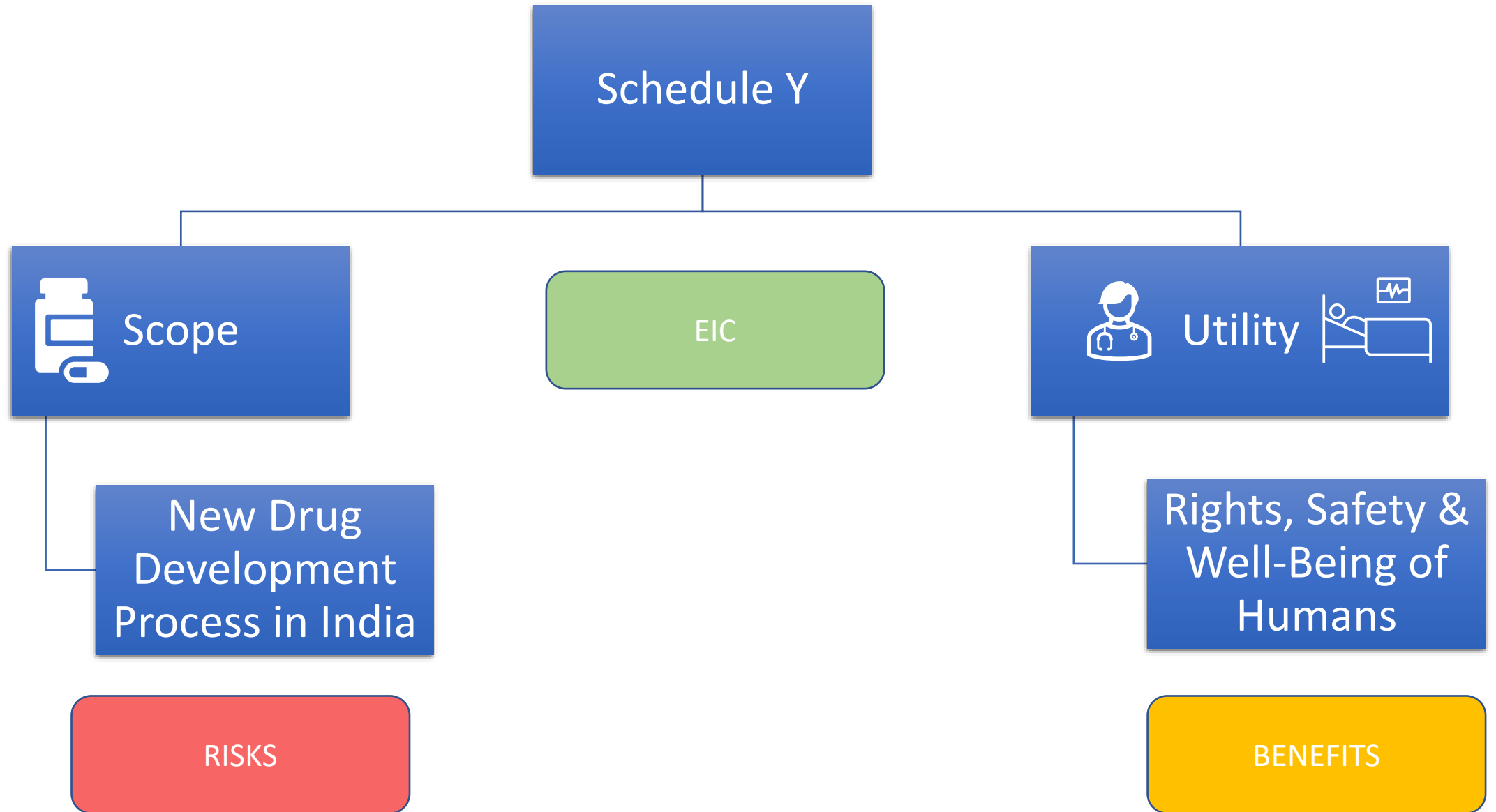


Schedule Y

Ms. Sonali Correa
Ph.D., M.Sc.

Introduction

- According to the Drugs and Cosmetics (11th Amendment) Rules, 2005 Ministry Of Health And Family Welfare (Department Of Health)
- In the Drugs and Cosmetics Rules, 1945 (hereinafter referred to as said rules),
- (1) in Part X-A, after rule 122-DA, the following shall be inserted, namely:-, 122-DAA,
- Definition of Clinical trial.- For the purpose of this Part, “Clinical trial” means a systematic study of new drug(s) in human subject(s) to generate data for discovering and/or verifying the clinical, pharmacological (including pharmacodynamic and pharmacokinetic) and/or adverse effects with the objective of determining safety and / or efficacy of the new drug.”
- (2) In the said rules for **Schedule Y**, the following Schedule shall be substituted, namely:-
“SCHEDULE Y [See rules 122A, 122B, 122D, 122DA, 122DAA and 122E]
- **Schedule Y.**
- **Refers to requirements and guidelines to be followed in order to attain permission of: Importing and/or Manufacturing New Drugs to market or To undertake clinical trials in India.**



Indian Drugs and Biologics Regulations

- Drug regulation in India is a complex process managed by law, mainly the Drugs and Cosmetics Act of 1940, and by multiple ministries, including the Ministry of Health and Family Welfare.
- The law creates a web of regulatory authorities to govern the process at both the central and the state level.
- At the central level, the Drugs and Cosmetics Act, 1940, has created the Central Drugs Standard Control Organisation (CDSCO), within which the Drugs Controller General of India (DCGI) is the key regulatory authority, acting under the advice of the Drug Technical Advisory Board (DTAB) and the Drug Consultative Committee (DCC).
- CDSCO operates through zonal offices spread across the country, which have designated roles in drug regulation, such as inspections, recalls, and market surveillance.
- CDSCO also has a role in overseeing the functioning of state authorities involved in drug regulation

- At the state level, there exist State Drug Regulatory Authorities (SDRAs), which are statutory bodies created under the Drugs and Cosmetics Act, 1940.
- Falling under the ambit of the respective Health Departments of each state, SDRAs are tasked with limited aspects of drug regulation.
- In practice, policy-setting tasks and permissions for manufacturing and licensing are handled by CDSCO, whereas some implementation appears to be handled by SDRAs under the oversight of CDSCO.
- SDRAs are often conjoined with the food regulation department under the Food and Drug Administration (FDA) in that state, which complicates the proper demarcation of regulatory responsibilities, and often adds to administrative confusion.
- With a posse of drug inspectors, the SDRAs seek to maintain a check on the quality of drugs and medical equipment being manufactured as well as standards regulating the points of sale.
- While only the central licensing authority may permit new drugs, state authorities regulate licensing of sale and manufacturing units within their jurisdiction.



Indian Drug market

The Indian drug market is unique in several ways: it is dominated by branded generics, mostly local companies, and a high volume-low pricing market.

This is besides the fact that the Indian market has the capacity for exponential growth, and is beginning to show signs of such development as well.

Other related factors have supported the growth of the pharmaceutical industry as well, including a burgeoning middle class with rising disposable incomes, improved access to medical facilities, and an increasing awareness of (and access to) health insurance.

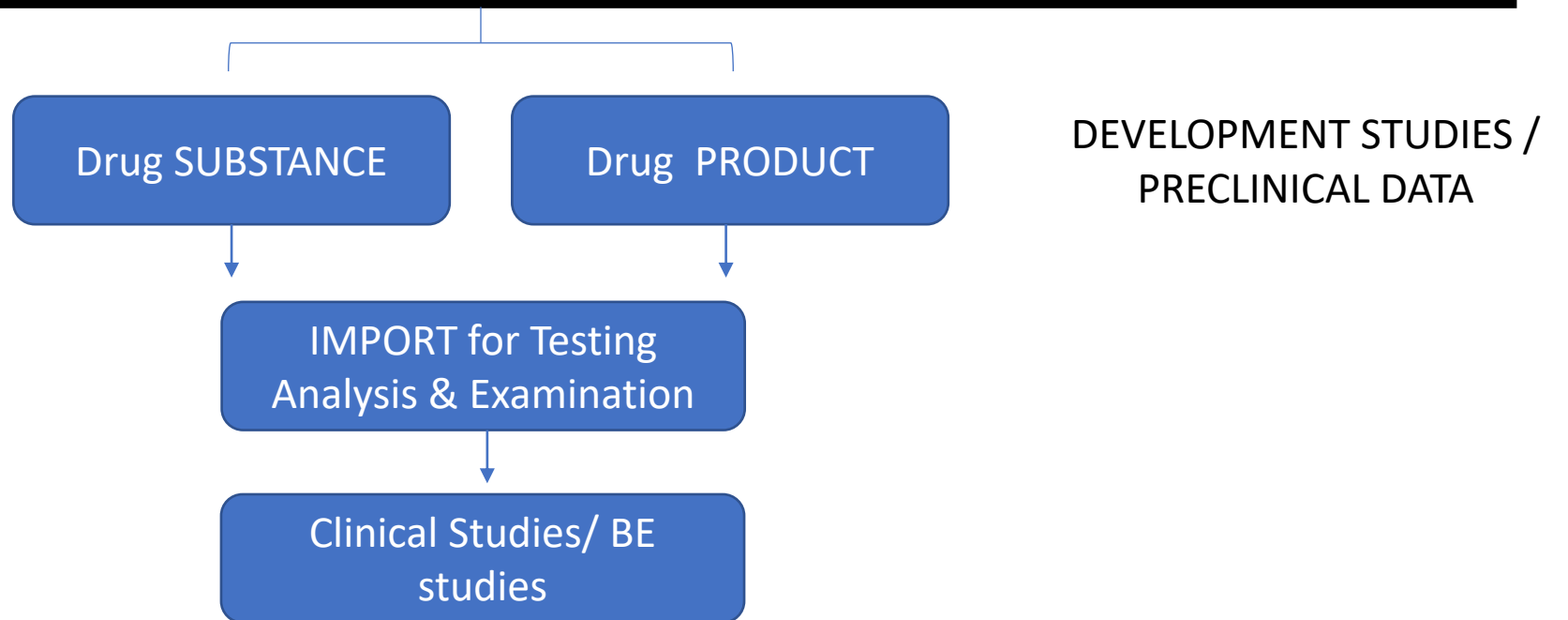
Process of drug regulation

- The DC Act entrusts CDSCO with the responsibility for the approval of new drugs, and the conduct of clinical trials in the country, as well as laying down the standards for drugs, controlling the quality of imported drugs, oversight over the SDRAs, and an advisory role in ensuring uniformity in the enforcement of the DC Act itself.
- CDSCO approves new drugs based on a combination of non-clinical data, clinical trial data (focusing on safety and efficacy) from abroad as well as in India, and the regulatory status of the drug in other countries.
- The law around new drug approvals is contained in **Rules 122 A, 122 B, 122D, 122 DA, 122 DAA, 122 DAB, 122 DAC, 122 DB, 122 DD and 122 E of Schedule-Y** of the DC Rules
- Applications for approval of New Drugs are evaluated by the 12 Subject Expert Committee (SEC) (formerly referred to as New Drug Advisory Committees (NDAC)), consisting of experts usually drawn from Government Medical Colleges and Institutes across India
- The drug approval process in practice has had its fair share of criticism, notably in the 2016 ban by the central government on fixed dose combinations (FDCs), with the government overruling prior approvals given by the DCGI and state licensing authorities, and without seeking advice from DCC and DTAB

- Besides approval, the other important regulatory roles are regarding licensing and inspections. Sections 22 and 23 of the DC Act give the Drug Inspectors (DI) the power to inspect premises manufacturing or selling drugs or cosmetics and take samples of any drug or cosmetic in exchange of its fair price and a written acknowledgement.
- In 2017, the DC Rules were amended, making it mandatory that before the grant of manufacturing license, the manufacturing establishment is to be inspected jointly by the Drug Inspectors of both the central government and the concerned state government. The amendment also made a similar joint inspection mandatory for manufacturing premises for not less than once every three years or as needed per the risk-based approach
- the DTAB has recommended amending the DC Act to authorize Licensing Authorities to issue stop-sale orders for drug retailers. Earlier, this power to issue stop-sale orders was available to the Licensing Authorities in cases of manufacturing non-compliances only

Scope of Schedule Y

Rule 122-A, 122-B, 122-E under DCR 1945



Decisions Based on BEST AVAILABLE SCIENTIFIC EVIDENCE

Scope of Schedule Y

HERBAL INDIAN SYSTEM OF Medicine & Homeopathy

BIOTECHNOLOGY PRODUCTS: RECOMBINANT PRODUCTS

DCGI
Drugs Controller
General of India

DBT
Dept of
Biotechnology

RDAC
Recombinant DNA Advisory
Committee

RCGM
review committee on genetic
manipulation

IBSC
Institutional Biosafety Committee

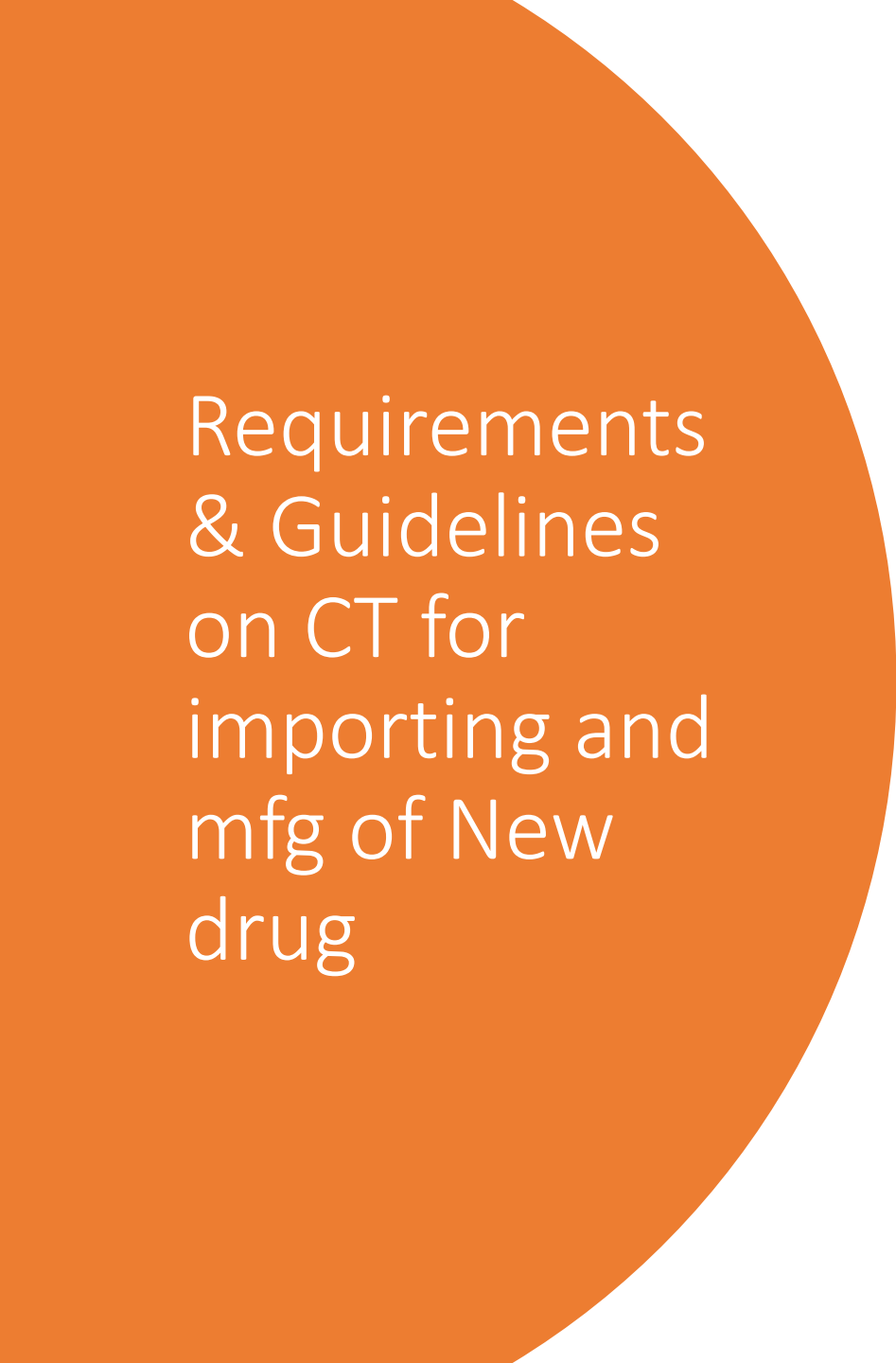
GEAC
Genetic Engineering Appraisal
Committee

APPROVAL BY STATE DRUG CONTROLLER

Purpose of Schedule Y

Schedule Y defines the clinical trials as the requirements and guidelines for import and manufacture of new drugs for sale or for clinical trials.

It describes the details of application process for conducting clinical trials; responsibilities of the sponsor, investigators and the Independent Ethics Committee.

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Requirements
& Guidelines
on CT for
importing and
mfg of New
drug

Application for permission

CT

Studies on Specific Population

PMS

Specific Studies: BA/BE studies

Implementation of Schedule Y

APPLICATION FORM 44 SUBMISSION			
RULE	122A	122B	122 DA
PURPOSE: Marketing Authorization	Import	Manufacture	Clinical study
Drug SUBSTANCE	FORM 45 A	FORM 46 A	
Drug PRODUCT	FORM 45	FORM 46	FORM 46
APPLICATION FORM 12 SUBMISSION			
PURPOSE	Import for Examination, Test and Analysis		
Drug PRODUCT	FORM 11		

Application for permission

- Application for permission to import or manufacture new drugs for sale or to undertake clinical trials shall be made in **Form 44** accompanied with following data in accordance with the appendices for permission
 - chemical and pharmaceutical information
 - animal pharmacology data
 - animal pharmacology
 - general pharmacological
 - pharmacokinetic data
 - animal toxicology data
 - human Clinical Pharmacology Data
 - Regulatory Status in other countries
 - Prescribing information
- Form 12: To import Study Drug for examination, test, or analysis

Clinical Trial

- **Approval for clinical trial**
 - Clinical trial on a new drug shall be initiated only after the permission has been granted by the Licensing Authority under rule 21 (b), and the approval obtained from the respective ethics committee(s).
- **Responsibilities of Sponsor**
 - Implementing & maintaining QA
 - Submit Status report to the Licensing Authority (LA)
 - SAE should be reported to LA within 14 calendar days
- **Responsibilities of Investigator**
 - Ensure adequate medical care is provided to volunteer
 - SAE and unexpected AE should be reported to the sponsor within 24h and to the EC within 7 working days
- **Informed consent**
 - Freely given informed written consent
 - Provide information about the study verbally
 - Non technical and comprehensible language

- **Responsibilities of EC**
 - Approval trial protocol to safeguard RSW of all volunteers and to protect RSW of all vulnerable subjects
 - Conduct ongoing review of trials
- **Human Pharmacology (Phase I)**
 - Safety and Tolerability
- **Therapeutic exploratory trials (Phase II)**
 - To evaluate the effectiveness of a drug for a particular indication or indications
 - To determine the common short-term side-effects and risks associated with the drug
 - To determine dosage for Phase III
- **Therapeutic confirmatory trials (Phase III)**
 - demonstration or confirmation of therapeutic benefits(s)
 - drug is safe and effective for use in the intended indication and recipient population and for market approval
- **Post Marketing Trials (Phase IV)**
 - performed after drug approval and related to the approved indication(s). These trials go beyond the prior demonstration of the drug's safety, efficacy and dose definition
 - include additional drug-drug interaction(s), dose-response or safety studies and trials designed to support use under the approved indication(s), e.g. mortality/morbidity studies, epidemiological studies etc

Studies in special populations

- Information supporting the use of the drug in children, pregnant women, nursing women, elderly patients, patients with renal or other organ systems failure, and those on specific concomitant medication is required to be submitted if relevant to the clinical profile of the drug and its anticipated usage pattern
- **Geriatrics**-Geriatric patients should be included in Phase III clinical trials (and in Phase II trials, at the Sponsor's option)
- **Paediatrics**- The timing of paediatric studies in the new drug development program will depend on the medicinal product, the type of disease being treated, safety considerations, and the efficacy and safety of available treatments. For a drug expected to be used in children, evaluations should be made in the appropriate age group. When clinical development is to include studies in children, it is usually appropriate to begin with older children before extending the trial to younger children and then infants.
- **Pregnant or nursing women**- Pregnant or nursing women should be included in clinical trials only when the drug is intended for use by pregnant/nursing women or fetuses/nursing infants and where the data generated from women who are not pregnant or nursing, is not suitable.

Post Marketing Surveillance

- Subsequent to approval of the product, new drugs should be closely monitored for their clinical safety once they are marketed. The applicants shall furnish Periodic Safety Update Reports (PSURs)
- The PSURs shall be submitted every six months for the first two years after approval of the drug is granted to the applicant. For subsequent two years - the PSURs need to be submitted annually

Special studies: Bioavailability/Bioequivalence Studies

- All bioavailability and bioequivalence studies should be conducted according to the Guidelines for Bioavailability and Bioequivalence.
- For drugs approved elsewhere in the world and absorbed systemically, bioequivalence with the reference formulation should be carried out wherever applicable
- Evaluation of the effect of food on absorption following oral administration should be carried out
- Dissolution and bioavailability data submitted with the new drug application must provide information that assures bioequivalence or establishes bioavailability and dosage correlations between the formulation(s) sought to be marketed and those used for clinical trials during clinical development of the product

Appendices of Schedule Y

Schedule Y is applicable for rules 122A, 122B, 122D, 122DA, 122DAA and 122E of the Drugs and Cosmetics Rules, 1945

Schedule Y had 2 appendixes - Appendix - 1, Appendix 1A.

The 8th amendment has incorporated a new appendix - Appendix 1B applicable for Phytopharmaceuticals.

Appendix - 1

DATA TO BE SUBMITTED ALONG WITH THE APPLICATION TO CONDUCT CLINICAL TRIALS/IMPORT/MANUFACTURE OF NEW DRUGS FOR MARKETING IN THE COUNTRY.

Appendix - 1 A

DATA REQUIRED TO BE SUBMITTED BY AN APPLICANT FOR GRANT OF PERMISSION TO IMPORT AND / OR MANUFACTURE A NEW DRUG ALREADY APPROVED IN THE COUNTRY

Appendix - 1 B

DATA TO BE SUBMITTED ALONG WITH APPLICATION TO CONDUCT CLINICAL TRIAL OR IMPORT OR MANUFACTURE OF A PHYTOPHARMACEUTICAL DRUG IN THE COUNTRY.

APPENDIX I

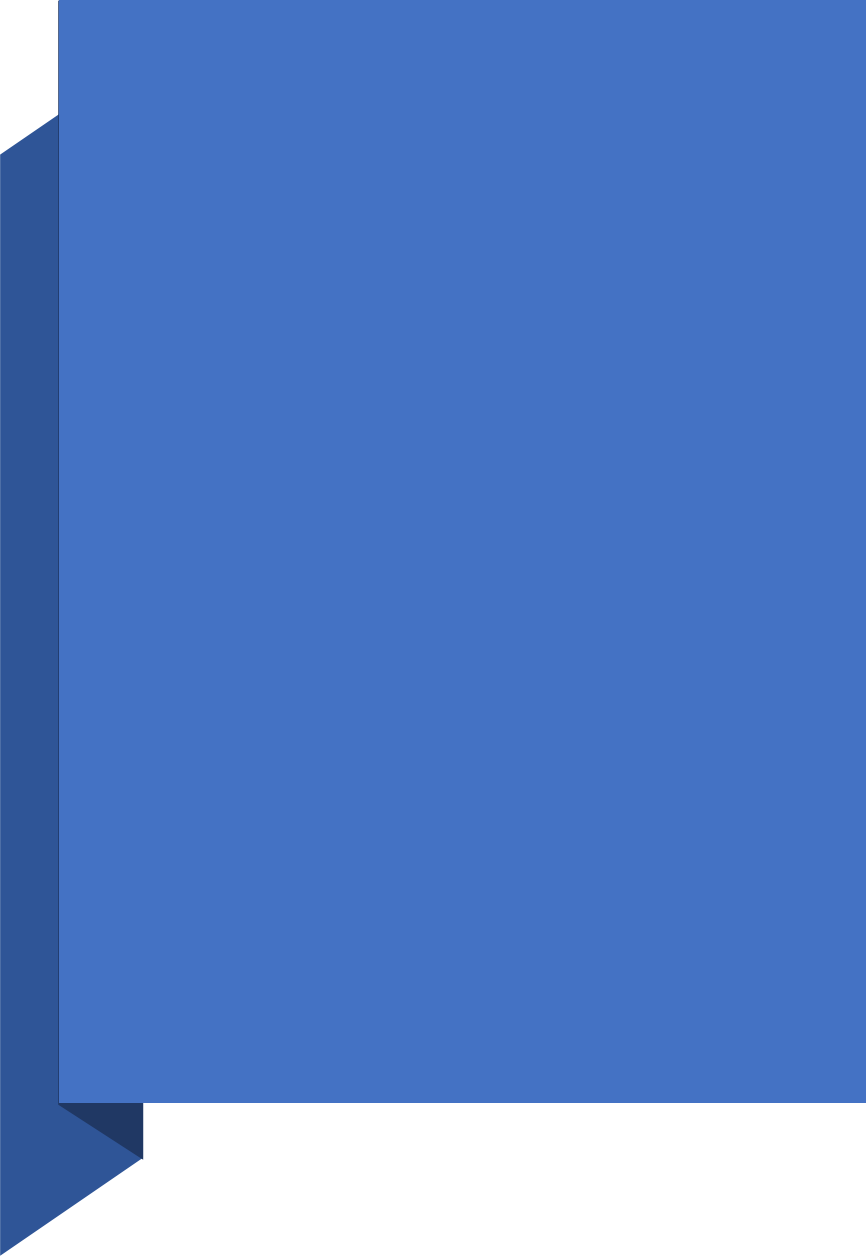
- DATA TO BE SUBMITTED ALONG WITH THE APPLICATION TO CONDUCT CLINICAL TRIALS / IMPORT / MANUFACTURE OF **NEW DRUGS FOR MARKETING** IN THE COUNTRY
- 1. Introduction
- 2. Chemical and pharmaceutical information
- 3. Animal Pharmacology (for details refer Appendix IV)
- 4. Animal Toxicology (for details refer Appendix III)
- 5. Human / Clinical pharmacology (Phase I)
- 6. Therapeutic exploratory trials (Phase II)
- 7. Therapeutic confirmatory trials (Phase III)
- 8. Special studies
- 9. Regulatory status in other countries
- 10. Prescribing information
- 11. Samples and Testing Protocol/s

APPENDIX I-A

- DATA REQUIRED TO BE SUBMITTED BY AN APPLICANT FOR GRANT OF PERMISSION TO IMPORT AND / OR MANUFACTURE A NEW DRUG **ALREAY APPROVED** IN THE COUNTRY.
- 1. Introduction
- 2. Chemical and pharmaceutical information
- 3. Marketing information
- 4. Special studies conducted with approval of Licensing Authority

APPENDIX – I-B

- **PART-I**
- 1. ***Data to be submitted by the applicant:***
 - A brief description or summary of the phytopharmaceutical drug giving the botanical name of the plant (including vernacular or scriptural name, wherever applicable), formulation and route of administration, dosages, therapeutic class for which it is indicated and the claims to be made for the phytopharmaceutical product.
- 2. **Human or clinical pharmacology information**
 - Published scientific reports in respect of pharmacological studies including human studies or clinical studies or epidemiological studies, relevant for the phytopharmaceutical drug intended to be marketed
 - Pharmacodynamic information
 - Monographs
- **PART-II**
Data generated by applicant:
- 3. **Identification, authentication and source of plant used for extraction and fractionation**
- 4. **Process for extraction and subsequent fractionation and purification**

- 
- **5 Formulation of phytopharmaceutical drug applied for**
 - **6. Manufacturing process of formulation**
 - **7. Stability data:**
 - **8. Safety and pharmacological information**
 - **9. Human studies**
 - **10. Confirmatory clinical trials**
 - **11. Regulatory status**
 - **12. Marketing information**
 - **13. Post marketing surveillance (PMS)**
 - **14. Any other relevant information**

Appendix II

STRUCTURE, CONTENTS AND FORMAT FOR CLINICAL STUDY REPORTS

- 1. Title Page
- 2. Study Synopsis (1 to 2 pages)
- 3. Statement of compliance with the 'Guidelines for Clinical Trials on Pharmaceutical Products in India
- 4. List of Abbreviations and Definitions
- 5. Table of contents
- 6. Ethics Committee
- 7. Study Team
- 8. Introduction
- 9. Study Objective:
- 10. Investigational Plan
- 11. Trial Subjects
- 12. Efficacy evaluation
- 13. Safety Evaluation
- 14. Discussion and overall Conclusion
- 15. List of References
- 16. Appendices

List of Appendices to the Clinical Trial Report

- a. Protocol and amendments
- b. Specimen of Case Record Form
- c. Investigators' name(s) with contact addresses, phone, email etc.
- d. Patient data listings
- e. List of trial participants treated with investigational product
- f. Discontinued participants
- g. Protocol deviations
- h. CRFs of cases involving death and life-threatening adverse event cases
- i. Publications from the trial
- j. Important publications referenced in the study
- k. Audit certificate, if available
- l. Investigator's certificate that he/she has read the report and that the report accurately describes the conduct and the results of the study.

Appendix III

ANIMAL TOXICOLOGY (NON-CLINICAL TOXICITY STUDIES)

- **1. General Principles**

For Phase I Clinical Trials

- **Systemic Toxicity studies**

- i. Single dose toxicity studies
- ii. Dose Ranging Studies
- iii. Repeat-dose systemic toxicity studies of appropriate duration to support the duration of proposed human exposure.

- **For Phase II Clinical Trials**

- Provide a summary of all the non-clinical safety data (listed above) already submitted while obtaining the permissions for Phase I trial, with appropriate references

- **For Phase III Clinical Trials**

- Provide a summary of all the non-clinical safety data (listed above) already submitted while obtaining the permissions for Phase I and II trials, with appropriate references.

- **For Phase IV Clinical Trials**

- Provide a summary of all the non-clinical safety data (listed above) already submitted while obtaining the permissions for Phase I, II and III trials, with appropriate references.

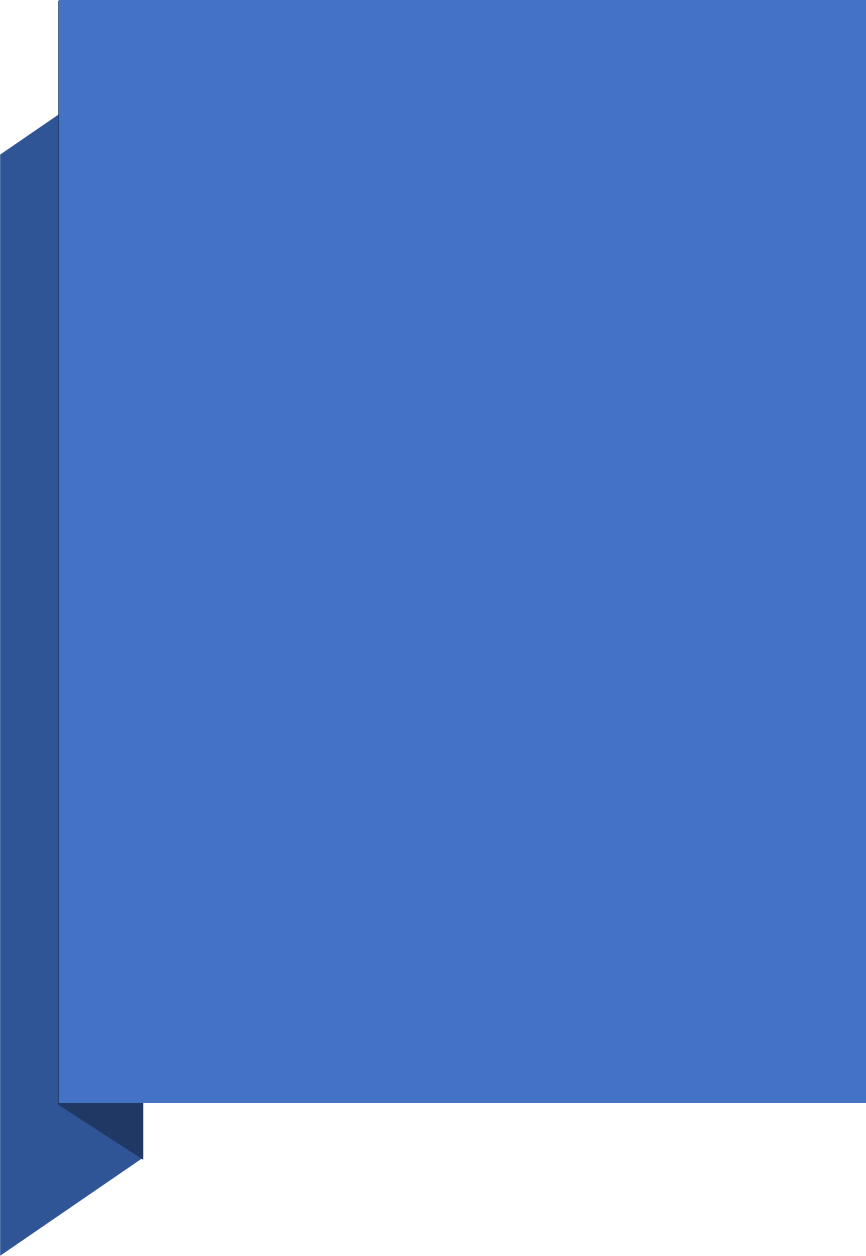
Appendix IV

- **ANIMAL PHARMACOLOGY**
- Specific and general pharmacological studies should be conducted to support use of therapeutics in humans.
- In the early stages of drug development enough information may not be available to rationally select design for safety assessment. In such a situation, a general approach to safety pharmacology studies can be applied.
- Safety pharmacology studies are studies that investigate potential undesirable pharmacodynamic effects of a substance on physiological functions in relation to exposure within the therapeutic range or above.

Appendix V

INFORMED CONSENT

- 1. Checklist for study Subject's informed consent documents
- 1.1. Essential Elements:
 - 1. Statement that the study involves research and explanation of the purpose of the research
 - 2. Expected duration of the Subject's participation
 - 3. Description of the procedures to be followed, including all invasive procedure and
 - 4. Description of any reasonably foreseeable risks or discomforts to the Subject
 - 5. Description of any benefits to the Subject or others reasonably expected from research. If no benefit is expected Subject should be made aware of this.
 - 6. Disclosure of specific appropriate alternative procedures or therapies available to the Subject.
 - 7. Statement describing the extent to which confidentiality of records identifying the Subject will be maintained and who will have access to Subject's medical records
 - 8. Trial treatment schedule(s) and the probability for random assignment to each treatment (for randomized trials)
 - 9. Compensation and/or treatment(s) available to the Subject in the event of a trial related injury
 - 10. An explanation about whom to contact for trial related queries, rights of Subjects and in the event of any injury
 - 11. The anticipated prorated payment, if any, to the Subject for participating in the trial

- 
- 12. Subject's responsibilities on participation in the trial
 - 13. Statement that participation is voluntary, that the subject can withdraw from the study at any time and that refusal to participate will not involve any penalty or loss of benefits to which the Subject is otherwise entitled
 - 14. Any other pertinent information
 - 1.2 Additional elements, which may be required
 - a. Statement of foreseeable circumstances under which the Subject's participation may be terminated by the investigator without the Subject's consent.
 - b. Additional costs to the Subject that may result from participation in the study.
 - c. The consequences of a Subject's decision to withdraw from the research and procedures for orderly termination of participation by Subject.
 - d. Statement that the Subject or Subject's representative will be notified in a timely manner if significant new findings develop during the course of the research which may affect the Subject's willingness to continue participation will be provided.
 - e. A statement that the particular treatment or procedure may involve risks to the Subject (or to the embryo or fetus, if the Subject is or may become pregnant), which are currently unforeseeable
 - f. Approximate number of Subjects enrolled in the study
 - 2. Format of informed consent form for Subjects participating in a clinical trial

Appendix VI

FIXED DOSE COMBINATIONS (FDCs)

- Fixed Dose Combinations refer to products containing one or more active ingredients used for a particular indication(s).
 - The **first group** of FDCs includes those in which one or more of the active ingredients is a new drug
 - The **second group** FDCs includes those in which active ingredients already approved/marketed individually are combined for the first time, for a particular claim and where the ingredients are likely to have significant interaction of a pharmacodynamic or pharmacokinetic nature [see rule 122E, item (c)]
 - For marketing permission, appropriate chemical and pharmaceutical data will be submitted
 - For any other such FDCs, clinical trials may be required. For obtaining permission to carry out clinical trials with such FDCs a summary of available pharmacological, toxicological and clinical data on the individual ingredients should be submitted, along with the rationale for combining them in the proposed ratio
 - The **third group** of FDCs includes those which are already marketed, but in which it is proposed either to change the ratio of active ingredients or to make a new therapeutic claim
 - The **fourth group** of FDC includes those whose individual active ingredients (or drugs from the same class) have been widely used in a particular indication(s) for years, their concomitant use is often necessary and no claim is proposed to be made other than convenience

Appendix VII

UNDERTAKING BY THE INVESTIGATOR

1. Full name, address and title of the Principal Investigator (or Investigator(s) when there is no Principal Investigator)
2. Name and address of the medical college, hospital or other facility where the clinical trial will be conducted: Education, training & experience that qualify the Investigator for the clinical trial (Attach details including Medical Council registration number, and / or any other statement(s) of qualification(s).
3. Name and address of all clinical laboratory facilities to be used in the study.
4. Name and address of the Ethics Committee that is responsible for approval and continuing review of the study.
5. Names of the other members of the research team (Co-or sub-Investigators) who will be assisting the Investigator in the conduct of the investigation(s).
6. Protocol Title and Study number (if any) of the clinical trial to be conducted by the Investigator.
7. Commitments:
8. Signature of Investigator with Date

Appendix VIII



ETHICS COMMITTEE

1. The number of persons in an Ethics Committee should have at least seven members. Ethics Committee should appoint, from among its members, a Chairperson (who is from outside the institution) and a Member Secretary. Other members should be a mix of medical/non-medical, scientific and non-scientific persons, including lay public, to reflect the different viewpoints.

For review of each protocol the quorum of Ethics Committee should be atleast 5 members with the following representations:

- (a) basic medical scientists (preferably one pharmacologist).
- (b) clinicians
- (c) legal expert
- (d) social scientist/ representation of non-governmental voluntary agency / philosopher/ ethicist / theologian or similar person
- (e) lay person from the community.

2. Format for Approval of Ethics Committee

Appendix IX

STABILITY TESTING OF NEW DRUGS

- Stability testing is to be performed to provide evidence on how the quality of a drug substance or formulation varies with time under the influence of various environmental factors
- Stability studies should include testing of those attributes of the drug substance that are susceptible to change during storage and are likely to influence quality, safety, and/ or efficacy.
- Validated stability-indicating analytical procedures should be applied. For long term studies, frequency of testing should be sufficient to establish the stability profile of the drug substance.

Appendix X

CONTENTS OF THE PROPOSED PROTOCOL FOR CONDUCTING CLINICAL TRIAL

1. Title Page
2. Table of Contents
 1. Background and Introduction
 2. Study Rationale
 3. Study Design
 - a. Overview of the Study Design: Including a description of the type of study (i.e., double-blind, multicentre, placebo controlled, etc.) a detail of the specific treatment groups and number of study Subjects in each group and investigative site, Subject number assignment, and the type, sequence and duration of study periods.
 - b. Flow chart of the study
 - c. A brief description of the methods and procedures to be used during the study.
 - d. Discussion of Study Design: This discussion details the rationale for the design chosen for this study.

Appendix XI

Data Elements for reporting serious adverse events occurring in a clinical trial

- 1. Patient Details
- 2. Suspected Drug(s)
- 3. Other Treatment(s)
- 4. Details of Suspected Adverse Drug Reaction(s)
- 5. Outcome
- 6. Details about the Investigator*

RECENT AMENDMENTS

- **Introduction of Rule 122DAB** – Specifying the procedures for payment of compensation to the subjects of the trial in cases of injury or death
- **Introduction of Rule 122DAC** – Specifying various conditions for conduct and inspection of clinical trials
- **Introduction of Rule 122DD** – Specifying the detailed guidelines for registration of Ethics Committee
- **Drugs and Cosmetics (Amendment) Bill 2013** – Yet to be introduced

- **Audio-Visual Recording of Consent**
 - Gazette Notification for Audio-Visual recording of informed consent
 - Audio-Video recording of informed consent has become mandatory
- **SAE Reporting and Compensation**
 - The process of SAEs reporting in case of injury/death has been revised in recent Schedule Y amendments and below is required information to be submitted along with the SAEs report.

- 
- **System of Pre-screening for submission of reports of SAEs to CDSCO**
 - Panel of expert for reviewing of SAE of death
 - Formula to determine the quantum of compensation in the cases of clinical trial related serious adverse events (SAEs) of deaths occurring during clinical trials
 - **Ethics Committee Registration**
 - System for the pre-screening of the applications for registration of ethics committees
 - IEC in respect of periodic review of ongoing clinical trials
 - List of Registered Ethics Committees State Wise

Difference between ICH GCP and Schedule Y

Parameters	United States	India
Regulatory bodies	USFDA	CDSCO
Clinical trial application	Investigational new drug application (IND)	Form 44 is an application made for getting approval to start clinical trial
Application fee	No Fee	Fees is required in Phase I,II,III i.e. 50000,25000,250000 respectively
Approval Timeline	30 days	16-18 weeks
Regulations	Code of federal regulations 21 CFR Part 312,50,54,56 have to follow	Schedule Y of the Drug and cosmetics act and rules 1945,
GCP guidelines	ICH GCP	Indian GCP
Compensation	Compensation according to informed consent 21 CFR Part 50	According to 122 DAB rule
Forms Required	FDA forms 1571,1572,and 3455	Form 44 and 12
Record Retention Time	Till 2 years	Maximum 3 years

Important forms and rules of drug and cosmetic act for clinical research

Important forms of D and C Act

Form 44	Application for grant of permission to import or manufacture a New Drug or to undertake clinical trial
Form 12	Application for license to import drugs for the purposes of examination, test or analysis
Form 11	License to import drugs for the purposes of examination, test or analysis (Validity is one year)
Form 4	Issue of import certificate (Validity is six months)
Form 1	Application form for issue of import certificate for import of narcotic drugs and psychotropic substance
Form 4	Issue of import certificate (Validity is six months)

Important rules of D and C Act

Rule 34	Application for license for examination, test or analysis
Rule 122-A	Application for permission to import new drug
Rule 122-D	Permission to import fixed dose combination
Rule 122-DA	Application for permission to conduct clinical trials for new drug/investigational new drug
Rule 122-DAA	Definition of clinical trial
Rule 122-E	Definition of new drug
Rule 55	Application for import certificate
Rule 56	Issue of Import certificate
Rule 122 DAB	Draft guidelines for registration of all clinical research Organizations involved in clinical trial in India. (valid for five years)

Regulatory bodies in india involved in drug regulation

Central Drugs Standard Control Organization (CDSCO)	Regulatory body controlled and governed by Directorate General of Health Services of Ministry of Health and Family Welfare which regulates drugs in India
Drugs Controller General India (DCGI)	Regulatory agency under the government of India whose function is to assess the quality, safety and efficacy of a drug
Indian Council of Medical Research (ICMR)	Apex body in India that formulates, co-ordinates and promotes biomedical research
Genetic Engineering Approval Committee (GEAC)	Apex body constituted in the Ministry of Environment and Forests; clinical trials involving the use of biotech products would be referred by DCGI to GEAC for recommendations
Department of Biotechnology (DBT) ^[54]	Apex body that oversees the new impetus to develop the field of modern biology and biotechnology through several R and D projects in areas of healthcare in India
Atomic Energy Regulatory Board (AERB) ^[55]	Regulatory authority that exercises regulatory control over the approval of new types of radiation equipment through Department of Atomic Energy (DAE), and for the registration/ commissioning of new radiation equipment, inspection and decommissioning of installations
Bhabha Atomic Research Centre (BARC) ^[56]	Apex body that oversees and approves all radiation related projects in India. DCGI refers all clinical trials that involve the use of radiopharmaceuticals to BARC for its expert opinion
Drugs Consultative Committee (DCC)	Provides technical guidance to the Central Drugs Standard Control Organization
Central Drugs Laboratory (CDL)	National statutory laboratory of the Indian government for quality control of drugs
Drugs Technical Advisory Board (DTAB)	Provides technical guidance to the CDSCO

Conclusion

- Schedule Y establishes a set of guidelines and requirements for clinical trials.
- Schedule Y defines the clinical trials as the requirements and guidelines for import and manufacture of new drugs for sale or for clinical trials. It describes the details of application process for conducting clinical trials; responsibilities of the sponsor, investigators and the Independent Ethics Committee.
- The Central Drugs Standard Control Organisation and its chairman Drug Controller general of India are bequeathed to protect the citizens from the marketing of unsafe medication. The startling findings, of the 59th report of the Parliamentary Standing Committee on Health and Family Welfare, have uncovered the lax standards followed by the regulatory authorities in India.
- Schedule Y was written with the generics industry in mind but increase entry of foreign pharmaceutical companies after the introduction of strict patent rules in the area of clinical research led the government to introduce many changes.
- The government recognized the importance of their regulation and thus developed Ethical and Regulatory Guidelines.
- The Indian Council of Medical Research (ICMR) issued the Ethical Guidelines for Biomedical Research on Human Subjects in 2000 and CDSCO released Indian Good Clinical Practice (GCP) guidelines in 2001.

ICMR Guideline

Ms. Sonali Correa

Ph.D., M.Sc.

Introduction

The ICMR, New Delhi, is one of the oldest medical research body in the world and apex body in India

It is the premier national agency for the

Planning

Formulation

Co-
ordination

Implementat
ion

Conduct or
promotion of
biomedical
research in India

- The council promotes biomedical research in the country through intramural as well as extramural research. Over the decades, the base of extramural research and also its strategies have been expanded by the council
- In 2007 the organization established the Clinical Trials Registry - India, which is India's national registry for clinical trials

ICMR Timeline

- In 1911, the Govt. of India set up the Indian Research Fund Association (IRFA) with the specific objective of sponsoring & coordinating medical research in the country
- After Independence several important changes were made in the organization & activities of IRFA
- It was redesignated as ICMR in 1949, with expanded scope of functions and responsibilities
- ICMR is funded through the Govt. of India under the Ministry of Health and Family Welfare
- The council not only plans and promotes medical research of emerging health problems of the country but also is expected to build strong national group of skilled professional biomedical scientists/ researchers for organization of such research

1911

IRFA (NOW ICMR) SET UP

Indian Research Fund Association (IRFA) is established by Sir Harcourt Butler on November 15.

1913

IJMR STARTS

The Indian Journal of Medical Research (IJMR) starts (now a leading peer-reviewed indexed journal of biomedical research).

1918

ICMR NIN

The 'Beri-Beri Enquiry' under Sir Robert McCarrison is set up (now known as National Institute of Nutrition or NIN, Hyderabad).



Beri Beri Enquiry in Pasteur Institute, Coonoor, birth place of National Institute of Nutrition.

1919

RESEARCH ON INDIGENOUS DRUGS

Research on indigenous drugs (under Col. R.N. Chopra at the Calcutta School of Tropical Medicine, Kolkata) initiated.

1927

MALARIA SURVEY OF INDIA

Central Malaria Organization known as "Malaria Survey of India" forms under the guidance of Lt. Col. S.R. Christophers with its first station set up in Karnal.

FIRST PUBLICATION

First publication of "Records of Malaria Survey of India" (later renamed as Journal of Malaria Institute of India then Indian Journal of Malariology, now Journal of Vector Borne Diseases).

1949

RENAMING OF IRFA AS ICMR

IRFA is renamed to the Indian Council of Medical Research.



Delivery of DEC medicated salt by vendors to the community for filariasis control.

1951

ICMR NIV

The Virus Research Centre is set up in Pune (now known as National Institute of Virology or NIV, Pune).



Renaming of Rockefeller Centre to National Institute of Virology.

1952

BOMBAY BLOOD GROUP

A new blood group, the Bombay blood group is discovered by Dr. H.M. Bhatia and his team in Bombay which saves numerous lives.

1953

ICMR AWARDS

The ICMR receives a grant from Col. Amir Chand for instituting a system that rewards research being conducted in Indian Medicine.

1954

ICMR NIRRH

The Contraceptive Testing Unit and the Reproductive Physiology Unit is set up in Bombay (now known as National Institute for Research in Reproductive Health or NIRRH, Mumbai).

1955

TB SURVEY

The ICMR organizes the first national wide tuberculosis survey.

1976

ICMR
NJILOMD

ICMR takes over the Central JALMA Institute for Leprosy at Agra, established by the Japanese Leprosy Mission for Asia- JALMA (now known as National JALMA Institute for Leprosy and Other Mycobacterial Diseases or NJILOMD, Agra).

ICMR
NIMS

Institute for Research in Medical Statistics (IRMS) is set up (now known as National Institute of Medical Statistics or NIMS, Delhi).

1977

ICMR
NIMR

The Malaria Research Centre is established at Delhi (now known as National Institute of Malaria Research or NIMR).

1979

ICMR
NICPR

The Cytology Research Centre is set up in New Delhi (now known as National Institute of Cancer Prevention and Research or NICPR, Noida).

1980

CHINGLEPUT
TRIAL

'Chingleput Trial' (the largest ever-trial) demonstrates inefficiency of BCG vaccine in adults.

HEPATITIS E
TRIAL

Human Hepatitis E Virus is discovered.

ETHICS
POLICY

Policy on ethical considerations involved in research on human subjects released.

1981

NATIONAL CANCER
REGISTRY
PROGRAMME

The National Cancer Registry Project starts with the establishment of Population and Hospital based Cancer Registries.

2018

**ICMR
RMRC-G**

Regional Medical Research Centre Gorakhpur (RMRC-G) is setup by upgrading the NIV field unit established in 2008.

2019

**RESEARCH
PLATFORM**

RESEARCH 'Regional Enabler for South East Asia Research Collaboration for Health' platform launched in partnership with WHO and 10 countries to combat emerging and re-emerging infectious diseases in SE Asia Region.

**DETECTION KIT FOR
NIPAH VIRUS**

Point of Care diagnostic test for Nipah Virus is developed.

2019

**NERA
INDIA**

Launched Malaria Elimination Research Alliance (NERA) India to fast track malaria elimination.

**E-CIGARETTES
BAN**

E-cigarettes ban Bill passed based on evidence report by ICMR.

2019

**ETHICS
GUIDELINES**

ICMR "National Ethical Guidelines in Biomedical and Health Research Involving Human Participants, 2017" became legally binding under New Drugs and Clinical Trial Rules, 2019.

**ETHICS
REGISTRY**

National Ethics Committee Registry for Biomedical and Health Research (NECRBHR) is set up also known as the NAITIK Portal.

2019

**INDIA AFRICA
PLATFORM**

Established India-Africa Health Sciences Collaborative Platform (IAHSP) to initiate and strengthen India-Africa strategic partnership in the areas of Health Research.

**AMR
HUB**

Antimicrobial Resistance Hub and National Repository of Antimicrobial Resistant Bacteria (NRAMRB) at ICMR-NICED is inaugurated in September in Kolkata.

Development of Ethical Guidelines

Ethics in clinical research focuses largely on identifying and implementing the acceptable conditions for exposure of some individuals to risks and burdens for the benefit of society at large.

Ethical guidelines for clinical research were formulated only after discovery of inhumane behaviour with participants during research experiments

Global perspective:

The World Medical Association (WMA) to develop a set of guidelines to safeguard the rights and well being of participants in clinical research post the unethical Willowbrook State Study and the Jewish Chronic Disease Hospital study.

The United States the ethical guidelines were setup after the discovery of the Tuskegee Syphilis Study

1982, the Council for International Organizations of Medical Sciences (CIOMS) [6] in association with World Health Organization (WHO) developed 'International Ethical Guidelines for Biomedical Research Involving Human Subjects'.

Indian Perspective

- The Indian Council of Medical Research (ICMR), in February 1980, released a 'Policy Statement on Ethical Considerations involved in Research on Human Subjects'.
- This was the first policy statement giving official guidelines for establishment of ethics committees (ECs) in all medical colleges and research centres.
- But as with other nations of the world, these guidelines were not respected by many researchers and India was not free of controversial research works.
- In 1970s and 1980s researchers at the Institute for Cytology and Preventive Oncology in New Delhi, carried out a study on 1158 women patients of different stages of cervical dysplasia or precancerous lesions of the cervix. These patients were left untreated to see how many lesions progressed to cancer and how many regressed. By the end of the study seventy one women had developed malignancies and lesions in nine of them had progressed to invasive cancer. Sixty-two women were treated only after they developed localised cancer.
- After the controversy about the study became public in 1997, the ICMR started developing 'Ethical Guidelines for Biomedical Research on Human Subjects' and finalised them in the year 2000.
- These are a set of guidelines which every researcher in India should follow while conducting research on human subjects. Although not a law, these guidelines have been put into force through Schedule Y. With the changing scenario in the research field and development of modern techniques, the guidelines were revised in 2006.

Objective of ICMR

Vision

- Translating Research into Action for Improving the Health of the Population.

Mission

- **Generate**, manage and disseminate new knowledge.
- **Increase focus** on research on the health problems of the vulnerable, the disadvantaged and marginalized sections of the society.
- **Harness and encourage** the use of modern biology tools in addressing health concerns of the country.
- **Encourage innovations and translation** related to diagnostics, treatment, methods/ vaccines for prevention.
- **Inculcate a culture of research in academia** especially medical colleges and other health research institutions by strengthening infrastructure and human resource.

Principles of ICMR

Principle of essentiality

- The research being carried out should be essential for the advancement of knowledge that benefits patients, doctors and all others in aspects of health care and also for the ecological and environmental well being of the planet.

Principles of voluntariness, informed consent and community agreement

- The research participant should be aware of the nature of research and the probable consequences of the experiments and then should make a independent choice without the influence of the treating doctor, whether to take part in the research or not. When the research treats any community or group of persons as a research participant, these principles of voluntariness and informed consent should apply to the community as a whole and also to each individual member who is the participant of the research or experiment.

Principle of non-exploitation

- Research participants should be remunerated for their involvement in the research or experiment. The participants should be made aware of all the risks involved irrespective of their social and economic condition or educational levels attained. Each research protocol should include provisions of compensation for the human participants either through insurance cover or any other appropriate means to cover all foreseeable and hidden risks.

Principle of privacy and confidentiality

- All the data acquired for research purpose should be kept confidential to prevent disclosure of identity of the involved participant and should not be disclosed without valid legal and/or scientific reasons.

Principle of precaution and risk minimisation

- Due care and caution should be taken at all stages of the research and experiment (from its beginning as a research idea, formulation of research design/ protocol, conduct of the research or experiment and its subsequent applicative use) to prevent research participant from any harm and adverse events. EC has to play an active role in risk minimization.

Principle of professional competence

- Clinical research should be carried out only by competent and qualified persons in their respective fields.

Principle of accountability and transparency

- The researcher should conduct experiments in fair, honest, impartial and transparent manner after full disclosure of his/her interests in research. They should also retain the research data, subject to the principles of privacy and confidentiality, for a minimum period of 5 years, to be scrutinized by the appropriate legal and administrative authority, if necessary.

Principle of the maximisation of the public interest and of distributive justice

- The results of the research should be used for benefit of all humans, especially the research participants themselves and/or the community from which they are drawn and not only to those who are socially better off.

Principle of institutional arrangements

- It is required that all institutional arrangements required to be made in respect of the research and its subsequent use or applications should be duly made in transparent manner.

Principle of public domain

- The results of any research work done should be made public through publications or other means. Even before publication, the detailed information of clinical trials should be made public before start of recruitment via clinical trial registry systems that allow free online access like: www.ctri.in/; www.actr.org.au/; www.clinicaltrials.gov/ or www.isrctn.org/.

Principle of totality of responsibility

- All those directly or indirectly connected with the research should take the professional and moral responsibility, for the due observance of all the principles, guidelines or prescriptions laid down in respect of the research.

Principle of compliance

- All those associated with the research work should comply by the guidelines pertaining to the specific area of the research.

Informed Consent

- A well-documented informed consent is the hallmark of any ethical research work.
- It is the responsibility of the investigator/researcher to obtain the informed consent of the prospective participant or in the case of an individual who is not capable of giving informed consent, the consent of a legal guardian.
- Informed consent respects individual's autonomy to participate or not to participate in research.
- Adequate information about the research is given in a simple and easily understandable vernacular language in a document known as the 'Participant/Patient Information Sheet' attached along with the 'Informed Consent Form (ICF)'

The patient information sheet should include:

- A statement that the study involves research; an explanation of the purpose of the research and the expected duration of the subject's participation;
- a description of the procedures to be followed and identification of any procedures which are experimental;
- a description of any reasonably foreseeable risks or discomforts to the subjects;
- a description of any benefits to the subjects or to others which may reasonably be expected from the research; trial treatment schedule(s) and the probability for random assignment to each treatment (especially in randomized placebo controlled trials);
- a disclosure of appropriate alternative procedures or courses of treatment, if any, that might be advantageous to the subjects;
- a statement describing the extent, if any, to which confidentiality of records identifying the subjects will be maintained; for research involving more than minimal risk, an explanation as to whether any compensation and an explanation as to whether any medical treatments are available if injury occurs and, if so, what they consist of, and where further information may be obtained;
- an explanation of whom to contact for answers to pertinent questions about the research and research subjects' rights, and whom to contact in the event of a research-related injury to the subjects;
- a statement that participation is voluntary and refusal to participate will involve no penalty or loss of benefits to which the subjects are otherwise entitled, also the subjects may discontinue participation at any time without penalty or loss of benefits.

The ICF should specify

- that the participant has read and understood the patient information sheet;
- no further permission is required to look into his health records for study purpose until his identity is not revealed;
- the results arising from the study can be used only for scientific purposes and he voluntarily agrees to take part in the study.

The ICF should have

- space for signature/thumb print of the participant, the principal investigator, a witness and a legally acceptable representative when required.
- The ICF with participant/patient information sheet should be approved by the EC before use.
- The ICF should have the sign or thumb impression of the prospective participant before start of the experiment.
- If the participant is illiterate, the document should have the signature of a witness, who has seen that the contents of the patient information sheet were adequately explained to the participant.
- If the participant is a minor or not capable of giving consent, a verbal assent should be taken from him and the consent form should be signed by his legally acceptable representative.
- If the treating physician of a prospective participant is also the investigator, the informed consent should be taken by any other neutral physician to prevent biased decision of the participant.

Informed consent if properly taken protects the rights of prospective participants and thus forms the basis of ethical research work.

Vulnerable Population

- Persons who are relatively or absolutely incapable of protecting their own interests are termed as vulnerable research population.
- The very poor, illiterate patients, children, individuals with questionable capacity to give consent (including psychiatric patients), prisoners, foetuses, pregnant women, terminally ill patients, students, employees, comatose patients, tribals and the elderly are examples of vulnerable population.
- Declaration of Helsinki states that ‘Medical research involving a underprivileged or vulnerable population or community is only justified if the research is responsive to the health needs and priorities of that population or community and if there is a reasonable likelihood that this population or community stands to benefit from the results of the research.’
- It is the responsibility of the EC to see whether the inclusion of vulnerable populations in the study is justifiable or the population is just being exploited to generate clinical data.
- To prevent even minor exploitation the EC should consult the representative of vulnerable population that is to be researched upon while reviewing the protocol.

Therapeutic Misconception

- The therapeutic misconception (TM) is a vexing ethical issue for obtaining valid informed consent.
- A patient coming to a physician may misinterpret and enrol in a research study thinking it to be routine medical care without understanding the experimental nature of the treatment given. He may misinterpret the information given about the research, such that he believes that aspects of the research will directly benefit him.
- It is important that investigators should make efforts to dispel the TM in order to promote ethical and valid informed consent.
- **ICFs should clarify the salient features of research:**
 - The purpose of randomized controlled trials (RCTs),
 - random selection of treatment, masking of treatment,
 - meaning and rationale of placebo,
 - restrictions on treatment flexibility and how treatment decision making differs in RCTs compared with routine medical care.
- To safeguard the ethical rights of the participants therapeutic misconception needs to be taken care of.

Post-Trial Access

- The concept of post trial access holds special importance for clinical research works in the less developed countries.
- Pharmaceutical companies from developed countries collect the clinical data for their new and experimental drugs from the population in less developed countries.
- Most of these drugs would never be used by the communities from where the experimental data are collected and here comes the importance of post trial access for safeguarding the rights of such communities.
- The Helsinki Declaration of WMA, 2000 states that at the end of the trial, every participant should be assured of access to the best proven prophylactic, diagnostic and therapeutic methods identified by the study.
- The Declaration of the WMA in 2004 reaffirmed its position that “it is necessary during the study planning process to identify post-trial access by study participants to prophylactic, diagnostic and therapeutic procedures identified as beneficial in the study or access to other appropriate care.
- Post-trial access arrangements or other care must be described in the study protocol so that ethical review committee may consider such arrangements during its review.” Therefore, whenever possible EC should consider such an arrangement in the a priori agreement.
- Sometimes more than the benefit to the participant, the community may be given benefit in indirect way through improving their living conditions, establishing counselling centres, clinics or schools and giving education on maintaining good health practices.

Ethics Committee

- The first appearance of need of ethics committee (EC) was made in Declaration of Helsinki in 1964, while in India it appeared in 1980 in the ICMR Policy Statement.
- EC also called as the Institutional Review Board or the Ethics Review Board stands as the bridge between the researcher and the ethical guidelines of the country
- The establishment of EC requires 5-15 members with at least one basic medical scientist (preferably one pharmacologist), one clinician, a legal expert, a social scientist / representative of NGO / philosopher or theologian and a lay person from the community.
- Every institute, where research is going on should have its own EC with its head preferably from outside the institute.
- Individuals carrying out research can approach to independent ECs. The decisions of EC should be taken only after quorum formation with a minimum of five members having at least one basic medical scientist, one clinician and one legal expert or retired judge.
- The ECs should have independence from political, institutional, professional, and market influences, in their composition, procedures, and decision-making.
- As there are no laws governing the registration, formation or working of ethics committees in India, each ethics committee should have their own standard operating procedures for proper functioning.

Responsibilities of EC:

- ECs are responsible for carrying out the review of proposed research before the commencement of the research.
- The basic responsibility of EC is to ensure an independent, competent and timely review of all ethical aspects of the project proposals received in order to safeguard the dignity, rights, safety and wellbeing of all actual or potential research participants.
- The scientific design and conduct of the study should also be reviewed at the outset as poor science is poor ethics. The appropriateness of the study design in relation to the objectives of the study, the statistical methodology (including sample size calculation) and the potential for reaching sound conclusions with the smallest number of research participants should be assessed.
- The EC should also look into matters like informed consent process, qualifications of principal investigator and supporting staff, adequacy of infrastructure and facilities, risk benefit ratio, plans to maintain confidentiality and plans for post trial access and compensations.
- They also need to ensure that there is regular evaluation of the ongoing studies that have received a positive decision. EC is the most important check point for promoting ethical research in the country.

General Ethical Issues

- **Benefit-risk assessment**
- The EC must decide about the type of review required (exempted, expedited, full committee) based on the type of risk involved.

Table 1: Types of risk

Type of risk	Definition/description
Less than minimal	Probability of harm or discomfort is nil or not expected.
Minimal risk	Probability of harm or discomfort anticipated in the research is not greater than encountered in routine life activities/ serious harm or adverse event is unlikely
Minor increase over minimal risk or Low risk	Increment in probability of harm or discomfort is only a little more than the minimal risk threshold. Such research should have a social value. Social risks, psychological harm and discomfort may also fall in this category.
More than Minimal/high risk	Probability of harm or discomfort anticipated in the research is invasive and greater than minimal risk or interventional study.

- **Privacy Confidentiality:**
- Information related certain diseases may be highly sensitive in nature with a lot of scope for stigmatization, discrimination, violence etc. Maintaining confidentiality of research related data and its publication is important to protect the privacy of individuals and avoid any discrimination against them.
- **Distributive Justice**
- Individuals or communities invited for research should be selected in such a way that the benefits and burdens of research are equitably distributed without leading to social, racial or ethnic inequalities.

- **Payment for participation**
- Participants should not be made to pay for any expenses incurred beyond routine clinical care and wherever possible may be given a reasonable amount to cover incidental expenses
- **Compensation for research-related harm:**
- Research participants who suffer direct physical, psychological, social, legal or economic harm as a result of participating in the research are entitled to free health care and referrals as needed. However, for research related Serious Adverse Events (SAE), appropriate financial compensation and insurance coverage be provided as per norms.
- Sponsor to include insurance coverage/other provision within budget. In investigator initiated research, investigator/institution must provide through insurance, corpus funds or grants
- SAEs should be reported to EC (including on non-working days) within 24 hours and a report on SAE relatedness (causality assessment) within 14 days for EC review regarding quantum and type of assistance.
- **Conflict of interest:**
- Implement procedures to declare and management conflict of interest (financial/nonfinancial) of researchers, EC members, institution, sponsor.
- **Community Engagement:**
- Engaging with the community in a culturally sensitive manner can improve public trust, help improve design, conduct and responsiveness to health needs.

- Various measures to educate the public or communities, proposed research, risks and benefits, persons to be contacted etc. should be undertaken.
- Efforts must also be made to prevent the infodemic or spread of fake information, or to sensationalize, make false promises or claims, or spread negativity or create a scare.
- Wherever possible, community representatives (e.g., Community advisory board) be involved in conceptualization, review, research, dissemination of results in such settings.
- **Post research access and benefit sharing:**
 - Efforts be made to communicate the research findings to the individuals/communities.
 - EC should consider the need for an a priori agreement between researchers and sponsors regarding post-research access of the community to successful interventions and benefit sharing if relevant.
- **Storage of Biological Material/ datasets:**
 - samples may be in the form of expectorated sputum, endotracheal aspirate, or Broncho alveolar lavage (BAL) etc. besides other body fluids, such as blood, plasma, dried blood spots, body fluids, urine, stool, tissues, organs - stored or prospectively collected. Storage of infectious samples requires adequate safeguards
 - A dataset is an organized collection of data and information maintained in physical and/or electronic/digital form ranging from small numbers to large numbers or whole population.
 - Samples/ datasets may be classified as anonymous (unidentified), anonymized or identifiable.
 - It may be useful to have repository of samples/ registry which can be further used to generate forecasts of trends and identify hotspots.

- Provide clarity on custodianship, obtain approval of the EC/ governance committee, appropriate written consent, maintain individual confidentiality and privacy.
- Collaboration in research
- Existing guidelines on collaboration for sharing biological samples, data and intellectual property including publication related issues will be applicable. Collaborations to address possible inequity of expertise, access between partnering institutions /funding relationships.
- Rapid data sharing while safeguarding the above is critical during public health emergency.
- An appropriate MoU and/or MTA to safeguard the interests of participants and ensure compliance (addressing issues of confidentiality, sharing of data, joint publications, etc).
- Biomedical and health research proposals involving foreign assistance and/or collaboration be submitted to Health Ministry's Screening Committee (HMSC) for approval before initiation.
- Role of Agencies/ Sponsors & Governance of Research:
- Central regulatory authority to undertake expeditious review process for clinical trials for new drugs/ compassionate use and ensure safety/efficacy monitoring processes.
- In case of an outbreak of infectious diseases, monitored emergency use of unregistered and experimental interventions (MEURI) may be approved with the following precautions:
- Thorough scientific review followed by an ethics review / locally or by national level EC
- Tackle public concerns and ensure oversight by a local EC.
- Use GMP products, make rescue medicines/supportive treatment accessible

- Meticulous documentation of therapeutic processes including adverse events
- Fast track research and possible sharing of data on safety and efficacy for further research
- Consent process is important and must be carried out with care.
- Community engagement and ensuring fair distribution of scarce supply
- Facilitate post-trial access of the successful investigational drug/ vaccine free of cost to the trial participants till the same is available in the market.

- **Ethics Review:**
- Researchers should submit research proposals in the ICMR Common Forms for Ethics Review as soft or hard copies enclosing required documents. (http://ethics.ncdirindia.org/Common_forms_for_Ethics_Committee.aspx)
- The EC should adopt/ include an SOP for Emergency Research review.
- Submission of e-copy of research protocol and relevant documents followed by their screening by Secretariat for completeness and categorization as exempt/ expedited review/ emergency full committee review depending on the urgency and need.
- Electronic documents may be accepted for review and timelines shortened for accelerated procedures.
- Agenda of meetings should be kept short, however, EC may meet more frequently for fast track review within in 24-48 hrs.
- The EC may plan a prior review by subject experts/obtain clarifications from researchers before the meeting or/ invite independent consultants (non-voting) or representative from a

Research Priorities

The Council's research priorities coincide with the national health priorities such as

- control and management of communicable diseases,
- fertility control,
- maternal and child health,
- control of nutritional disorders,
- developing alternative strategies for health care delivery,
- containment within safety limits of environmental and occupational health problems,
- research on major noncommunicable diseases like cancer, cardiovascular diseases, blindness, diabetes and other metabolic and haematological disorders, mental health research and drug research (including traditional remedies).

All these efforts are undertaken with a view to reduce the total burden of disease and to promote health and well-being of the population.

Organization of ICMR

- The Governing Council of ICMR is presided over by the Union Minister for Health & Family Welfare, GoI.
- It is assisted in scientific and technical matters by a Scientific Advisory Board comprising of eminent experts in different biomedical disciplines.
- The Board, in its turn, is assisted by a series of Scientific Advisory Groups, Scientific Advisory Committees, Expert Groups, Task Forces and Steering Committees etc. which evaluate and monitor different research activities of the Council.

The ICMR promotes biomedical research in the country through intramural as well as extramural research.

1. Intramural research is carried out currently through the Council's 32 Research Institutes/ Centres/Units. These include:
 - a) 21 mission-oriented national institutes located in different parts of India that address the research on specific areas such as tuberculosis, leprosy, cholera and diarrhoeal diseases, viral diseases including AIDS, malaria, kala-azar, vector control, nutrition, reproduction, immunohematology, oncology, medical statistics, etc;
 - b) 6 Regional Medical Research Centres that address regional health problems, and also aim to strengthen or generate research capabilities in different geographic areas of the country; and
 - c) 5 Units/Centres dealing with food & drug toxicology, viral diseases, handling microorganisms of highly infectious nature, prenatal diagnosis for neonatal retardation etc and supply of various animal models and feeds for experimental purposes.

2. Over the decades, the base of extramural research and also its strategies have been expanded by the Council. Extramural research is promoted by ICMR through:

- a) Setting up of Centres for Advanced Research in different research areas around existing expertise and infrastructure in selected departments of Medical Colleges, Universities and other non-ICMR Research Institutes.
- b) Task force studies which emphasize a time-bound, goal-oriented approach with clearly defined targets, specific time frames, well-standardized and uniform methodologies, and often a multi-centric structure.
- c) Open-ended research on the basis of applications for grants-in-aid received from scientists in non-ICMR Research Institutes, Medical colleges, Universities etc. located in different parts of the country.

3. In addition to research activities, the ICMR encourages human resource development in biomedical research through:

- a) Research Fellowships
- b) Short-Term Visiting Fellowships
- c) Short-Term Research Studentships
- d) Various Training Programmes and Workshops conducted by ICMR Institutes and Headquarters

4. For retired medical scientists and teachers

- the Council offers the position of Emeritus Scientist to enable them to continue or take up research on specific biomedical topics. The Council also awards prizes to Indian scientists, in recognition of significant contributions to biomedical research. At present, the Council offers 38 awards, of which 11 are meant exclusively for young scientists (below 40 years).

List of ICMR facilities

- National Institute of Nutrition (NIN), Hyderabad
- National Animal Resource Facility for Biomedical Research (NARF-BR), Hyderabad
- National Institute for Research in Tuberculosis (NIRT), Chennai
- National Institute of Epidemiology (NIE), Chennai
- National Institute of Cancer Prevention and Research (NICPR), Noida
- National Institute of Malaria Research (NIMR), Delhi
- Rajendra Memorial Research Institute of Medical Sciences (RMRIMS), Patna
- National Institute for Research in Reproductive Health (NIRRH), Mumbai
- National Institute of Virology (NIV), Pune
- National Institute of Traditional Medicine (NITM), Belgavi
- Microbial Containment Complex (MCC), Pune
- National AIDS Research Institute (NARI), Pune
- National Institute of Occupational Health (NIOH), Ahmedabad
- National Institute of Pathology (NIP), Delhi
- National Institute of Medical Statistics (NIMS), Delhi
- Vector Control Research Centre (VCRC), Puducherry
- National Institute of Cholera and Enteric Diseases (NICED), Kolkata
- National Institute for Research in Tribal Health (NIRTH), Jabalpur
- National Center for Disease Informatics and Research (NCDIR), Bengaluru
- Bhopal Memorial Hospital and Research Center (BMHRC), Bhopal
- National Institute for Research in Environmental Health (NIREH), Bhopal
- National JALMA Institute for Leprosy & Other Mycobacterial Diseases (NJILOMD), Agra
- Centre for Research in Medical Entomology (CRME), Madurai
- National Institute of Immunohaematology (NIIH), Mumbai
- Enterovirus Research Centre (ERC), Mumbai
- Genetic Research Centre (GRC), Mumbai
- National Institute for Implementation Research on Non-Communicable Diseases (NIIRNCD), Jodhpur
- Regional Medical Research Center (RMRC), Port Blair
- Regional Medical Research Center (RMRC), Bhubaneswar
- Regional Medical Research Centre, Dibrugarh
- ICMR Virus Unit (IVU), Kolkata
- Institute of Reproductive Medicine (IRM), Kolkata

Conclusion

- For research to be conducted ethically we need to follow these twelve general principles laid down by the ICMR.
- In order to follow the principles we should be aware about the informed consent process, vulnerable population, therapeutic misconception, post trial access and structure and role of ethics committees.
- These concepts hold special importance in developing countries like ours, as most of the research participants are uneducated and economically backwards.

1932

HYGIENE AND PUBLIC HEALTH CENTRE

The Governing Body of IRFA sets up the Institute of Hygiene and Public Health at Kolkata.

1937

NUTRITIVE VALUE OF INDIAN FOOD

The Nutritive Value of Indian Foods and Planning of Satisfactory Diets" is published for the first time for India.

1941

THE FIRST EVER BIOMEDICAL RESEARCH FELLOWSHIP

The first-ever biomedical research fellowship scheme is initiated by IRFA aimed at nurturing health research in India.

1942

TRANSMISSION CYCLE OF KALA- AZAR

Transmission cycle of the parasite of Kala-azar is elucidated, which helps in prevention and control of the disease.

1944

RECOMMENDED DIETARY ALLOWANCES

Recommended Dietary Allowances (RDA) are made for the first time for the Indian context and are updated every 10 years.



Diet Survey in Progress

1949

FILARIASIS CONTROL

National Programme for the control of Filariasis in association with the Medical Research Institute of India is initiated.

1956

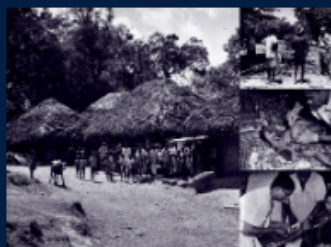
ICMR NIRT

The Tuberculosis Chemotherapy Centre is set up in Madras (now known as National Institute for Research in Tuberculosis or NIRT, Chennai).

1957

KYASANUR FOREST DISEASE

Kyasanur forest disease (commonly called Monkey Fever) is discovered in the Sagar-Soraba district of Karnataka.



The Kyasanur Forest Fever virus: from discovery to creating vaccine.

1957

ICMR NII

The Blood Group Reference Centre is established at Bombay (now known as National Institute of Immunohaematology or NII, Mumbai).

1959

HOME-BASED TREATMENT OF TB

Demonstrated for the first time in the world that home-based treatment of TB is as successful as hospital-based treatment.

1959

ICMR NARFBR

ICMR takes over the Laboratory of Animal Information Services Centre (LAISC) established in the campus of the Cancer Research Institute, Bombay, which then shifts to Hyderabad in 1976 and renamed as NCLAS in 1992 (now functioning as National Animal Resource Facility for Biomedical Research or NARFBR, Hyderabad since 2016).

1960

HEADQUARTERS OFFICE

The ICMR Headquarters office moves to its newly constructed building in An Nagar, New Delhi.

1962

ICMR NICED

The Cholera Research Centre is established at Calcutta (now known as National Institute of Cholera and Enteric Diseases (NICED), Kolkata).

1964

DIRECTLY OBSERVED TREATMENT, SHORT COURSE (DOTS)

First ever pioneering study demonstrated that a fully supervised twice weekly regimen of streptomycin and isoniazid was as effective as unsupervised daily administration of PAS and isoniazid, laying the foundation of Directly Observed Treatment, Short Course (DOTS).

1965

ICMR NIP

The Indian Registry of Pathology is set up at New Delhi (now known as National Institute of Pathology or NIP).

1966

ICMR NIOH

Occupational Health Research Institute is set up at Ahmedabad (now known as National Institute of Occupational Health or NIOH).

1967

SINGH'S CELL LINE

Cell culture from mosquito tissues is established (known as Singh's cell line and used all over the world in the study of arboviruses).

CHANDIPURA VIRUS

Chandipura Virus is discovered as a causal agent for human encephalitis.

1968

IRON-FOLATE SUPPLEMENT

Iron-folate supplement is recommended to pregnant women (Later implemented in the national programme in 1970).

1969

PROTEIN
ENERGY
MALNUTRITION

ICMR dispels the myth that protein deficiency is the main reason for malnutrition and highlights the calorie gap as the actual bottle neck in Protein Energy Malnutrition (PEM).

1970

NIGHT
BLINDNESS

National Prophylaxis Programme against Nutritional Blindness due to Vitamin A Deficiency is initiated after a series of trials by ICMR.

HUMAN LEUKOCYTE
ANTIGEN
DEPARTMENT

Establishment of first Human Leukocyte Antigen (HLA) department in India to provide tissue typing facilities for renal transplantation facilities at Blood Group Research Centre (BGRC), Mumbai.

1971

HOME
AVAILABLE FLUIDS

The demonstration of Oral Rehydration Therapy (ORT) to prevent mortality due to diarrhea is carried out. Home available fluids (HAF) such as Sherbat (salt, sugar, lemon, either singly or in combination) or tender coconut water and pressed rice water is found to be effective as well.

1972

NATIONAL NUTRITION
MONITORING
BUREAU

ICMR sets up a chain of Regional Centres of National Nutrition Monitoring Bureaus, in different parts of India.

1973

GOITER

ICMR demonstrates iodized salt controls endemic Goiter prevalence.

INA
ANTIGEN

The Ina blood group antigen was discovered by Dr. Badakere and his team at NIIH, and consequently, the Indian blood group system. Antibodies against this antigen carry clinical significance.

1975

ICMR
VCRC

The Vector Control Research Centre (VCRC) is established at Pondicherry.

1981

ICMR
RMRIMS

ICMR takes over the Rajendra Memorial Research Institute of Medical Sciences (RMRIMS) at Patna to research on vectorborne diseases especially Kala Azar.

ICMR
RMRC-B

Regional Medical Research Centre at Bhubaneswar (RMRC-B) is set up.

1982

ICMR
RMRC-NE

Regional Medical Research Centre at Dibrugarh (RMRC-NE) is set up.

1983

ICMR
RMRC-PB

Regional Medical Research Centre at Port Blair (RMRC-PB) is set up.

1984

FIRST
GENETIC ATLAS

The first genetic atlas of Indian tribes is compiled.

1984

ICMR
NIRTH

Regional Medical Research Centre for Tribal Health at Jabalpur is set up (now known as National Institute of Research in Tribal Health OR NIRTH).

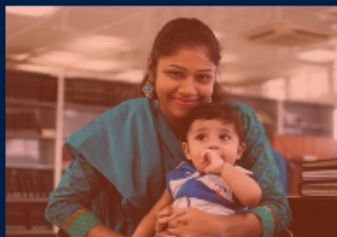
ICMR NIIRNCD
(FORMERLY DMRC)

Desert Medicine Research Centre at Jodhpur is set up (now known as National Institute of Implementation Research in Non- Communicable Diseases or NIIRNCD).

1986

TEST TUBE BABY

ICMR partners in India's scientifically established test tube baby at IRR (now NIRRH), Mumbai.



Test tube baby, Harsha with her child.

1986

HIV SERO- SURVEILLANCE

ICMR demonstrates the presence of HIV infection in India in 1980s and supports initiation of country-wide serosurveillance.

BRUGIA FILARIASIS

Shertalai Project: Demonstrated control of Brugian filariasis through IDVC methods.

MALARIA CONTROL

Demonstrated bioenvironmental control of Malaria.

1990

PG COURSE IN MEDICAL ENTOMOLOGY

Initiated M.Sc. in Public Health Entomology – the only course of its kind offered in India.

1992

NARI ICMR

National AIDS Research Institute (NARI) is established at Pune.

1993

NEW PHAGE TYPING SCHEME

A new phage typing scheme for V. cholerae biotype El Tor strain is developed.

NEW CHOLERA STRAIN

A new toxigenic strain of V. cholerae 0139 is detected and characterized.

LEPTOSPIROSIS

Identified Andaman Fever as Leptospirosis, first reported in Andaman Islands among convicts during 1929.

1994

SALT FORTIFICATION

Double fortification of cooking salt with iron and iodine is achieved.

1994

ICMR

NIE

National Institute of Epidemiology (NIE), Chennai is established.

1996

NEW

LEPTOSPIRA
SEROGROUP

Isolated and identified new serovar of new serogroup of Leptospira for first time causing pulmonary hemorrhages.

2002

KALA AZAR

TREATMENT

Miltefosine is validated as treatment for Kala Azar.

2003

POLIO

RESEARCH

ICMR contributes significantly to Polio Research and Global Polio Eradication Program and helps in achieving Polio free status in India.

2004

IMMUNE

CHROMATOGRAPHIC
DIPSTICK KIT

Supports development of Immune-chromatographic dipstick kit for the rapid diagnosis of cholera.

PPE FOR

TOBACCO
HARVESTERS

Personal protective equipment (PPE) for tobacco harvesters is produced. Seamless knitted nylon gloves are found to be most suitable, durable, and feasible for the workers.

2004

INDIGENOUS MAC-ELISA

TESTING KITS

Indigenous development of MAC-ELI kits against dengue, chikungunya and Japanese encephalitis viruses for diagnostic use in National Vector Borne Disease Control Program (NVBDCP) is carried out.

2006

LYMPHATIC
FILARIASIS

Policy change in the national programme for Lymphatic Filariasis elimination to replace the single drug therapy with combination therapy.

ACUTE
FLACCID
PARALYSIS

Identified acute flaccid paralysis for the first time that resembles Guilleain-Barre syndrome - a complication of chikungunya, carrying implications for treatment.

2006

ICMR
NITM

Regional Medical Research Centre (RMRC), Belgaum is set up (now known as National Institute of Traditional Medicine or NITM, Belagavi).

2007

DHR

Department of Health Research (DHR) is created with support from ICMR in September.

CTRI

Clinical Trial Registry of India CTRI (CTRI), a free and online public record system for registration of clinical trials being conducted in India is launched.

2008

SCHOOL OF PUBLIC HEALTH
CHENNAI

School of Public Health is established at National Institute of Epidemiology, Chennai.

2010

ICMR
NIREH

National Institute for Research in Environmental Health (NIREH) is established at Bhopal.

2011

WEST NIL
VIRUS

Lineage 1 West Nile Virus (highly pathogenic lineage with outbreak potential in Western countries) is detected for the first time in India, Kerala.

2011

ICMR NCDIR

National Centre for Disease Informatics and Research (NCDIR) is set up at Bangalore.

2012

BHOPAL MEMORIAL HOSPITAL

Bhopal Memorial Hospital and Research Centre, Bhopal is transferred to ICMR for providing clinical services to Bhopal Gas Tragedy victims.

PATHOGEN DETECTION KIT

A new test for detection of pathogens in food is developed and transferred to industry.

2013

BSL 4 LAB



The first high end biosafety level-IV laboratory (BSL) facility is established in Pune for epidemic preparedness and management.

MRU

ICMR initiates establishment of networks of Multi-Disciplinary Research Units (MRUs), Model Rural Health Research Units (MRHRUs), Viral Research & Diagnostic Laboratories (VRDLs).

2013

JENVAC VACCINE

Indigenous vaccine 'JENVAC' for Japanese Encephalitis is launched.

AMR SURVEILLANCE

Antimicrobial Resistance Surveillance Network established.

2015

VITAMIN A AND IRON ESTIMATION KITS

Dried Blood Spot (DBS) – collection kit for sub-clinical deficiency of Vitamin A and an ELISA-based kit to estimate iron in the blood are launched.

2016

MICRONUTRIENT MIX FOR ANAEMIA

ICMR develops a micronutrient mix (containing seven micronutrients) to fight against anaemia.

TB CONSORTIUM

India TB Research Consortium established.

2016

WHO-FCTC KNOWLEDGE HUB ON SMOKELESS TOBACCO

National Institute of Cancer Prevention and Research becomes seventh knowledge hub of WHO-FCTC (Framework Convention on Tobacco Control), globally.

2017

DISEASE BURDEN ESTIMATES

The India State Level Disease Burden Report is released which was the first comprehensive state wise disease burden estimation.

NEW DIAGNOSTIC KITS

ICMR has launched three new diagnostic kits for the detection of Crimean-Congo haemorrhagic fever (CCHF) Sheep and Goat, Crimean Congo haemorrhagic fever (CCHF) in Cattle, Japanese Encephalitis virus (JEV) from Mosquitos.

2017

OCCUPATIONAL SAFETY

Personal cooling garments for workers operating in high temperatures are developed.

2018

SAMRAT ASHOK TROPICAL DISEASE RESEARCH CENTRE

Samrat Ashok Tropical Disease Research Centre is established at Rajendra Memorial Research Institute of Medical Sciences, Patna.

VIRAL OUTBREAK CONTAINMENT

Successfully contained Zika, Nipah and Canine Distemper Virus.

2019



Brought out the special issue of IJMR titled "Gandhi and Health @150" to commemorate the 150th birth anniversary of Mahatma Gandhi.

2020

COVID-19 PANDEMIC

SARS-CoV-2 is first reported in India by NIV Pune.

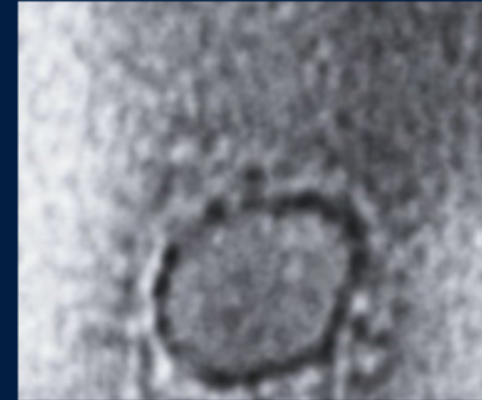
Isolated the virus.

Developed RT-PCR & ELISA for diagnosis.



Laboratory Investigations in National Institute of Virology, Pune.

2020

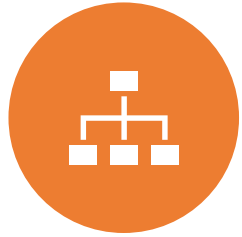


Transmission electron microscopy image of SARS-CoV-2.

Project Management in CT

Ms. Sonali Correa

Ph.D., M.Sc



CLINICAL TRIAL MANAGEMENT IS MOST SIMPLY DEFINED AS THE PROCESS THAT AN ORGANIZATION FOLLOWS TO ENSURE THAT QUALITY (DEFINED AS MINIMIZED RISKS AND CLEAN DATA) IS DELIVERED EFFICIENTLY AND PUNCTUALLY



IT REFERS TO A STANDARDS-DRIVEN PROCESS THAT A PROJECT MANAGER INITIATES AND FOLLOWS IN ORDER TO SUCCESSFULLY MANAGE CLINICAL TRIAL SITES, CLINICAL RESEARCH ASSOCIATES, AND WORKFLOW BY USING CLINICAL TRIAL MANAGEMENT TOOLS OR SOFTWARE



GOOD PROJECT MANAGEMENT IN CLINICAL TRIALS REQUIRES A DETAILED UNDERSTANDING OF BUSINESS PROCEDURES, BEING AWARE OF COMMERCIAL AND STRATEGIC PRIORITIES, AND AN AWARENESS OF KEY PERFORMANCE INDICATORS AND TRENDS.



THE KEY AREAS OF EXPERTISE FOR CLINICAL TRIAL PROJECT MANAGERS ARE COMMUNICATION, RELATIONSHIPS, INFLUENCE, PLANNING, ATTENTION TO DETAIL, AND FOLLOWING INSTRUCTIONS.



STRONG TEAMWORK SKILLS ARE ALSO ESSENTIAL WHEN WORKING ON PROJECTS AS WELL AS PEOPLE MANAGEMENT SKILLS

Clinical Trial Management

Need

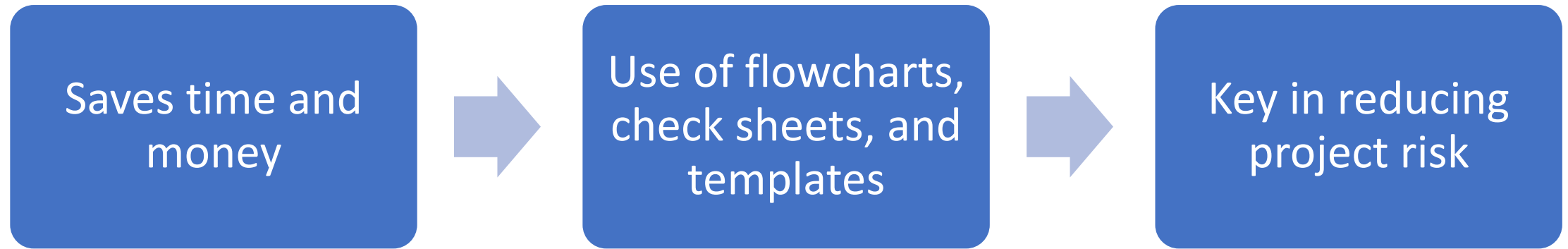


Prolonged timelines
and heavy costs
related to large trials
have prompted a
new focus on more
efficient clinical trial
management

The diagram consists of two large circles connected by a right-pointing triangle. The left circle is orange and contains text about the current state of clinical trial management. The right circle is grey and contains text about a potential future state. The triangle is also orange and points from the left circle to the right circle, indicating a flow or transition.

It is possible to
dramatically reduce
the total cost of a
clinical trial by 60% -
90% without
compromising the
scientific validity of
the results

Standardization



Flexibility

- Flexibility needed for differences in projects
- Situations that will arise amidst the projects

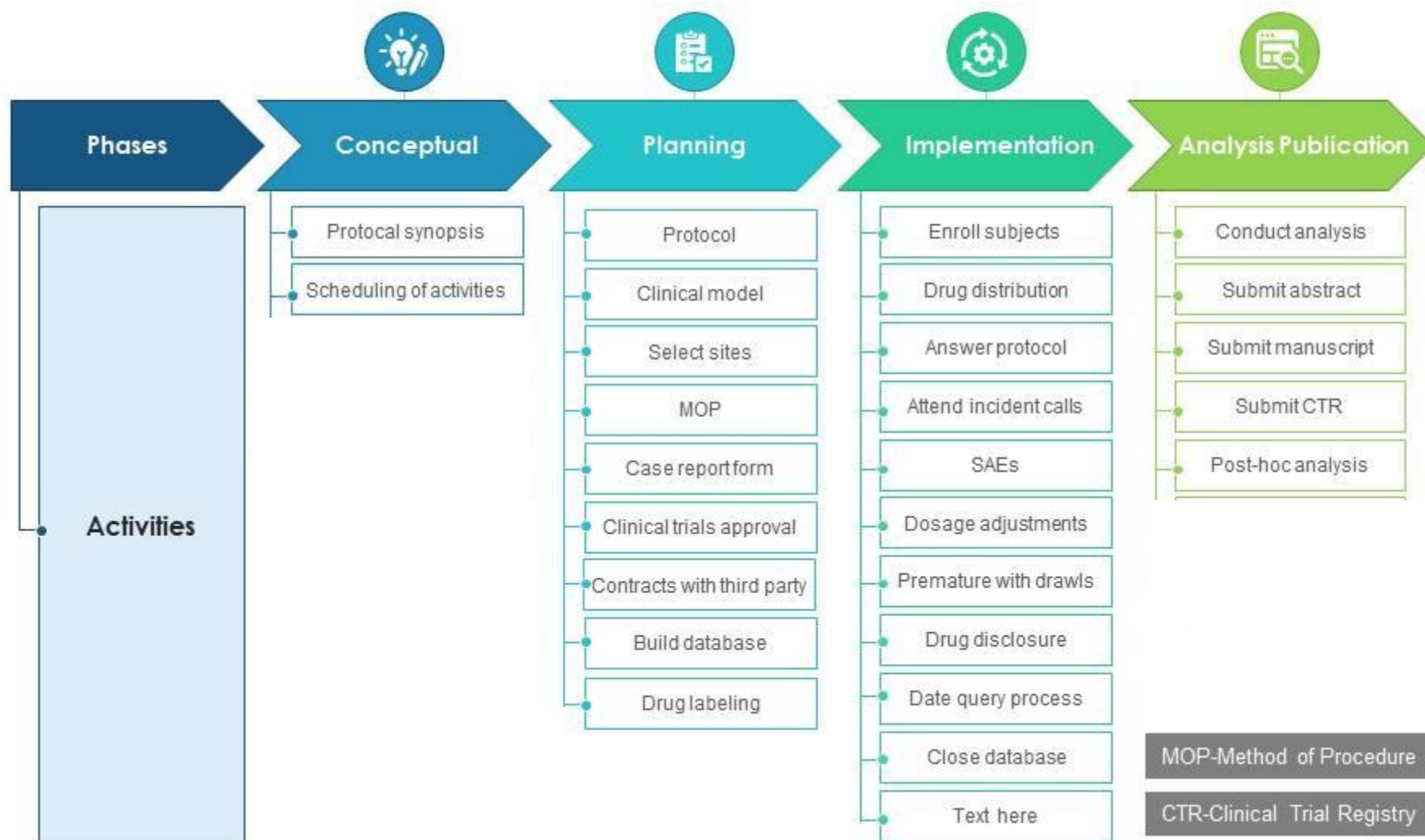
Constraints



Project Management Constraints

Project Management Process for Clinical Trial

This slide shows the project management process for clinical trial which includes four different phases such as conceptual, planning, implementation and analysis publication with activities like building database, making adjustments, conducting primary and secondary analysis, etc.





Conceptual & Planning Phase



Conceptual/Planning Phase- Key Activities

- Protocol/Consent Selection
- Confidentiality Agreement
- ClinicalTrials.gov registration
- Scope of Work
- Creation of Recruitment Materials
- Budget
- Vendor Identification
- Creation of Study Plans and Work Ins.
- Case Report Creation
- Database Build
- TrialMaster File (General and for Sites)
- Steering Committee Selection
- Site Selection by Steering Committee
- Site Selection Survey
- Data Safety Monitoring Board

Standard Operating Procedures (SOPs)

- It is important to develop site-specific Research SOP's, utilize the ones you have as a research site and use these as a jumping off point for clinical trials management.
- An SOP will describe and provide guidance on the development and management of clinical trials. Your sponsor may require an audit of your SOPs.
- **Include policies on (if applicable):**
 - Study Start-Up (TrialMaster Files, Protocol and Consent Development, etc.)
 - Database Build and Validation (RedCap is not Part 11 compliant so may need to research another
 - database system for your trial if required by sponsor)
 - Database Lock
 - Safety Monitoring
 - Training
 - Quality Assurance
 - Audit Procedures

Confidentiality Agreements

You may be provided a confidentiality agreement by your sponsor to use or you may need to create one for your site(s).

This may be needed in order for you or your sites to receive additional information:

- Protocol Synopsis
- Schedule of Activities
- Per Subject Fee
- In general, once there is an executed subcontract; there is no need to have new staff sign the

Confidentiality Agreement, as staff employed at the site are bound by confidentiality terms in the subcontract.



Using the Protocol Synopsis, Schedule of Activities, Scope of Work, and other related documents, develop a budget, which shall describe the costs of activities performed during the course of the planned agreement.



Present this budget to your Sponsor and work with Sponsor to negotiate mutually acceptable compensation for the activities requested and an appropriate timeline for payments (milestone payments vs. at specific time points vs. a combination of the two)



When applicable, the budget shall include a proposed Per-Subject Fee (PSF) to be paid to the participating clinical sites (if applicable)

Budgets

Contracts

- **Overview of the Process**
- Clinical Trial Agreement (CTA) negotiated and executed between CRO and Sponsor
- **If a multi-center study:**
- Site Contracts have flow down language from the sponsor's contract
- Terms and Conditions of the subcontract are prepared
- Study team identifies selected sites
 - Provides list of PIs, coordinators, administrators at each site to ORPA
- Office of Research and Project Administration (ORPA) sends out subcontract to sites
 - Subcontracts negotiated by ORPA – timeline can be 1-6 months
- Signed subcontract must be returned before the site can enroll subjects

General Trial Master File

PM Responsibility:

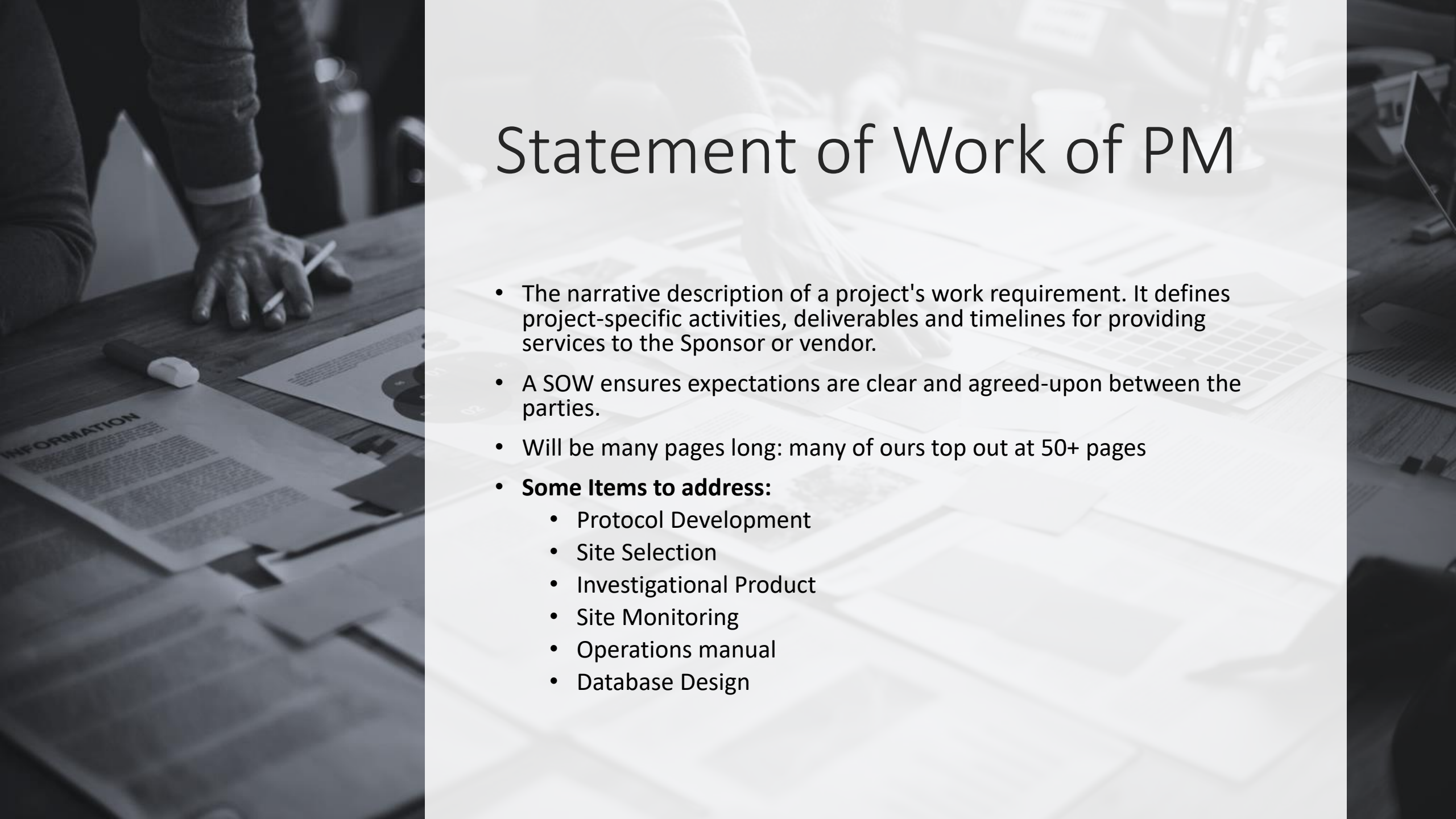
Collecting, handling, and safeguarding regulatory study documents throughout the lifecycle of a study, and conducting ongoing file review throughout each study phase, including review for accuracy and consistency between documents.

A Master Tracker should be maintained to track key site documents.

Tabs and columns can be added to or removed from the Master Tracker Template based on discussions with Sponsor and study-specific needs.

TMF should include but is not limited to the following folder structure:

- Trial Management
- Central Trial Documents
- Regulatory
- IRB
- Site management (if applicable)
- IP and Trial Supplies
- Safety Reporting
- Central and Local Testing
- DATA Management
- Statistics



Statement of Work of PM

- The narrative description of a project's work requirement. It defines project-specific activities, deliverables and timelines for providing services to the Sponsor or vendor.
- A SOW ensures expectations are clear and agreed-upon between the parties.
- Will be many pages long: many of ours top out at 50+ pages
- **Some Items to address:**
 - Protocol Development
 - Site Selection
 - Investigational Product
 - Site Monitoring
 - Operations manual
 - Database Design

ID	Task/Description	SPONSOR	PRINCIPAL INVESTIGATOR	CTCC	BIostatistics	VENDORS	Comment
A. Distribution of Case Report Forms (CRFs) or eSource Document Worksheets							
1	Ship sufficient CRF Source Documents or eSource Document Worksheets to sites						EDC will be utilized. Source Docs prepared and provided to sites by CTCC
B. Distribution of Operations Manual							
1	Perform power analysis to determine sample size estimates. Distribute the Operations Manual to site investigators and coordinators, Study Group and Sponsor staff prior to initiation of the study.						
C. Data Management							
1	Finalize a Data Management Plan which will include an SAE reconciliation plan.						SPONSOR must approve formally
2	Develop Clinical Data Management System						SPONSOR must approve formally
3	Develop additional software manuals incidence reporting, study notification, Monitor tracking, RAF Process, clinically significant lab, eMedWatch, Payment Tracking, Query Resolution Tracking, Other Metrics Report						SPONSOR must approve formally
4	Set up (Video, MRI, PET) repository SFTP site and transfer process						Sponsor to contract with vendor to receive and read FILES during trials. At conclusion of trials, SPONSOR will provide HSG
5	Helpdesk support and IT training for all sites						
6	Move Database to Production.						
7	Conduct validation including Production Qualification Run (PQR).						
8	Provide read-only access to the CTDMS.						
9	Provide access to standardized or study specific reports						
10	Initiate live data entry.						

Transfer of Obligations

The responsibility for performing study-related activities and obtaining regulatory documents should be negotiated with the Sponsor through the SOW and documented per the Transfer of Obligation prior to study initiation.

The Transfer of Obligation should be placed in the Trial Master File. Example: If the Sponsor is responsible for regulatory document collection, they must forward a list to the PM noting the documents they will collect.

Study Timeline

Study timeline

- Populate with study milestones
- Serves as a reference for the study team, sponsor, vendor
- One document – easy to update
- Aids the department in determining how to adequately staff the project

Can be created in:

- Microsoft Project (pictured)
- Word
- Excel
- Tableau

Task Mode	Task Name	Duration	Start	Finish	Predecessors	Comments
	MS PROJ TEMPLATE_ABBREVIATED	773 days	Tue 1/1/19	Thu 1/13/22		Adjust dur & pred
	CONCEPTUAL PRE-STUDY	20 days	Mon 4/1/19	Fri 4/26/19		
	RFP to Final Proposal Submitted	0 days	Mon 4/1/19	Mon 4/1/19		
	Final Agreement Signed	0 days	Fri 4/26/19	Fri 4/26/19	3FS+1 mon	
	EARLY PLANNING	162 days	Wed 1/2/19	Tue 8/20/19		
	Kick-off meeting	1 day	Wed 1/2/19	Wed 1/2/19		If available
	RSRB Approval	1 mon	Wed 7/24/19	Tue 8/20/19	4FS+3 mons	
	IMPLEMENTATION	192 days	Tue 1/1/19	Wed 10/2/19		
	Sites Selected, under contract, IRB approved	6 wks	Wed 8/21/19	Wed 10/2/19	4,7	
	Database Development	0 days	Tue 1/1/19	Tue 1/1/19		When contracted
	Investigational Material Shipped to 1st Activated Site(s)	0 days	Tue 1/1/19	Tue 1/1/19		
	ENROLLMENT & TREATMENT PERIOD	531 days	Thu 9/19/19	Tue 10/19/21		
	ENROLLMENT PHASE	251 days	Thu 9/19/19	Sun 9/13/20		
	First Subject Screening	0 days	Thu 9/19/19	Thu 9/19/19	9FS+2 wks	
	First Subject Enrollment/Randomized	0 days	Sat 10/19/19	Sat 10/19/19	14FS+30 edays	
	25% Enrolled/Randomized	0 days	Wed 12/18/19	Wed 12/18/19	15FS+2 emons	
	50% Enrolled/Randomized	0 days	Tue 3/17/20	Tue 3/17/20	15FS+5 emons	
	75% Enrolled/Randomized	0 days	Mon 6/15/20	Mon 6/15/20	15FS+8 emons	
	100% Enrolled/ Randomized (LSFTx)	0 days	Sun 9/13/20	Sun 9/13/20	15FS+11 emons	
	TREATMENT PHASE (LSFT to LSLT)	0 days	Mon 8/23/21	Mon 8/23/21		When appropriate
	100% Subjects complete treatment (LSLTx)	0 days	Mon 8/23/21	Mon 8/23/21	19FS+12 mons	
	F/U VISITS POST TX TO STUDY COMPLETION (LSLV)	0 days	Tue 10/19/21	Tue 10/19/21		When appropriate
	100% Subjects complete final study visit (LSLV)	0 days	Tue 10/19/21	Tue 10/19/21	21FS+2 mons	
	CLOSEOUT	50 days	Wed 11/3/21	Thu 1/13/22		
	Monitoring completed	2 wks	Wed 11/3/21	Tue 11/16/21	23FS+2 wks	
	All subjects secured	2 wks	Wed 11/24/21	Thu 12/9/21	25FS+1 wk	
	Post DB lock c/o visits (preparedness)	2 wks	Fri 12/31/21	Thu 1/13/22	30FS+1 wk	
	DATABASE LOCK FOR FINAL ANALYSIS	20 days	Thu 12/2/21	Thu 12/30/21		
	All vendor data transferred to DB	0 days	Thu 12/2/21	Thu 12/2/21	25FS+2 wks	
	Database lock	0 days	Thu 12/23/21	Thu 12/23/21	26FS+2 wks, 29FS+2 wks	
	Data Transfer to Biostats	0 days	Thu 12/30/21	Thu 12/30/21	30FS+1 wk	

Site Selection – High Level

The PM or designee will send the following to the potential investigative sites:

- Site Interest letter - a letter introducing the potential study
- Protocol synopsis and Schedule of Study Activities
- Site Feasibility Survey information
- Request for Investigator's and Coordinator's signed and dated current (as of date of the confidentiality statement is sent) Curricula Vitae (CV's) or Biographical sketch (if federally funded).
- Request for Human Subject Protection Program (HSPP) Certification for those study personnel having direct patient contact. (This may also be obtained at a later date once the site is chosen.)
- Request for Investigator's and Coordinator's Good Clinical Practice (GCP) training certification
- Conflict-of-Interest, if required by Sponsor and/or study group

Selected Sites

Once a site has been selected and has agreed to perform the study the following may be sent to the site:

- Follow-Up Site Letter
- Critical Activities Checklist
- Final protocol
- Model Informed Consent Form
- Investigator's Brochure or Package Insert (for approved investigational product)
- Conflict of Interest Guideline
- Conflict of Interest Disclosure
- Any other required items will be sent to the site for completion
- IRB submission, review, and approval.

Implementation Phase

SIGNIFIES THE START OF PARTICIPANT ENROLLMENT

- Develop an agenda and meeting template that is approved by Sponsor
- Determine the types of meetings (Ops, Team, Sponsor), frequency, and attendees

[illegible]

Data Transfer Agreements

- All data transfers, consideration is given so that any variables subject to The Health Insurance Portability and Accountability Act of 1996 (HIPAA) or other privacy regulations and other potentially identifying information are transferred only when necessary and are provided via a secure mechanism.
- The method of data transfer as well as the appropriate documentation should be agreed upon by both the sending and receiving parties to assure an accurate and complete transfer
- DTS should contain, but not be limited to, the following components:
 - Version Control/Summary of Changes
 - Contact Information
 - General description of the type of data that is being transferred
 - Data File Structure (ASCII, SAS, etc.)
 - Data File Transfer (specifications regarding, frequency, dates as applicable, cumulative/incremental, blinded/unblinded, file naming conventions, delivery method, documentation, confirmation of receipt)
 - Test transfer information, if applicable
 - Information on communication regarding issues with the transfer, or data quality issues
 - Identification of the recipient of the data
 - Data File Contents (data dictionary, or other documentation that describe the data files, variable mappings, etc.)
 - Signature of those approving the DTS



Data Safety Monitoring Board

- The blinded and un-blinded statisticians are involved in the creation and review of SMC charter.
- The data may be requested in an aggregate form, grouped by treatment but with treatment unknown (e.g., blue or green arm), or fully un-blinded to the treatment.

Types of data may include, but are not limited to:

- Adverse Events
- Serious Adverse Events
- Study drug discontinuations, suspensions
- Concomitant medications
- Safety Laboratories
- Vital signs
- Occurrences of un-blinding by study sites
- Overall enrollment and enrollment by site
- Number of subjects that have withdrawn or completed
- Protocol violations

Closeout Phase

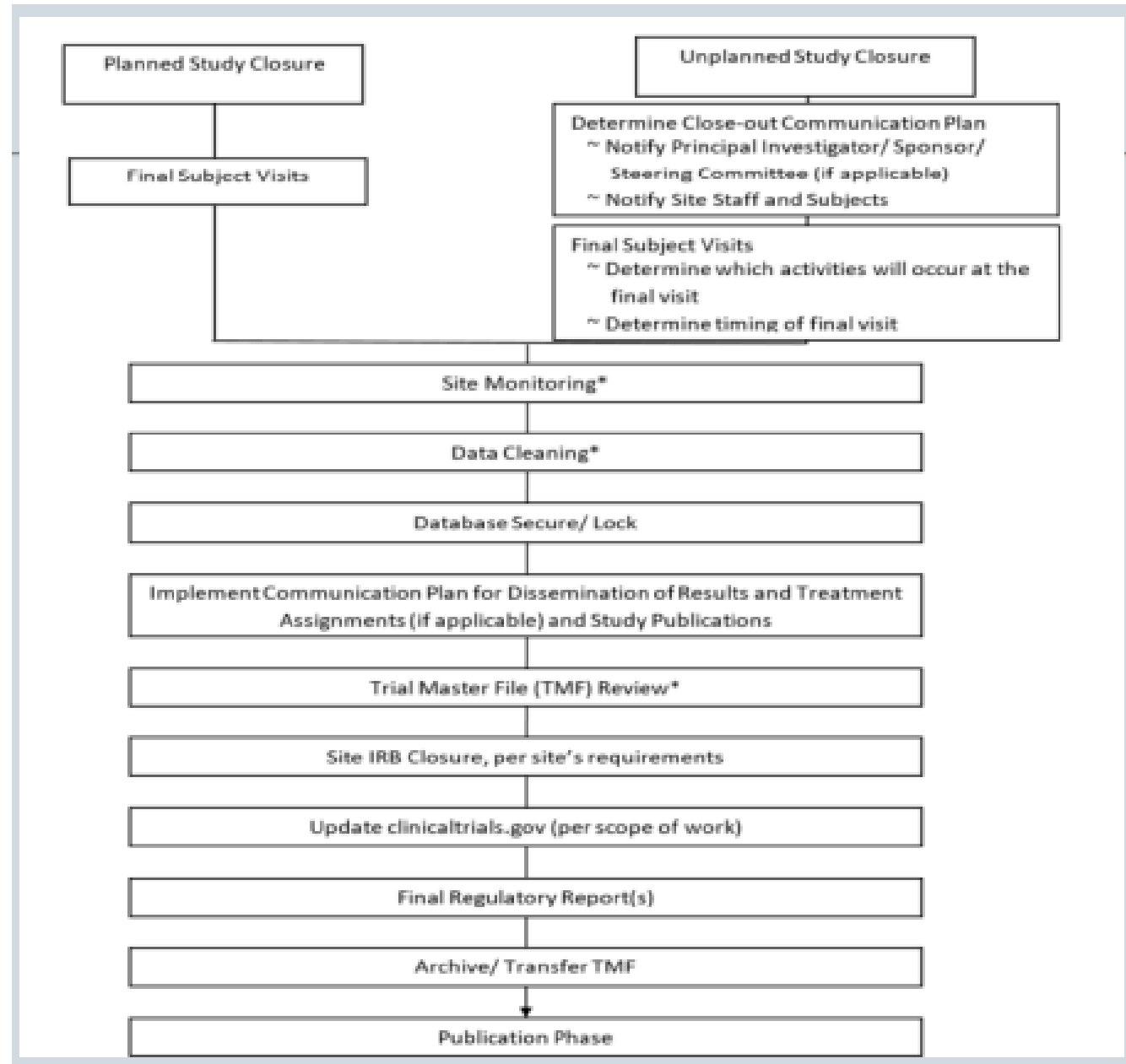
CLOSEOUT ACTIVITIES



Key Closeout Activities

- Closeout communication plan
- Treatment assignment disclosures
- Monitoring
- Survey of IRB/IEC Closure Requirements
- SAE reconciliation
- TMF reconciliation
- Final DSMB/SMC meeting
- IP reconciliation & disposition
- Return of study supplies/equipment
- Final site payments
- Final regulatory reports
- Database closeout
- Data analysis
- Publication
- Final study summary

Closeout Tasks



Closeout Checklist/Timeline

Track tasks, projected completion dates, actual completion dates and responsible parties

Projected Completion Day/Date	Actual Completion Day/Date	Task Category	Task	Description/Comments	Responsible Party
Day 0		Data Management	Last Subject Last Visit (LSLV) Completed		
Day 1		Equipment	Notify Vendors of Closeout		
Day 1		Regulatory	Update to ClinicalTrials.gov	Update Study Status	
Day 1		Regulatory	Notify Rochester IRB (RSRB) of Study Status	Update Study Status	
Day 30		Site Monitoring	Last On-site Monitoring Visit		
Day 45		Site Monitoring	Review of final site monitor visit trip reports	Review and approve final site visits reports submitted by site monitor.	
Day 45		Data Management	Final Data Queries Issued	Per CTCC Data Management Plan data is reviewed and queries issued to resolve any data discrepancies.	
Day 45		Safety	Ensure all SAE follow-up is complete		
Day 46		Study Drug	Complete site investigational product (IP) accountability		



Survey of IRB/IEC Closure Requirements

- Does your IRB/IEC have different levels/tiers of closure (i.e., closed to enrollment but open for administrative purposes)?
- Does the Template Treatment Assignment Letter to subjects require your IRB's approval before sending the letter with the treatment assignment to the subject?
- Are you required to keep the study open with your IRB/IEC until you provide the treatment assignments to subjects?
- Are you required to keep your IRB/IEC open for submission of the primary abstract?
- Are you required to keep your IRB/IEC open for submission of the primary manuscript?

Monitoring closeout visits

- Occurs when all study subjects have completed their last study visits per the protocol
- Early site closeout may occur in some instances
- Sites that close prior to database lock may need to re-open to address queries that arise during preparation for database lock and study closure.
- **Schedule and plan closeout monitoring visits across sites**
 - Final source verification
 - Close any open action items
 - Final IP accountability/authorization for destruction or return
 - Review of Trial Master File and archiving procedures
 - Final review of lab sample shipment to central lab/biorepository
 - Management/return of equipment & study supplies



Closeout Communication Plan

- Communication of results to sites
- Communication of results to participants, families & advocacy groups
- Teleconferences/webinars & letters
- Public release

Treatment assignment disclosures

- Facilitate providing all sites with subject treatment disclosure information to be provided to study subjects.
- Treatment assignment letter- template letter to be provided to each subject.
- IRB/IEC review and approval may be required.
- Following database lock and unblinding, each site is sent a list of treatment assignments.
- Each participant should be provided with the approved letter with their treatment assignment included.
- If a subject does not want to know their treatment assignment, document when the information was offered to the subject and when they declined.



TMF Reconciliation and Archiving



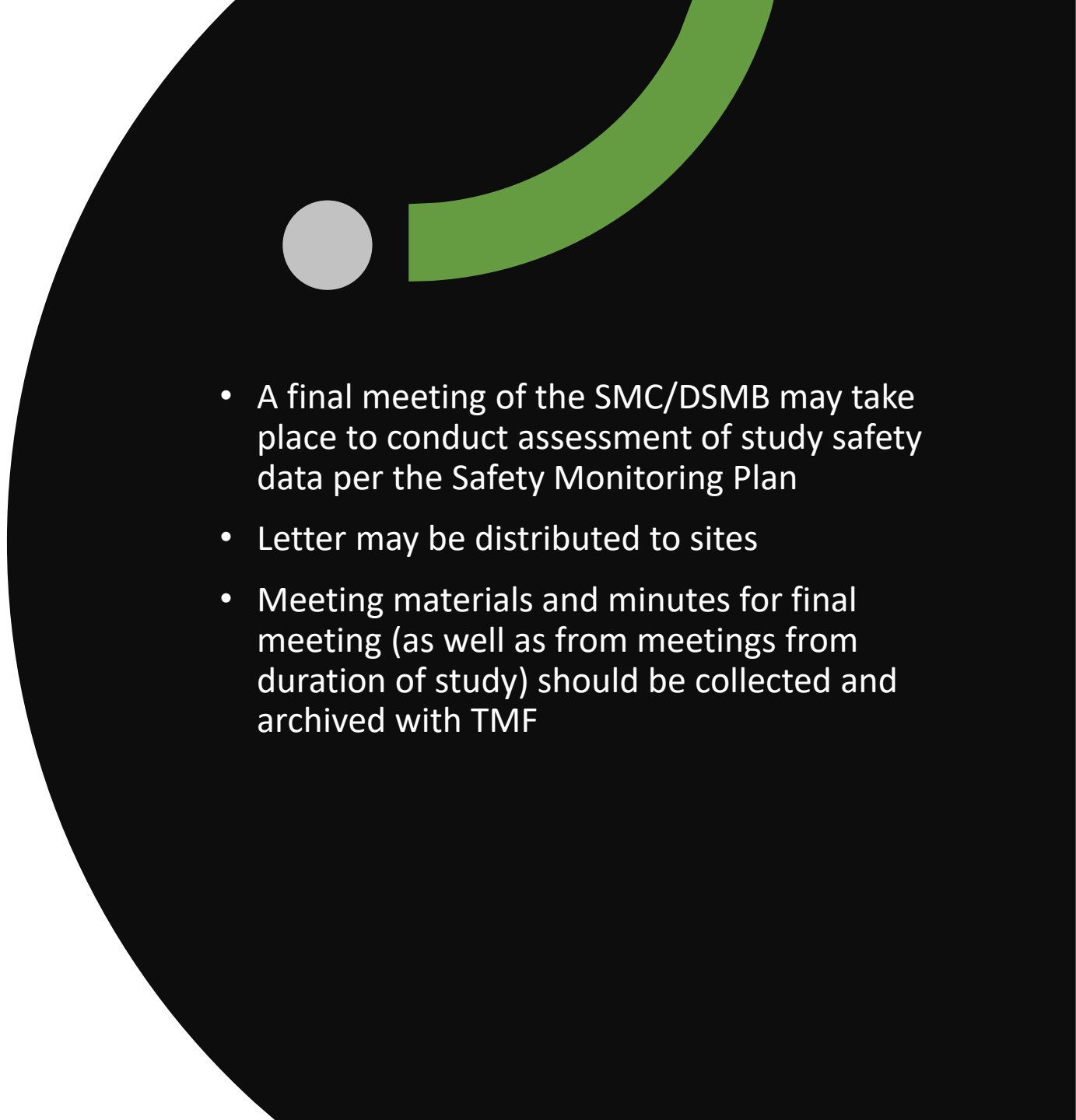
- Coordination Center TMF site files reconciled with site essential documents
 - Instructions for site file review
 - Site Archiving Form
- General and site files are archived according to the Project Management Plan

Data Management



- Last subject last visit (LSLV) completed
- Final data queries issued/resolved
- Final vendor data transfers (eg, central lab, EKG, etc)
- Final data review
- Database lock
- Data transfer to Biostatistics
- Statistical analysis

Final DSMB/SMC meeting

- 
- A final meeting of the SMC/DSMB may take place to conduct assessment of study safety data per the Safety Monitoring Plan
 - Letter may be distributed to sites
 - Meeting materials and minutes for final meeting (as well as from meetings from duration of study) should be collected and archived with TMF



Final Reconciliations

- Final investigational product reconciliation and disposition
 - Documentation of IP accountability and destruction for all distributed IP
- SAE reconciliation
- Biospecimens/digital reconciliation
- Reconciliation specimens (may include clinical labs, biospecimens, video recordings and repositories)

Other Closeout Activities

- **Study Supplies/Equipment**
 - All study-related equipment and supplies are returned
- **Final regulatory reporting**
 - Final reports submitted to applicable regulatory authorities per FDA and ICH guidelines, or funding agencies.
- **Administrative**
 - Update status of sites and personnel in study systems/remove accesses to EDC system and others
- **Final site payments**
 - May be held if there are outstanding activities that haven't been completed (eg, site hasn't returned equipment or submitted biospecimens)

Final Study Summary

- Study Group (if applicable)
- Sponsor
- IND number (if applicable)
- Study name/Acronym
- Protocol number
- Version date of original protocol and protocol amendment(s)
- Primary outcome measure(s)
- Duration of treatment
- Patient population
- Final number of sites
- Number of enrolling investigators
- Enrollment period
- Enrollment goal met (Yes/No)
- Number of subjects enrolled
- Investigational Product(s) (IP)
- IP manufacturer
- IP packager
- IP distribution center
- Personnel

Publications

The background features decorative curved lines in the corners. In the top-right corner, there is a thick, multi-layered arc transitioning from blue to green. In the bottom-left corner, there is a similar thick, multi-layered arc transitioning from green to blue.

Federal regulations and NIH policy require the responsible party to report results information for the clinical trial no later than 1 year after the Primary Completion Date

- The **Primary Completion Date** is the "date that the last subject was examined or received an intervention to collect final data for the primary outcome measure" of the study

Scientific information is submitted as four separate modules:

- Participant Flow
- Baseline Characteristics
- Outcome Measures and Statistical Analyses
- Adverse Events
 - Results are reported by study arm or comparison group

Participant Flow: numbers of participants who started, completed, and dropped out of each period of the study based on the sequence in which interventions were assigned

Baseline Characteristics: demographics, such as age and gender, and study-specific measures

Outcome Measures and Statistical Analyses: prespecified Primary Outcome and Secondary Outcome measures

- May also include other prespecified outcomes, post hoc outcomes, and any appropriate statistical analyses

Adverse Events:

- Tabular summary of all anticipated and unanticipated Serious adverse events
- Tabular summary of anticipated and unanticipated other adverse events exceeding a specific frequency threshold

Administrative Information: point of contact for study results, etc

PROTOCOL IN CT





DEFINITION

- Every clinical investigation begins with the development of a clinical protocol.
- The protocol is a document that describes how a clinical trial will be conducted (the objective(s), design, methodology, statistical considerations and organization of a clinical trial,) and ensures the safety of the trial subjects and integrity of the data collected.
- The protocol details exactly how the study should be conducted and therefore sets uniform standards for study conduct, allowing researchers to combine data from multiple sites to reach valid conclusions.
- Protocol review involves an assessment of the protocol to ensure that the study can be successfully completed considering the available resources and capabilities.

Protocol development process

- The protocol demonstrates the guidelines for conducting the trial.
- It illustrates what will be made in the study by explaining each essential part of it and how it is carried out.
- It also describes the eligibility of the participants, the length of the study, the medications and the related tests.
- The protocol should outline the rationale for the study, its objective, the methodology used and how the data will be managed and analysed.
- It should highlight how ethical issues have been considered, and, where appropriate, how gender issues are being addressed.

The actual writing of a protocol is a team effort with contributions from a

- medical expert
- a statistician
- a pharmacokinetics expert
- the clinical research coordinator
- the **project manager**

Who all provide input to the medical writer to produce the final document.



Protocol is
written by

The protocol language should be understood by

- Other physicians
- Nurses/ CRAs
- IRB Members
- Scientific reviewers
- IC for a lay person

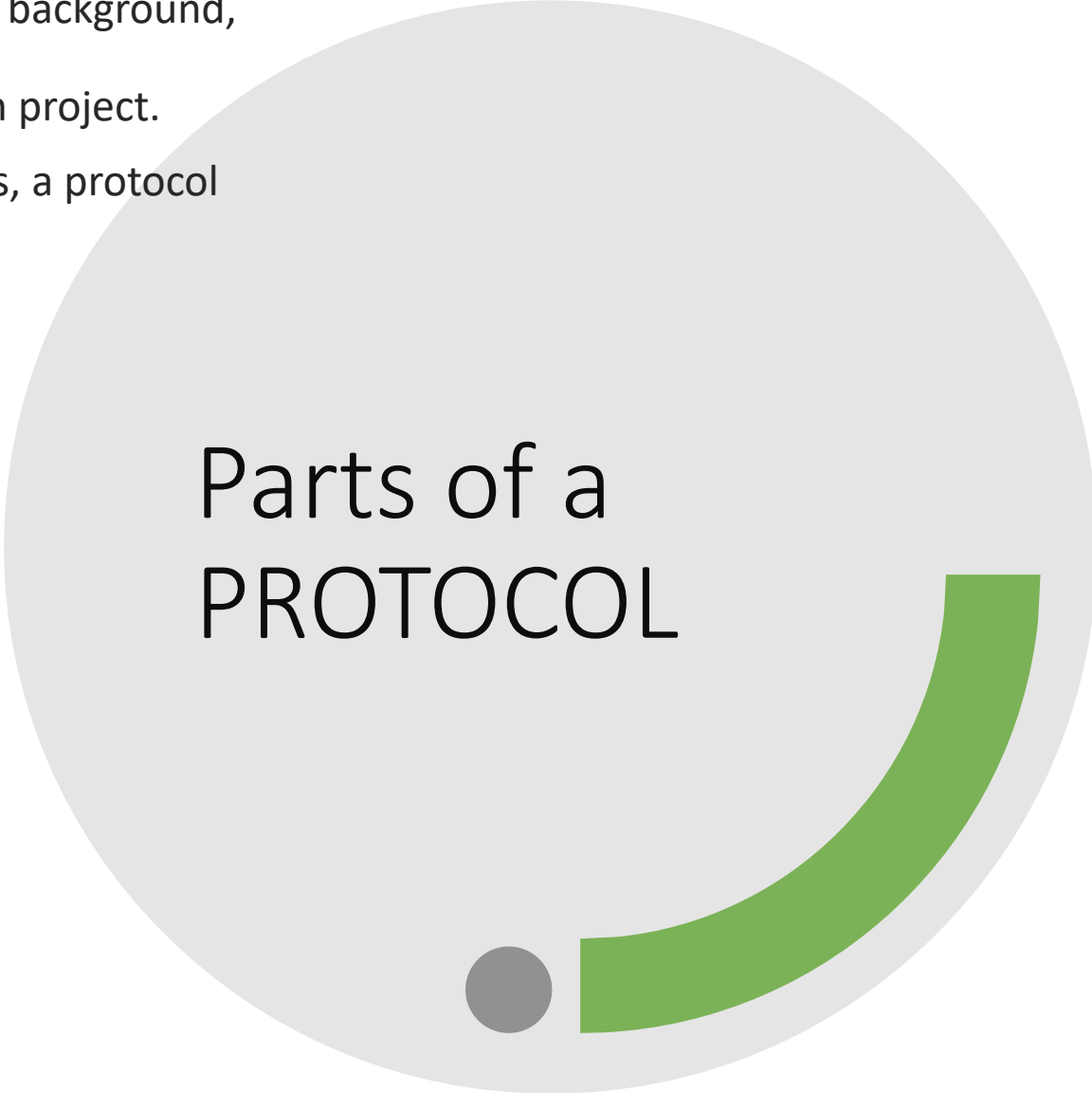


Protocol is read
by

A research protocol is a document that describes the background, rationale, objectives, design, methodology, statistical considerations, and organization of a clinical research project.

According to the ICH Good Clinical Practice guidelines, a protocol should include the following topics:

- Title Page (General Information)
- Background Information
- Objectives/Purpose
- Study Design
- Selection and Exclusion of Subjects
- Treatment of Subjects
- Assessment of Efficacy
- Assessment of Safety
- Adverse Events
- Discontinuation of the Study
- Statistics
- Quality Control and Assurance
- Ethics
- Data handling and Recordkeeping
- Publication Policy
- Project Timetable/Flowchart
- References
- Supplements/Appendices



Parts of a PROTOCOL

General information

- **Details of the protocol:** Protocol title, protocol identifying number, date, and any amendment(s)
- Details of the **sponsor, investigator(s), and sponsor's medical expert**
- Details of the **person(s) authorized to sign the protocol**
- Details of the **qualified physician** who is responsible for trial-site related medical decisions (if other than investigator)
- Details of the **clinical laboratory(ies) and other medical and/or technical department(s) and/or institutions** involved in the trial

Background information

- **Details of the investigational product(s)**
- **Summary of findings from nonclinical studies and clinical trials**
- **Summary of the known and potential risks and benefits** to human subjects
- **Description of and justification** for the route of administration, dosage regimen, and treatment period(s)
- Description of the **population** to be studied
- References to **literature and data** that provide background for the trial

A detailed description of

- Objectives of the trial
- Purpose of the trial

Purpose and Objectives



Study design and End-points

- Description of the **type/design** of trial to be conducted and a **schematic diagram** of trial design, procedures, and stages
- **End-points** of the study
- Measures taken to **minimize/avoid bias** (randomization, blinding etc.)
- Description of the **trial treatment(s)** (including packaging and labelling)
- **Duration** of participation (including sequence and duration of each period and follow-up)
- **Discontinuation criteria**

Selection and withdrawal of Subjects

- **Inclusion** criteria
- **Exclusion** criteria
- **Withdrawal** criteria along with procedures specifying:
 - When and how to withdraw subjects from the trial/ investigational product treatment
 - Type and timing of the data to be collected for withdrawn subjects
 - Whether and how subjects are to be replaced
 - Follow-up for subjects withdrawn from investigational product treatment/trial treatment

Study procedures & Schedule

- **Name of the product to be used**
- **Dose and dosing schedule**
- **Route of administration**
- **Treatment period** (including follow-up)
- **Medication(s)/treatment(s)** permitted (including rescue medication) and not permitted before and/or during the trial
- **Procedures** for monitoring subject compliance

Assessment of Safety & Efficacy

- Specification of **efficacy and safety parameters**
- **Methods and timing** for assessing, recording, and analyzing these parameters
- **For safety:**
 - Procedures for eliciting reports of and for recording and reporting **adverse events**
 - Type and duration of the **follow-up of subjects after adverse events**

Statistics

- **Statistical methods employed**
- Number of **subjects planned to be enrolled**
 - Reason for choice of sample size, including calculations of the power of trial and clinical justification
- **Level of significance** to be used
- **Criteria for termination** of trial
- Procedure for accounting for **missing, unused, and spurious data**
- Procedures for reporting **any deviation(s)** from the original statistical plan
- **Selection of subjects** to be included in the analyses

Human Subject Protection

- This section includes discussion of:
 - Subject selection and exclusion
 - Proposed methods of patient recruitment
 - Minority representation
 - Recruitment (or exclusion) of special subjects, including vulnerable subjects
 - Lists of potential risks and benefits, including justification for risks

Informed consent

- Informed consent involves providing a potential participant with:
- adequate information to allow for an informed decision about participation in the clinical investigation.
- facilitating the potential participant's understanding of the information.
- an appropriate amount of time to ask questions and to discuss with family and friends the research protocol and whether you should participate.
- obtaining the potential participant's voluntary agreement to participate.
- continuing to provide information as the clinical investigation progresses or as the subject or situation requires.
- To be effective, the process must provide sufficient opportunity for the participant to consider whether to participate. (21 CFR 50.20.)
- FDA considers this to include allowing sufficient time for participants to consider the information and providing time and opportunity for the participant to ask questions and have those questions answered.
- The investigator (or other study staff who are conducting the informed consent interview) and the participant should exchange
- information and discuss the contents of the informed consent document. This process must occur under circumstances that minimize the possibility of coercion or undue influence. (21 CFR 50.20.)

21CFR part 50.20

§ 50.20 General requirements for informed consent.

Except as provided in §§ 50.23 and 50.24, no investigator may involve a human being as a subject in research covered by these regulations unless the investigator has obtained the legally effective informed consent of the subject or the subject's legally authorized representative. An investigator shall seek such consent only under circumstances that provide the prospective subject or the representative sufficient opportunity to consider whether or not to participate and that minimize the possibility of coercion or undue influence. The information that is given to the subject or the representative shall be in language understandable to the subject or the representative. No informed consent, whether oral or written, may include any exculpatory language through which the subject or the representative is made to waive or appear to waive any of the subject's legal rights, or releases or appears to release the investigator, the sponsor, the institution, or its agents from liability for negligence.

- A central part of the informed consent process is the informed consent document. The Food and Drug Administration (FDA) does not dictate the specific language required for the informed consent document, but does require certain basic elements of consent be included.

Before enrolling in a clinical trial, the following information must be given to each potential research subject:

- A statement explaining that the study involves research.
- An explanation of the purposes of the research.
- The expected length of time for participation.
- A description of all the procedures that will be completed during enrollment on the clinical trial.
- Information about all experimental procedures that will be completed during the clinical trial.
- A description of any predictable risks.
- Any possible discomforts (e.g., injections, frequency of blood test etc.) that could occur as a result of the research.
- Any possible benefits that may be expected from the research.
- Information about any alternative procedures or treatment (if any) that might benefit the research subject.

A statement describing:

- the confidentiality of information collected during the clinical trial,
- how records that identify the subject will be kept
- the possibility that the FDA may inspect the records.
- For research involving more than minimal risk information including an explanation as to whether any compensation or medical treatments are available if injury occurs, what they consist of, or where more information may be found. questions about the research, research subjects' rights, injury related to the clinical trial.
- Research subject participation is voluntary,
- Research subjects have the right to refuse treatment and will not losing any benefits for which they are entitled,
- Research subjects may choose to stop participation in the clinical trial at any time without losing benefits for which they are entitled.

When Appropriate, one or more of the following elements of information must also be provided in the informed consent document:

- A statement that the research treatment or procedure may involve unexpected risks (to the subject, unborn baby, if the subject is or may become pregnant).
- Any reasons why the research subject participation may be ended by the clinical trial investigator (e.g., failing to follow the requirements of the trial or changes in lab values that fall outside of the clinical trial limits).
- Added costs to the research subject that may result from participating in the trial.
- The consequence of leaving a trial before it is completed (e.g. if the research and procedures require a slow and organized end of participation).
- A statement that important findings discovered during the clinical trial will be provided to the research subject.
- The approximate number of research subjects that will be enrolled in the study.

A potential research subject must have an opportunity to:

- read the consent document,
- ask questions about anything they do not understand.
- Usually, if one is considering participating in a clinical trial, he or she may take the consent document home to discuss with family, friend or advocate.

An investigator should only get consent from a potential research subject if:

- enough time was given to the research subject to consider whether or not to participate
- the investigator has not persuaded or influenced the potential research subject.
- The information must be in language that is understandable to the research subject.

Informed consent may not include language that:

- the research subject is made to ignore or appear to ignore any of the research subject's legal rights,
- releases or appears to release the investigator, the sponsor, the institution, or its agents from their liability for negligence.

<https://www.fda.gov/media/88915/download>

Protocols informed consent must provide

- Statement that the study involves research
- Purpose of the research and the length of the study
- Description of risks and benefits
- Discussion of alternative therapies
- Confidentiality policy
- Compensation for injury
- Contact for further questions/information
- Statement of voluntary participation

Other components of a Protocol

- **Direct Access to Source Data/Documents**
- **Quality Control and Quality Assurance**
- **Ethics**
- **Data Handling and Recordkeeping**
- **Financing and Insurance**
- **Publication Policy**
- **Supplements**

Information

- <https://methods.cochrane.org/bias/resources/study-protocols>

Case Report Form

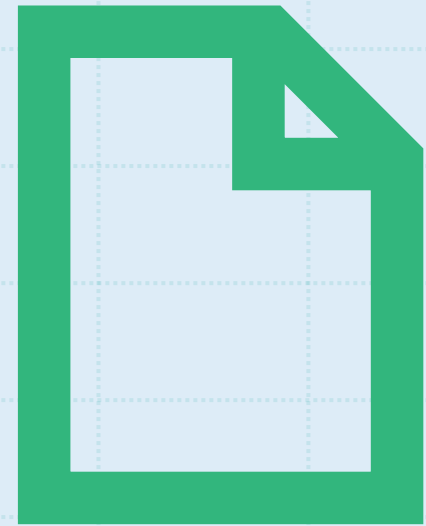
Ms. Sonali Correa

Ph.D., M.Sc.



Case Report Form

- **A Case Report Form (CRF) is an electronic or paper document which is used in a clinical trial to record the protocol and required information about each participant.**
- It enables efficient and complete data collection, processing, analysis and reporting.
- It is the main tool that Investigators use to collect information from clinical trial participants which is why there needs to be one completed CRF for each participant.
- CRF extracts most important data from source documents for data analysis/ entry. Source Documents are defined as original documents, data and records e.g. hospital records, x-rays, subject diaries, laboratory notes, ECG.





A printed, optical or electronic document designed to record all of the protocol-required information to be reported to the sponsor on each trial subject (ICH E6 GCP Guidelines)

An informative and well-structured CRF simplifies database design and data validation processes as well as manipulation of data during statistical analysis



The Sponsor-Investigator puts together the CRF Completion Guidelines and ensures that the site staff have received training before collecting, entering and reviewing data.

All users are given a unique username and password to gain access to the CRF and the trial database.

The Principal Investigator assigns permissions to each user which includes **'data entry' or 'data view only'**.

Permission to access the data must be in accordance with the consent given by participants.

The CRF should be completed soon after each participant's initial visit which ensures that information can be followed up on while the participant is easily accessible.

All the data should be collected with users in mind and the requirements set out by regulatory agencies. The questions should be as clear and concise as possible to prevent any duplication.

Relationship to Protocol

The clinical trial protocol determines what kind of data should be collected - all the information found in the CRF should be specified by the protocol.

Any data that will not be analyzed shouldn't be included in the CRF.

Elements of a CRF

- **A Case Report Form consists of three main parts: a header, a safety-related module and an efficacy-related module.**

- **Header**

The header must contain key identification information including a study number, a site number and a participant ID number.

- **Safety Modules**

Consists of the key safety analysis requirements found in the protocol which includes demographic information, adverse events, medical history, physical exam, deaths, drop-outs and eligibility confirmation.

- **Efficacy Modules**

The protocol outlines the elements which are required in the efficacy modules. The elements of this module include the key efficacy endpoints of trial, additional tests to measure efficacy, how lesions will be measured and required diagnostics.

CRF Development Process

The development of the CRF begins as soon as the development process begins.



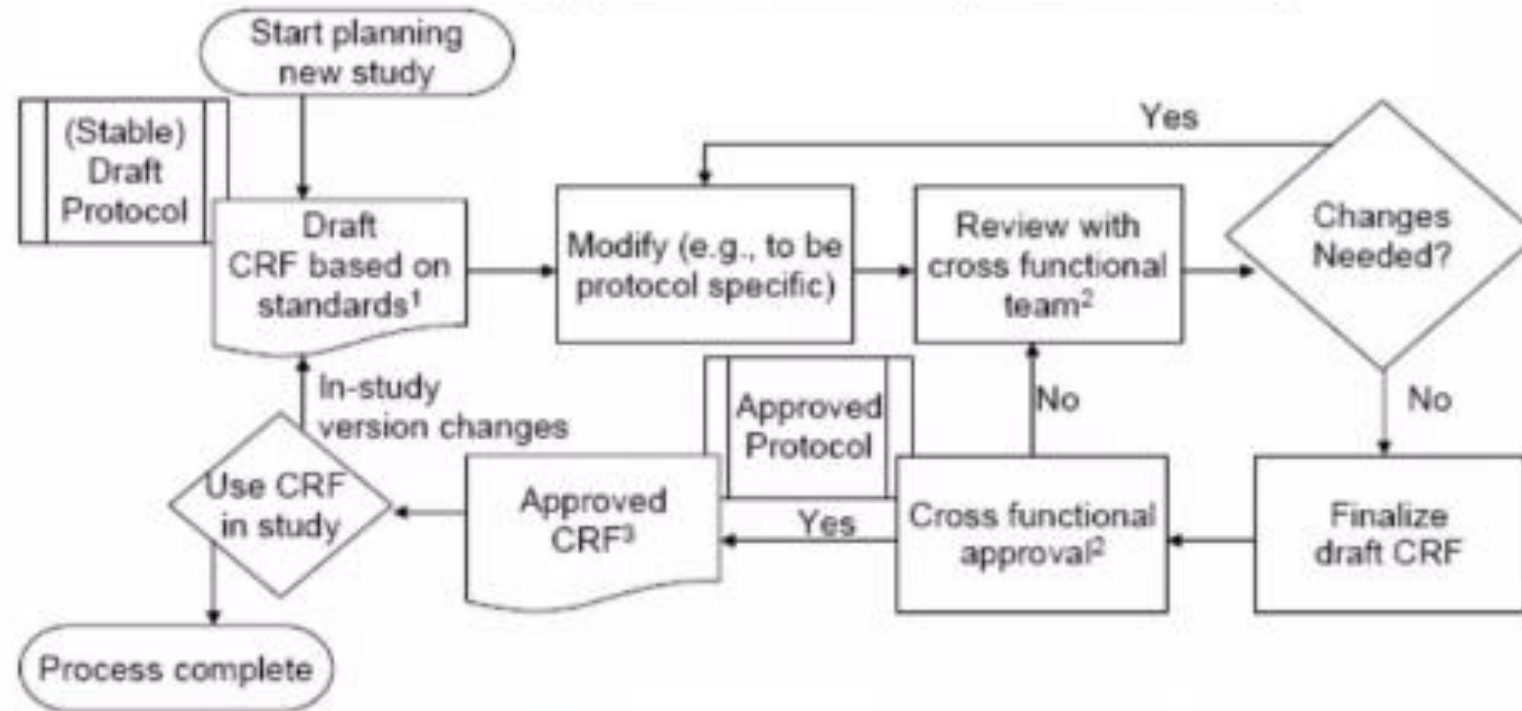
The responsibility of the design varies between Clinical Research Organizations, Clinical Research Associates, Data Managers, Research Nurses and Database Development.



The only data collected for the CRF should be what is required by the protocol- nothing more, nothing less. An interdisciplinary review is needed for every CRF:

- Each company has its own process for review.
- Should include the relevant staff members involved in the conduct, analysis and reporting of the clinical trial.

Suggested CRF Development Workflow



Designing of CRFs

- An effective CRF allows for components to be reused across the clinical trial, saving time and money.
- A poorly designed CRF is one that is missing requested data or has unnecessary information meaning that it requires edits and modifications throughout the clinical trial.

Table 1: Illustrating a well-designed and poorly designed data fields imparting the significance of visual cues to help the site personnel to understand the format

Poorly designed	Well designed
Date of visit: ____	Date of visit: □□/□□/□□□□ (DD/MM/YYYY)
Blood pressure: ____ / ____	Blood pressure: □□□/□□□ (mmHg)
Pulse: ____	Pulse: □□□ (beats/min)
Temperature: ____	Temperature: □□.□ (°C)
Respiration: ____	Respiration: □□ (/min)

CRF Completion

- **For a CRF to be completed, it needs to have:**
- Completed sections including headers with identifying items.
- Relevant modifications.
- A record of adverse events and specific documentation for any serious adverse events.
- The data needs to be collected from the relevant source documentation (medical records) and entered into the CRFs by study personnel. Only designated members of the clinical trial staff are allowed to record or edit data in the CRFs.

General Guidelines

- Entries in the CRF to be made using preferably Black ink ballpoint pen
- Ensure all entries are accurate, legible to others and verifiable with the source data
- In case of correction
 - Do not overwrite or erase!
 - Do not use correction materials (whiteners, cello tapes, and bleaching)
 - Write the correct entry nearby
 - Date and initial the correction (see the example given below)

PATIENT DETAILS: RKN ^{PK}_{03/12/2012}

Initials: <u>RKN</u> (e.g. SKN for S.K. Nagraj)	Serial No: <u>01</u>	Gender: ☒ M ☐ F	Age: (Yr): <u>25</u>
Diagnosis: <u>Tinea Corporis</u>			
Duration of illness: _____			

- Acceptable abbreviations are the following:
 - Unknown: UNK or UK
 - Not Done: ND
 - Not Applicable: NA.
- In questions where some of the possible options listed has/have to be selected, it should be Ticked (as shown below).

Gender	Male ☒	Female ☐
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Important Tips for Completion

The CRF Completion Guidelines should be followed to ensure that each CRF is accurate and completed in a timely manner.

Make sure the appropriate protocol, Investigator, and participant identifying information is included in the header.

Ensure the data is entered in the correct data field (provide boxes or lines to make it clear where the answers should be entered).

Use the appropriate units of measurement.

Be consistent.

Double-check spelling and grammar.

Watch for transcription errors.

Use the comment section to expand on any answer (be as concise as possible).

Use consistent formats, font style and font sizes.

Visual cues (boxes) which indicate the place and the format of data to be recorded.

Use check boxes.

Separate columns with thick lines.

Use precoded answer sets (male/female, yes/no and mild/moderate/severe, etc) wherever possible.

DEMOGRAPHY	
Date of birth (DD/MM /YYYY)	
Gender	Male ☐ 1 Female ☐ 2
Height (cm)	.
Weight (kg)	.
Smoker	Yes ☐ 1 No ☐ 2
Family history	Yes ☐ 1 No ☐ 2

- answers are coded in order to simplify the data collection. When codes are used to obtain an answer for a question, consistency in codes should be maintained throughout the CRF booklet and there should not be any variation in the answer for the same question
- For example, yes/no answers are coded as 1 = yes and 2 = no (preferred coding) as shown in Figure 4. If the codes are assigned in this order, the same order should be practiced throughout the CRF. Nowhere in the same CRF “1” should be coded for “no” and “2” should be coded for “yes”.

Electronic CRF (eCRF)

Ensures data integrity.

Provides an audit trail for data changes.

Ensures data preservation (automatic back-ups).

Essential identification information (protocol ID, site code and participant ID) will be generated from the first page to all other pages ensuring no duplication.

Linking data between 2 or more pages is easy.

There's always the threat of losing valuable data due to software malfunction or if the eCRFs aren't backed up successfully.

- **Confidentiality and Security**

- To protect confidential participant and clinical trial data, eCRFs must be stored on secure servers with restricted access, while paper CRFs must be stored in a secure location.
- Regular backups and archiving for future copies must be utilized.

- **Freezing the Database**

- Once the data has been finalized, it is necessary to lock the database which protects the integrity of the collected data and prevents any unauthorized edits from being made.



Input Team

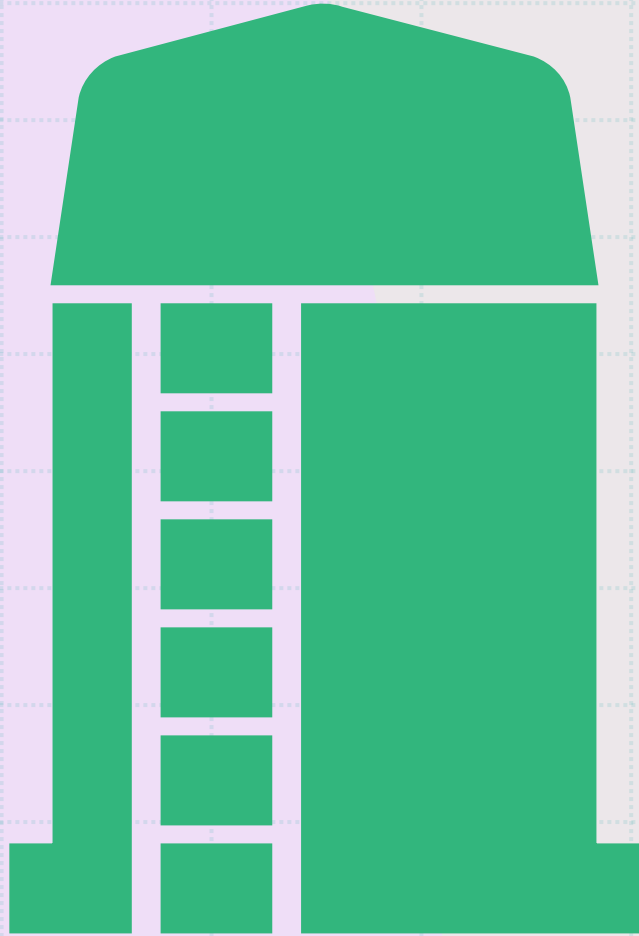
- **Input By:**
- Sponsors team Under Sponsor
- Investigators
- Data Managers
- Clinical investigation site Staff
- Medical person
- Barcode readers (e.g. That are used to record medicatins or devices)
- Specialty role - related to particular organ or disease ie. Consulting team (e.g Radiologist reporting on computed tomography (CT) scan)

Output Team

- Output to
- CRO
- Sponsor
- IRB/IEC
- FDA

Review Team

- Project Clinician
- Lead CRA
- Lead Statistician
- Lead Programmer
- Lead Data Manager
- Finalized draft CRF or CRF book approved by IRB/IEC after approving protocol and it is printed, accordingly stored and then distributed to research sites.

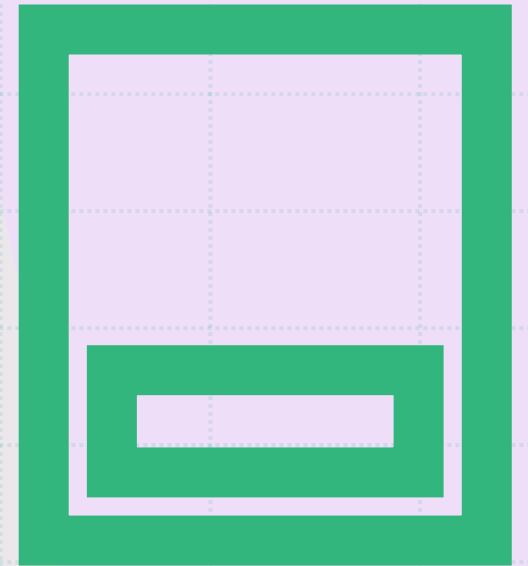


Data Management Minimum Requirements

- CRFs are not developed in a silo, majority of data is typically captured using
- CRFs or Forms and is integral to study success and part of the data
- management plan. Minimum requirements for data management:
 - **Traceability**
 - **Data Quality**

A few notes about Modality and Patient- Reported Outcomes

- **Paper:** Greater potential for missing data, particularly if poorly designed and not administered. Use common format (date fields, headers, response options, shading), page numbering (e.g., 1 of 5), large font, avoid referential questions
- **EDC:** Build in edit checks, picture fields, calendar pop-up, better for referential questions (dependent on whether CRF is paper or electronic) source or paper)
- **PRO:** Content should be clear and understandable to the subject population



ADVERSE EVENT		
Did any adverse event occur after the last visit?—Indicator Question		Yes ☐ 1 No ☐ 2
If "Yes", please provide the information in the adverse events (AE) page (page no. XX) and give details below:—Skip		
AE page number	AE serial number	Did any unscheduled visit happen after the last visit? (Please provide details on page no. YY)—Skip
☐ ☐	☐ ☐	Yes ☐ 1 No ☐ 2
☐ ☐	☐ ☐	Yes ☐ 1 No ☐ 2
☐ ☐	☐ ☐	Yes ☐ 1 No ☐ 2

Figure 1: A sample case report form (CRF) page. An adverse event page of CRF is depicted showing codes, and skips questions

Schedule Evaluation and Events (SOEE)

Construct/Domain	Measures	Visit (Days from Hospital Discharge Date)				
		SCRN	BASE (1:14)	1 Mo (28:55)	2 Mo (56:83)	3 Mo (84:168)
Participant Characteristics						
Consent/Eligibility	Date of Consent, Inclusion/Exclusion Criteria including toxicology (urine) and safety (blood)	X				
Demographics & SES	Age, Sex, Race, Marital/Partner Status, Health Insurance and Living Situation (homeless)	X	X			X
Health Characteristics						
Co-Morbidities	Charleson Index (EMR)	X	X	X	X	X
Mental Health	Depression (PHQ)	X	X			X
Substance Use, Motives, Consequences, Cravings, Readiness and Biomarkers						
Alcohol Severity	AUDADIS	X				X
Alcohol Use	Timeline Follow-back (TLFB)		X			X
Consequences	Short Inventory of Problems – Revised (SIP-2R)		X			X
Intervention						
NTX Route	Randomization Group (XR-NTX – PO-NTX)		X			
Compliance						
Medications	# of RW-MM visits with nurse or PO med mailing		X	X	X	X
Visit	Research Visits		X			X
Safety						
Adverse Events	Patient reported side effects, problems during RW-MM visits –opioid meds	X	X	X	X	X
Labs	Liver Tests (ALT, AST, GGT*), Pregnancy Test (SCRN as indicated)*	X		X	X	X

Conclusion

- Case report form design is the initial step in translating the protocol into standard questionnaires and is paramount to a successful clinical trial.
- Standard CRF should be designed in such a way that it helps the collection of consistent and valid data, ultimately resulting in submission of data to regulatory authorities and its acceptance.
- Regardless of the time and effort spent conducting the trial, the correct data points (response to a CRF question/data is entered) must be collected; otherwise, a meaningful analysis may not be possible. Therefore, a sound SAP should be used as a tool to develop and judge the adequacy of the CRF, which should be available to guide on what data points need to be captured on the CRF
- To avoid future amendments it is important to have design principles in mind well in advance before CRF designing is initiated.
- These standard guidelines will contribute in preparing a well-designed CRF for data acquisition

Investigator Brochure (IB)



IB

The Investigator's Brochure (IB) is a comprehensive compilation of clinical and nonclinical data on the investigational product (drug, supplement, device or other product) maintained by a drug developer or investigator that contains the body of information about the investigational product obtained before and during a drug trial.

The IB is a document of critical importance throughout the drug development process and is updated with new information as it becomes available.

Purpose:

Is to compile data relevant to studies of the investigational product in human subjects gathered during preclinical and other clinical trials to provide the investigator with information necessary for the management of study conduct and study subjects throughout a clinical trial.

- **An IB facilitates understanding of the rationale for, and their compliance with, many key features of the protocol, such as:**
 - Dose (of the study drug)
 - Dose frequency and dosing interval
 - Methods of administration
 - Safety monitoring procedures
- Information in the IB should be presented in a concise, simple, objective, balanced, and non-promotional form that enables a clinician, or potential investigator, to understand it and make his/her own unbiased risk-benefit assessment of the appropriateness of the proposed trial.
- When the investigational product is marketed and its pharmacology is widely understood by medical practitioners, an extensive IB may not be necessary. In these cases, a basic product information brochure, package leaflet or an expanded background information section in the trial protocol may be permitted by regulatory authorities as an alternative, provided that it includes current, comprehensive, and detailed information on all aspects of the investigational product that might be of importance to the investigator.
- The IB should be reviewed at least annually and revised as necessary in compliance with a sponsor's written procedures. When new information relevant to the clinical trial becomes available, it should be communicated to the investigators, and possibly to the Institutional Review Boards (IRBs) prior to a scheduled revision of the IB.
- Generally, the sponsor is responsible for ensuring that an up-to-date IB is made available to the investigator(s) and the investigators are responsible for providing the up-to-date IB to the responsible IRBs/IECs.

INVESTIGATOR'S BROCHURE: ICH E6 (R2) Good clinical practice

- **7.2.1 Title Page**

- Sponsor's name, identity of investigational product including research number, chemical or approved generic name, trade name and release date.
- Edition number and date if the current Investigator's Brochure replaces a previous one.

- **7.2.2 Confidentiality Statement**

- A statement which reminds Investigators and other recipients to treat the Investigator's Brochure as a confidential document and an important resource for the Investigators team and the Institutional Review Boards (IRBs) and Independent Ethics Committee (IEC).

- **7.3 Contents of the Investigator's Brochure**

- **7.3.1 Table of Contents**

- **7.3.2 Summary**

- Not longer than 2 pages.
- Highlighting significant physical, chemical, pharmaceutical, pharmacological, toxicological, pharmacokinetic, metabolic and clinical information available that is relevant to the state of clinical development of the investigational product.

- **7.3.3 Introduction**

- An introduction which includes a chemical name (generic and trade names), all active ingredients, the investigational product's pharmacological class and expected position within this class, the rationale for performing research with the investigational product and the anticipated prophylactic, therapeutic or diagnostic indications.
- The general approach to follow when evaluating an investigational product.

- **7.3.4 Physical, Chemical and Pharmaceutical Properties and Formulation**

- A description of the investigational product substances including the chemical and or structural formulae and the brief summary of the relevant physical, chemical and pharmaceutical properties.
- For safety measures, a description of the formulations to be used including excipients (a substance formulated with the active ingredient of a medication) should be provided and justified if clinically relevant.
- Instructions for the storage and handling of the dosage should be provided.
- Structural similarities to other known compounds should be mentioned.

- **7.3.5 Non-Clinical Studies**

- Introduction
- Results of the relevant pharmacology, toxicology, pharmacokinetic and investigational product metabolism studies should be provided in a summary form.
- The summary should address the methodology used, the results, and discussion of the relevance of findings on the investigated therapeutic, and possible unfavourable and unintended effects in humans.

- **Information Provided May Include the Following, as Appropriate, if Known/Available:**

- Species tested
- Number and sex of animals in each group
- Unit dose (milligram/kilogram)
- Dose interval
- Route of administration
- Duration of dosing
- Information on systemic distribution
- Duration of post-exposure follow-up
- Results, including the following aspects
 - - Nature and frequency of pharmacological or toxic effects
 - - Severity or intensity of pharmacological or toxic effects
 - - Time to onset of effects

- - Reversibility of effects
- - Duration of effects
- - Dose response

- A table and list format should be used whenever possible to make the presentation easier to understand.

- The following sections should include the most important findings from the studies, including the dose response of observed effects, the relevance to humans and any aspects to be studied in humans.

- If applicable, the effective and non-toxic dose findings in the same animal species should be compared (therapeutic index should be discussed).

- The relevance of this information to the proposed human dosing should be addressed.

- Whenever possible, comparisons should be made in terms of blood/tissue levels rather than on a mg/kg basis.

- **Non-Clinical Pharmacology**

- A summary of the pharmacological aspects of the investigational product, and, where appropriate, its significant metabolites studied in animals, should be included.
- The summary should incorporate studies that assess potential therapeutic activity (efficacy models, receptor binding and specificity) as well as those that assess safety (special studies to assess pharmacological actions other than the intended therapeutic effects).

- **Pharmacokinetics and Product Metabolism in Animals**

- A summary of the pharmacokinetics and biological transformation and disposition of the investigational product in all species studied should be given.
- The discussion of the findings should address the absorption and the local and systemic bioavailability of the investigational product and its metabolites and their relationship to the pharmacological and toxicological findings in animal species.

- **Toxicology**

- A summary of the toxicological effects found in relevant studies conducted in different animal species should be described under the following headings where appropriate:
- Single dose
- Repeated dose
- Carcinogenicity (the ability to produce cancer)
- Special studies (irritancy and sensitisation)
- Reproductive toxicity
- Genotoxicity (mutagenicity)

7.3.6 Effects in Humans

• Introduction

- A thorough discussion of the known effects of the investigational product(s) in humans should be provided, including information on pharmacokinetics, metabolism, pharmacodynamics, dose response, safety, efficacy, and other pharmacological activities.
- Where possible, a summary of each completed clinical trial should be provided.
- Information should also be provided regarding results of any use of the investigational product(s) other than from in clinical trials, such as from experience during marketing.

• **A. Pharmacokinetics and Product Metabolism in Humans**

- A summary of information on the pharmacokinetics of the investigational product(s) should be presented, including the following, if available:

- Pharmacokinetics (including metabolism, as appropriate, and absorption, plasma protein binding, distribution and elimination).
- Bioavailability of the investigational product (absolute, where possible, and/or relative) using a reference dosage form.
- Population subgroups (e.g. gender, age, and impaired organ function).
- Interactions (e.g. product-product interactions and effects of food).
- Other pharmacokinetic data (e.g. results of population studies performed within clinical trial(s)).

B. Safety and Efficacy

- A summary of information should be provided about the investigational product/product's (including metabolites, where appropriate) safety, pharmacodynamics, efficacy and dose response that were obtained from preceding trials in humans (healthy participants and/or patients). The implications of this information should be discussed.
- In cases where a number of clinical trials have been completed, the use of summaries of safety and efficacy across multiple trials by indications in subgroups may provide a clear presentation of the data.
- Tabular summaries of adverse drug reactions for all the clinical trials (including those for all the studied indications) would be useful.
- Important differences in adverse drug reaction patterns/incidences across indications or subgroups should be discussed.
- The Investigator's Brochure should provide a description of the possible risks and adverse drug

reactions to be anticipated on the basis of prior experiences with the product under investigation and with related products.

- A description should also be provided of the precautions or special monitoring to be done as part of the investigational use of the product(s).

C. Marketing Experience

- The Investigator's Brochure should identify countries where the investigational product has been approved and marketed.
- Any important information arising from the marketed use should be summarised (e.g. formulations, dosages, routes of administration, and adverse product reactions).
- Should identify all the countries where the investigational product did not receive approval/registration for marketing or was withdrawn from marketing/registration.

• **7.3.7 Summary of Data and Guidance for the Investigator**

- Provide an overall discussion of the nonclinical and clinical data and should summarise the information from various sources on different aspects of the investigational product(s), wherever possible.
- The Investigator can be provided with the most detailed interpretation of the available data and with an assessment of the implications of the information for future clinical trials.
- Where appropriate, the published reports on related products should be discussed to help the Investigator to anticipate adverse drug reactions or other problems in clinical trials.

Investigator's Brochure regulated

Regulatory authorities (such as the European Medicines Agency (EMA) and national competent authorities (NCAs)) require an up-to-date IB for any medicine being studied. An IB is submitted to the regulatory authorities along with the Clinical Trial Application (CTA), and regulatory authorities also review any updates to the IB to ensure that it is accurate, complete, and impartial.

In some cases, for instance if the investigational product already has a marketing authorisation (MA) and its pharmacology is widely understood by medical practitioners, an extensive IB may not be necessary. In such cases, the Summary of Product Characteristics (SmPC) – or, where permitted by regulatory authorities, a package leaflet or labelling – may be used instead as an appropriate alternative to an IB. However, the substitute must provide current, comprehensive, and detailed information on all aspects of the investigational product that might be of importance to the investigator.

If a marketed product is being studied for a new use (a new indication), an IB specific to that new use should be prepared. When relevant new information is available, the investigators and Research Ethics Committees (RECs) should be told – where possible, before this new information is included in the revised IB.

The ICH E6 guideline specifies that an IB should include information on the investigational product itself as well as on its use in non-clinical and clinical studies, together with a section providing guidance for the investigator on the use of the drug.

Preparation of IB

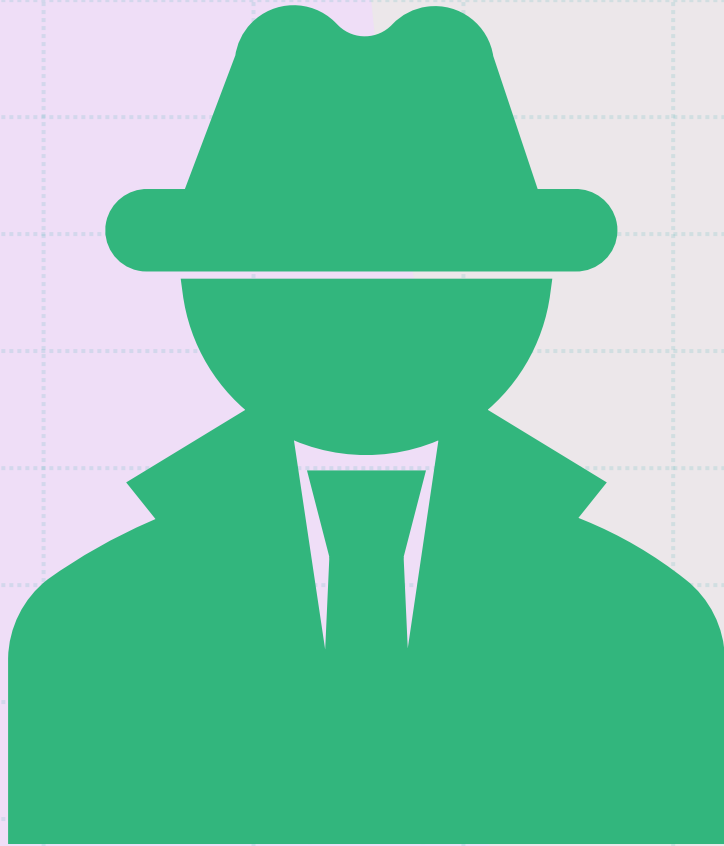
- **the sponsor**
- Investigator's Brochures are prepared by the sponsor, who also controls the distribution of the document. The IB for a clinical trial is submitted to national competent authorities (NCAs) for approval along with the Clinical Trial Application (CTA).

Conclusion

- An IB contains a “Summary of Data and Guidance for the Investigator” section, of which the overall aim is to “provide the investigator with a clear understanding of the possible risks and adverse reactions, and of the specific tests, observations, and precautions that may be needed for a clinical trial.
- This understanding should be based on the available physical, chemical, pharmaceutical, pharmacological, toxicological, and clinical information on the investigational product(s). Guidance should also be provided to the clinical investigator on the recognition and treatment of possible overdose and adverse drug reactions that is based on previous human experience and on the pharmacology of the investigational product”.
- The sponsor is responsible for keeping the information in the IB up-to-date.
- The IB should be reviewed annually and must be updated when any new and important information becomes available, such as when a drug has received marketing approval and can be prescribed for use commercially.
- Owing to the importance of the IB in maintaining the safety of human subjects in clinical trials, and as part of their guidance on Good Clinical Practice (GCP), the U.S. Food and Drug Administration (FDA) has written regulatory codes and guidances for authoring the IB, and the International Conference on Harmonisation (ICH) has prepared a detailed guidance for the authoring of the IB in the European Union (EU), Japan, and the United States (US)



Selection of Investigator & Site



Trial Planning Phase

- Feasibility & Investigator Selection follows the Confirm Sponsor station and precedes the Contracts & Agreements station. This process occurs in parallel with Funding Secured, Trial Master File, and Unique Trial Number.
- Feasibility & Investigator Selection is good practice and is relevant to all trials. This station is part of the 'trial planning phase' group of stations.

- According to ICH GCP investigator may be considered as a person proficient in education, training, and experience who will take the major restraint for the good conduct of the trial and should satisfy all the requirements described by applicable regulatory authorities and must comply with GCP.
- Investigators are very much important for the primary conduct of the study, whose primary importance is to conduct research while protecting the rights, safety, and well-being of the participants.
- Investigators have many important roles in conducting ethical research, taking responsibility for the informed consent process, giving the statement of the investigator, checking of investigational product, reporting adverse events, maintaining accurate records.
- A selection of the investigator for the conduct of research is a complex process

Investigator Selection

- For multi-site trials, the careful selection and evaluation of investigator sites is critical for the successful completion of a trial within budget, timelines and to ensure the generation of high quality data.
- **Factors that should influence investigator site selection include:**
 - Interest in the research question
 - Experience and qualifications of the investigator
 - Sufficient staff to conduct the study and their experience and qualifications
- **Availability of suitable patient population:**
 - Anticipated rate of patient recruitment (determined through feasibility assessments)
 - Conflicting studies (competing for the patient population and potentially introducing recruitment bias)
 - Adequate time to conduct and oversee the trial

Investigator Selection

- **Selection is Based on**
 - Education
 - Training
 - Experience
- **The investigator is obligated to provide a copy of:**
 - Their CV/ relevant documents requested by the Sponsor the ethics committee, the CRO or other regulatory authorities.
 - They should be aware and clearly understand the time and other resource demands the study is likely to make in order to ensure they are available throughout the course of the Study
 - Should be able to demonstrate potential for recruiting the required number of suitable subjects
 - Should have sufficient time to properly conduct and complete the study within the agreed upon duration

Investigator Selection

- They should have an adequate number and qualified staff and facilities for conducting the trial
- **The Investigator MUST be familiar with**
 - Safety
 - Efficacy
 - Appropriate Use of the IP in the investigation as described in the protocol
 - IB
 - Other information sources provided by the SPONSOR
- **Should be compliant and thorough with**
 - GCP guidelines
 - SOPs
 - Applicable regulatory requirements

Site Selection

- Site selection for a clinical trial is a comprehensive process. It is akin to a site's formal interview for study participation.
- Site selection is a rigorous process in conducting CT successfully, efficiently and in compliance with regulatory guidelines
- There are several factors to be considered when selecting a site but the utmost important factor is to select one that can pool high quality data
- If high- quality sites are selected, the trial will likely finish on time, on budget, and with clinically sound findings
- This broad statement encompasses:
 - The experience and qualifications of site staff
 - The availability of suitable patients who meet the protocol inclusion criteria
 - The ability to perform the required clinical assessments.

Site selection requires due effort on the part of the Sponsor to ensure that the proposed trial site meets all considerations for a successful clinical trial

Site Selection

- Step 1: Define Site Requirements and Selection Criteria
- One of the most cited inefficiencies in the industry regarding site selection is the lack of information about the protocol when starting the process
- The first step to selecting appropriate sites for a study is to identify key site criteria from the study design as this data provides the fundamentals that will guide site selection.
- Important points to define before the site identification and selection can start include participant population profile, staffing and facilities requirements, enrollment targets and period, study duration, desired geographical area (if applicable), and target start up timelines.
- Staff Qualifications: staff availability, specialty, credentials, experience in clinical research and the studied indication, including their performance as it relates to regulatory compliance.

- Facilities and Equipment: adequate facility space, drug/device storage space and security, types of source documents, and equipment needed for the study.
- Site Profile and Timelines: site types (e.g., hospitals or clinics, academic centers, Veteran Affairs facilities, non-profit, government, and private sites), site's Institutional Review Board (IRB) meeting timeframe, and typical contract negotiation timeline.
- Population Profile and Access: eligible participants' availability and proximity, disease/condition incidence, ongoing trials recruiting similar patients, and recruitment capabilities including resources for conducting outreach.
- Past Performance: clinical trial experience including in trials with similar enrollment timelines, enrollment target, and complexity, and past enrollment rates.
- Competition: concurrent trials in the same indication or targeting the same population profile that are ongoing or scheduled to start during study conduct.

- Facilities and Equipment: adequate facility space, drug/device storage space and security, types of source documents, and equipment needed for the study.
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- Population Profile and Access: eligible participants' availability and proximity, disease/condition incidence, ongoing trials recruiting similar patients, and recruitment capabilities including resources for conducting outreach.
- Past Performance: clinical trial experience including in trials with similar enrollment timelines, enrollment target, and complexity, and past enrollment rates.
- Competition: concurrent trials in the same indication or targeting the same population profile that are ongoing or scheduled to start during study conduct.
- This initial step should involve defining as many criteria and sub-criteria as necessary to answer the study and sponsor business needs in terms of timelines and finances.
- It's also important to prioritize and define the non-negotiable items as some criteria are absolute must-haves while others may be flexible.
- Assigning a weight to each criterion supports a better evaluation of the sites' strengths and weaknesses and allows for an objective comparison.

- Step 2: Identify Sites and Gather Initial Information
- Once the criteria are defined, site identification can start. Sites can be identified through diverse paid and free tools such as:
 - The sponsor's internal database of previously utilized sites
 - External database through a Contract Research Organization (CRO) that has its own list of potential sites
 - A Site Network Organization that owns a network of sites, oftentimes with various experience
 - Online and offline directories
 - Publications of recent clinical trials in the studied indication
 - Clinicaltrials.gov postings
 - Word-of-mouth and references
- The most helpful tools for pre-selecting sites are the ones that provide filtering functionalities (e.g. site size, location, specialty, etc.), and detailed datasets about past experience and performance which provide initial insights on the site's compatibility for the study.

- Once a potential site is identified, the first step is to send the PI an inquiry for clinical trial participation which includes a study synopsis to assess initial interest and request their CV.
- If the PI is interested and the site is new to the sponsor or CRO, the next step is to set up a Non-Disclosure Agreement (NDA) before divulging further study details.
- Once completed, the study protocol can be shared with the PI and a survey that includes specific questions that address each of the pre-defined criteria can be provided to the site.
- Questions should be unambiguous and generate answers that can be scored. The questionnaire can be submitted and answered through an online survey system.
- This allows to establish specific rules (e.g., create mandatory fields), obtain standardized answers that can be more easily compared, and produces results that can be easily exported to a spreadsheet and analyzed.

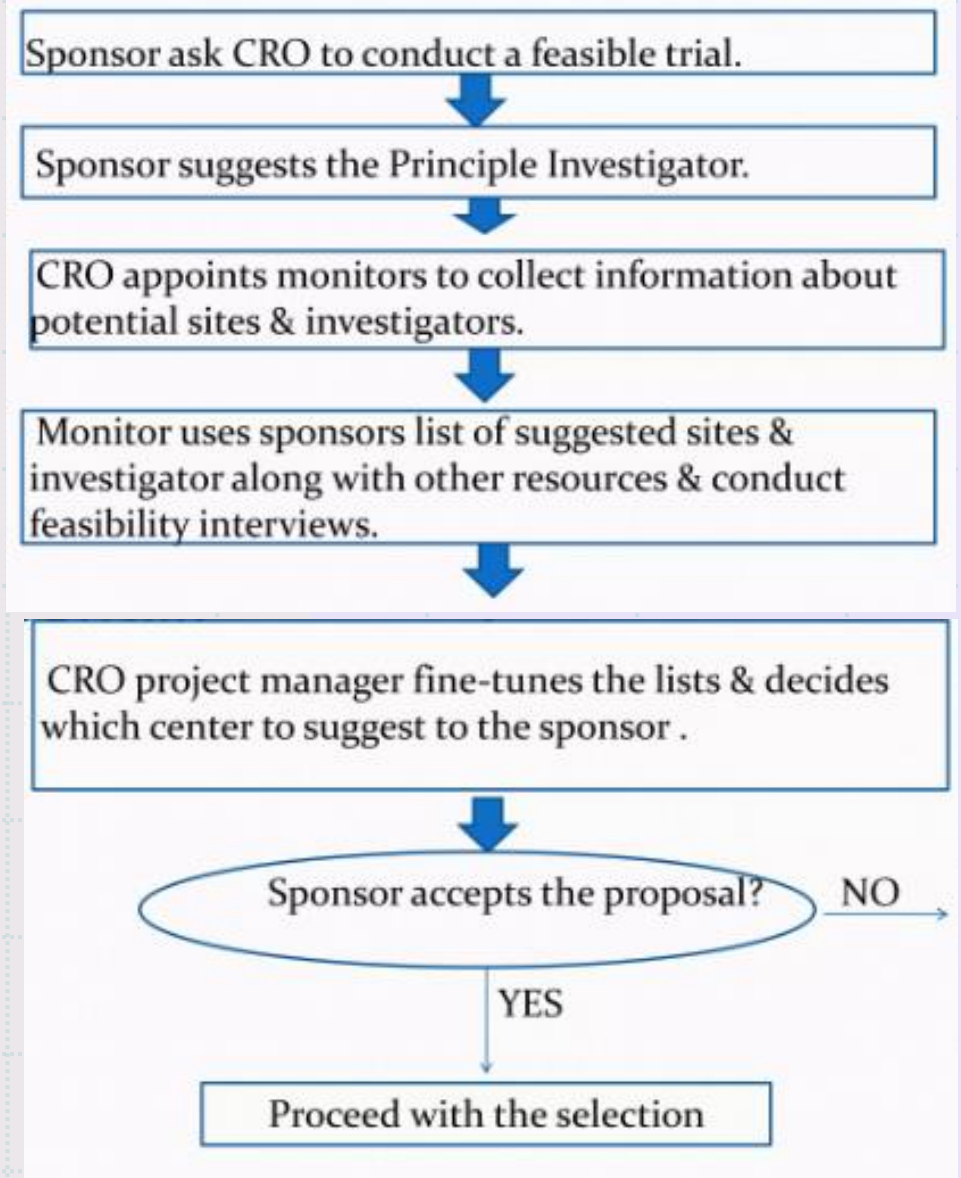
- Step 3: Evaluate and Select the Sites
- Studies report that 11% of sites selected for a study are in fact never activated, the main reason being due to contracting issues
- It's important to involve site administration staff in charge of contracting and budgeting early in the conversation, to understand their timeline and expectations.
- Setting expectations from the start will ensure more efficient start-up and study conduct. In addition, if the sponsor does not have a pre-defined cost per patient, the site should provide a forecast budget that includes study setup cost, study conduct charges, closeout fees, and overhead percentage.
- Providing a standard budget template to the sites will ensure that all line items are accounted for and facilitate budget comparison.
- Prestudy visits (PSVs), also called Site Selection Visits (SVSs) or Site Assessment Visits (SAVs), are then conducted by Site Monitors, especially if the site is new to the sponsor or indication, or if key site staff has changed since their last participating trial.
- A meeting is scheduled to establish a relationship with the PI and site staff, explain the study in more detail to the staff including study objectives, overall design, procedures, and endpoints, and communicate the sponsor expectations.
- The visit is essential for the sponsor to verify that the investigator has the personnel, motivation, time, subject population (if possible, as demonstrated by review of the site's patient database), and facilities and equipment to adequately conduct the study.

- After all the information is gathered, reviewed, and evaluated, an objective comparison and decisions can be made. Additional support in the areas of Good Clinical Practice (GCP) training, subject recruitment tools (advertising, etc.) may be considered for sites with otherwise adequate qualifications.
- After such full consideration, the final sites are notified of their selection for the trial. Subsequently, Clinical Trial Agreements (CTAs) are sent to each site, and documents in support of 21 CFR 312.53 can be formally prepared and organized.

Conclusion

- To select the optimal sites for a given study, an unbiased and systematic process that consists of conducting an analysis of the site's compatibility with the study needs, must take place.
- Proper planning to define study-mandated evaluation criteria, and using the right tools to gather information and facilitate communication, are absolute musts to make the right decisions.
- However, essential, proper site selection is not the sole factor for site performance and study success.
- External forces and challenges often affect study conduct (e.g., protocol design such as unclear or difficult-to-meet inclusion/exclusion criteria, supply chain delays, cultural and technological considerations such as in the case of trials conducted in resource-challenged areas, etc.) and require additional planning and mitigation strategies

Process of Site & Investigator Selection





Clinical trial Stakeholders, CRO & SMO

CT stakeholders

- The stakeholders of clinical research industry are subjects, sponsor, investigator and team, government agencies, Regulatory authorities, monitor, Institutional review board, Contract research organization, Independent ethics committee, Academic institutions, public and private sectors whose contribution significantly influence the clinical research process
- All Stakeholders are bounded by three basic responsibilities – Ethical, Scientific and Administrative Responsibilities..

- **Sponsors are primary stakeholders** who initiates, organizes, finances clinical research. Sponsors can be an individual or company or an organization. Usually, Sponsors are pharmaceutical company, biotechnology companies, Academic Institutions.
- **Investigator and investigator team** is appointed/hired by sponsor.
- The Investigator team is constituted by co-investigator, sub-investigator, clinical research coordinator, study nurse, pharmacist, laboratory personnel, research assistant and subject recruiter.
- All team members should be qualified, experienced and trained to conduct assigned clinical research duties under supervision of principal investigator.
- Clinical Research should be conducted in accordance with ethical principles that have their origin in declaration of Helsinki and consistent with good clinical practice, standard operating procedures and Applicable Regulatory Requirements.

- **Regulatory Authorities** are the stakeholders responsible for giving permission to conduct clinical research in the country.
- The sponsor submits the investigational new drug application or new drug application to the regulatory authorities for review and approval. Once the sponsor obtains approval to initiate and continue the research, regulatory authorities conduct audits and inspection upon serious non compliance, major violations.
- Regulatory Authorities are also responsible for ensuring protection of public health by safeguarding rights, safety and wellbeing of trial subjects
- **Institutional review board or independent ethics committee** is stakeholder at the clinical research site. Trial should be initiated at the site only after approval from the ethics committee. Ethics committee is responsible for protecting rights safety and well being of trial subjects, Review of ongoing clinical trials, compensation and payments to trial subjects.
- Public and private entities

Stakeholder and Responsibilities

sino	stake holder	Responsibilities
1.	Sponsor/ CRO	▪ Trial planning and allocation of resources ▪ Data monitoring & management ▪ Ensuring compliance with GCP guidelines ▪ Publication of trial result
2.	Investigator	▪ Study conduct/ enrolment of study subjects ▪ Medical care of subjects ▪ Ensure GCP guidelines
3.	IRB / IEC	▪ trial permission / approval ▪ Review of trial process ▪ Trial termination / suspension ▪ Ensure GCP guidelines
4.	Regulatory authority	▪ trial permission /approval ▪ Review of trial process ▪ Trial termination / suspension ▪ Ensure GCP guidelines ▪ inspection
5.	Study subject	▪ Voluntary participation & protocol compliance

CRO

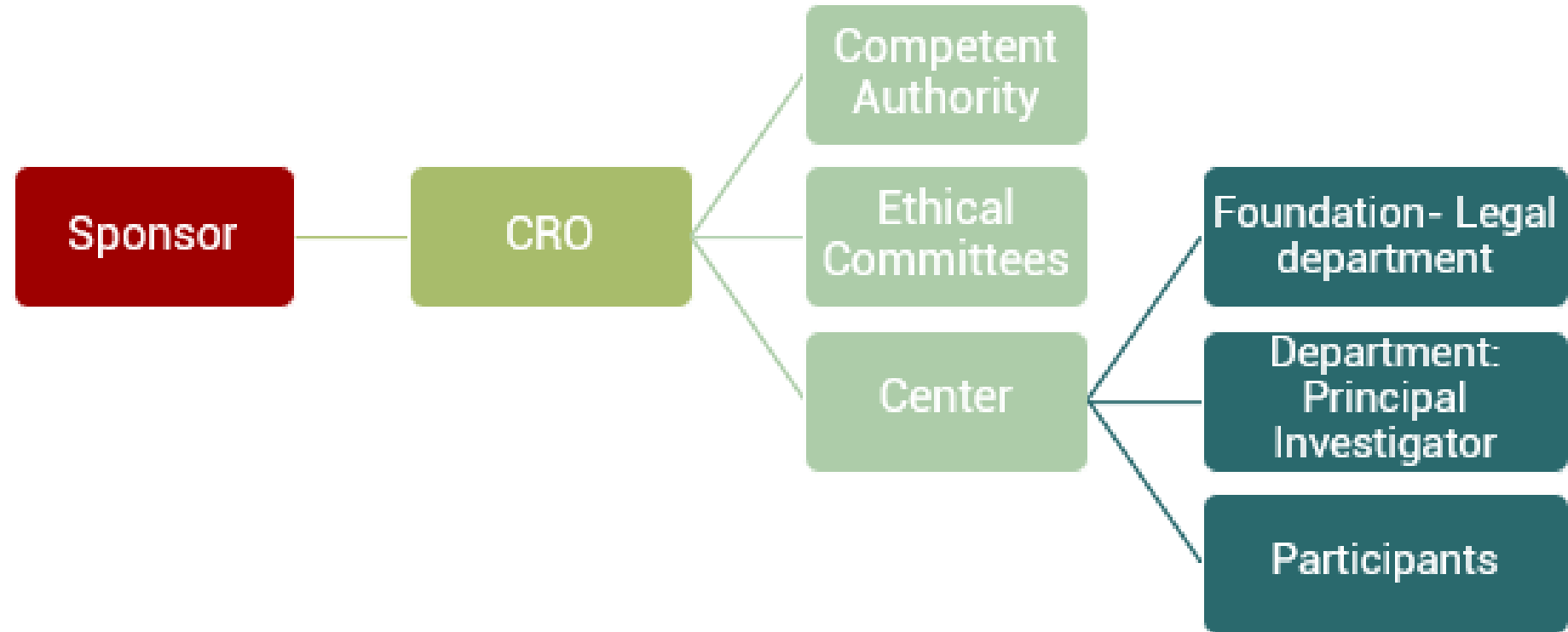
- A CRO (Contract Research Organization) is a company that provides clinical trial management services for the pharmaceutical, biotech, and medical device industries.
- Although there are different types of CROs and diverse levels of specialization (distinct therapeutic areas for instance), typical CRO services include regulatory affairs, site selection and activation, recruitment support, clinical monitoring, data management, trial logistics, pharmacovigilance, biostatistics, medical writing, and project management, among others.
- In a clinical trial, CROs are hired by sponsors to perform a set of tasks, taking various technical and administrative responsibilities on the sponsor's behalf.

Purpose of CRO

- The main role of the CRO is to plan, coordinate, execute, and supervise the processes involved in the development of a clinical trial, being a central contact point between the sponsor and other trial actors (e.g. ethics committees, regulatory agencies, vendors, and hospitals).

Roles and Responsibilities

- CROs are key players in clinical research, since they have the knowledge and the capabilities needed for the proper development of a clinical study. They help sponsors by reducing their workload, while ensuring trial quality and compliance with national and international standards.
- At the same time, many CROs supply innovative technological tools to increase efficiency in the study processes, which translates into cost reductions.
- Without doubt, CROs play a crucial role in the success of a clinical trial. Sponsors should carefully assess the particular needs of their projects, and look for the CRO that best meets their technical requirements and budget.



Outsourcing

- Tactical partnerships are less mature relationships with a contract research organization (CRO). They are driven by an immediate client need, and often established to maximize cost savings. They are often entered into through a competitive bidding process and are designed to satisfy short-term goals.
- Individual studies or selected activities
- strategic partnerships tend to evolve over time. They generally involve greater investment in the project by the CRO, which might be a larger operation able to provide more comprehensive list of services. Cost savings does not drive these outsourcing relationships as much as the established trust that develops between the client and the vendor. Strategic partnerships are therefore considered more mature vendor-client relationships; they are ongoing partnerships that often yield greater quality and satisfaction.
- Sponsor conducts P1&2 and outsources larger and routine studies to CRO

Steps to Success

- Determine accurate study specifications
- Select right CRO
- Manage the study

Selection of CRO

- Capability
- Compatibility
- Cost

Managing Sponsor CRO relationship

- Clarify the roles & responsibilities
- Define and use “performance metrics” to measure study progress
- Ensure efficient communication

A CRO's Services

- CROs offer a large range of services, including the following:
- Site selection and set-up
- Site management
- Regulatory affairs
- Monitoring
- Data management
- Logistics management
- Pharmacovigilance
- Biostatistics
- Medical reporting and writing
- Project management
- Chemistry and biological expertise
- Data entry and validation
- Data processing and management
- Clinical trial participant recruitment
- Medicine and disease coding
- Safety and efficiency reports
- Marketing help
- Statistical analysis of reports and plans
- Study report review and submission
- Database design and build

CRO roles

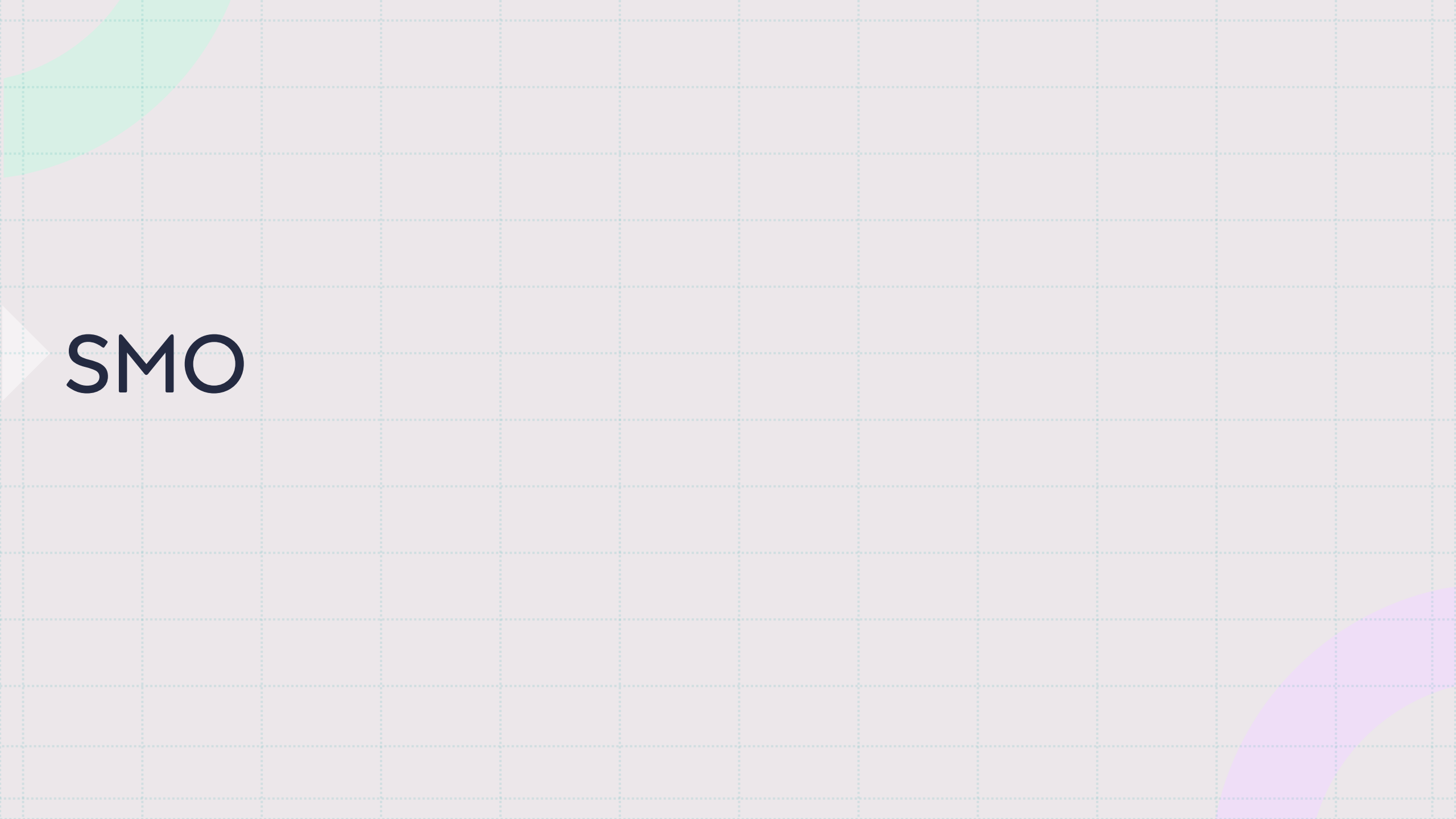
- The main roles found in a CRO include:

- Project managers
- Research assistants
- Site contract managers
- Quality managers
- Clinical research associates
- Clinical trial assistants
- Regulatory affairs managers
- Heads of clinical operations
- Data managers
- Statistical programmers
- Biostatisticians

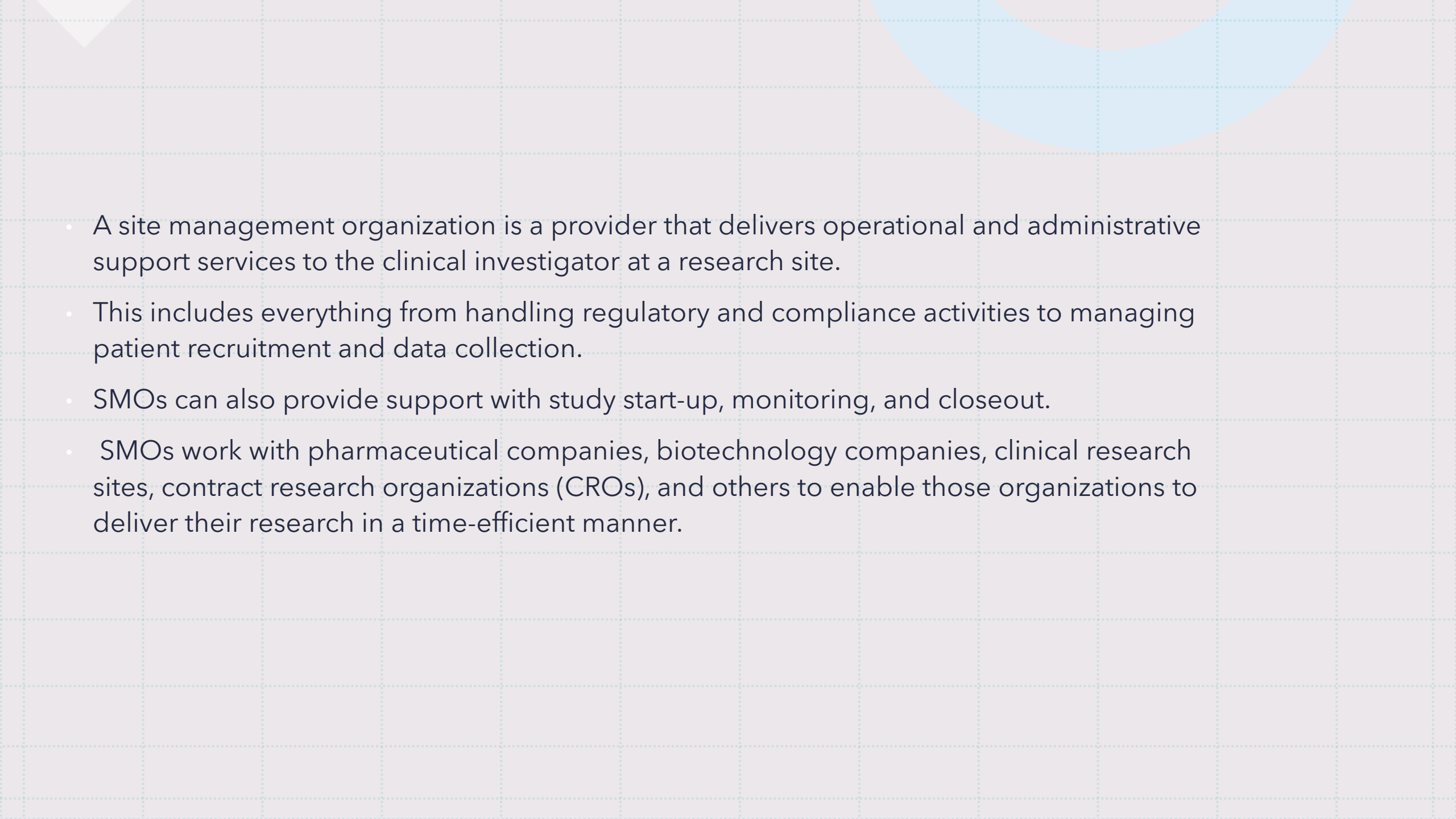
Benefits

There are many advantages of outsourcing work to a CRO, including:

- Pharmaceutical companies can put all their focus on the development process of the clinical trial, while the CRO does all the other work, including all the stages of clinical trials.
- CROs have experience and previously established systems to deal with project management, data collection and patient care and can efficiently complete the work.
- Hiring a CRO saves time and money. With a CRO by the pharmaceutical company's side, the clinical trial sponsor will lower the overall costs of a clinical trial.
- CROs are ready to begin the work as soon as a contract is signed.
- CROs will have experience and awareness of the international medical and research markets and knowledge of the different regulations in each country.
- Strategic alliances with CROs build trust and leads to an open dialogue around clinical trial details and priorities. This allows research teams and pharmaceutical companies to be able to resolve any issues with the assistance of a diverse business culture and operational systems.
- CROs have knowledge of the best clinical trial investigators and sites in a specific location which allows for high participant enrollment numbers.



SMO

- 
- A site management organization is a provider that delivers operational and administrative support services to the clinical investigator at a research site.
 - This includes everything from handling regulatory and compliance activities to managing patient recruitment and data collection.
 - SMOs can also provide support with study start-up, monitoring, and closeout.
 - SMOs work with pharmaceutical companies, biotechnology companies, clinical research sites, contract research organizations (CROs), and others to enable those organizations to deliver their research in a time-efficient manner.

Benefits

- **Faster patient recruitment** – SMOs leverage networks of qualified patient pools and faster follow-up to recruit and retain patients more efficiently than in-house teams.
- **Improved data management** – With the technological infrastructure and data handling processes in place to manage both clinical and regulatory data, SMOs limit delays that can sometimes crop up with study and regulatory data addressing inefficiencies.
- **Better focus for research teams** – A common pitfall that clinical research teams can face is spending more time cleaning data than analyzing data. With the grunt work outsourced to SMOs, research teams have more time to analyze meaningful data.
- **Shorter study timelines and lower costs** – SMOs provide end-to-end clinical trial-related services, from setup to monitoring, compliance activities, and closeout, to shorten clinical research timelines and stick to budgets.

Services

- SMOs can reduce trial timelines and costs and lead to more efficient and accurate trial results through a full range of services, including:
- Documentation preparation (informed consent forms (ICF), recruitment materials, contracts, legal agreements, etc.)
- Managing patient recruitment
- Monitoring
- Identifying and/or managing investigators and sub-investigators.
- Managing and fulfilling staff requirements and recruitment.
- Preparing Institutional Review Boards (IRB) submissions.
- Assisting with trial startup, site initiation, and closeout operations

Site management organizations are an integral part of clinical trials.

- Clinical trials are the pathway to new advancements in medicine reaching patients. The goal of bringing new treatments to market is to provide those suffering from medical ailments and diseases the opportunity to live happy, healthier lives.
- There are many people involved in clinical trials, all of which play an essential role in the trial's overall success. One of these key players is the site management organization. They are the bridge between the sponsor and the sites that are conducting research. The SMO oversees every aspect of running the trial at all site locations. They have a hand in everything from recruiting participants and supervising staff to managing the physical space where the study takes place — whether it's a lab, an office building, or some other location.
- Like CROs, SMOs alleviate some of the headaches of running a clinical trial of any scale (small or large). SMOs are integral for a trial to achieve accelerated timelines, streamlined processes, improved data management, and more accurate results.

Difference between CRO & SMO

Site management organizations (SMOs) differ from CROs in several ways, including that they:

Are not explicitly regulated or defined by the Food and Drug Administration (FDA).

Provide services specific to the clinical study sites.

Do not assume any of the sponsor's regulatory obligations listed in the FDA's investigational new drug application regulations.

May have a network of clinical research sites which they manage.

SMO services may include:

Identifying or managing investigators and sub-investigators.

Hiring study staff.

Preparing submissions to Institutional Review Boards.

Assisting with:

project feasibility;

patient recruitment;

study site initiation and close-out operations; or

other site-specific study activities.

Conclusion

- Clinical studies can be fraught with challenges, but like CROs, SMOs offer the capability to outsource the most challenging work to service providers who specialize in operational complexity.
- They can rein in excess costs and allow research teams to finish on time.
- As with any industry, there is room for improvement regarding collaboration between sites and sponsors – but SMOs have proven time and again to be valuable partners who can smoothly deliver superior trials by providing support at every stage of development.

Clinical Data Management

MS. SONALI CORREA
PH.D., M.SC.



Introduction

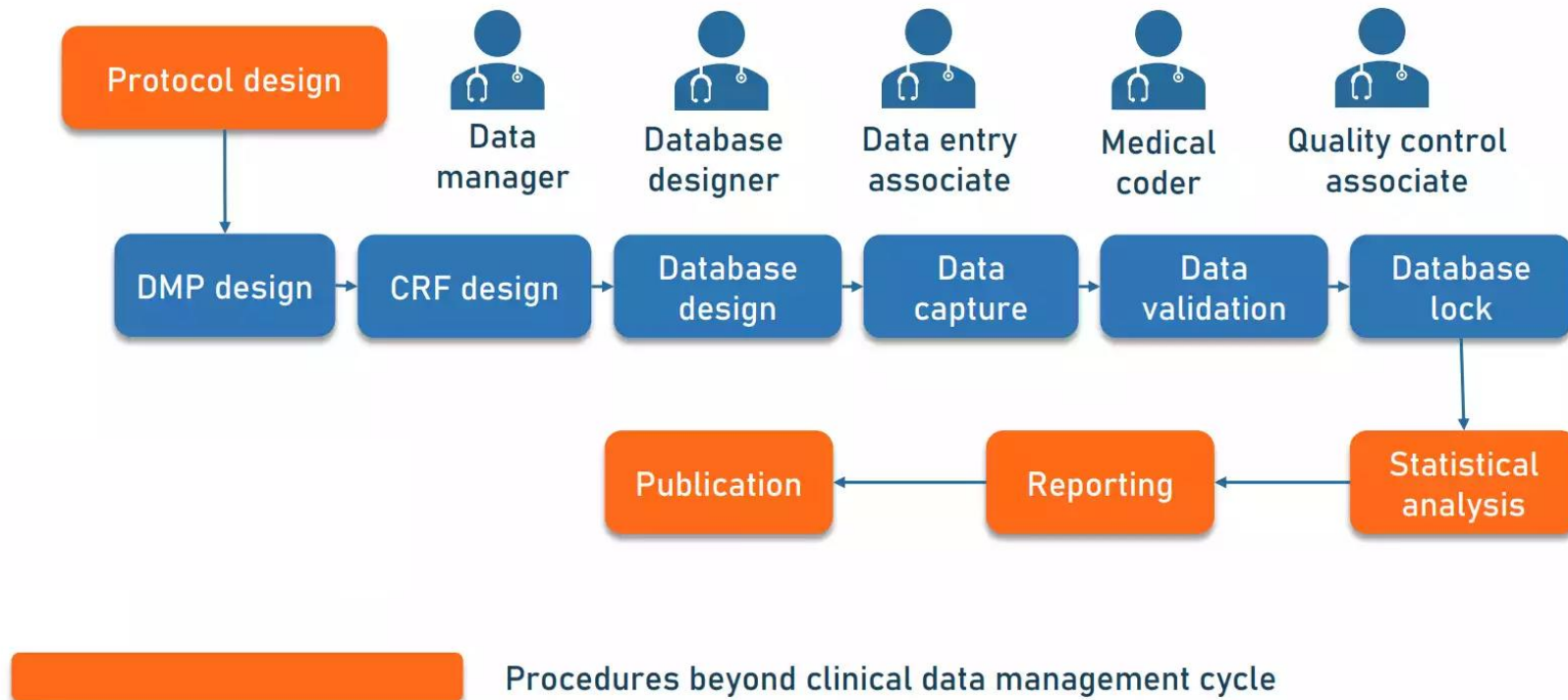
- Clinical data management is a critical process in clinical research, which leads to generation of **high-quality, reliable, and statistically sound data from clinical trials. Clinical data management ensures collection, integration and availability of data at appropriate quality and cost.**
- **Goal** is to gather as much such data for analysis as possible that adheres to federal, state, and local regulations.
- The field of CDM has come about due to demands from both the pharmaceutical industry and the regulatory authorities.
- As the drive to “fast-track” the development of pharmaceutical products continues to accelerate, regulatory entities have responded by requiring quality-assurance standards to be met in collecting the data used in the drug evaluation process.

CDM & CT Stakeholders

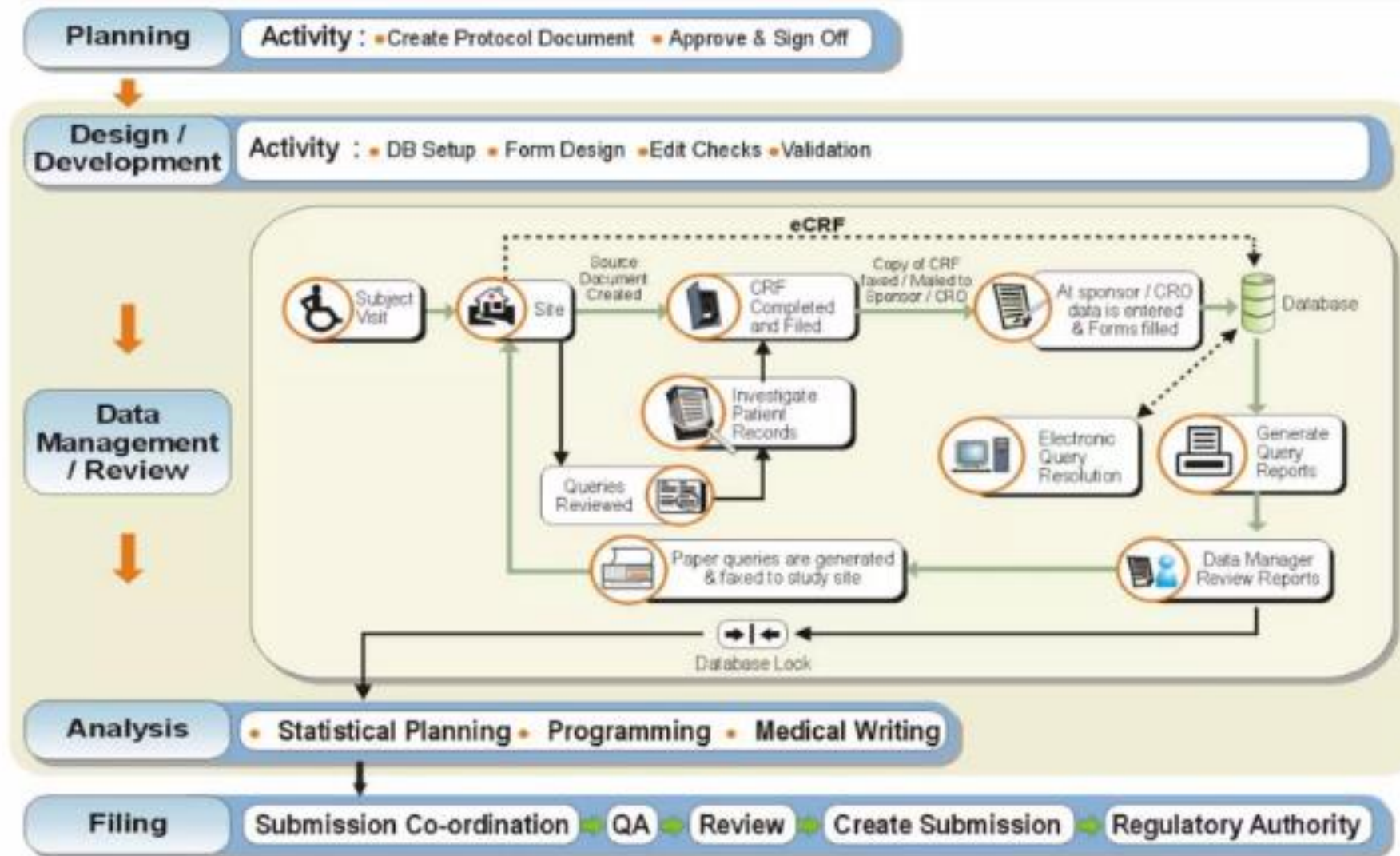
- the CDM process helps keep key clinical trial stakeholders on the same page :
- 1. Sponsors — pharmaceutical companies, institutions, and other organizations that initiate, monitor, and finance the trial.
- 2. CROs (control research organizations) — research entities hired by the sponsor to plan and run the investigation.
- 3. Sites — centers coordinating data collection from trial participants.
- CDM plays a crucial role in evaluating the safety and effectiveness of drugs, diets, medical devices, digital therapeutics tools, and other types of treatment, diagnostics, or methods to prevent health problems. If properly handled, it significantly reduces time required for a new medical product launch.

Workflow

DATA FLOW IN CLINICAL TRIALS AND CLINICAL DATA MANAGEMENT STAGES



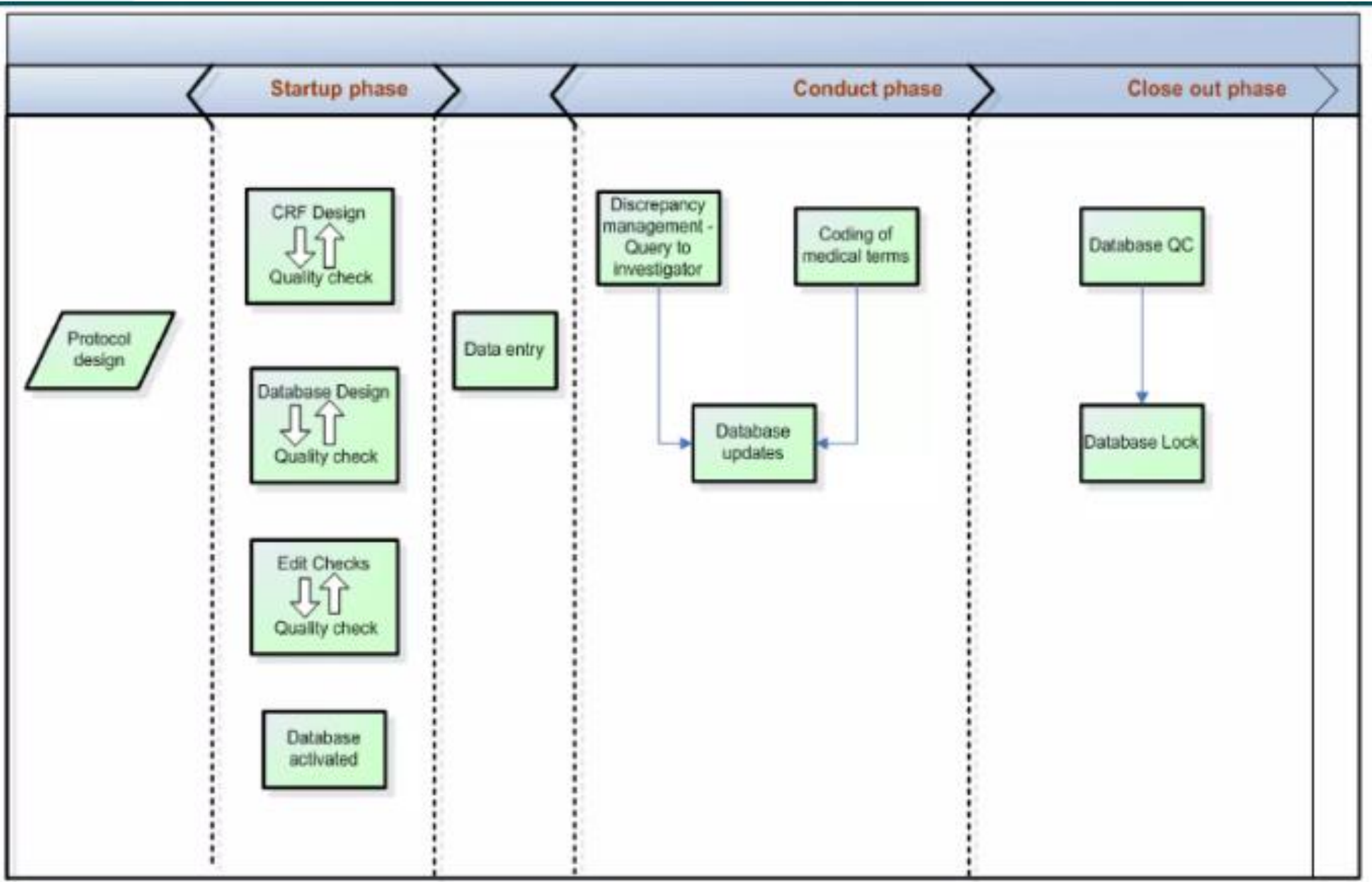
Clinical trial System Management overview



How Does Clinical Data Management (CDM) Work?

- To maintain data integrity, the CDM process starts at the start of a clinical trial, even before the study protocol is finalized.
- The CDM team designs a case report form (CRF) and defines the data fields.
- CRFs specify the data type to be collected, the units of measurement, and CRF completion guidelines. Variables are using annotated coded terms.
- A data management plan (DMP) is then developed as a guide, including a description of the trial's CDM activities.
- Next, databases are built to support CDM tasks with corresponding compliance tools.
- Finally, testing is done before using the plan with actual clinical trial data. CFR tracking, data entry, validation, discrepancy management, medical coding, and database locking are subsequent steps in the process.

- 
- Case report forms may be used to collect data by paper or electronic means; however, as technology has continued to evolve, the trend towards electronic data collection has followed suit.
 - Furthermore, many pharmaceutical companies have adopted remote data entry or e-CRF as a time-saving measure.



Roles Involved in CDM

Roles	Responsibilities
Data manager	supervises the CDM process
Database programmer or designer	performs the CRF annotation, creates the study database, enables data validation, designs data entry screens, and performs edit checks using dummy data
Medical coder	codes variations such as adverse events and medical history
Clinical data analyst	designs the CRF, prepares the filling instructions, develops discrepancy protocols
Quality control associate	checks the accuracy of data entry and performs data audits
Data entry associate	tracks the receipt of CRF pages and enters data into a database
Clinical research associate	serves as a go-between for the sponsors funding the research and the clinical participants and employees

Data management plan design



Experts in charge: **data manager, database designer**



A data management plan or DMP is a document detailing all procedures, tasks, milestones, and deliverables throughout the CDM lifecycle.



It gives a roadmap on how to work with information and handle possible risks. Another important function is to clearly communicate what happens in the course of the trial to each stakeholder.

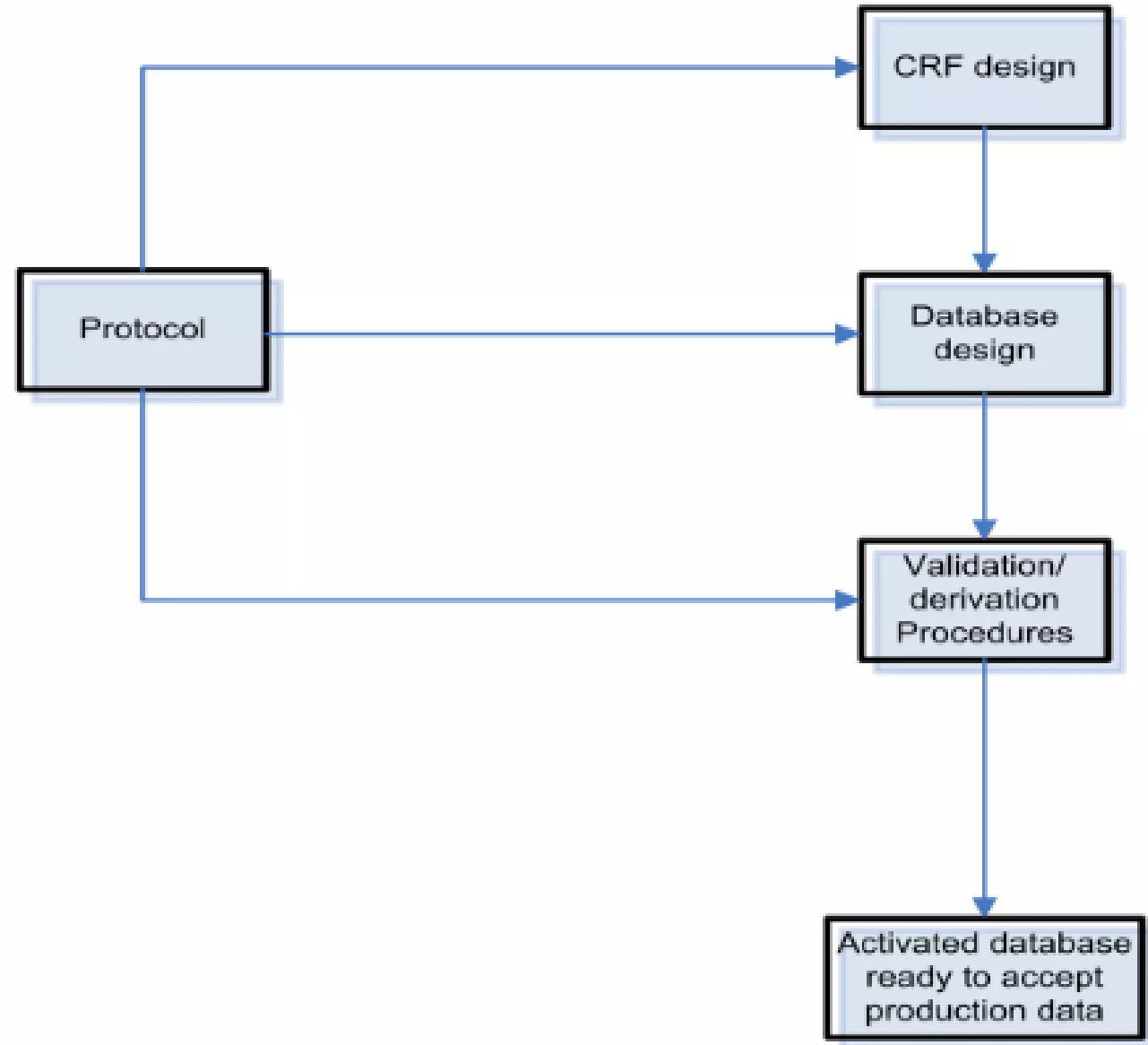


The DMP must be ready at a trial design stage, before the first participant is enrolled.



This will ensure that data will be collected in the correct format and properly organized. However, the plan is not something immutable: It has to be updated across the trial, capturing any changes that influence data management.

- ***The DMP typically describes the following aspects:***
- *data to be gathered from trial participants,*
- *existing data that can be integrated,*
- *data formats,*
- *metadata and its standards,*
- *storage and backup methods,*
- *security measures to protect confidential information,*
- *data quality procedures,*
- *responsibility assignments across team members,*
- *access and sharing mechanisms and limitations,*
- *long-time archiving and preservation procedures,*
- *the cost of data preparation and archiving, and*
- *compliance with relevant regulations and requirements.*

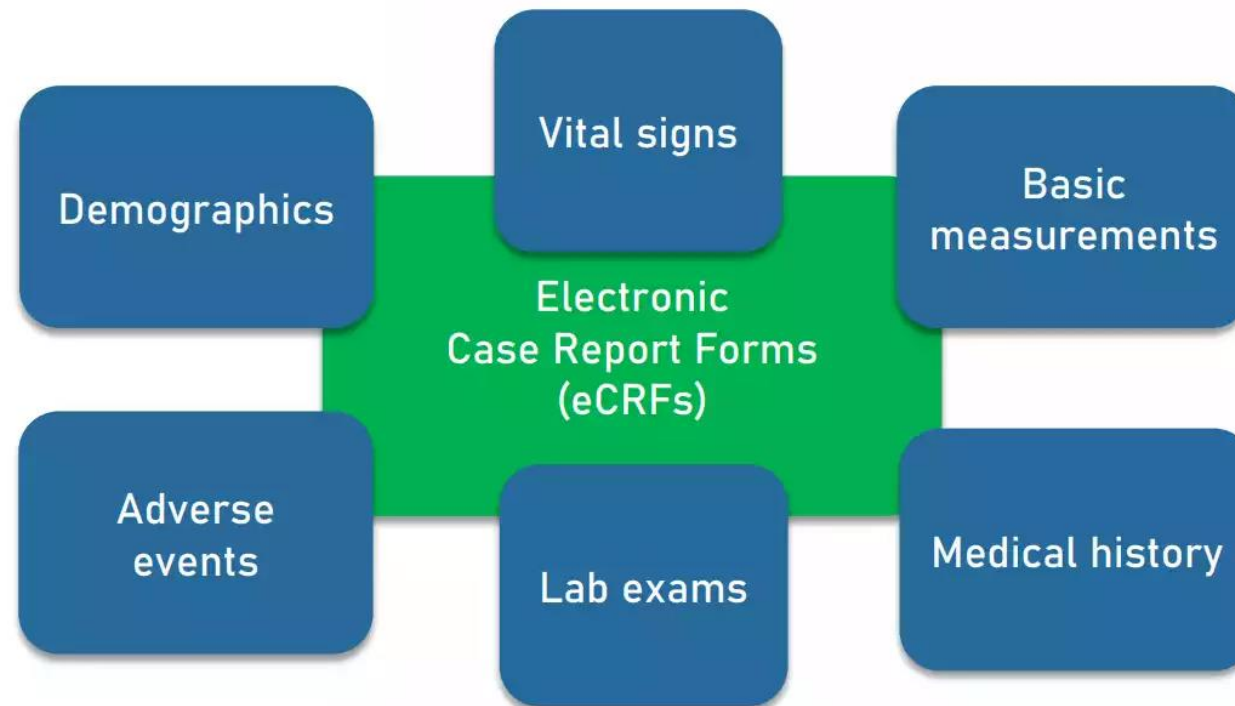


eCRF form design

- *Experts in charge: data manager, database designer*
- The case report form is a printed or electronic questionnaire for collecting data from study participants and reporting it to trial sponsors. The document is created specifically for each research project in accordance with
- the trial protocol, and recommendations of the Clinical Data Acquisitions Standards Harmonization (CDASH).
- They are developed by Clinical Data Interchange Standards Consortium to streamline industry-wide data exchange.

Structure of eCRF

eCRF STRUCTURE



Paper CRF

e-CRF

Design of eCRF

Well-designed case report forms collect only data necessary for the particular study avoiding any redundancy. The fields to be filled in may include

- demographics (age, gender),
- basic measurements (height, weight),
- vital signs (blood pressure, temperature, etc.) captured at various time points,
- lab exams,
- medical history,
- adverse events, and
- more, based on the research requirements.

Data managers create data entry screens and eCRF layouts in collaboration with a database programmer.

The design usually goes through several review cycles before finalization.

Clinical trial database design

- *Experts in charge: database designer, data manager*
- Databases are the clinical software applications, which are built to facilitate the CDM tasks to carry out multiple studies.
- Generally, these tools have built-in compliance with regulatory requirements and are easy to use. “System validation” is conducted to ensure data security, during which system specifications, user requirements, and regulatory compliance are evaluated before implementation.
- A clinical trial database is a set of data collected during the study and organized in rows and columns.
- It's designed with the CRF structure in mind. In other words, the database incorporates a questionnaire schema of the case report forms.

- 
- To preserve storage space, some questionnaire data can be coded into meaningful categories.
 - In this case, database specialists create detailed descriptions of decoding or how to map codes into elements of CRFs.
 - Before launching into the production (research) environment, the database is tested with dummy data in a secure, non-study site.

Clinical trial database design

Combined Coding Approach

Question	Answer Coded	Respondent	Answer Numeric	Answer Free Text	Validation Flag
01010100001C	0002	1	-	-	1
01010200011T		1	-	sister	1
01010100002C	0007	1	-	-	1
01020100003M	0033	1	-	with help	1
01020100004M	0008	1	5	-	1
01020100004M	0009	1	35	-	1
01020100004M	0010	1	40	-	1
01020100004M	0033	1	20	reading	1
01020100005C	0013	1	-	-	0
01020100005C	0014	1	-	-	1
01030100006C	0016	1	-	-	1
01030200061M	0018	1	-	-	1
01030300062M	0033	1	-	voices	1

Source: Vera Pomerantseva

Table 3. Elements were combined according to the schema suggested on Figure 2.

CRF Annotations

- An annotated CRD is generally defined as a blank CRF with marking, or annotations that coordinated each data point in the form with its corresponding dataset name.
- Essentially, an annotated CRF communicates where the data collected for each question is stored in the database
- CDM annotates the CRFs by establishing variable names for each item to be entered.
- Reviewed by CMD and Statistician

	SITEID	SUBJID	USUBJID		
CIV			VISIT 1, DAY 1	Site/Subject Number:	Page 1 of 33
			Date of Visit: DD / MON / YYYY		VISDAT
DS	INFORMED CONSENT				
	NO study related activities may take place before the patient has signed the Informed Consent form.				
	Date informed consent was obtained: DD / MON / YYYY		DVSTDAT		
DM	Time informed consent was obtained: : 24hr clock		DSSTIM		
	DEMOGRAPHICS				
	Date of Birth:		BRTHDAT		

Data collection

- Data collection is done using the CRF that may exist in the form of a paper or an electronic version.
- The traditional method is to employ paper CRFs to collect the data responses, which are translated to the database by means of data entry done in-house.
- These paper CRFs are filled up by the investigator according to the completion guidelines.
- In the e-CRF-based CDM, the investigator or a designee will be logging into the CDM system and entering the data directly at the site.
- In e-CRF method, chances of errors are less, and the resolution of discrepancies happens faster.
- Since pharmaceutical companies try to reduce the time taken for drug development processes by enhancing the speed of processes involved, many pharmaceutical companies are opting for e-CRF options (also called remote data entry).

CRF tracking

- The entries made in the CRF will be monitored by the Clinical Research Associate (CRA) for completeness and filled up CRFs are retrieved and handed over to the CDM team.
- The CDM team will track the retrieved CRFs and maintain their record.
- CRFs are tracked for missing pages and illegible data manually to assure that the data are not lost.
- In case of missing or illegible data, a clarification is obtained from the investigator and the issue is resolved.

Electronic data capture in clinical trials

Experts in charge: clinicians, data manager, data entry associate, medical coder

- CRFs are the main instrument of data capturing in clinical trials. The information for report forms is traditionally gathered by clinicians or data entry associates from participants, when they visit medical facilities.
- In recent years medical sites stopped to be the primary point of data capturing. Nowadays, details for trials are also extracted from
- Electronic Health Records (EHRs),
- medical devices (blood pressure monitors, ECG recording machines, and others),
- a laboratory information management system (LIMS), and
- patient-reported outcomes (PROs) or any descriptions of health conditions that come directly from patients, without mediation and interpretation from medical experts.

Electronic data capture in clinical trials

- With paper-based questionnaires, the data is manually entered into printed versions and then inserted into the database.
- In the case of eCRFs, information gets into the computer from the start, and some fields can be populated automatically, with data transmitted from medical devices or EHRs by robotic process automation (RPA) tools.
- All information that made it to the forms and then to the clinical database must go through the data validation process.

Data validation: edit checks, source data verification, and data anonymization

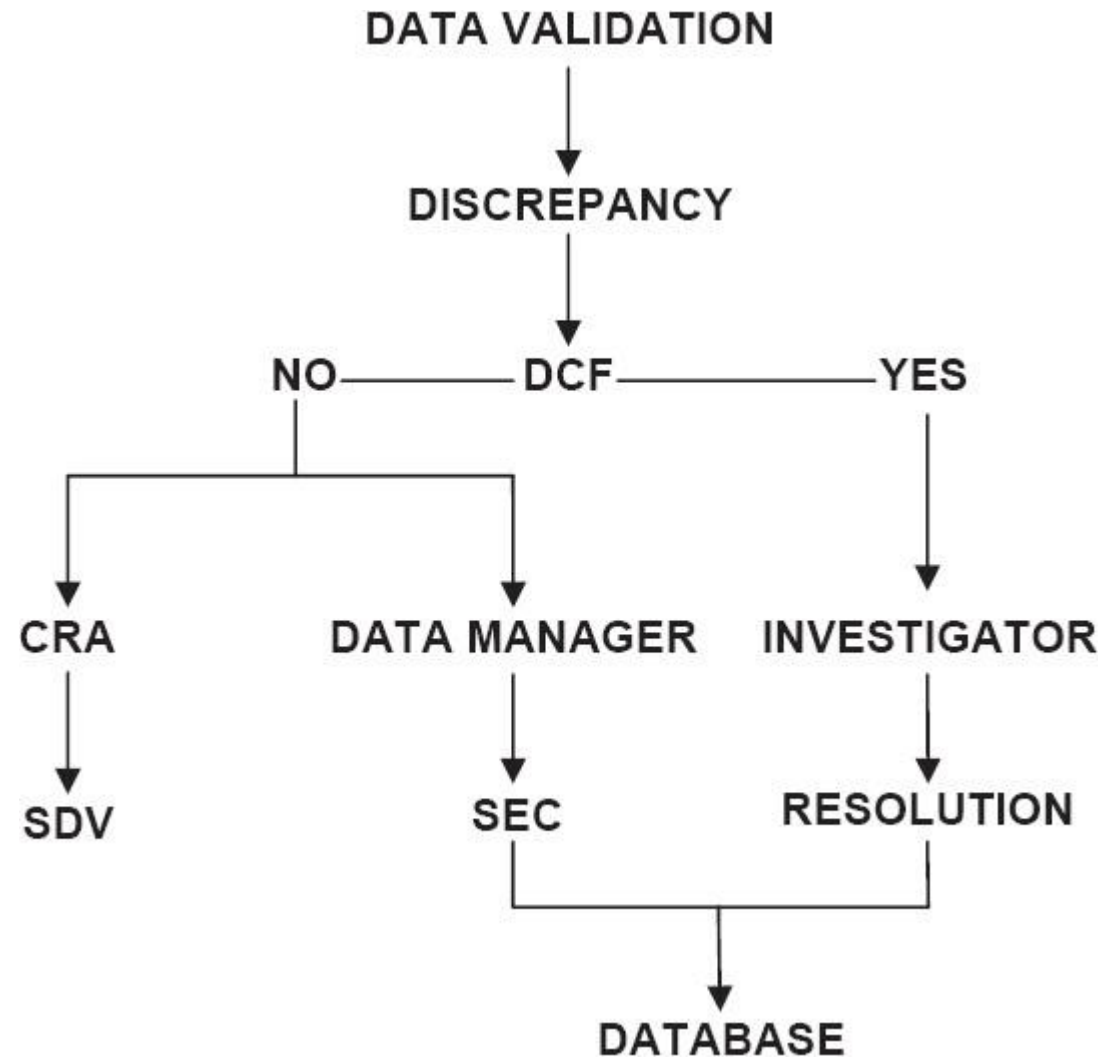
Experts in charge: data manager, database designer, quality control associate

- Clinical data validation is a series of quality tests to ensure accuracy, consistency, legibility, and integrity of information. This includes the following steps.
- **Electronic edit checks.** Edit checks are created by a database designer and embedded into eCRFs to automatically compare inputs against numerical and logical criteria. This prevents unlikely values from appearing in the document. Say, a check for the body temperature field may flag all entries lower than 95 and higher than 105 if the system expects measurements in Fahrenheit.
- **Source data verification (SDV).** SDV is a process of checking CRF entries against original medical records and other source files. The aim of this step is to confirm that an eCRF contains all relevant information and truly represents a participant's profile.
- **Data anonymization.** Before submission to sponsors, clinical data must be de-identified to comply with the Health Insurance Portability and Accountability Act (HIPAA). This means removing all elements of protected health information (PHI) that can link the document to a particular person.

Discrepancy management

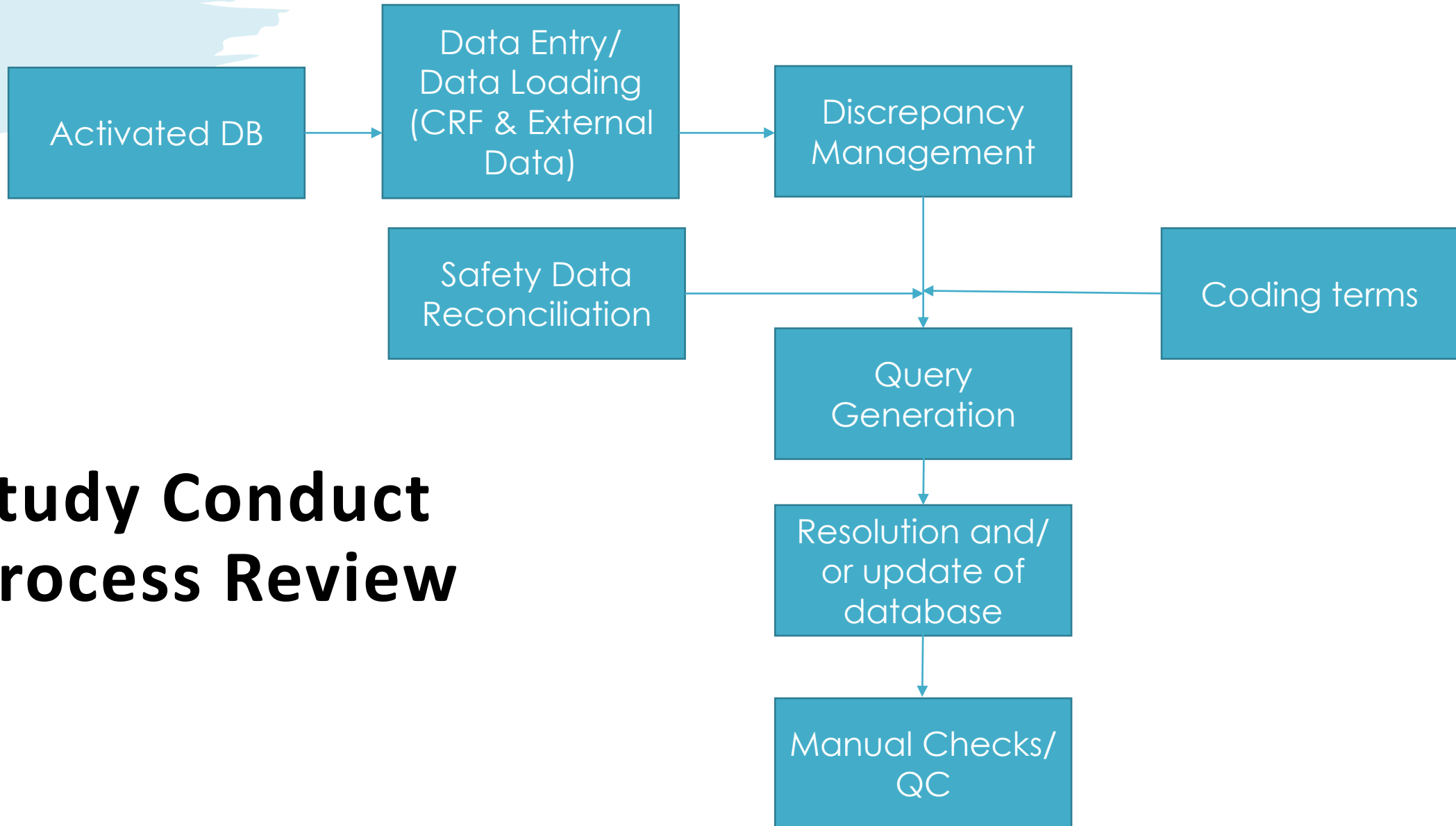
- This is also called query resolution.
- Discrepancy management includes reviewing discrepancies, investigating the reason, and resolving them with documentary proof or declaring them as irresolvable.
- Discrepancy management helps in cleaning the data and gathers enough evidence for the deviations observed in data.
- Almost all CDMS have a discrepancy database where all discrepancies will be recorded and stored with audit trail.
- Based on the types identified, discrepancies are either flagged to the investigator for clarification or closed in-house by Self-Evident Corrections (SEC) without sending DCF to the site. The most common SECs are obvious spelling errors.
- For discrepancies that require clarifications from the investigator, DCFs will be sent to the site.
- The CDM tools help in the creation and printing of DCFs.

- Investigators will write the resolution or explain the circumstances that led to the discrepancy in data.
- When a resolution is provided by the investigator, the same will be updated in the database.
- In case of e-CRFs, the investigator can access the discrepancies flagged to him and will be able to provide the resolutions online.
- The CDM team reviews all discrepancies at regular intervals to ensure that they have been resolved.
- The resolved data discrepancies are recorded as 'closed'.



Discrepancy management (DCF = Data clarification form, CRA = Clinical Research Associate, SDV = Source document verification, SEC = Self-evident correction)

- 
- This means that those validation failures are no longer considered to be active, and future data validation attempts on the same data will not create a discrepancy for same data point.
 - But closure of discrepancies is not always possible. In some cases, the investigator will not be able to provide a resolution for the discrepancy. Such discrepancies will be considered as 'irresolvable' and will be updated in the discrepancy database.
 - Discrepancy management is the most critical activity in the CDM process. Being the vital activity in cleaning up the data, utmost attention must be observed while handling the discrepancies.



Study Conduct Process Review

Clinical Data coding: Medical coding

- Clinical coding is the process whereby information written in the patient notes is translated into coded data and entered into hospital information systems.
- Medical coding helps in identifying and properly classifying the medical terminologies associated with the clinical trial.
- For classification of events, medical dictionaries available online are used.
- Technically, this activity needs the knowledge of medical terminology, understanding of disease entities, drugs used, and a basic knowledge of the pathological processes involved.
- Functionally, it also requires knowledge about the structure of electronic medical dictionaries and the hierarchy of classifications available in them.
- Adverse events occurring during the study, prior to and concomitantly administered medications and pre-or co-existing illnesses are coded using the available medical dictionaries.
- Medical Dictionary for Regulatory Activities (MedDRA) is used for the coding of adverse events as well as other illnesses and World Health Organization–Drug Dictionary Enhanced (WHO-DDE) is used for coding the medications.

- 
- These dictionaries contain the respective classifications of adverse events and drugs in proper classes.
 - Other dictionaries are also available for use in data management (eg, WHO-ART is a dictionary that deals with adverse reactions terminology).
 - Some pharmaceutical companies utilize customized dictionaries to suit their needs and meet their standard operating procedures.
 - Medical coding helps in classifying reported medical terms on the CRF to standard dictionary terms in order to achieve data consistency and avoid unnecessary duplication.
 - For example, the investigators may use different terms for the same adverse event, but it is important to code all of them to a single standard code and maintain uniformity in the process. The right coding and classification of adverse events and medication is crucial as an incorrect coding may lead to masking of safety issues or highlight the wrong safety concerns related to the drug.

Patient: Meredith Dewietz
Central Clinic

DOB: 07/31/1938
Admit Date:

Sections

- Discovery
- Health
- Pharmacy
- Account
- Registration
- Encounters
- Insurance
- Scheduling
- Correspondence
- Claims
- Ledger

Claim Edit

Medicare Part A and B 4558971

Appointment Data

DATE OF APPOINTMENT PT C

Diagnosis or Nature of Illness or Injury

Line # ICD10 Co

Procedure/Service/Supplies

From	To	Place of Service	CPT/HCPCS
------	----	------------------	-----------

Add New Procedure

Date(s) of Service: 05/19/2021 - 05/19/2021

Place of Service: Central Clinic

Rendering Provider: Manuel Bachman, MD

CPT/HCPCS: Please select

Modifier 1: 494

Modifier 2: 49465 Contrast injection(s) for radiological evaluation of existing gastrostomy, duodenostomy, jejunostomy, gastro-jejunostomy, or cecostomy (or other colonic) tube, from a percutaneous approach including image documentation and report

Modifier 3: 49446 Conversion of gastrostomy tube to gastro-jejunostomy tube, percutaneous, under fluoroscopic guidance including contrast injection(s), image documentation and report

Modifier 4:

Charge:

Days or Units:

Diagnosis Pointer: Select a diagnosis pointer

Cancel Add

In this medical coding example, the coder receives a patient record on abscess drainage and needs to translate the information regarding the indications, medication and procedure into medical codes according to intervention radiology CPT codes.

First, she identifies the procedure performed, which is "CT guided retroperitoneal peripancreatic fluid collection drainage." Then, she reviews the indications to learn that the CT images show a mass that potentially represents an abscess. Next, she finds out that moderate sedation was used.

Lastly, she learns that the surgeon aspirated purulent fluid, which was sent to the lab for evaluation. The medical coder then translates all the steps involved in this procedure into medical codes using her knowledge in human physiology, medical terminology and the Current Procedural Terminology (CPT) coding process, which are skills you can gain from [medical coding training](#). She uses the CPT codes 49405-59 to represent the various steps and supplies used in the image-guided fluid collection drainage process.



Patient: Sally Ziemer
General Hospital

DOB: 01/09/1985 36 yo F
Admit Date: 05/19/2021 11:31

MR#: 900-144561
 PCN, Full Code Status

Sections

Discovery

Health

Pharmacy

Account

Registration

Encounters

Insurance

Claim Edit

Appointment Data

DATE OF APPOINTMENT	PT CONDITION IS RELATED TO	DATE OF CURRENT	ACCEPT ASSIGNMENT

Diagnosis or Nature of Illness or Injury

Line #	ICD10 Code	Description
1	R18.8	Other ascites
2	K82.8	Other specified diseases of gallbladder

Add Diagnosis

In this medical coding example, the coder receives a patient record for a cholecystostomy exam. To translate the details of this procedure into billing codes, he needs coding knowledge about ultrasound and fluoroscopic guidance in intervention radiology.

First, he identifies the procedure as "ultrasound-guided cholecystostomy drainage." Then, he reviews the indication to learn that the patient has cholecystitis. Next, he sees that moderate sedation was applied and monitored by ICU nurses. Lastly, he read the details of the procedure to learn the ultrasound-guided percutaneous cholecystostomy was performed successfully. A complete ultrasound was obtained for diagnostic paracentesis.

The medical coder then translates the complex procedure into billing codes using his knowledge in human physiology, medical terminology, CPT coding and the International Classification of Diseases (ICD) coding practices. He uses two CPT codes (47490 and 49083), two ICD codes (575.8 and 789.59) and two ICD-10 codes (K82.8 and R18.8) to represent the various steps and medications involved in this procedure.

CPT codes

- The Current Procedural Terminology (CPT®) codes offer doctors and health care professionals a uniform language for coding medical services and procedures to streamline reporting, increase accuracy and efficiency.
- CPT codes are also used for administrative management purposes such as claims processing and developing guidelines for medical care review.
- There are various types of CPT codes:
- Category I: These codes have descriptors that correspond to a procedure or service. Codes range from 00100–99499 and are generally ordered into sub-categories based on procedure/service type and anatomy.
- Category II: These alphanumeric tracking codes are supplemental codes used for performance measurement. Using them is optional and not required for correct coding.

- Category III: These are temporary alphanumeric codes for new and developing technology, procedures and services. They were created for data collection, assessment and in some instances, payment of new services and procedures that currently don't meet the criteria for a Category I code.
- Proprietary Laboratory Analyses (PLA) codes: Recently added to the CPT code set, these codes describe proprietary clinical laboratory analyses and can be either provided by a single ("solesource") laboratory or licensed or marketed to multiple providing laboratories that are cleared or approved by the Food and Drug Administration (FDA)). This category includes but is not limited to Advanced Diagnostic Laboratory Tests (ADLTs) and Clinical Diagnostic Laboratory Tests (CDLTs), as defined under the Protecting Access to Medicare Act of 2014 (PAMA).
- How are CPT codes created and maintained?
- For more than 50 years, CPT has been the primary medical language used to communicate across health care, enabling seamless processing and advanced analytics for medical procedures and services. The CPT code set is constantly updated by the CPT Editorial Panel with insight from clinical and industry experts to reflect current clinical practice and the latest innovations to help improve the delivery of care.
- The CPT® Editorial Panel, appointed by the AMA Board of Trustees, is responsible for maintaining and updating the CPT code set.

ICD 10 codes

- International Classification of Diseases, Tenth Revision (ICD-10) is a system used by physicians to classify and code all diagnoses, symptoms and procedures for claims processing.
- It was chiefly designed by the World Health Organization, with the U.S. version being created by the Centers for Medicare & Medicaid Services (CMS) and the National Center for Health Statistics (NCHS) to better align with the country's health care infrastructure.
- The AMA provides the latest medical diagnosis code resources and training materials to help physicians and their teams understand what is an ICD 10 code

- A00-B99 Certain infectious and parasitic diseases
- C00-D49 Neoplasms
- D50-D89 Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism
- E00-E89 Endocrine, nutritional and metabolic diseases
- F01-F99 Mental, Behavioral and Neurodevelopmental disorders
- G00-G99 Diseases of the nervous system
- H00-H59 Diseases of the eye and adnexa
- H60-H95 Diseases of the ear and mastoid process
- I00-I99 Diseases of the circulatory system
- J00-J99 Diseases of the respiratory system
- K00-K95 Diseases of the digestive system
- L00-L99 Diseases of the skin and subcutaneous

tissue

- M00-M99 Diseases of the musculoskeletal system and connective tissue
- N00-N99 Diseases of the genitourinary system
- O00-O9A Pregnancy, childbirth and the puerperium
- P00-P96 Certain conditions originating in the perinatal period
- Q00-Q99 Congenital malformations, deformations and chromosomal abnormalities
- R00-R99 Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified
- S00-T88 Injury, poisoning and certain other consequences of external causes
- U00-U85 Codes for special purposes
- V00-Y99 External causes of morbidity
- Z00-Z99 Factors influencing health status and contact with health services

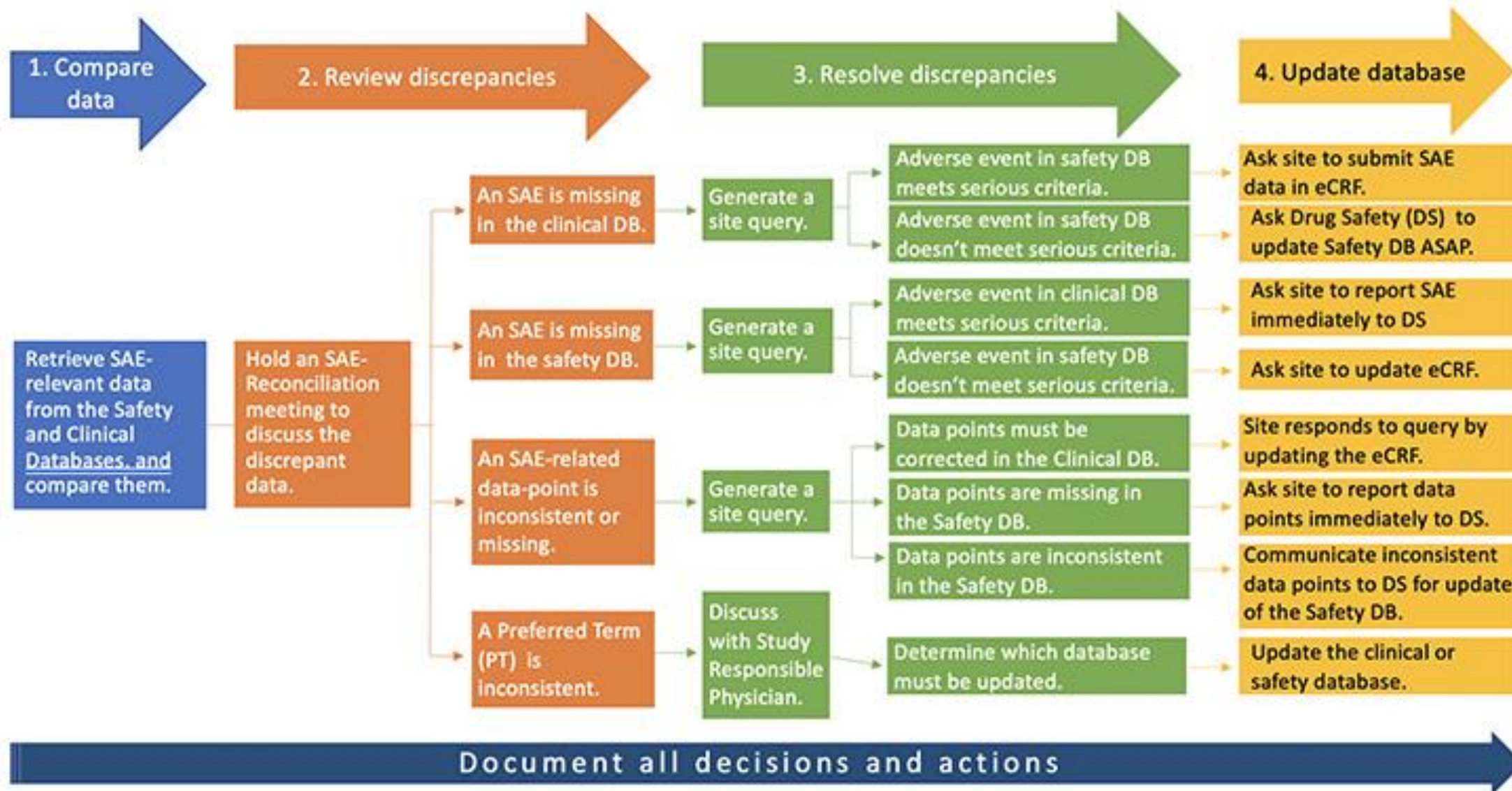
Healthcare Common Procedure Coding (HCPCS)

- The Centers for Medicare and Medicaid Services (CMS) developed this coding system to help with processing insurance claims for Medicare or other providers.
- Coders use them to note medical procedures, products, supplies, and services.

SAE data reconciliation

- The process of comparing key safety data variables between the drug or device safety SAE database and the clinical database in order to identify any discrepancy, determine whether a discrepancy is acceptable or not and, if acceptable, document the discrepancy.
- During the process of SAE data reconciliation, verbatim descriptions, coding terms, dates and other information are frequently found to be different.
- Sometimes, whole AEs are missing on one or the other side.
- Some data types must have an identical value and others have a margin of tolerance, but all must be compared, weighted, clarified and if needed corrected before the study is over.

The SAE Reconciliation Process



Roles and Responsibilities of Safety Data Manager

- Extracting listings from Safety Database and Clinical Database
- Reconciling mandatory/agreed upon SAE data points
- Communicating internally with coders to resolve coding issues in Study Database
- Checking internally with Data Entry to resolve data issues
- Raising SDARF (Safety Database Amendment Request Form) for missing/inconsistent SAE data in Safety Database
- Raising query to site in Database (DQF) when a field/s of SAE data is/are missing/inconsistent on both Databases (*Once resolution received from site confirms data in eCRF is correct, prepare and send an SDARF to safety team asking to update safety database.*)
- Sending mail to Monitor when data in between clinical and safety is missing/discrepant
- Preparing SRS log to capture discrepancies and actions taken Between clinical and safety
- Filling the monthly on-going Reconciliation form & Final SAE Reconciliation Form
- Tracking all request forms sent and filling the corresponding tracker on a regular basis

- Process:
- The SAE reconciliation process is typically driven by a Data Manager who, on a periodic basis, reviews data in the clinical database and compares defined variables with the corresponding records in the safety database.
- Data Managers collaborate with several functions including Drug Safety, Biostatistics and Clinical Development for the resolution of these discrepancies.
- SAE-Reconciliation meetings are often necessary in order to discuss the discrepancies and determine if they are acceptable or decide on the appropriate course of action.
- Often, queries are sent to the sites for clarification, in other occasions the study physician or the data manager must make a decision.
- Finally, all discussions, options and decisions must be clearly documented and filed in the Trial Master File (TMF) for future inspection by health authorities.



SAE discrepancy

- A mismatch between the safety database and the clinical database identified during the reconciliation process.
- A discrepancy can be for example:
 - An SAE present in one database but missing in the other one
 - Inconsistent SAE associated data between the databases
 - Missing SAE associated data in one of the databases
 - Mismatched SAE preferred term
- All the identified discrepancies, actions taken to address the discrepancies, and the status relevant to the action taken are recorded in the SAE manual or electronic reconciliation tool.

SAE reconciliation listing

- File listing, generally in Excel format, used to review and document all discrepancies, acceptable or not, found during SAE data reconciliation.
- SAE reconciliation process
 - The SAE reconciliation process can be summarized in the four following steps:
 - Retrieve and compare data.
 - Review the discrepancies.
 - Resolve the discrepancies.
 - Update database.

What is reconciled?

TACO FIELDS

T: adverse event term
A: Action Taken
C: Causality
O: Outcome

Adverse Event Term - refers to a singular and specific medical term for an AE listed on the AEM Form. When you reconcile the Event Term field, you should examine the Term, Onset and Cessation Date

Action Taken - refers to any change in study drug dose in response to an SAE reported on the AEM Form.

Causality - refers to the most likely cause, in the Investigator's opinion, of an AE reported on the AEM Form.

Clinical Outcome - refers to the Outcome of an AE reported on the AEM Form.

Additional FIELDS

Case ID
Patient ID
Date of Death

Non - TACO FIELDS

Demography
Total Daily Dose
Therapy Date

Demography Module, but not be limited to the following:

- Subject Identification
- Weight
- Date of Birth
- Gender
- Race

Discontinuation Module, but not be limited to the following:

- Subject Identification
- Primary reason for discontinuation being an event
- Cause of hospitalization
- Cause of death listed on the death certificate
- Autopsy Result

Concomitant medications Module, but not be limited to the following:

- Subject Identification
- Medication name
- Start Date
- Stop date or Ongoing
- Indication

Documents

- Data correction Request form (DCRF)
- SAE Reconciliation Spreadsheet (SRS)
- Ongoing SAE Reconciliation Form
- Final SAE REconciliation Form
- Coding Mismatch Tracker
- Recommended key data elements for inclusion in expedited reports for SA drug reactions as per ICH GCP

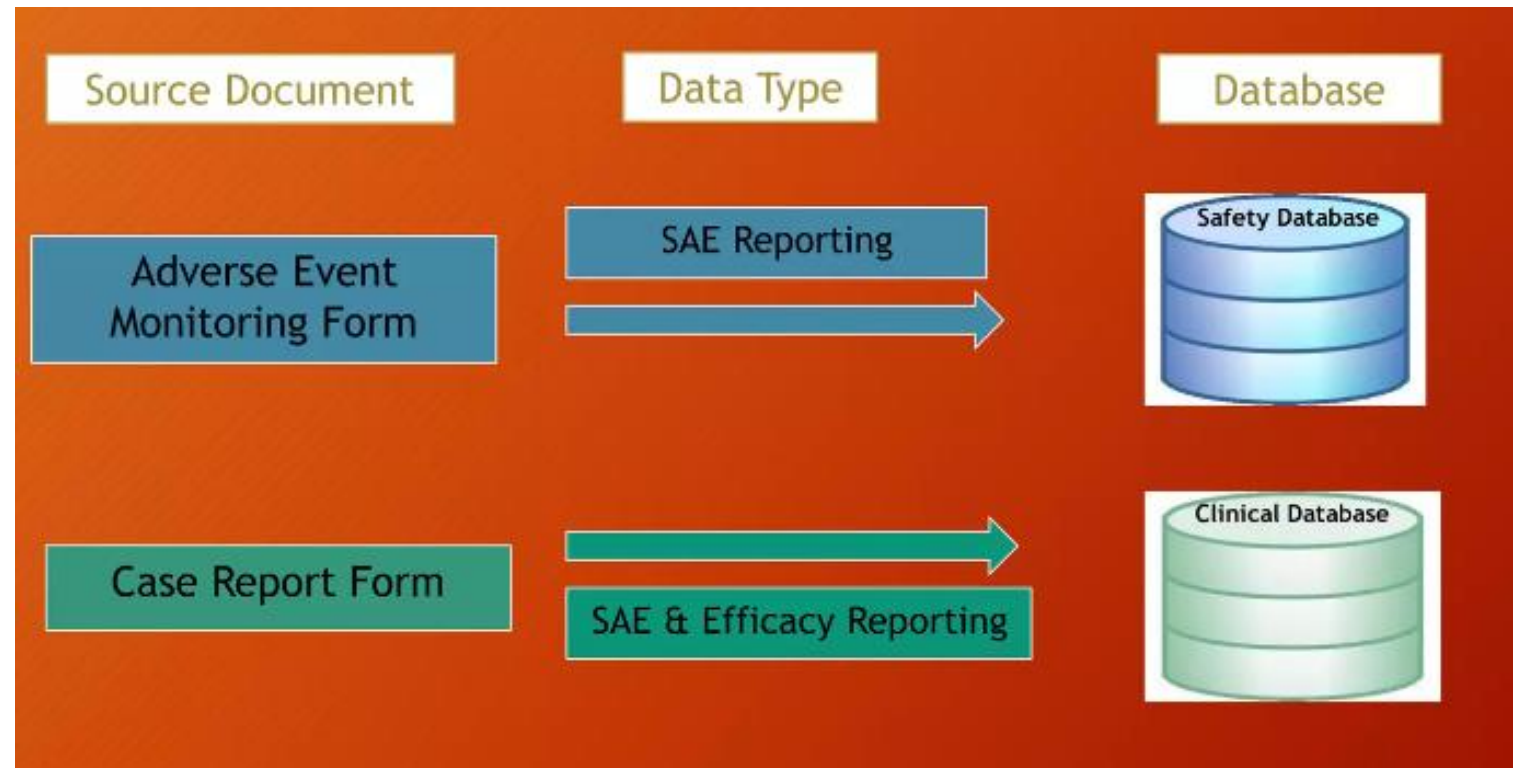
Examples of Databases

- **Clinical DB**

- Rave
- InForm
- Oracle Clinical

- **Safety DB**

- ARGUS
- ARISG
- AERS
- SAFIRE



Items to be reconciled during SAE reconciliation

The following items in the clinical study database (based on CRF data) and ARGUS database (based on SAE forms) should be reconciled unless otherwise stated.

Exact: Items must match exactly.

Consistent: Items do not have to match exactly but there must be a justifiable reason for non-exact match and the data must be comparable, not contradictory.

Adverse Event Item	REMARK	TYPE OF MATCH
CRF data	SAE report	
Study number	Study ID	Exact
Patient number	Patient ID	Exact
Adverse Event (as reported)	Event description	Consistent
Start of event	Onset date	Consistent
End of event	Stop date	Consistent
Severity	Intensity	Exact
Seriousness criterion	AE serious	Consistent
Relationship to Study Drug	Investigator causality	Consistent
Outcome / Date of Death on Mortality Report	Outcome of event/Death (seriousness criteria)	Consistent

Recon Plan Examples

Concomitant Medication item	REMARK	TYPE OF MATCH
Medication Name	All drugs present in the DSP database (suspect product, concomitant product) must be present in the CRF database but not vice versa. (DSP only collects information on drugs taken by the subject 1 month before onset event)	Consistent
Start and stop dates, dosage and frequency		Exact

Medical History item	REMARK	TYPE OF MATCH
Medical Condition	All conditions present in the CRF database must be present in the safety database but not vice versa. Verbatim term should match exact. Reconciliation of other information is not necessary.	Consistent

Study drug item	REMARK	TYPE OF MATCH
Start and stop dates		Exact
Dosage and frequency	Only if data available in the CRF.	Exact

Table 1: Patient and Case Identifiers

Field	Match
Demographics	1:1 match
Event Term	Rational match verbatim to preferred terms
Onset Date	Rational match first onset of symptoms versus date event becomes serious
Outcome	Rational match to subject summary or final page
Resolved Date	Rational match to outcome and summary pages
Causality	1:1 match
Study Drug Start Date	1:1 match
Study Drug Stop Date	1:1 match
Date of Death	1:1 match
Cause of Death	Rational match to causality
Action Taken regarding Study Drug	Action taken in response to SAE; rational check to dosing data

Database lock and data archiving

Experts in charge: data manager, database designer

- On the study completion, the database is locked so that no changes can be done to the information.
- After that, clean data is submitted to stakeholders for statistical analysis, reporting and, finally, publication of the results.
- All these steps are beyond the clinical data management workflow.
- All essential documents and trial supplies must be archived for at least three years.
- This enables post-trial monitoring and evaluation as well as trial reconstruction to facilitate further research.

Clinical data management software

- Spreadsheets and common office programs are obviously not enough to address all the challenges of clinical data management.
- You need software capable of handling large amounts of documents and customized for medical studies — **clinical data management system (CDMS), also called electronic data capture (EDC) systems**.
- CDMSs can be specifically tailored for psychiatry studies, medical device clinical trials, drug development, or other cases.
- However, all of them have common features covering basic data management operations.

- 21 CFR part 11 compliance. Title 21 of the CFR (the Code of Federal Regulations) regulates food and drugs produced or consumed across the USA. Specifically, its 11th part sets rules for information systems used by companies subject to the FDA (the Food and Drug Administration). Compliance with this document means that the technology has built-in mechanisms for data security and traceability — namely
 - access controls ensuring that only authorized users, under unique IDs (electronic signatures) are able to enter the system and work with data;
 - forced periodical password resets; and
 - audit trails, or keeping chronological records of all operations and changes to the database.

All these precautions are taken to guarantee that the data produced by the system can be trusted.

- **eCRF designer.** This module provides a set of templates and a drag-and-drop functionality to construct eCRF layouts and data entry fields in accordance with standards. Custom forms are then saved to be reused in future projects. Also, the designer allows for programming edit checks.
- **Query management.** In terms of clinical trials, a query is a request for clarification from trial sponsors to researchers. Such requests are made during data review, before database lock, and aimed at resolving errors and inconsistencies. The query management feature facilitates communication among data managers, sponsors, and other stakeholders. It helps faster resolve all questions.
- **Monitoring.** The monitoring capabilities include but are not limited to scheduling, adverse event tracking, and automatic notifications to sponsors.
- **Medical coding support.** The component automates code search, mapping across coding systems, and error checking.
- **Data import and export.** Some systems are equipped with powerful data integration tools that facilitate multisite studies. The data export feature, in turn, automatically transforms information from the trial database into formats applicable for statistical analysis and required by trial sponsors and regulatory authorities

CDMSs accommodating all the above-mentioned features

CLINICAL DATA MANAGEMENT SOFTWARE

	Coverage	Pricing model	Pros & Cons
IBM Clinical Development	3,000+ studies	Individual plan	✓ Reliability ✓ Coding with Watson AI ✗ Archaic UI ✗ High price
Oracle Clinical Research Suite	1,000+ studies	Individual plan	✓ Stability ✓ Oracle integrations ✗ Slow data entry ✗ High price
Castor EDC	8,500+ studies	Based on study needs	✓ Ease of setup and use ✓ Quick support ✓ Affordability ✗ Limited functionality
TrialKit	7,000+ studies	Based on number of features	✓ Ease of use ✓ Smooth wearable support ✗ Hard learning curve ✗ Limited functionality

IBM Clinical Development: a reliable platform for decentralized studies

- **Used by 3000+ clinical trials**
- **Pricing model individual plan**
- **Pros:** reliability, coding with Watson AI
- **Cons:** archaic UI, high price
- IBM Clinical Development (ICD) is an end-to-end cloud-based system allowing for data capturing from various sources and targeting large-scale, decentralized studies. Its capabilities extend beyond the trial data management cycle, covering analytical reports.
- The strong points of ICD are reliability, a large library of pre-built forms, ease of use, and a flexible, modular structure. You can choose and pay for only those features that you really need. The medical coding part is supported by the AI power of IBM Watson.
- Keep in mind, though, that you will need a programmer to set up a trial. Among other cons admitted by users are archaic UI, slow customer support, and high price of ownership, which makes the system unattainable for startup projects.

Oracle Clinical Research Suite the best support for paper-based studies

- **Used by 1000+ studies**
- **Pricing model: individual plan**
- **Pros:** stability, integrations with Oracle products
- **Cons:** slow data entry, high price
- One of the oldest CDMs on the market comes as a combination of three integrated subsystems
 - Oracle Clinical automating trial protocol design, data validation, and reporting;
 - Oracle Remote Data Capture, an EDC solution with layout editor to generate collection forms; and
 - Thesaurus Management System for standardizing clinical terminology.
- Running on the top of the Oracle database, the suite demonstrates great stability and is recognized as one of the best systems for paper-based data collection. It seems to be a natural choice for companies already using other Oracle products.

At the same time, its data entry part involves repetitive and tedious manual operations. Also, Oracle isn't meant for small companies with limited budgets.

Castor EDC: an affordable way to speed up trial builds

- **Used by 8500+ studies**
- **Pricing model: based on study needs**
- **Pros:** ease of setup and use, quick support, affordability
- **Cons:** limited functionality
- Castor EDC became popular across 90+ countries due to its proven ability to significantly shorten the duration of a clinical study build. Intuitive UI simplifies creating eCRF forms, assigning study roles, and joining new users.
- The data can be smoothly integrated from various sources — EHR systems, medical devices, wearables, etc. Should any problems arise, you'll get help from an expert in no more than 30 minutes.
- Relatively low price makes the platform affordable for small companies. Quite expectedly, it lacks some features and export formats that larger and more expensive systems typically have.

TrialKit: an intuitive CRF designer for virtual studies

- Used by 7000+ studies
- Pricing model: based on number of features
- Pros: ease of use, smooth integrations with wearables
- Cons: hard learning curve, limited functionality,
- Aimed at decentralized virtual studies, TrialKit comes in two versions — web-based and mobile.
- Due to a drag-and drop CRF builder and a library of ready-to-use templates, companies can create eCRFs with no programming skills and launch trials in days instead of weeks. In addition to EHRs, the tool easily integrates data from wearables like Fitbit, Apple Watch etc.
- As for cons, it takes time to learn how to work with the platform. Another major complaint relates to limitations in functionality.

Choose and adopt a CDMS: best practices

- **Check compatibility and integration options.** Make sure that the intended CDMS smoothly links to other platforms you'll use during the trial — namely, an EHR system, LIMS, and others. In case of compatibility issues, you 'll need to find a tech partner with system integration expertise.
- **Learn more about the level of support offered by a vendor.** This includes training programs, the quality of user documentation provided, and speed of response in case of any issues. Also, check if the provider will help you with the system deployment, integrations, and development of new features once you need them.
- **Make sure that all parties are satisfied.** Before finalizing your choice and implementing a new platform, try to get feedback from all intended users — CROs, sponsors, staff and technical support at the research site. Take advantage of a free trial or demo version so that everybody could test the user interface and share their experience.

- 
- **Start from the core functionality.** Redundant features not only inflate your budget but create additional complexities for staff who will have to get familiar with a new system. Later, you can expand functionality as needed and as your project grows by adding new modules from the same provider, integrating third-party tools or using custom development.
 - **Test and validate the system against the eCRF.** Once electronic forms are designed, you need to run user acceptance testing (UAT.) End users like data entry associates, clinicians, and researchers should determine if they feel comfortable with the eCRF structure and if the form contains all needed fields.
 - **Take your time to set up a database.** Quite naturally, trial sponsors want to get the system run in no time. However, it's important to balance speed and risk reduction. Take your time to thoroughly design and test a database before the study starts. Software changes during the trial might be costly and impact data validity.

Regulations, Guidelines, and Standards in CDM

- Akin to other areas in clinical research, CDM has guidelines and standards that must be followed.
- a need to follow good practices in CDM and maintain standards in electronic data capture.
- These electronic records have to comply with a Code of Federal Regulations (CFR), 21 CFR Part 11

Part 11, Electronic Records; Electronic Signatures - Scope and Application

- This guidance is intended to describe the Food and Drug Administration's (FDA's) current thinking regarding the scope and application of part 11 of Title 21 of the Code of Federal Regulations; Electronic Records; Electronic Signatures (21 CFR Part 11)
- This document provides guidance to persons who, in fulfillment of a requirement in a statute or another part of FDA's regulations to maintain records or submit information to FDA, have chosen to maintain the records or submit designated information electronically and, as a result, have become subject to part 11.
- Part 11 applies to records in electronic form that are created, modified, maintained, archived, retrieved, or transmitted under any records requirements set forth in Agency regulations.
- As an outgrowth of its current good manufacturing practice (CGMP) initiative for human and animal drugs and biologics, FDA is re-examining part 11 as it applies to all FDA regulated products.

- FDA has
- (1) spoken about part 11 at many conferences and met numerous times with an industry coalition and other interested parties in an effort to hear more about potential part 11 issues;
- (2) published a compliance policy guide, CPG 7153.17: Enforcement Policy: 21 CFR Part 11; Electronic Records; Electronic Signatures; and (3) published numerous draft guidance documents including the following:
 - 21 CFR Part 11; Electronic Records; Electronic Signatures, Validation
 - 21 CFR Part 11; Electronic Records; Electronic Signatures, Glossary of Terms
 - 21 CFR Part 11; Electronic Records; Electronic Signatures, Time Stamps
 - 21 CFR Part 11; Electronic Records; Electronic Signatures, Maintenance of Electronic Records
 - 21 CFR Part 11; Electronic Records; Electronic Signatures, Electronic Copies of Electronic Records

SUBPART A – GENERAL PROVISIONS

- Part 11 applies to all electronic records that fall under FDA regulations.
- If an organization can prove to an auditor that their electronic records/signatures are as trustworthy as paper records/ink signatures, the FDA will accept electronic instead of paper.
- The FDA will accept electronic submission instead of paper IF those submissions
 - 1) adhere to Part 11 requirements and
 - 2) are included among the types of documents that the FDA accepts electronically.

SUBPART B – ELECTRONIC PROVISIONS

- Organizations using electronic records must establish and document procedures and controls that ensure the following qualities in their electronic records:
- Authenticity
- Integrity
- Confidentiality (when appropriate)
- Irrefutability (i.e., no way to deny that a record is genuine)
- The following topics must be addressed in documented procedures and controls: computer systems validation (CSV), record rendering, document storage and record retention, system access, audit trails, workflows, authority checks, device checks, personnel qualifications, personnel accountability, and document control.
- Systems that fall into the category of “Open” (as defined in Subpart A) require additional procedures/ controls.
- Electronic signatures must include the printed name of the signer, the date and time of the signature, and the meaning of the signature.
- Electronic signatures must be forever linked to their respective records.

SUBPART C – ELECTRONIC SIGNATURES

- Organizations that wish to use electronic signatures must inform the FDA in writing prior to making the switch.
- Each individual who will be using an electronic signature must
 - 1) have their identity confirmed and
 - 2) use a unique signature that has never been and will never be used by another individual.
- There are specific design requirements for electronic signatures that are biometric (e.g., fingerprint scan) and those that are not (e.g., user ID and password).
- For electronic signatures that make use of user IDs and passwords/passcodes, there are specific requirements for passwords and for passcode-generating devices

• **GENERAL PROVISIONS: 11.1 – SCOPE**

- The purpose of Part 11 is to ensure that electronic records and electronic signatures can be trusted as much as paper records and ink signatures.
- All electronic records that are used for regulated purposes are subject to Part 11.
- One clarification made – a paper record that is transmitted electronically (e.g., as an email attachment) is NOT subject to Part 11.
- If an organization can prove, typically via computer system validation, that its electronic signatures comply with Part 11, the FDA will accept electronic signatures instead of ink. One exception is noted – if some other regulation specifically requires ink, that regulation supersedes Part 11.
- If an organization can prove that its electronic records comply with Part 11, the FDA will accept electronic records instead of paper. One exception is noted – if some other regulation specifically requires paper, that regulation supersedes Part 11.
- The proof required in the previous two letters (c and d) must be maintained in such a way that the FDA can inspect it (i.e., documentation is king).
- There are a few obscure types of records that are excluded from Part 11, because they fall under other regulations, but the vast majority need to comply

- **GENERAL PROVISIONS: 11.2 – IMPLEMENTATION**

- For regulated records that are NOT submitted to the FDA, an organization can use electronic instead of (or in addition to) paper, as long as it can prove that its electronic records comply with Part 11.
- For regulated records that ARE submitted to the FDA, an organization can use electronic instead of paper as long as these two conditions are met:
 - 1. It can prove that its electronic records comply with Part 11.
 - 2. The FDA is capable of accepting those types of records electronically.
- The types of e-records that the FDA accepts are listed in public docket No. 92S-0251.
- If there is any doubt as to whether a record can be submitted electronically, contact the receiving unit at the FDA before attempting submission.

- **ELECTRONIC RECORDS: 11.10 – CONTROLS FOR CLOSED SYSTEMS**

- An organizations using electronic records must document the procedures it follows and the controls it has in place for ensuring that their electronic records have these qualities:
 - Authenticity
 - Integrity
 - Confidentiality (when appropriate)
 - Irrefutability (i.e., no way to deny that a record is genuine)
- The documented procedures and controls must address the following topics

- **Validation:** How an organization proves (to itself and auditors) that the data in a computer system can be trusted.
- **Rendering Records:** How an organization makes sure that all electronic records that an auditor might want to see and/or copy can be provided in a language/format that humans (not just computers) can understand.
- **Document Storage & Record Retention:** How an organization protects documentation and keeps it readily available for as long as it's required to be stored.
- **System Access:** How an organization ensures that only the right people have access to each computer system.
- **Audit Trails:** How an organization ensures that a complete history of an electronic record is automatically captured by a computer system, retained in the system for the right amount of time, and viewable by humans.
- **Workflows:** How an organization makes sure that electronic workflows in computer systems function correctly.
- **Authority Checks:** How an organization limits user access (system level and record level) and verifies that the users performing functions in the system are authorized to do so

- **Device Checks:** How an organization verifies that equipment being used for regulated purposes is functioning properly. **Personnel Qualifications:** How an organization makes sure only trained and qualified people perform functions on or within the system.
- **Personnel Accountability:** How an organization holds individuals accountable for the integrity of their actions related to electronic records and electronic signatures.
- **Document Control:** How an organization controls documents related to system operation and maintenance and preserves the complete history of changes made to these documents.
- **ELECTRONIC RECORDS: 11.30 – CONTROLS FOR OPEN SYSTEMS**
 - Any time an electronic record is signed, the following information must be visible and associated with the signature:
 - Printed name of signer
 - Date and time of signature
 - Meaning of signature (e.g., content is accurate, format is correct, data calculations were verified)
- **ELECTRONIC RECORDS: 11.70 – SIGNATURE/RECORD LINKING**
 - Any kind of signature (ink or electronic) executed to an electronic record must remain connected to that record forever. It can't be removed, covered over, erased, transferred, etc.

- **ELECTRONIC SIGNATURES: 11.100 – GENERAL REQUIREMENTS**

- Each person must have a unique electronic signature that has never been and never will be used by anyone else.
- Before someone can use an electronic signature, his/her identity must be verified.
- Before an organization implements electronic signatures, it must notify the FDA of its intention and state that it will consider electronic signatures to be as legally binding as ink signatures.
- The first step in the process is to write and mail the FDA a paper letter signed with ink signatures.

- **ELECTRONIC SIGNATURES: 11.200 – ELECTRONIC SIGNATURE COMPONENTS AND CONTROLS**

- They must be made up of at least two distinct parts (i.e., user ID and password).
- The first time a user signs a record after logging into a system, the system must require them to enter ALL of the parts of their signature (i.e., user ID and password).
- Electronic signatures can only be used by the individuals to whom they are assigned.
- However, if an individual's electronic signature must be used by someone it's not assigned to, the system must require at least two people to work together to do so.
- Electronic signatures that are biometric (e.g., fingerprint scan, retinal scan) can only be used by the individuals to whom they are assigned.

- **ELECTRONIC SIGNATURES: 11.300 – CONTROLS FOR IDENTIFICATION CODES/PASSWORDS**

- For electronic signatures that make use of identification codes (i.e., user IDs) and passcodes/passwords, the following controls need to be in place:
- No two users can have the same combination of user ID and password – each combination must be unique – and each user ID can only be assigned to one individual, ever (no re-use allowed).
- Passwords must be checked, recalled, or changed from time to time.
- If a passcode token/device is lost or stolen, it must be deauthorized and a secure replacement must be issued.
- Unauthorized attempts to access user IDs or passwords/passcodes must be detected and reported to the appropriate person/group in the organization for investigation.
- Passcode tokens must be tested before they are issued for use and periodically while in use to make sure they're functioning correctly.

Society for Clinical Data Management (SCDM) publishes the Good Clinical Data Management Practices (GCDMP) guidelines,

- A document providing the standards of good practice within CDM.
- GCDMP was initially published in September 2000 and has undergone several revisions thereafter. The July 2009 version is the currently followed GCDMP document.
- GCDMP provides guidance on the accepted practices in CDM that are consistent with regulatory practices.
- Addressed in 20 chapters, it covers the CDM process by highlighting the minimum standards and best practices.
- The Good Clinical Data Management Practices (GCDMP®) standard provides a reference to clinical data managers in their implementation of high quality Clinical Data Management processes and is used as a guidance tool for clinical data managers when preparing for CDM training and education.

Good Clinical Practice (GCP):

- **Trial management; data handling, record keeping (2.10, 5.5.3 a-d)**
- All clinical trial information should be recorded, handled, and stored in a way that allows its accurate reporting, interpretation and verification

Trial management, data handling, and record keeping

- **5.5.1** The sponsor should utilize appropriately qualified individuals to supervise the overall conduct of the trial, to handle the data, to verify the data, to conduct the statistical analyses, and to prepare the trial reports.
- **5.5.2.** The sponsor may consider establishing an independent data-monitoring committee (IDMC) to assess the progress of a clinical trial, including the safety data and the critical efficacy endpoints at intervals, and to recommend to the sponsor whether to continue, modify, or stop a trial. The IDMC should have written operating procedures and maintain written records of all its meetings.
- **5.5.3.** When using electronic trial data handling and/or remote electronic trial data systems, the sponsor should:
 - a) Ensure and document that the electronic data processing system(s) conforms to the sponsor's established requirements for completeness, accuracy, reliability, and consistent intended performance (i.e. validation).
 - (b) Maintains SOPs for using these systems
 - (c) Ensure that the systems are designed to permit data changes in such a way that the data changes are documented and that there is no deletion of entered data (i.e. maintain an audit trail, data trail, edit trail).

- d) Maintain a security system that prevents unauthorized access to the data.
- e) Maintain a list of the individuals who are authorized to make data changes (see 4.1.5 and 4.9.3).
- f) Maintain adequate backup of the data.
- **Safety reporting (4.11)**
 - 4.11.1. All serious adverse events (SAEs) should be reported immediately to the sponsor except for those SAEs that the protocol or other document (e.g., Investigator's Brochure) identifies as not needing immediate reporting. The immediate reports should be followed promptly by detailed, written reports. The immediate and follow-up reports should identify subjects by unique code numbers assigned to the trial subjects rather than by the subjects' names, personal identification numbers, and/or addresses
- **Subject and data confidentiality (2.11; 5.5.3 g))**
 - **2.11.** The confidentiality of records that could identify subjects should be protected, respecting the privacy and confidentiality rules in accordance with the applicable regulatory requirement(s).
 - **5.5.3 g.** When using electronic trial data handling and/or remote electronic trial data systems, the sponsor should:
 - g) Safeguard the blinding, if any (e.g. maintain the blinding during data entry and processing)

- **Quality control, Record and report (4.9.1; 4.9.3; 5.1.3)**
- **4.9.**The investigator/institution should maintain adequate and accurate source documents and trial records that include all pertinent observations on each of the site's trial subjects. Source data should be attributable, legible, contemporaneous, original, accurate, and complete. Changes to source data should be traceable, should not obscure the original entry, and should be explained if necessary (e.g., via an audit trail).
- **4.9.1.** The investigator should ensure the accuracy, completeness, legibility, and timeliness of the data reported to the sponsor in the CRFs and in all required reports.
- **4.9.3.** Any change or correction to a CRF should be dated, initialed, and explained (if necessary) and should not obscure the original entry (i.e. an audit trail should be maintained); this applies to both written and electronic changes or corrections (see 5.18.4 (n)). Sponsors should provide guidance to investigators and/or the investigators' designated representatives on making such corrections. Sponsors should have written procedures to assure that changes or corrections in CRFs made by sponsor's designated representatives are documented, are necessary, and are endorsed by the investigator. The investigator should retain records of the changes and corrections.
- **5.1.3** Quality control should be applied to each stage of data handling to ensure that all data are reliable and have been processed correctly.

- 
- **Clinical trial/study reports** (5.23.2)
 - **5.23.2.** The CRFs are designed to capture the required data at all multicentre trial sites. For those investigators who are collecting additional data, supplemental CRFs should also be provided that are designed to capture the additional data.
 - **Monitoring**
 - **5.4.1.** The sponsor should utilize qualified individuals (e.g. biostatisticians, clinical pharmacologists, and physicians) as appropriate, throughout all stages of the trial process, from designing the protocol and CRFs and planning the analyses to analyzing and preparing interim and final clinical trial reports.
 - **5.5.4** If data are transformed during processing, it should always be possible to compare the original data and observations with the processed data.

Consortiums in CDM

- Two such standards created by the **Clinical Data Interchange Standards Consortium (CDISC)** are particularly vital to CDM:
- the Study Data Tabulation Model Implementation Guide for Human Clinical Trials (**SDTMIG**), which is currently mandated for use by the U.S. Food and Drug Administration (FDA).
- Its counterpart is the Clinical Data Acquisition Standards Harmonization (**CDASH**) created a standard format for collecting data across studies so that data submissions can be more easily tracked and reviewed.

CDMS

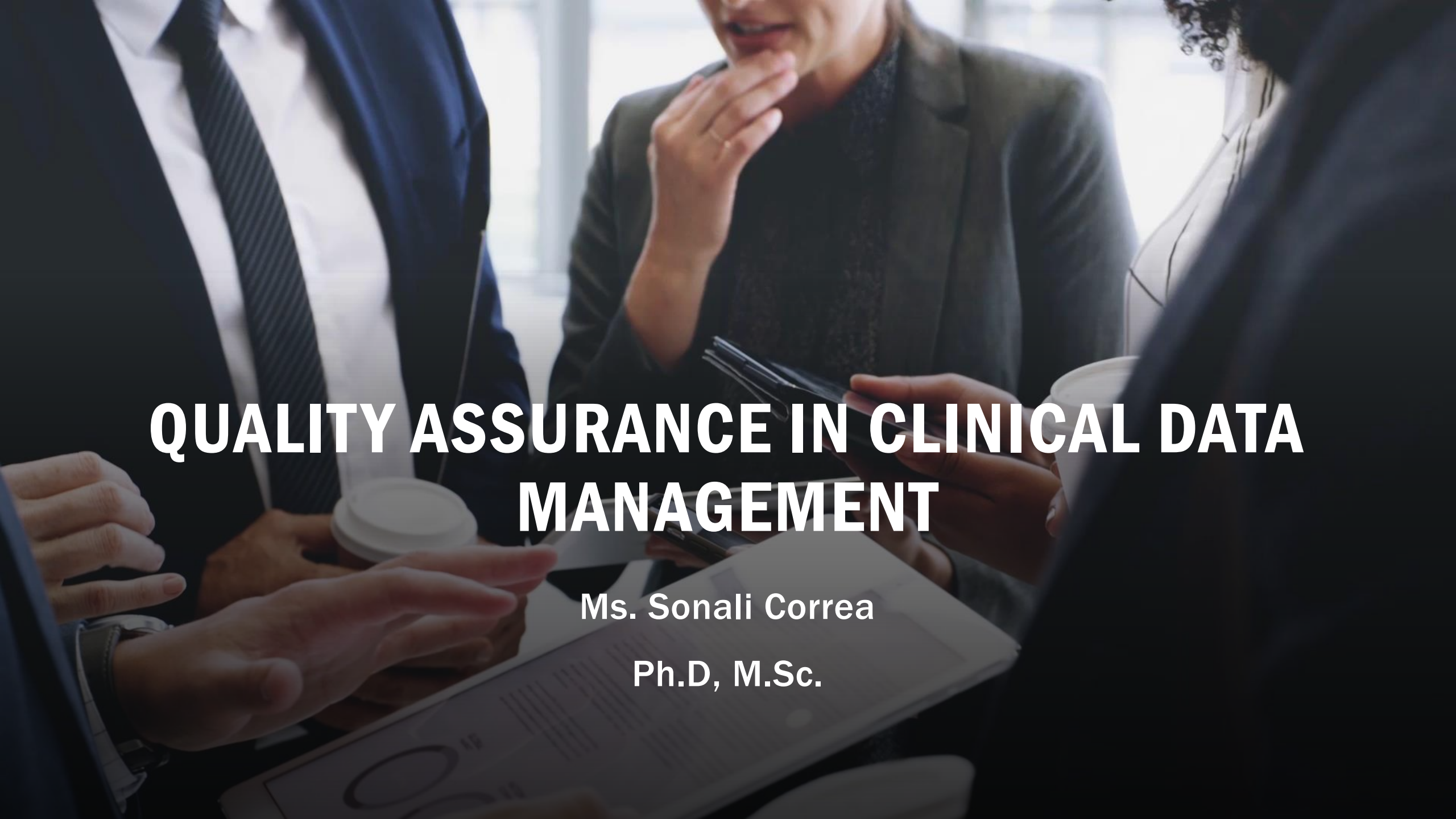
- Specialized software applications are used in CDM to create audit trails that allow discrepancies to be minimized even in large and complex clinical trials.
- Oracle Clinical,
- Rave,
- eClinical suite,
- Clintrial,
- Macro.
- Clinical data management systems (CDMS) are especially vital in trials conducted across medical centers in which an enormous amount of data is produced.

Open Source Tools

- CDMSs can be customized and may be tailor-made in the case of large, multinational pharmaceutical companies looking for tools that address the needs of their specific companies.
- In addition, open-source tools such as
 - TrailDB
 - open CDMS
 - OpenClinica
 - PhOSCo

are freely available and can be just as effective.

- Set standards guide the work of CDM professionals because, as mentioned above, the information is used in the pharmaceutical industry to evaluate medicine.
- In addition, the Code of Federal Regulations (CFR), 21 CFR Part 11, provides the compliance standards to which CDM systems must adhere.

A blurred background image of a business meeting. Several people in professional attire (suits, blouses) are gathered around a table. One person is holding a laptop, another is holding a pen, and there are coffee cups on the table. The scene is brightly lit, likely from a window in the background.

QUALITY ASSURANCE IN CLINICAL DATA MANAGEMENT

Ms. Sonali Correa

Ph.D, M.Sc.

QUALITY ASSURANCE

QA is a systematic and independent assessment of all the clinical trial activities and records.

It checks for following aspects:

Generation	G
recording	R
Analysis	A
reporting	R

of the clinical data are in accordance with the protocol, Standard Operating Procedures (SOPs) and Good Clinical Practices (GCPs).

- It is a measure of excellence
- Meeting customer expectations
- Superiority to competitors
- The role of the Quality Assurance (QA) group in this process is ensuring that the clinical data presented and interpreted in the clinical study report reflect a true picture of what took place in the trial.
- The presence of a QA group is perceived as of marginal importance, certainly in the view of senior management.



DEFINING THE TERMINOLOGY

Audit: A systematic and independent examination of trial related activities and documents to determine whether the evaluated trial related activities were conducted, and the 'clinical' data were recorded, analysed and accurately reported according to the protocol, sponsor's standard operating procedures (SOPs), GCP, and the applicable regulatory requirement(s)

Quality: the total set of characteristics of a product or service that affect its ability to satisfy a customer's stated or implied needs.

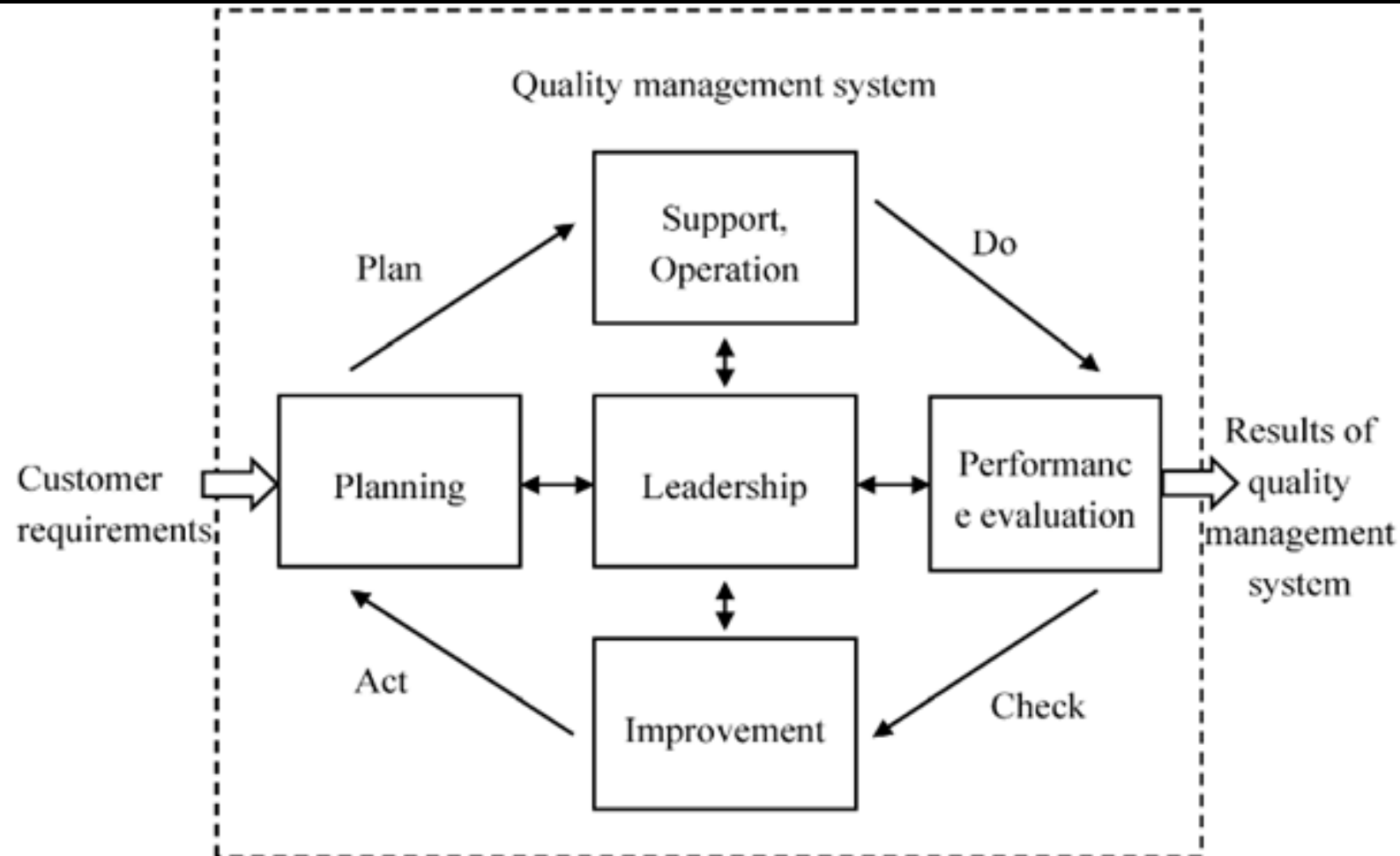
Quality system: the organizational structure, responsibilities, procedures, processes, and resources for implementing quality management.

Quality assurance (QA): the systematic and independent examination of all trial-related activities and documents. These audits determine whether the evaluated activities were appropriately conducted and that the data were generated, recorded, analyzed, and accurately reported according to protocol, standard operating procedures (SOPs), and good clinical practices (GCPs).

Quality control (QC): periodic operational checks within each functional department to verify that clinical data are generated, collected, handled, analyzed, and reported according to protocol, SOPs, and GCPs.



INTERNATIONAL STD OF QUALITY FRAMEWORK



GCP

Good Clinical Practice (GCP) is a set of internationally recognized ethical and scientific quality standard for conducting CT wrt to the following aspects

- Design
- Conduct
- Performance
- Monitoring
- Auditing
- Recording
- Analyses
- Reporting

ICH E6 (R2) GOOD CLINICAL PRACTICE

1.46 Quality Assurance (QA)

All those **planned and systematic actions** that are established to ensure that the trial is performed and the data are generated, documented (recorded), and reported in compliance with Good Clinical Practice (GCP) and the applicable regulatory requirement(s).

1.47 Quality Control (QC)

The **operational techniques and activities** undertaken within the quality assurance system to verify that the requirements for quality of the trial-related activities have been fulfilled

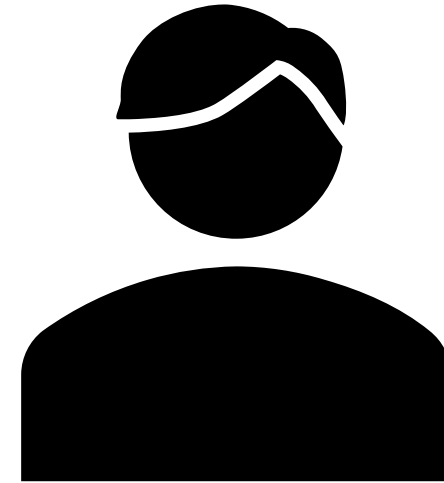


RESPONSIBILITIES OF A SPONSOR

The sponsor is responsible for ensuring that the clinical study is conducted in accordance with the protocol, GCP, and applicable regulatory requirements.

Specifically, the sponsor is responsible for:

Implementing a system to manage quality throughout all stages of the trial process.



5.0 Quality Management

The sponsor should **implement a system to manage quality** throughout all stages of the trial process.

Sponsors should focus on trial activities essential to ensuring human subject protection and the reliability of trial results.

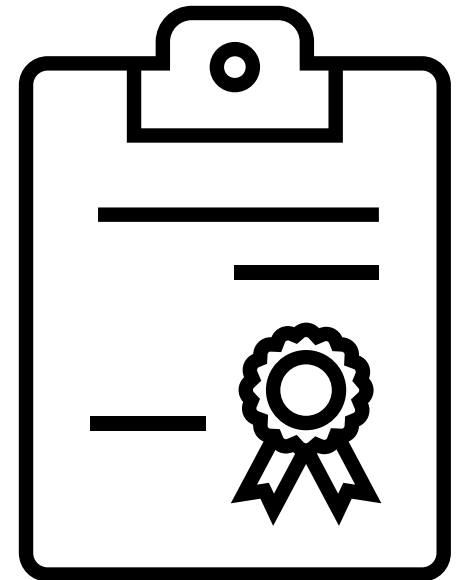
Quality management **includes the design of efficient clinical trial protocols and tools and procedures for data collection and processing, as well as the collection of information** that is essential to decision making.

5.1 Quality Assurance and Quality Control

5.1.1 The sponsor is responsible **for implementing and maintaining quality assurance and quality control systems with written SOPs to ensure that trials are conducted and data are generated, documented (recorded), and reported** in compliance with the protocol, GCP, and the applicable regulatory requirement(s).

The aim:

- Providing assurance of protection of rights, safety and well being of trial subjects and credibility of CT results
- Also applicable to CROs, Vendors or other delegated service providers
- However the sole responsibility lies with the **SPONSOR**



5.1.2 The sponsor is responsible for **securing agreement** from all involved parties to ensure direct access (see 1.21) to all trial related sites, source data/documents , and reports for the purpose of monitoring and auditing by the sponsor, and inspection by domestic and foreign regulatory authorities.

5.1.3 Quality control should **be applied to each stage of data handling** to ensure that all data are reliable and have been processed correctly.

5.1.4 Agreements, made by the sponsor with the investigator/institution and any other parties involved with the clinical trial, **should be in writing**, as part of the protocol or in a separate agreement

OBJECTIVES OF QA



QUALITY ASSURANCE

- This means in practical terms that clinical data management must have written procedures in place which will allow an independent group to audit against the actual processes taking place in the handling of clinical data.
- The documentation detailing what has happened during that trial should be present. The same group may have already reviewed the CRF, the protocol, and in some cases have trained the Investigator and his/her staff.
- The auditors may have visited the sites to audit the actual process of collecting clinical data by the Investigator and his/her staff.
- Independent QA should be built into the clinical data management system and carried out concurrently with other clinical data management activities.

QUALITY ASSURANCE GROUP

- QA auditing is the responsibility of an independent group which reviews clinical data at defined times to assure that procedures have been followed in accordance with approved quality procedures and that the quality of the clinical data is acceptable
- The benefits of such a group auditing early in the process are that ambiguities or inconsistencies can be identified, documented and, more importantly, acted upon before impacting too heavily on the final product
- QA group spent its limited resources auditing at investigator sites with occasional audits 'in-house' to review the final report
- QA were undertaking a QC role and may have been seen and used as a safety net, involved only at the end of the process when it was too late to make a constructive difference to a project
- The QA group should not hold up the clinical data flow; they should be auditing at intervals throughout the trial
- The QA group should not hold up the clinical data flow; they should be auditing at intervals throughout the trial, to avoid pitfalls and non compliance with GCP

AUDIT

- All key aspects of an audit are covered by this definition.
- All audits should be defined by an audit plan and the main scope should be recorded in the resulting audit report.
- Key questions will have been listed before the audit has commenced and the auditor should be encouraged to restrict his/her attention to the audit's scope
- too much rigidity can create the 'tick list' mentality which allows the CRF to match the clinical database listings but does not note that its storage is under a leaking roof; that is a lack of flexibility in the auditing process can mean that serious deficiencies are missed or fail to be addressed
- The European Network of GCP Auditors and other GCP Experts have published an Optional Guideline for GCP Compliance and Quality Systems Auditing which provides a basis to conduct any audit

QA ACTIVITIES

- The QA activities to be conducted during a specific clinical trial are included in a QA audit plan.
- These activities include the number of investigator sites, selection criteria, and vendors to be audited, such as labs and drug packaging and distribution providers.
- This plan also specifies what internal processes of the study will be audited from initial study design, site and data management, statistical analysis, and the final CSR.
- It specifies audit team members and auditees for each study stage, as well as the standards against which the audit will be conducted, such as the protocol, CRF completion guidelines, SOPs, ICH/GCP guidelines, and FDA regulations.

activities that QA helps define:

- requirements;
- design and prototyping;
- programming and code reviews;
- version control;
- continuous integration/continuous delivery (CI/CD);
- configuration management;
- test planning and execution;
- release management;
- deployment and integration;
- customer acceptance and feedback.

- Audits must also consider the standards of countries other than the United States, such as the recently adopted EU Clinical Trial Directives 2001/20/EC and 2005/28/EC
- A thorough QA audit plan also clearly states the documents to be provided by the auditee, as well as the location, date, and expected duration of the audits.
- Preparation for QA audits should include review of the approved protocol and amendments, SOPs (both general and study-specific), any specialized training associated with the study, annotated CRFs, and the statistical analysis plan
- Internal process audits are another important QA responsibility. Internal audits review all the drug development processes employed across several studies to determine if there are systemic problems.
- This includes a review of employee training, compliance with SOPs and regulatory requirements, and documented evidence that QC was appropriately conducted on the output of each internal process, as well as the final deliverable to a client.

QUALITY CONTROL

- This means that clinical data management must have documented evidence of what activities have been carried out throughout the trial to ensure the quality of the clinical data.
- QC tasks are the responsibility of the personnel handling the clinical data. In some cases, it is the actual group designated within clinical data management to conduct these tasks.
- It must be remembered that QA auditing is not QC and the responsibility for fully checking transcription, calculations, interpretation and reporting must remain with operational clinical monitors and clinical data management personnel—‘the experts’.
- The QA group is responsible for taking ‘snapshots’ of the study at critical times and places. The information obtained will allow data management to extrapolate findings forming a picture of the quality and integrity of the data on which to base decisions.

QUALITY PLANNING

The first step of quality management is planning. You need to take the time to identify your goals and what you want your baseline to be.

You should determine what your quality standards are, the requirements necessary to meet these standards, and what procedures will be used to check that these criteria are being met. In this planning stage, you will want to consider:

What your stakeholder's expectations and priorities are, if applicable

What your company's definition of success is

What legal standards or requirements are in place that must be abided by

Who will handle each role in the quality management process (supervision, testing, etc.)

How often processes will be evaluated for improvement

The quality plan describes how the quality control and quality assurance processes will be applied throughout the clinical trial.

It definitively defines the various quality-related tasks in the study. A quality plan documents specific quality practices, resources, and activities relevant to a specific project. This includes both operational QC and QA activities.

THE QUALITY CHALLENGE

The ongoing challenge in managing the quality of clinical data is to continually monitor data collection procedures and data management practices at every level of the study. This includes:

- ensuring that data generated during the study reflect what is specified in the protocol (case report form [CRF] vs. protocol)
- comparing data in the CRF and data collected in source documents for accuracy (CRF vs. source documents)
- ensuring that the data analyzed are the data recorded in the CRF (database vs. CRF).

Quality surveillance continues after the trial has ended and plays an important role in ensuring that:

- data presented in tables, listings, and graphs (TLGs) correctly match data in the database (TLGs vs. database)
- data reported in the clinical study report (CSR) are the data analyzed (CSR vs. TLGs)
- all aspects of the data management processes are compliant with SOPs and GCPs.

OPERATIONAL QC

It is critical that trial managers develop a QC plan for each key operational stage of the study that defines standards against which QC will be conducted, including:

- sampling plan to be used (if applicable)
- data source to be used for QC at each operational stage
- metrics to be documented
- acceptable quality levels
- appropriate methods to report and distribute results.

During the study design phase, QC personnel provide an independent review of the approved proposed protocol.

The QC plan includes comparison of the study's CRF to the objectives set forth in the protocol to ensure that it is designed to collect all necessary data.

A requirement to review CRF completion guidelines is also an element of the QC plan.

QC involves a number of tasks, including:

- **identifying product errors or blemishes;**
- **testing against quality requirements set forth during planning;**
- **ensuring conformance with regulations or industry standards;**
- **creating and maintaining test reports;**
- **documenting bugs that either must be fixed before product release or can wait until the next iteration;**
- **determining areas where quality or the user experience could be improved.**

For overall site management, a complete QC plan addresses the following:

- investigator selection and qualifications
 - experience in conducting clinical trials
 - experience with the specific indication
 - not on the FDA's restricted or debarred lists
 - adequate staff and facilities
 - personal involvement
- source document verification
 - medical records
 - lab data
 - progress notes
 - diagnostic tests
- compliance with regulations
 - 21 CFR 11, 50, 54, 56, 312
 - EU Clinical Trial Directives 2001/20/EC and 2005/28/EC
 - ICH/GCP Consolidation Guidelines (ICH-EG).
- study conduct (monitoring)
 - subject informed (signed informed consent form)
 - subject's eligibility (inclusion/exclusion)
 - protocol compliance
 - adverse events (AEs) and concomitant medication
 - drug accountability and storage
- query resolution
 - completed data clarification forms
- During the data management process, the accuracy of the initial data entry is verified by an independent entry of the same data and a subsequent comparison of both sets of data for nonagreement.
- The reality of the data is checked with a preprogrammed logic check program and a subsequent manual review.
- The database entries are then QC'd versus the CRFs.
- The TLGs that are generated as part of a statistical analysis of the data are also inspected to ensure their accuracy, as is any text in a CSR that refers to the TLGs.

SITE MANAGEMENT METRICS

Internal audits of the site selection and management processes ensure that qualified investigators are selected, that they have adequate facilities and adequately trained staff, and that the study was conducted in compliance with the protocol and all appropriate regulations.

Several metrics commonly evaluated by internal process audits after the study has begun include:

- percentage of monitoring visits completed on time
- percentage of evaluable subjects (no protocol violations)
- percentage of serious adverse events (SAEs) reported within 24 hours to an Institutional Review Board (IRB) and sponsor
- percentage of properly executed informed consent forms
- number of queries/CRF pages reviewed
- number of missing data entries/CRF pages reviewed.

COMPUTER SYSTEMS VALIDATION

Computer systems validation examines all aspects of the data handling computer systems (hardware and software) to ensure the accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records.

This includes initial installation and procedures that document how changes to a computer system are justified, approved, and implemented.

The validation process begins with examining user requirements, the results of the initial hardware installation qualification (IQ) tests, the operational qualification (OQ) tests, and the qualification and training of user personnel.

The user acceptance test results (Performance Qualification) are then compared to the user requirements to ensure that these requirements are met. Having assurance that the data handling computer system is validated, data can then be entered.

DATA MANAGEMENT QC

Since an average error rate for keying text or numbers is about 1 per 300 keystrokes, the entered data is QC'd by having an independent data entry person enter the same data.

Both sets of data are compared electronically, and discrepancies are resolved by a senior data entry person.

After all of the data has been entered and all discrepancies and questions resolved, the database is QC'd by comparing the database to the CRFs from which the data was entered.

DATA MANAGEMENT METRICS

Examples of data management metrics for QA are:

- percentage of database errors
- percentage of queries manually generated
- time from last patient out to database lock
- number of times a locked database is opened.

DATA MANAGEMENT QA

Data entry and the database QC process are other critical areas of the data management process that are audited by QA personnel.

The audits review the documented evidence that shows the data accuracy and integrity were verified and checked manually, independently, and programmatically to ensure the data were logical.

These audits also ensure that all data queries are resolved and that the overall database QC review was conducted according to the QC SOP.

STATISTICAL ANALYSIS QC

After a study database has undergone a QC review, it is exported into a **SAS** (statistical analysis system) to develop analytical programs that create data **TLGs** that are to be included in a **CSR**.

The **TLGs** are QC'd and validated by having independent programmers create programs for the same **TLGs**, and all discrepancies are then resolved.

STATISTICAL ANALYSIS QA

QA of the statistical analysis process ensures SAS programs are validated for the generation of all TLGs by checking that all the requirements were met and boundary conditions were tested.

QA also verifies that the SAP was developed according to the processes defined in the SOPs and that all statistical analysis plans are approved by the appropriate authority.

In addition to reviewing the statistical analysis process, QA also inspects a predetermined sample of TLGs.

Numbers are checked against database listings, and tables are reviewed against format requirements specified in the SAP.

The QA report will document the following information:

percentage of TLGs with numerical or formatting errors

percentage of SAS programs adequately validated

time from database lock to final TLGs.

STUDY SITE AUDITS

The QA group conducts site audits throughout the course of a trial to assess protocol and regulatory compliance, to ensure that the safety and welfare of subjects are addressed, and to confirm that problems reported by study monitors have been resolved.

QA's criteria for site selection include:

- high patient enrollment
- high staff turnover
- abnormal number of AEs (high and low)
- high or low subject enrollment rates that are unexpected given the research site's location and demographics.

Site audits ensure adequate documentation of case histories (source documents), such as medical records, progress notes, hospital charts, drug accountability records, ECGs, laboratory test results, SAEs, and informed consents.

Audits examine whether all clinical tests were performed at the time specified in the study protocol, and review specimen collection, storage and shipping packages (if applicable), and the timeliness of review of clinical test results.

QA site audits evaluate the timeliness of entering data into a CRF, and examine the accuracy of the data by comparing them to their respective source documents mentioned above.

Audits also ensure all investigational product received by a site is adequately accounted for.

QbD

What is Quality by Design (QbD)?

Quality by Design (QbD) is a strategic process for development and manufacturing.

It is meant to ensure that the intended performance of a final drug product is as expected – both in terms of purity and efficacy.

To achieve this requires well-described objectives, and proper risk management.

Quality by Design includes a thorough understanding of the technical process, the variables that affect the process and the performance envelope in which those variables remain in order to achieve the objectives.

The advantages of QbD benefit both the consumer (since the product will be consistently safe and effective) and the manufacturer (since quality and costs will be well understood, controlled and predictable).

RISK-BASED QUALITY MANAGEMENT SERVICES (RBQM)

RBQM approach helps clients identify, capture, and manage data quality issues at the earliest stages of emergence before potential risks escalate into liabilities.

This risk-based monitoring approach forms the basis of a fit-for-purpose monitoring strategy, using study risk assessments, complexity, country requirements, site monitoring and capabilities as key considerations.

A continuum of models includes:

Central monitoring

Remote monitoring variations

Traditional onsite monitoring



Quality-by-design (study design and planning)

Optimal study and monitoring strategy based on critical data, critical processes, and risks.

Risk-based study execution

Quality execution of centralized/remote/onsite monitoring; continual risk evaluation, adaptation and reporting

ADVANCED ANALYTICS

AI techniques complement clinical development and therapeutic knowledge to offer an unprecedented level of sensitivity when characterizing trends of importance or potential compliance issues.

Preventable quality issues can be avoided with agile decision-making based on real-time, centralized data

Known and unknown risks can be rapidly detected on both critical and non-critical data, reducing 'noise' by filtering out less relevant data

Monitoring resources can be prioritized on data insights rather than the traditional 100% Source Data Verification (SDV) approach

Scores of well-established benefits related to risk based quality management adoption include:

4x lower error rate in critical data in a head-to-head comparison of RBQM to traditional 100% SDV approach¹

Up to 40% faster database lock timelines compared to non-RBQM trials

20% reduction in SDV

45% reduction in the number of missing pages in RBQM trials vs non-RBQM trials

QUALITY TOLERANCE LIMITS

A QTL is a level, threshold, or value associated with a parameter which is critical to quality.

QTLs are set for risks identified at the trial level.

A deviation from a threshold during the conduct of the trial may indicate a systematic issue that could impact participants' safety or reliability of trial results.

QTLs act as controls for risk and are part of the Clinical Trial Quality Risk Management process.

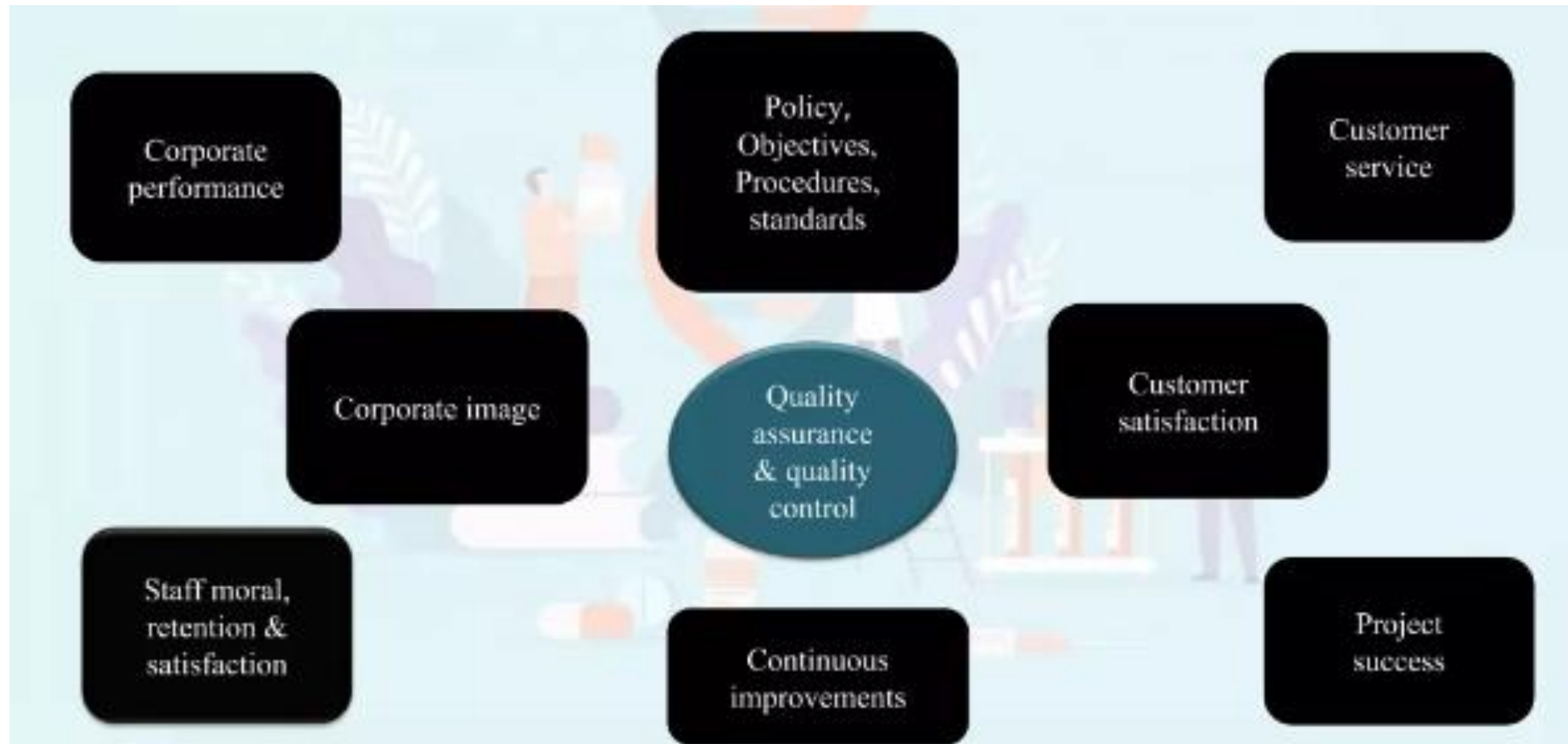
This QTL Framework includes a process for defining, monitoring, and reporting QTLs, which corresponds to the set-up, conduct, and closeout phases of a clinical trial



DIFFERENCE

	QC	QA
Focus	Product quality at a given moment	Project process
Character	Reactive (defects the issues when there're already in)	Preventive (blocks the issues by detecting vulnerabilities)
Starting point	Require Presentment gathering stage in each iteration	Project planning stage (most efficient) Any point in the SDLC when the project is in trouble
Tools and measures	Various types of testing Testing metrics Test reports	Quality metrics Reviews audits

IMPORTANCE



CASE STUDY

CS 1: ASCEND-HF, STUDY DESIGN

- RCT, Double Blind, in acute Heart Failure (HF) patients
- 3500 Nesiritide, 3500 Placebo
- Randomize <48 hours from hospitalization, <24 hours from IV Rx for HF
- Co-Primary Endpoint Dyspnea index at 6/24hrs
- Co-Primary Endpoint 30-day Death/HF rehospitalization and secondary endpoints
- All-cause Mortality at 180 days

CS 1: ASCEND-HF, BUILDING & DEFINING QUALITY

○ Building

- Building Establish principles for quality operations/data before trial begins
- Integrate principles throughout trial operations
- Communicate expectations to sites and trial teams
- Implement surveillance plans and provide feedback
- Adhere to pragmatic principles
 - Efficient, effective (& hopefully economical)

○ Defining

- Enrolled the right participants according to the protocol with adequate consent?
- Did participants receive the assigned treatment and did they stay on the treatment?
- Was there complete ascertainment of primary and secondary efficacy data?
- Was there complete ascertainment of primary and secondary safety data?
- Were there any *major GCP related issues*?

Is the primary efficacy and safety data complete + correct?



- Establish aggressive targets/alarms for data clean
 - >90% clean for baseline, 30 day, 180 day forms
 - No late visits, open/answered queries, missing items > 30 days
- Weekly review of data status and radar reports
 - Sites out of compliance are contacted
- Focused monitoring on at least 50% of re-hospitalized patients
 - Since the primary endpoint includes re-hospitalization for HF, monitoring was weighted to include a high percentage of patients with potential endpoints.

Is the primary efficacy and safety data complete + correct?



- Monthly review of operations reports aggregate event and safety data with trial leadership/EC/SC
 - Identified lower than expected incidence of use of ACE/ARB, beta blockers and other guideline based meds for ADHF
 - Con-med action plan implemented involving additional database queries to ensure sites were capturing data appropriately as well as additional site education
 - Identified lower than expected event rate in central Europe region and at the recommendation of the Executive Committee stopped enrollment in this region

Is the primary safety data complete + correct?



- DSMC charter designed to empower committee to use resources (i.e. blinded statistician/independent group preparing data) to assess safety
- Frequent DSMC safety reviews (approx every 1000 enrolled)
 - Committee conducted 6 reviews of the data and recommended continuation of trial at each review

Were there any *major* GCP related issues?



What may be "*major*"

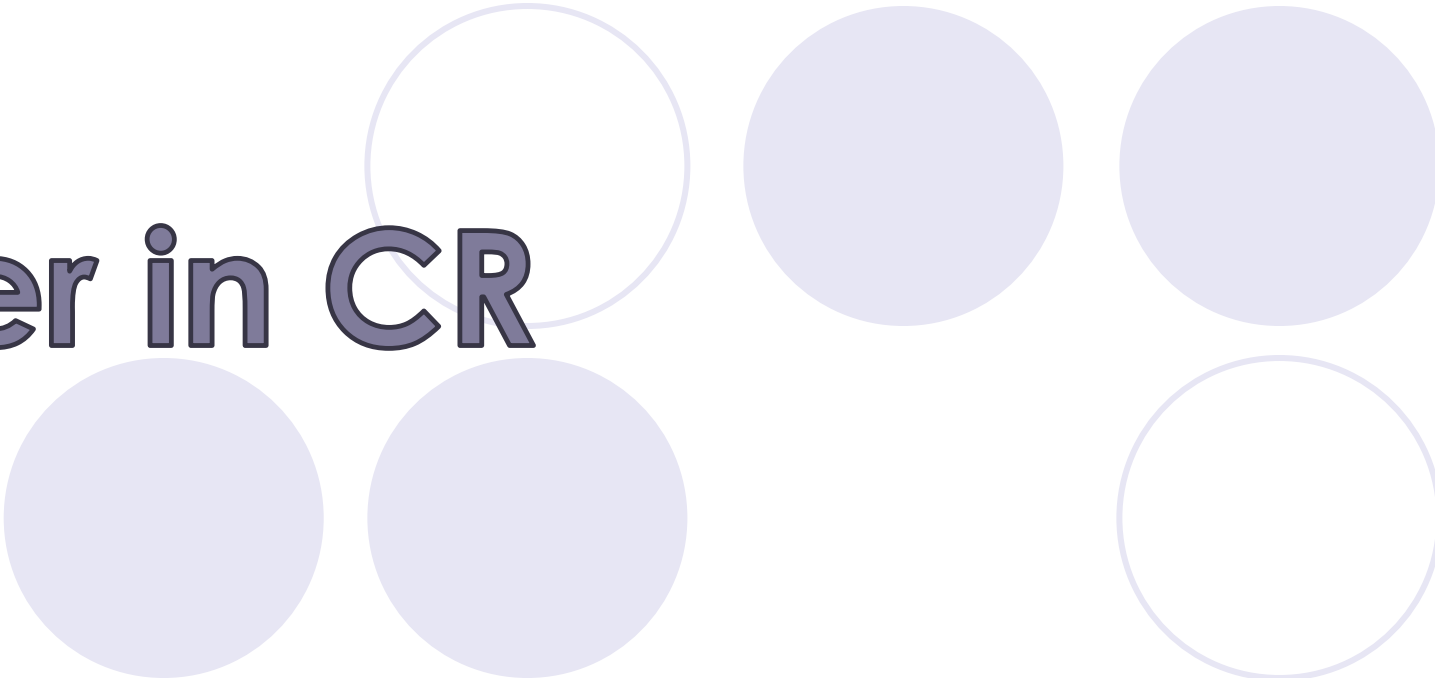
- »Indications that source may not belong to participant
- »Participants treated without obtaining consent at all

What probably isn't "*major*"

- »Missing initials on ICF page
- »Follow up visits that fall outside protocol window

No Major issues identified

Career in CR

A decorative graphic consisting of five circles arranged in a cross pattern. The top and bottom circles are white with a light purple outline. The left, right, and center circles are solid light purple.

Why Clinical Research As A Career

- ◉ The clinical research industry is a highly lucrative and expanding field. The global clinical trials market has been estimated at \$46.8 billion in 2019.
- ◉ As the push for new vaccines and therapeutics continues get stronger, the field is expected to grow even more in value. Experts predict that the global market will hit \$69.9 billion by 2027. The future in clinical research is bright, and it is one that you will want to be a part of.

Working In Clinical Research

- ◉ **1.You like to a job that's flexible**

- ◉ Don't like working in a cubical? How about heading to the airport every morning instead? If you like a job that keeps you moving, then becoming a Clinical Research Associate (CRA) and working in clinical trials might be the right move for you.
- ◉ CRAs, contrary to what most people believe, don't collect data or interact with patients. A CRA's day to day job is to travel between different research sites and verify data transcription (i.e data management). CRAs can also become part of the project management as a clinical trial manager of an entire trial.
- ◉ They are also called "monitors" and a part of "regulatory affairs", because it is also their job and clinical experience to ensure that every site is following proper compliance and protocols.
- ◉ There are two types of CRAs: home base or in-house. Home base CRAs work remotely. That means they work and travel from home. Most CRAs work for contract research organizations who are hired by sponsors to conduct their multi-site trials.
- ◉ If you get tired of working home base, you can become an in-house CRA. In-house CRAs stay in one site and work together with a home base CRA to keep each other updated with what is happening at their site.

Working In Clinical Research

◉ 2. You like working with people

- ◉ Have you ever been told that you are a people person with great communication skills? If talking to someone about how you can change their health for the better sounds like something you'd enjoy, you should definitely look into becoming a Clinical Research Coordinator (CRC).
- ◉ CRCs are the backbones to every project. They conduct patient visits, input source documents into the electronic data capture (EDC), and ensure that every trial is following compliance. They are also responsible for handling regulatory documents and updating the Principal Investigator (PI) with trial results.
- ◉ CRCs conduct a variety of tasks, all of which impact the progress and development of the trial. Every successful clinical trials team needs is a good CRC. So, if you have strong interpersonal skills and know how to stay organized, you will be an indispensable part of the team.

Working In Clinical Research

◉ 3. You are detail-oriented and tech-savvy

- ◉ Are you a self-proclaimed techie? Perhaps you've dabbled in coding, pick up computer programs easily, and maybe even have a background in IT. Technology is the future. If you think you have a knack for organizing data, you should look into becoming a Clinical Trial Assistant (CTA).
- ◉ CTAs, also known as Clinical Research Assistants, manage the Trial Master File (TMF). They file, archive, and maintain trial documents and study files. They are also responsible for closing inquiries from the CRA, as well as providing administrative support to the research team. Every important step in clinical research, pre-clinical research, study startup, site management, needs a dependable CTA.
- ◉ While most jobs in clinical research require some understanding of technology, it is especially important for the CTA to know what they are doing when it comes to managing trial documents and study files. In addition, it is equally important that the CTA is organized and knows how to pay attention to detail.

Working In Clinical Research

- ◉ **4.Means you like a good salary...with room for promotion**
- ◉ Though there are many career paths within clinical research, most people begin their career as CTAs or CRCs in entry level positions. Depending on your skill set and what kind of experiences you can bring to the table, either position will help you get your feet through the door.
- ◉ According to salary.com, a CTA's average salary in 2020 is \$63,000. They generally earn between a range of \$54,300 and \$73,000, and are provided with benefits such as healthcare and social security.
- ◉ If being more hands-on in the research process appeals to you, you might be a good fit for a CRC. Similarly, CRCs are making an average of \$63,117 in 2020. Most make between a range of \$54,210 and \$72,902, plus employee benefits.
- ◉ While numbers tend to fluctuate between cities and states, there's no denying that these are great salaries for an entry position. Since according to one of the largest global job recruitment sites, Glassdoor, the average base salary in America is \$40,000. Starting with an annual salary of \$60,000 is considered uncommon and on the high end of the spectrum.
- ◉ After one to two years of experience on the job, most companies provide CTAs and CRCs opportunities of professional development and promotion. Many become Clinical Research Associates, or CRAs.
- ◉ Indeed reports that the average salary of a CRA with one to two years of experience is \$72,358. After building at least 6 years experience, a seasoned CRA should expect to earn \$110,102 a year. If you would like to make more money, you can consider becoming an individual contractor CRAs. They can earn up to \$300,000 in a year.

Indian perspective



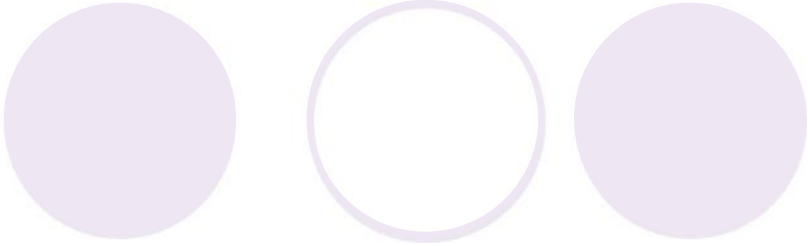
- ◉ CRA salary in India ranges between ₹ 1.0 Lakhs to ₹ 6.0 Lakhs with an average annual salary of ₹ 2.4 Lakhs. Salary estimates are based on 241 latest salaries received from CRAs.
- ◉ Clinical Research Coordinator salary ranges between ₹ 1.2 Lakhs to ₹ 4.0 Lakhs with an average annual salary of ₹ 2.5 Lakhs. Salary estimates are based on 329 latest salaries received from Clinical Research Coordinators.
- ◉ Clinical Data Manager salary ranges between ₹ 2.5 Lakhs to ₹ 8.0 Lakhs with an average annual salary of ₹ 4.6 Lakhs.

Why Do You Want To Work In Clinical Research?

- ◉ 5. You are a science professional wanting to change careers and don't want to go back to school
- ◉ In clinical research, experience is often valued over degrees. Rather what you didn't study in college, hiring managers are more interested in what you have done in the past and how they can integrate you into their company.
- ◉ This includes getting certified through clinical research courses, but more so what you learn from the courses you take. CCRPS offers the most in-depth CRA and CRC training so that there's tons to talk about during the interview and a working bank of knowledge during the first few months of the job.
- ◉ While graduate programs can help point you in the right direction, you don't need a master's degree to succeed in clinical research. In fact, certain positions don't even require a bachelor's or associate's degree; they have certification in clinical research.
- ◉ Applying to CRC and CTA positions are one of the most common segways into higher positions in clinical research. CRCs don't need a bachelor's or associate's degree to get their foot in the door. While both CTA and CRA positions require a bachelor's degree, they don't have to be in the life sciences.

Career Guide

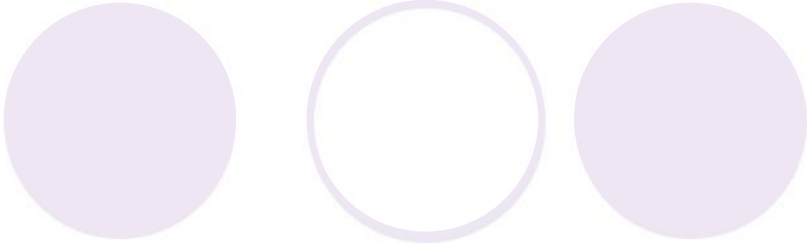
- ◉ Clinical Research Associate
- ◉ Clinical Research Coordinator
- ◉ Drug Safety Monitor
- ◉ Clinical Trial Assistant
- ◉ Clinical Research Nurse
- ◉ Medical Monitor
- ◉ Principal Investigator
- ◉ ICH GCP



<https://ccrps.org/clinical-research-blog/tag/hierarchy+in+clinical+research+organization>

Certification

- ◉ Clinical Research Associate
- ◉ Clinical Research Coordinator
- ◉ Clinical Trial Assistant
- ◉ Pharmacovigilance
- ◉ ICH GCP
- ◉ Medical Monitor
- ◉ Principal Investigator



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